

**Department of Information and Communication Technology**  
Faculty of Engineering and Technology  
Islamic University, Kushtia-7003, Bangladesh  
Curriculum for Integrated B.Sc. Engg. Degree for 2021-2022 Session

### **Vision of the University**

To provide for instructions in theology and other fields of Islamic Studies and comparative jurisprudence and other branches of study at graduate and post-graduate level as the University may think fit, and prepare for the advancement and dissemination of knowledge including post-doctoral research and training.

### **Mission of the University**

#### **Quality Policy**

### **Vision of the Department**

The Department of Information and Communication Technology (ICT) aims to discover engineering knowledge and novel information technologies, thereby improving technology-based graduates skills to build digital Bangladesh.

### **Mission of the Department**

The mission of the ICT Department is to provide competence-based graduates and carry out research in science and technology for socio-economic development. The mission of the department is:

|           |                                                                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M1</b> | To produce a core of skilled human beings and long-term IT professionals through high-quality IT education combined with other engineering fields, committed to serving the country and also the domestic or overseas IT industries, with ethics. |
| <b>M2</b> | To introduce new knowledge, new technologies and innovative applications such as the Internet of Things, data science, block chain, artificial intelligent etc., to cover the concept of the fourth industrial revolution.                        |
| <b>M3</b> | To provide sufficient facilities/guidelines and environment for graduates to be highly interested in advanced knowledge and research.                                                                                                             |

### **Program Education Objectives (PEO)**

The department of ICT at IU has adopted the educational objectives are:

|             |                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PEO1</b> | Graduates will be able to understand and apply basic science, physical science, mathematics, and statistics to solve the difficult problems in ICT fields.                                                                 |
| <b>PEO2</b> | Graduates will be able to exploit their theoretical and practical knowledge in order to identify appropriate design specifications and solutions for the systems, components and/or processes that can meet IT needs.      |
| <b>PEO3</b> | Graduates will be able to develop sustainable technologies and computing applications to provide a better place for mankind, and they will take on ethical responsibilities related to the use of the invented technology. |

### **PEO to Mission Statement Mapping**

| <b>Mission Statement</b> | <b>PEO1</b> | <b>PEO2</b> | <b>PEO3</b> |
|--------------------------|-------------|-------------|-------------|
| <b>M1</b>                | High        | High        | Medium      |
| <b>M2</b>                | Low         | High        | High        |
| <b>M3</b>                | Low         | Medium      | High        |

**Correlation:** 3-High, 2-Medium, 1-Low

**Program Learning Outcomes (PLO)**

|              |                                                                                                                                                                                                                                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PLO1</b>  | <b>Engineering knowledge:</b> Apply basic knowledge of mathematics, science and engineering to solve complex problems in information technology.                                                                                                                                                                       |
| <b>PLO2</b>  | <b>Problem analysis:</b> Identify, analyze and formulate IT-based complex engineering problems                                                                                                                                                                                                                         |
| <b>PLO3</b>  | <b>Design/Development of solutions:</b> Design new systems, components, tools or methods to solve IT-based complex engineering problems that meet specific technical requirements                                                                                                                                      |
| <b>PLO4</b>  | <b>Investigation:</b> Consider experiment design, data analysis, and interpretation to investigate complex IT problems.                                                                                                                                                                                                |
| <b>PLO5</b>  | <b>Modern tool usage:</b> Create, select and apply appropriate techniques, resources and modern computer engineering and ICT tools.                                                                                                                                                                                    |
| <b>PLO6</b>  | <b>The engineer and society:</b> Apply reasoning informed by contextual knowledge to develop the IT for society benefits.                                                                                                                                                                                              |
| <b>PLO7</b>  | <b>Environment and sustainability:</b> Understand the feasibility study of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge for sustainable development.                                                                                                        |
| <b>PLO8</b>  | <b>Ethics:</b> Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice                                                                                                                                                                              |
| <b>PLO9</b>  | <b>Individual work and teamwork:</b> Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.                                                                                                                                                               |
| <b>PLO10</b> | <b>Communication:</b> Communicate effectively about complex engineering activities with the engineering communities and with society at large scale. More particularly, be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions. |
| <b>PLO11</b> | <b>Project management and finance:</b> Demonstrate knowledge and understanding of IT based engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.                                                                      |
| <b>PLO12</b> | <b>Life-long learning:</b> Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.                                                                                                                                   |

**Mapping of PEOs to PLOs**

| PEO/PLO     | PLO1 | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 | PLO7 | PLO8 | PLO9 | PLO10 | PLO11 | PLO12 |
|-------------|------|------|------|------|------|------|------|------|------|-------|-------|-------|
| <b>PEO1</b> | 3    | 3    | 2    | 1    | 1    | 1    | 1    | 2    | 2    | 1     | 1     | 2     |
| <b>PEO2</b> | 2    | 3    | 3    | 3    | 3    | 3    | 1    | 1    | 1    | 1     | 2     | 2     |
| <b>PEO3</b> | 2    | 3    | 2    | 3    | 3    | 2    | 3    | 3    | 3    | 3     | 3     | 3     |

**Correlation:** 3- Strong, 2-Medium, 1-Low

## Curriculum for Integrated B.Sc. Engg. Degree

## Overall Marks and Credits distribution for B.Sc. Engg. in ICT (Session: 2021-22)

| Course Type              | Course Name                           | Marks | Total Marks | Marks (%)  | Credits    |
|--------------------------|---------------------------------------|-------|-------------|------------|------------|
| <b>Humanities</b>        | Sociology                             | 100   | 700         | 9.33       | 11.0       |
|                          | Economics                             | 100   |             |            |            |
|                          | Communicative English                 | 200   |             |            |            |
|                          | Industrial Management and Accountancy | 100   |             |            |            |
|                          | Ethics and Cyber Law                  | 100   |             |            |            |
|                          | Non-credit course                     | 100   |             |            |            |
| <b>Basic Science</b>     | Physics                               | 100   | 500         | 6.67       | 14.0       |
|                          | Mathematics                           | 300   |             |            |            |
|                          | Statistics                            | 100   |             |            |            |
| <b>Major Engineering</b> | Theoretical                           | 3000  | 6300        | 84         | 87.0       |
|                          | Laboratory                            | 2700  |             |            | 39.0       |
|                          | Project                               | 100   |             |            | 3          |
|                          | Field work                            | 100   |             |            | 2          |
|                          | Viva-voce                             | 400   |             |            | 4          |
| <b>Total</b>             |                                       |       | <b>7500</b> | <b>100</b> | <b>160</b> |

## Courses in Brief

Session: 2021-2022

### 1<sup>st</sup> Year 1<sup>st</sup> Semester-2022

| Course Code  | Course Title                                                     | Marks      | Credits     | Contact hours/week |
|--------------|------------------------------------------------------------------|------------|-------------|--------------------|
| PHY-1101     | Physics                                                          | 100        | 3           | 3                  |
| ICT-1101     | Discrete Mathematics                                             | 100        | 3           | 3                  |
| ICT-1102     | Basic ICT Laboratory                                             | 100        | 2           | 4                  |
| EEE-1101     | Electricity and Electromagnetism                                 | 100        | 3           | 3                  |
| EEE-1102     | Electricity and Electromagnetism Laboratory                      | 100        | 1.5         | 3                  |
| MATH-1101    | Calculus and Differential Equations                              | 100        | 3           | 3                  |
| HUM-1101     | Communicative English                                            | 100        | 2           | 2                  |
| HUM-1102     | Communicative English Laboratory                                 | 100        | 1           | 2                  |
| NCC-01       | Bangladesh Studies (for Alim Background and Non-Muslim Students) | 100        | 0           | 2                  |
| NCC-02       | Islamic Studies (for Muslim & HSC background students)           | 100        | 0           | 2                  |
| ICT-1000     | Viva-voce                                                        | 0          | 0           | 0                  |
| <b>Total</b> |                                                                  | <b>900</b> | <b>18.5</b> | <b>27</b>          |

### 1<sup>st</sup> Year 2<sup>nd</sup> Semester-2022

| Course Code  | Course Title                               | Marks      | Credits     | Contact hours/week |
|--------------|--------------------------------------------|------------|-------------|--------------------|
| ICT-1201     | Electronic Devices and Circuits            | 100        | 3           | 3                  |
| ICT-1202     | Electronic Devices and Circuits Laboratory | 100        | 1.5         | 3                  |
| ICT-1203     | Structured Programming Language            | 100        | 3           | 3                  |
| ICT-1204     | Structured Programming Language Laboratory | 100        | 1.5         | 3                  |
| ICT-1205     | Numerical Methods and Analysis             | 100        | 2           | 3                  |
| ICT-1206     | Numerical Methods and Analysis Laboratory  | 100        | 1.5         | 3                  |
| MATH-1201    | Geometry and Complex Variables             | 100        | 3           | 3                  |
| HUM-1201     | Sociology                                  | 100        | 2           | 2                  |
| ICT-1000     | Viva-voce                                  | 100        | 1           | 2                  |
| <b>Total</b> |                                            | <b>900</b> | <b>18.5</b> | <b>25</b>          |

### 2<sup>nd</sup> Year 1<sup>st</sup> Semester-2023

| Course Code | Course Title        | Marks | Credits | Contact hours/week |
|-------------|---------------------|-------|---------|--------------------|
| ICT-2101    | Digital Electronics | 100   | 3       | 3                  |

|              |                                          |            |           |           |
|--------------|------------------------------------------|------------|-----------|-----------|
| ICT-2102     | Digital Electronics and DLD Laboratory   | 100        | 1.5       | 3         |
| ICT-2103     | Data Structure and Algorithm             | 100        | 3         | 3         |
| ICT-2104     | Data Structure and Algorithm Laboratory  | 100        | 1.5       | 3         |
| ICT-2105     | Object Oriented Programming I            | 100        | 3         | 3         |
| ICT-2106     | Object Oriented Programming Laboratory I | 100        | 1.5       | 3         |
| ICT-2107     | Signals and Systems                      | 100        | 3         | 3         |
| ICT-2108     | Signals and Systems Laboratory           | 100        | 1.5       | 3         |
| MATH-2101    | Vector Analysis and Linear Algebra       | 100        | 3         | 3         |
| ICT-2000     | Viva-voce                                | 0          | 0         | 0         |
| <b>Total</b> |                                          | <b>900</b> | <b>21</b> | <b>27</b> |

**2<sup>nd</sup> Year 2<sup>nd</sup> Semester-2023**

| Course Code  | Course Title                              | Marks       | Credits     | Contact hours/week |
|--------------|-------------------------------------------|-------------|-------------|--------------------|
| ICT-2201     | Database Management Systems               | 100         | 3           | 3                  |
| ICT-2202     | Database Management Systems Laboratory    | 100         | 1.5         | 3                  |
| ICT-2203     | Web Technology                            | 100         | 3           | 3                  |
| ICT-2204     | Web Technology Laboratory                 | 100         | 1.5         | 3                  |
| ICT-2205     | Communication Systems                     | 100         | 3           | 3                  |
| ICT-2206     | Communication Systems Laboratory          | 100         | 1.5         | 3                  |
| STAT-2201    | Statistics for Communication Engineering  | 100         | 2           | 2                  |
| ICT-2207     | Object Oriented Programming II            | 100         | 2           | 3                  |
| ICT-2208     | Object Oriented Programming Laboratory II | 100         | 1           | 3                  |
| HUM-2201     | Economics                                 | 100         | 2           | 2                  |
| ICT-2000     | Viva-voce                                 | 100         | 1           | 2                  |
| <b>Total</b> |                                           | <b>1100</b> | <b>21.5</b> | <b>30</b>          |

**3<sup>rd</sup> Year 1<sup>st</sup> Semester-2024**

| Course Code  | Course Title                            | Marks      | Credits   | Contact hours/week |
|--------------|-----------------------------------------|------------|-----------|--------------------|
| ICT-3101     | Wave Propagation and Antenna            | 100        | 3         | 3                  |
| ICT-3102     | Wave Propagation and Antenna Laboratory | 100        | 1.5       | 3                  |
| ICT-3103     | Microwave Engineering                   | 100        | 3         | 3                  |
| ICT-3104     | Microwave Engineering Laboratory        | 100        | 1.5       | 3                  |
| ICT-3105     | Data Communication                      | 100        | 3         | 3                  |
| ICT-3106     | Data Communication Laboratory           | 100        | 1.5       | 3                  |
| ICT-3107     | Computer Networking                     | 100        | 3         | 3                  |
| ICT-3108     | Computer Networking Laboratory          | 100        | 1.5       | 3                  |
| ICT-3000     | Viva-voce                               | 0          | 0         | 0                  |
| <b>Total</b> |                                         | <b>800</b> | <b>18</b> | <b>24</b>          |

**3<sup>rd</sup> Year 2<sup>nd</sup> Semester-2024**

| Course Code | Course Title                       | Marks | Credits | Contact hours/week |
|-------------|------------------------------------|-------|---------|--------------------|
| ICT-3201    | Microprocessor and Microcontroller | 100   | 3       | 3                  |

|              |                                               |            |             |           |
|--------------|-----------------------------------------------|------------|-------------|-----------|
| ICT-3202     | Microprocessor and Microcontroller Laboratory | 100        | 1.5         | 3         |
| ICT-3203     | Optical Communications                        | 100        | 3           | 3         |
| ICT-3204     | Optical Communications Laboratory             | 100        | 1.5         | 3         |
| ICT-3205     | Computer Architecture and Organization        | 100        | 3           | 3         |
| ICT-3207     | Operating Systems                             | 100        | 3           | 3         |
| ICT-3208     | Operating Systems Laboratory                  | 100        | 1.5         | 3         |
| HUM-3201     | Ethics and Cyber Law                          | 100        | 2           | 3         |
| ICT-3000     | Viva-voce                                     | 100        | 1           | 2         |
| <b>Total</b> |                                               | <b>900</b> | <b>19.5</b> | <b>26</b> |

#### 4<sup>th</sup> Year 1<sup>st</sup> Semester-2025

| Course Code  | Course Title                                        | Marks       | Credits   | Contact hours/week |
|--------------|-----------------------------------------------------|-------------|-----------|--------------------|
| ICT-4101     | Information Theory and Coding                       | 100         | 3         | 3                  |
| ICT-4102     | Information Theory and Coding Laboratory            | 100         | 1.5       | 3                  |
| ICT-4103     | Digital Image Processing                            | 100         | 3         | 3                  |
| ICT-4104     | Digital Image Processing Laboratory                 | 100         | 1.5       | 3                  |
| ICT-4105     | System Analysis and Software Engineering            | 100         | 3         | 3                  |
| ICT-4106     | System Analysis and Software Engineering Laboratory | 100         | 1         | 3                  |
| ICT-4107     | Computer Graphics and Visualizations                | 100         | 3         | 3                  |
| ICT-4108     | Computer Graphics and Visualizations Laboratory     | 100         | 1         | 3                  |
| ICT-4109     | Cryptography and Network Security                   | 100         | 3         | 3                  |
| ICT-4110     | Cryptography and Network Security Laboratory        | 100         | 1         | 3                  |
| ICT-4002     | Project                                             | 0           | 0         | 0                  |
| ICT-4000     | Viva-Voce                                           | 0           | 0         | 0                  |
| <b>Total</b> |                                                     | <b>1000</b> | <b>21</b> | <b>30</b>          |

#### 4<sup>th</sup> Year 2<sup>nd</sup> Semester-2025

| Course Code | Course Title                                            | Marks | Credits | Contact hours/week |
|-------------|---------------------------------------------------------|-------|---------|--------------------|
| ICT-4201    | Artificial Intelligence and Machine Learning            | 100   | 3       | 3                  |
| ICT-4202    | Artificial Intelligence and Machine Learning Laboratory | 100   | 1.5     | 3                  |
| ICT-4203    | Wireless Communications                                 | 100   | 3       | 3                  |
| ICT-4204    | Wireless Communications Laboratory                      | 100   | 1.5     | 3                  |
| ICT-4205    | Telecommunication Engineering                           | 100   | 3       | 3                  |
| ICT-4207    | Software Project Management                             | 100   | 2       | 3                  |
| HUM-4201    | Industrial Management Accountancy                       | 100   | 2       | 2                  |
| ICT-4208    | Field Work/Industrial Attachment                        | 100   | 2       | 4                  |

|              |           |             |           |           |
|--------------|-----------|-------------|-----------|-----------|
| ICT-4002     | Project   | 100         | 3         | 6         |
| ICT-4000     | Viva-Voce | 100         | 1         | 2         |
| <b>Total</b> |           | <b>1000</b> | <b>22</b> | <b>32</b> |

**B.Sc. Engg. Session: 2021-2022**  
**1<sup>st</sup> Year 1<sup>st</sup> Semester**

**PHY-1101: Physics**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Basic Physics at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course is an introduction to a broad range of topics in Modern Physics. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Physics classes. The course covers Special Relativity, Quantum Mechanics, Atomic Physics, Statistical Physics, Solid State Physics, Heat and Thermodynamics and Properties of Matter.

**Course Objectives:** To provide knowledge on Fundamental physics including Heat and Thermodynamics, properties of matter and also on modern physics including theory of relativity, quantum theory, photo-electric effect, nuclear reactions etc.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                         |
|-------------|---------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course            |
| <b>CLO2</b> | <b>Discuss</b> about heat and thermodynamics in physics |
| <b>CLO3</b> | <b>Analyze</b> the properties of matter                 |
| <b>CLO4</b> | <b>Understand</b> about modern physics                  |
| <b>CLO5</b> | <b>Explain</b> about nuclear reactions                  |

| Content of the Course  |                                                                                                                                                                                                           | Time Frame              | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                                                           |                         | CLO            | PLO         |                                                                                                        |                                                                       |
| <b>Course Overview</b> | <b>To comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Fundamental and Modern Physics<br><b>To grasp</b> the content and policies for the class | <b>Week-1<br/>L-1-3</b> | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |                      |                      |                                                                                             |                                                                                                                         |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|----------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Heat and Thermodynamics</b> | <p><b>To discuss</b> about Principles of temperature measurements, Platinum resistance thermometer, Thermoelectric thermometer, Pyrometer, Kinetic theory of gases</p> <p><b>To analyze</b> Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy, Brownian motion, van der waal's equation of state</p> <p><b>To classify and apply</b> thermodynamics laws</p> <p><b>To calculate</b> Efficiency of heat engines, Carnot theorem, Entropy and Disorder, Thermodynamic functions, Maxwell relations, Clausius-Clapeyron equation, Gibbs phase rule.</p> | <b>Week-2,3<br/>L-4-9</b>     | <b>CLO2</b>          | <b>PLO1<br/>PLO2</b> | Students will discuss with a group. To apply the knowledge to solve the different problems. | #Interactive discussion<br>#Lecture discussion with multimedia.                                                         |
| <b>Exam Review</b>             | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-4<br/>L-10-12</b>     |                      |                      | Lectures, Questions and Answers                                                             |                                                                                                                         |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-5<br/>L-13-15</b>     |                      |                      | Quiz and Tutorial Exam. Will be taken                                                       |                                                                                                                         |
| <b>Properties of Matter</b>    | <p><b>To discuss</b> States of matter, Elastic properties of solids, Coefficients of elasticity, Energy calculation, Flow of liquids, Equation of continuity, Laminar and turbulent flow, Reynolds number &amp; its significance, Bernoulli's theorem and its application, Viscosity, poiseulles equation, Motion in a viscous medium, Determination of coefficient of viscosity, Surface tension, Surface tension as a molecular phenomenon, Surface tension and surface energy, Capillarity and angle of contact, Quincke's method.</p>                                              | <b>Week-6,7<br/>L-16-21</b>   | <b>CLO2<br/>CLO3</b> | <b>PLO2<br/>PLO3</b> | Students will identify, analyze and formulate complex engineering problems.                 | #White board discussion<br>#Lecture discussion with multimedia                                                          |
| <b>Exam Review</b>             | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-8<br/>L-22-24</b>     |                      |                      | Lectures, Questions and Answers                                                             |                                                                                                                         |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-9<br/>L-25-27</b>     |                      |                      | Presentation and Tutorial Exam. Will be taken                                               |                                                                                                                         |
| <b>Modern Physics</b>          | <p><b>To analyze</b> Michelson-Morley's experiment, Galilean transformation, Special theory of relativity &amp; its consequences, Quantum theory of Radiation.</p> <p><b>To discuss</b> Photo-electric effect, Compton effect, wave particle duality, Interpretation of Bohr's postulates, Radioactive disintegration, Properties of nucleus, Nuclear reactions, Fission, Fusion, Chain reaction, Nuclear reactor.</p>                                                                                                                                                                 | <b>Week-10,11<br/>L-28-33</b> | <b>CLO4<br/>CLO5</b> | <b>PLO3<br/>PLO3</b> | Students will apply reasoning informed by contextual knowledge to understand modern physics | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about modern physics to motivate students |



|                            |                            |                            |  |  |                                            |  |
|----------------------------|----------------------------|----------------------------|--|--|--------------------------------------------|--|
| <b>Exam. Review</b>        | <b>To make an overview</b> | <b>Wee-12<br/>L-34-36</b>  |  |  | Lectures, Questions and Answers            |  |
| <b>Exam.</b>               |                            | <b>Week-13<br/>L-37-39</b> |  |  | Assignment and Tutorial Exam will be taken |  |
| <b>Recess before Exam.</b> |                            | <b>Week-14,15</b>          |  |  |                                            |  |
| <b>Sem. End Exam.</b>      | <b>Start</b>               | <b>Week-16</b>             |  |  |                                            |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. *Perspective of Modern Physics*, Arthur Beiser, McGraw Hill
2. *Concepts of Modern Physics*, Arthur Beiser, Tata McGraw Hill

#### **Reference Books and Other Materials**

3. *Heat and Thermodynamics*, N. Subrahmanyam and Brij Lal, S. Chad & Co. Ltd
4. *Fundamentals of Physics*, Halliday, Resnick & Walker, John Wiley & Sons Inc
5. *Lecture Sheet and Presentation Materials*

**ICT-1101: Discrete Mathematics**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Basic Algebra and Calculus**Course Type:** Theory

**Course Summary:** To provide knowledge to transform natural language statements to propositional statements. To express a logic sentence in terms of predicates, quantifiers, and logical connective, perform modulo arithmetic and computer arithmetic. To apply rules of inference, mathematical induction, counting, advanced counting techniques, relations, graph, and tree data structures to solve problems.

**Course Objectives:** Ideas of mathematical logic, concepts of set theory, mathematical induction, counting, advanced counting techniques, relations, graph, and tree data structures. To impart the mathematical concepts required to computer science.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                                             |
| <b>CLO2</b> | <b>Explore</b> different types of discrete structure and mathematical logic.                                             |
| <b>CLO3</b> | <b>Understand</b> about number theory and some algorithms.                                                               |
| <b>CLO4</b> | <b>Apply</b> counting and advanced counting techniques to solve real-world problem                                       |
| <b>CLO5</b> | <b>Solve</b> a variety of basic recurrence relations and analyze a problem to determine underlying recurrence relations. |
| <b>CLO4</b> | <b>Know</b> about how to generate tree and graph for solving real life problems.                                         |

| Course Content         |                                                                                                                                                                                                                                                               | Time Frame              | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Outcomes                                                                                                                                                                                                                                             |                         | CLO            | PLO  |                                                                                                        |                                                                       |
| <b>Course Overview</b> | <p><b>To comprehend</b> the mission and vision of the Department.</p> <p><b>To acquire</b> general idea about the course Fundamental and about discrete mathematics and numerical methods.</p> <p><b>To grasp</b> the content and policies for the class.</p> | <b>Week-1<br/>L-1-3</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                   |            |                                                                                                                     |                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------|------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Discrete Structure</b>        | <b>To analysis mathematical logic:</b> Proposition and notation, Connectives, proposition formulas and truth tables, Conditional and Bi-conditional, tautologies, equivalence of formulas, duality law, set operations, normal forms, ordering and uniqueness of normal forms, methods of proof, Predicate calculus- predicates, statement function, variables and quantifiers, predicate formulas, free and bound variables, the universe of discourse, Rules of Inference- universal specification, existential generalization and universal generalization, Mathematical induction. | <b>Week-2<br/>L-4-6</b>     | CLO2              | PLO2       | Students will give brief introduction about discrete structure                                                      | #Socialized recitation<br>#Show presentation to motivate the students                          |
| <b>Elements of Number Theory</b> | <b>To explain</b> division in the integers, prime numbers, the fundamental theorem of Arithmetic, division algorithm, modular arithmetic, Euclidean algorithm, data encryptions using prime number.                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-3<br/>L-7-9</b>     | CLO3              | PLO3       | Students will discuss with a group. To apply the knowledge to solve the different real-time number theory problems. | #Interactive discussion<br>#Lecture discussion with number theory.                             |
| <b>Exam. Review</b>              | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-4<br/>L-10-12</b>   |                   |            | Lectures, Questions and Answers                                                                                     |                                                                                                |
| <b>Exam.</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-5<br/>L-13-15</b>   |                   |            | Quiz and Tutorial Exam. Will be taken                                                                               |                                                                                                |
| <b>Counting and Relations</b>    | Counting arguments, the pigeonhole principle, permutations and combinations, Finite probability space, events, discrete probability, probability theory, solving recurrence relations, divide-and-conquer relations, generating functions, relations and their properties, n-ary relations and their applications, representing relations, closure relations, equivalence relations                                                                                                                                                                                                    | <b>Week-6,7<br/>L-16-21</b> | <b>CLO4, CLO5</b> | PLO2, PLO4 | Students will identify, analyze and formulate complex counting, advanced counting, and relations related problems.  | #White board discussion<br>#Lecture discussion with counting, advanced counting, and relations |
| <b>Exam. Review</b>              | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-8<br/>L-22-24</b>   |                   |            | Lectures, Questions and Answers                                                                                     |                                                                                                |
| <b>Exam.</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-9<br/>L-25-27</b>   |                   |            | Presentation and Tutorial Exam. Will be taken                                                                       |                                                                                                |

|                            |                                                                                                                                                                                                                 |                               |      |              |                                                                                        |                                                                    |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------|--------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Graphs and Trees</b>    | To discuss graph terminology, paths and circuits, connectedness, matrix representation of graphs, isomorphism of graphs, trees, rooted trees, path lengths in rooted trees, tree representation and traversing. | <b>Week-10,11<br/>L-28-33</b> | CLO6 | PLO2<br>PLO3 | Students will identify, analyze and formulate complex graph and tree related problems. | #White board discussion<br>#Lecture discussion with graph and tree |
| <b>Exam. Review</b>        | <b>To make an overview</b>                                                                                                                                                                                      | <b>Wee-12<br/>L-34-36</b>     |      |              | Lectures, Questions and Answers                                                        |                                                                    |
| <b>Exam.</b>               |                                                                                                                                                                                                                 | <b>Week-13<br/>L-37-39</b>    |      |              | Assignment and Tutorial Exam will be taken                                             |                                                                    |
| <b>Recess before Exam.</b> |                                                                                                                                                                                                                 | <b>Week-14-15</b>             |      |              |                                                                                        |                                                                    |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                                                                                                                                                                                    | <b>Week-16</b>                |      |              |                                                                                        |                                                                    |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

**Text Books**

1. *Discrete Mathematics and Its Applications*, Kenneth H. Rosen, WCB/ McGraw-Hill
2. *Introductory Methods of Numerical Analysis*, S S Sastry

**Reference Books and Other Materials**

3. *Discrete Mathematical Structures*, Bernard Kolman, Robert C. Busby, Sharon Ross, Prentice Hall of India.
4. *Discrete Mathematical Structures with Applications to Computer Science*, J.P. Tremblay, R. Manohar, Tata McGraw Hill Publishing Company Ltd.
5. *Numerical Mathematical Analysis*, J B Scarborough.
6. *Object Oriented Numerical Methods*, V Rajaraman.

**ICT-1102: Basic ICT Laboratory**

Full Marks:100

Credit: 2

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** Basic ICT at Higher Secondary Level**Course Type:** Practical/Laboratory work**Course Summary:** The practical knowledge on computer systems and IT tools are highly significant to develop expertise on IT field. The course covers different hardware and software of computer systems, Microsoft offices, Picture/Image processing and different Web Technology.**Course Objectives:** This course addresses essential foundation of ICT applications.**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                                           |
|-------------|-----------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course              |
| <b>CLO2</b> | <b>Assemble</b> the Computer System                       |
| <b>CLO3</b> | <b>Demonstrate</b> and Apply Microsoft Offices            |
| <b>CLO4</b> | <b>Apply</b> IT Tools and Software for Picture Processing |
| <b>CLO5</b> | <b>Technique</b> to use different Web Technology          |

| Course Content                      |                                                                                                                                                                      | Time Frame                | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                                                                   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Outline                             | Specific Topics                                                                                                                                                      |                           | CLO            | PLO         |                                                                                                        |                                                                                                        |
| <b>Course Overview</b>              | To acquire general idea about the ICT Laboratory<br>To supply the experimental list for this Laboratory.                                                             | <b>Week-1<br/>L-1-3</b>   | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab.         |
| <b>Assemble the Computer System</b> | To identify different hardware of a computer system<br>To assemble the computer system<br>To install different software to a computer system for working environment | <b>Week-2,3<br/>L-4-9</b> | <b>CLO2</b>    | <b>PLO1</b> | Students will discuss with a group.<br>To apply the knowledge to identify the different hardware.      | #Interactive discussion<br>#Show the video to assemble and install the software of a computer system.. |
| <b>Exam. Review</b>                 | <b>To make an overview</b>                                                                                                                                           | <b>Week-4<br/>L-10-12</b> |                |             | Demonstrate, Questions and Answers                                                                     |                                                                                                        |
| <b>Exam.</b>                        |                                                                                                                                                                      | <b>Week-5<br/>L-13-15</b> |                |             | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                                                        |

|                                                           |                                                                                                                                                                                                                                       |                               |                      |                       |                                                                                      |                                                                                    |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|-----------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Demonstrate and Apply Microsoft Offices</b>            | <b>To classify</b> the software<br><b>To create and apply</b> the MS Word, Excel and Power point file<br><b>To create</b> spreadsheet and work sheet using appropriate software<br><b>To solve</b> the given problems using MS Excel. | <b>Week-6,7<br/>L-16-21</b>   | <b>CLO2C<br/>LO3</b> | <b>.PLO2<br/>PLO3</b> | Students will demonstrate /present and solve the problems.                           | #White board discussion<br>#Lecture discussion with multimedia                     |
| <b>Exam Review</b>                                        | <b>To make an overview</b>                                                                                                                                                                                                            | <b>Week-8<br/>L-22,23</b>     |                      |                       | Lectures, Questions and Answers                                                      |                                                                                    |
| <b>Exam.</b>                                              | <b>PLO5</b>                                                                                                                                                                                                                           | <b>Week-8,9<br/>L-24,25</b>   |                      |                       | Presentation and Tutorial Exam. Will be taken                                        |                                                                                    |
| <b>Apply IT Tools and Software for Picture Processing</b> | <b>To apply</b> the appropriate software to edit the picture /Image.<br>.                                                                                                                                                             | <b>W-9<br/>L-26-27</b>        | <b>CLO4</b>          | <b>PLO3</b>           | Students will create and edit the Picture/ Image.                                    |                                                                                    |
| <b>Technique to use different Web Technology</b>          | <b>To apply</b> the different techniques of Web Technology for the given specific purposes.                                                                                                                                           | <b>Week-10,11<br/>L-28-33</b> | <b>CLO5</b>          | <b>PLO2<br/>PLO3</b>  | Students will apply contextual knowledge to understand and apply the web technology. | #Lecture discussion with multimedia<br>#Show video about different web technology. |
| <b>Exam. Review</b>                                       | <b>To make an overview</b>                                                                                                                                                                                                            | <b>Wee-12<br/>L-34,35,36</b>  |                      |                       | Lectures, Questions and Answers                                                      |                                                                                    |
| <b>Exam.</b>                                              |                                                                                                                                                                                                                                       | <b>Week-13<br/>L-37-39</b>    |                      |                       | Presentation and Tutorial Exam. will be taken                                        |                                                                                    |
| <b>Recess before Exam.</b>                                |                                                                                                                                                                                                                                       | <b>Week-14,15</b>             |                      |                       |                                                                                      |                                                                                    |
| <b>Sem. End Exam.</b>                                     | <b>Start</b>                                                                                                                                                                                                                          | <b>Week-16</b>                |                      |                       |                                                                                      |                                                                                    |

**Assessment Pattern (According FET Ordinance)****CIE: (Continuous Internal Evaluation-20Marks)**

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |

|                 |    |
|-----------------|----|
| <b>Apply</b>    | 15 |
| <b>Analyze</b>  | 00 |
| <b>Evaluate</b> | 05 |
| <b>Create</b>   | 00 |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. William, Sawyer, Hutchinson: *Using Information Technology-A Practical Introduction to Computers and Communications*, McGraw-Hill

**Reference Books and Other Materials**

2. V Rajaraman: *Fundamentals of Computers*
3. Lab. Guide and Lab. Manuals

**EEE-1101: Electricity and Electromagnetism**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** PHY-1101**Course Type:** Theory

**Course Summary:** This course to develop expertise on electrical engineering. The knowledge on electrical circuits and filter circuits are highly significant for communication engineering. The course covers Circuit Theory, Transients, AC Fundamentals, and Filter Circuits.

**Course Objectives:** To provide exposure to basic electrical engineering concepts to the students. As well as the students will have sound knowledge of the nature of electricity, emf wave and how they relate to everyday phenomena.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                                                                                                                                        |
| <b>CLO2</b> | <b>Define</b> and explain the meaning/function of charge, current, voltage, power, energy, resistors (R), and the fundamental principles of Ohm's law, KVL and KCL including an understanding of electrical safety. |
| <b>CLO3</b> | <b>Understand</b> the behavior of inductances (L) and capacitances (C).                                                                                                                                             |
| <b>CLO4</b> | <b>Analysis</b> electric circuit theorem.                                                                                                                                                                           |
| <b>CLO5</b> | <b>Understand</b> the phenomena of AC power and AC circuit analysis.                                                                                                                                                |
| <b>CLO6</b> | <b>Apply</b> the filter circuits                                                                                                                                                                                    |
| <b>CLO7</b> | <b>To describe</b> the concepts in electromagnetism that applied in real-life situations.                                                                                                                           |

| Course Content |                 | Time Frame | Alignment with |     | Suggested Activities | Teaching Strategy(s) |
|----------------|-----------------|------------|----------------|-----|----------------------|----------------------|
| Outline        | Specific Topics |            | CLO            | PLO |                      |                      |

|                             |                                                                                                                                                                                                                                                                                                                                         |                           |                      |                               |                                                                                                        |                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|-------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Course Overview</b>      | <b>To comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Fundamental and Modern Physics<br><b>To grasp</b> the content and policies for the class                                                                                                                               | <b>Week-1<br/>L-1</b>     | <b>CLO1</b>          | <b>PLO1</b>                   | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Circuit Fundamentals</b> | <b>To define and classify</b> the circuit.<br><b>To state</b> voltage and current divider rule.<br><b>To explain</b> the open and shorts circuits in a series and parallel resistors.                                                                                                                                                   | <b>Week-1<br/>L-2</b>     | <b>CLO2</b>          | <b>PLO1<br/>PLO2</b>          | Students will discuss with a group. To apply the knowledge to explain the different Circuits.          | #White board discussion<br>#Lecture discussion with multimedia.       |
| <b>Circuit Theory</b>       | <b>To state and explain</b> Kirchhoff's voltage and current Law. Kirchhoff's Current Law, Circuit<br><b>To analyze</b> Branch, Mesh, and Nodal current methods.<br><b>To analyze and prove</b> Superposition, Thevenin's, Norton's, Maximum Power Transfer, and Delta and Star Conversion theorems.                                     | <b>Week-2<br/>L-3-5</b>   | <b>CLO3<br/>CLO4</b> | <b>PLO2<br/>PLO3</b>          | Student will apply the knowledge to analyze different network theorems.                                | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Electromagnetism</b>     | Electric fields, charges, dipoles, Gauss's law, magnetic force, relation between magnetism and electricity, Amperes circuital law, Biot-Savart law, force between two parallel conductors, Faraday's Law, Maxwell's equations, traveling and standing Waves, generating electromagnetic waves, plane EM Waves, electric dipole EM waves | <b>Week-3<br/>L-6-8</b>   | <b>CLO7</b>          | <b>PLO1<br/>PLO2<br/>PLO4</b> | Student will study and apply the knowledge of electromagnetism.                                        | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam Review</b>          | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                              | <b>Week-3<br/>L-9</b>     |                      |                               | Lectures, Questions and Answers                                                                        |                                                                       |
| <b>Examination</b>          |                                                                                                                                                                                                                                                                                                                                         | <b>Week-4<br/>L-10-11</b> |                      |                               | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                       |



|                                        |                                                                                                                                                                                                                                                                                                                  |                               |                      |                      |                                                                                                              |                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Transients</b>                      | <b>To classify</b> the Transients.<br><b>To calculate</b> Transient in R-L Circuits (D.C), Short Circuit Current, Time Constant, R-L Circuits (A.C), R-C Series Circuits (D.C), R-C Series Circuits (A.C).<br><b>To calculate</b> Double Energy Transients.                                                      | <b>Week-4-5<br/>L-12-15</b>   | <b>CLO3<br/>CLO4</b> | <b>PLO3</b>          | Students will identify, analyze and formulate complex engineering problems.                                  | #White board discussion<br>#Lecture discussion with multimedia |
| <b>A.C Fundamentals</b>                | <b>To explain</b> sinusoidal AC Voltage Generation,<br><b>To calculate</b> Alternating Voltage & Current equation, Average and RMS Value of AC Voltage and Current, The Derivative, Response of R,L,C to AC Quantities, Average, Apparent and Reactive Power.<br><b>To show</b> the phase relation of sine wave. | <b>Week-6-7<br/>L-16-19</b>   | <b>CLO5</b>          | <b>PLO3<br/>PLO4</b> | Students will understand the sinusoidal signal and the calculate the equations.                              | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam Review</b>                     | <b>To make an overview</b>                                                                                                                                                                                                                                                                                       | <b>Week-7-8<br/>L-20-22</b>   |                      |                      | Lectures, Questions and Answers                                                                              |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                  | <b>Week-8<br/>L-23-24</b>     |                      |                      | Presentation and Tutorial Exam. Will be taken                                                                |                                                                |
| <b>Series and Parallel AC Circuits</b> | <b>To show</b> the pharos Diagram of Series RL, RC & RLC Circuits,<br><b>To calculate</b> Active and Reactive Components of Circuit Current, Admittance & Susceptance (B) of Parallel RL,RC,RLC Circuit<br><b>To find Out</b> Resonance, Bandwidth and Q-factor of RL,RC,RLC Circuits.                           | <b>Week-9-10<br/>L-25-29</b>  | <b>CLO5</b>          | <b>PLO3<br/>PLO4</b> | Students will apply reasoning informed by contextual knowledge to the different equations of circuit theory. | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Filters</b>                         | <b>To define and characterization</b> Symmetrical Networks<br><b>To define and explain</b> the different types of filter circuits.                                                                                                                                                                               | <b>Week-10-11<br/>L-30-33</b> | <b>CLO6</b>          | <b>PLO4</b>          | Students will classify and apply the filter circuits.                                                        | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>                    | <b>To make an overview</b>                                                                                                                                                                                                                                                                                       | <b>Wee-12<br/>L-34-36</b>     |                      |                      | Lectures, Questions and Answers                                                                              |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                  | <b>Week-13<br/>L-37-39</b>    |                      |                      | Assignment and Tutorial Exam will be taken                                                                   |                                                                |
| <b>Recess before Exam.</b>             |                                                                                                                                                                                                                                                                                                                  | <b>Week-14-15</b>             |                      |                      |                                                                                                              |                                                                |
| <b>Sem. End Exam.</b>                  | <b>Start</b>                                                                                                                                                                                                                                                                                                     | <b>Week-16</b>                |                      |                      |                                                                                                              |                                                                |

**Assessment Pattern (According FET Ordinance)****CIE: (Continuous Internal Evaluation-30Marks)**

| Bloom's Category<br>Marks:30 | Tutorial/In-<br>course<br>(10) | Assignment<br>(05) | Quizz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|--------------------------------|--------------------|----------------------------|------------------------|
| Remember                     |                                |                    | (05)                       |                        |
| Understand                   | (05)                           |                    |                            |                        |
| Apply                        |                                |                    |                            | (10)                   |
| Analyze                      |                                |                    |                            |                        |
| Evaluate                     | (05)                           |                    |                            |                        |
| Create                       |                                | (05)               |                            |                        |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 05   |
| Understand          | 05   |
| Apply               | 10   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 05   |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. *Introductory Circuit Analysis, R L Boylestad, Universal Book Stall.*
2. *Electrical Technology ( Vol I), B.L. Theraja and A.K. Theraja, S. Chand & Co. Ltd.*

**Reference Books and Other Materials**

3. *Physics Part II, Resnick & Haliday*
4. *Network Analysis, Van Valkenberg*
5. *A textbook of Optics, N. Subrahmanyam and BrijLal*

**EEE-1102: Electricity and Electromagnetism Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** EEE-1101**Course Type:** Practical/Laboratory work

**Course Summary:** The practical knowledge on electrical engineering are highly significant to develop expertise on IT field. The course covers basic circuit theory, conversion of circuit, ac characterization R-L, R-C, R-L-C series parallel circuits and design and apply the filter circuits.

**Course Objectives:** To provide practical knowledge of design and implement electric circuits.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                          |
| <b>CLO2</b> | <b>Gain</b> the practical understanding of different circuit theories |
| <b>CLO3</b> | <b>Design</b> and implement some common electric circuits.            |

| Course Content | Time | Alignment with | Suggested | Teaching |
|----------------|------|----------------|-----------|----------|
|----------------|------|----------------|-----------|----------|

| Outline                                        | Specific Topics                                                                                                                                                                                 | Frame                        | CLO                  | PLO                   | Activities                                                                                             | Strategy(s)                                                                                            |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|-----------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Course Overview</b>                         | <b>To acquire</b> general idea about the ICT Laboratory<br><b>To supply</b> the experimental list for this Laboratory.                                                                          | <b>Week-1<br/>L-1-3</b>      | <b>CLO1</b>          | <b>PLO1</b>           | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab.         |
| <b>Circuit Analysis</b>                        | <b>To state and verify</b> voltage and current divider rule<br><b>To verify</b> KCL and KVL<br><b>To draw</b> series and parallel circuit and calculate voltage and current in open/short mode. | <b>Week-2-3<br/>L-4-9</b>    | <b>CLO2</b>          | <b>PLO1<br/>POL2</b>  | Students will discuss with a group. To apply the knowledge to identify the different hardware.         | #Interactive discussion<br>#Show the video to assemble and install the software of a computer system.. |
| <b>Exam Review</b>                             | <b>To make an overview</b>                                                                                                                                                                      | <b>Week-4<br/>L-10-12</b>    |                      |                       | Demonstrate, Questions and Answers                                                                     |                                                                                                        |
| <b>Examination</b>                             |                                                                                                                                                                                                 | <b>Week-5<br/>L-13-15</b>    |                      |                       | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                                                        |
| <b>Circuit Theory Analysis and Application</b> | <b>To covert</b> different circuits<br><b>To state and prove</b> the different circuit theorems                                                                                                 | <b>Week-6-7<br/>L-16-21</b>  | <b>CLO2<br/>CLO3</b> | <b>.PLO2<br/>PLO3</b> | Students will demonstrate /present and solve the problems.                                             | #White board discussion<br>#Lecture discussion with multimedia                                         |
| <b>Exam Review</b>                             | <b>To make an overview</b>                                                                                                                                                                      | <b>Week-8<br/>L-22-23</b>    |                      |                       | Lectures, Questions and Answers                                                                        |                                                                                                        |
| <b>Exam.</b>                                   |                                                                                                                                                                                                 | <b>Week-8,9<br/>L-24-25</b>  |                      |                       | Presentation and Tutorial Exam. Will be taken                                                          |                                                                                                        |
| <b>Series-Parallel ac Circuit Analysis</b>     | <b>To covert</b> series to parallel and parallel to series circuit<br><b>To calculate</b> different properties of series and parallel R-L, R-C, R-L-C circuits.                                 | <b>W-9<br/>L-26-27</b>       | <b>CLO2<br/>CLO3</b> | <b>.PLO3</b>          | Students will create and edit the Picture/ Image.                                                      |                                                                                                        |
| <b>To Design and Apply Filter Circuits</b>     | <b>To draw and calculate</b> the efficiency of different filter circuits.                                                                                                                       | <b>Week10-11<br/>L-28-33</b> | <b>CLO2<br/>CLO3</b> | <b>PLO2P<br/>LO3</b>  | Students will apply contextual knowledge to understand and apply the web technology.                   | #Lecture discussion with multimedia<br>#Show video about different web technology.                     |
| <b>Exam. Review</b>                            | <b>To make an overview</b>                                                                                                                                                                      | <b>Wee-12<br/>L-34-36</b>    |                      |                       | Lectures, Questions and Answers                                                                        |                                                                                                        |
| <b>Exam.</b>                                   |                                                                                                                                                                                                 | <b>Week-13<br/>L-37-39</b>   |                      |                       | Presentation and Tutorial Exam. will be taken                                                          |                                                                                                        |
| <b>Recess before Exam.</b>                     |                                                                                                                                                                                                 | <b>Week14-15</b>             |                      |                       |                                                                                                        |                                                                                                        |
| <b>Sem. End Exam.</b>                          | <b>Start</b>                                                                                                                                                                                    | <b>Week-16</b>               |                      |                       |                                                                                                        |                                                                                                        |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| Bloom's Category<br>Marks:30 | Lab. Performance<br>(05) | Lab. Note Book<br>(05) | Continuous Viva-<br>Voce<br>(10) | Lab. Attendance<br>(10) |
|------------------------------|--------------------------|------------------------|----------------------------------|-------------------------|
| Remember                     |                          |                        | (10)                             |                         |
| Understand                   |                          |                        |                                  |                         |
| Apply                        |                          | (05)                   |                                  | (10)                    |
| Analyze                      |                          |                        |                                  |                         |
| Evaluate                     | (05)                     |                        |                                  |                         |
| Create                       |                          |                        |                                  |                         |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 10   |
| Understand          | 00   |
| Apply               | 15   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 00   |

#### **Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

#### **Learning Resources**

##### ***Text Books***

1. *Introductory Circuit Analysis, R L Boylestad, Universal Book Stall.*
2. *Electrical Technology ( Vol I), B.L. Theraja and A.K. Theraja, S. Chand & Co. Ltd.*

##### ***Reference Books and Other Materials***

3. *Physics Part II, Resnick & Haliday*
4. *Network Analysis, Van Valkenberg*
5. *A textbook of Optics, N. Subrahmanyam and Brij Lal*
6. *Lecture Sheet and Presentation Materials*
7. *Lab. Guide and Lab. Manuals*

#### **MATH-1101: Calculus and Differential Equations**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Basic Mathematics at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course is an introduction to a broad range of topics in Mathematics. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Mathematics classes. The course covers Differential Calculus, Integral Calculus, Differential Equations, Explicit Solution of Higher-Order Linear Differential Equations, and Series Solutions of Linear Differential Equations.

**Course Objectives:** To provide knowledge of differentiation, different techniques of differentiation and its applications, integration, different techniques of integrations and its applications.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                    |
| <b>CLO2</b> | <b>Find</b> the derivatives and integration of mathematical expressions.        |
| <b>CLO3</b> | <b>Formulate</b> different phenomena using derivatives and integration.         |
| <b>CLO4</b> | <b>Model</b> of different phenomena through the form of differential equations. |
| <b>CLO5</b> | <b>Solve</b> the differential equations.                                        |

| Course Content                |                                                                                                                                                                                                                                                                | Time Frame                  | Alignment with       |             | Suggested Activities                                                                                 | Teaching Strategy(s)                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------|-------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                       | Specific Topics                                                                                                                                                                                                                                                |                             | CLO                  | PLO         |                                                                                                      |                                                                       |
| <b>Course Overview</b>        | <b>Comprehended</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course<br>Fundamental Mathematics<br><b>To grasp</b> the content and policies for the class                                                           | <b>Week-1<br/>L-1-3</b>     | <b>CLO1</b>          | <b>PLO1</b> | Students will give brief introduction about themselves. Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Differential Calculus</b>  | <b>To review</b> of function, Limits, and continuity,<br><b>To understand</b> the concepts of differentiation, Successive differentiation, partial differentiation, applications of differentiation.                                                           | <b>Week-2-3<br/>L-4-9</b>   | <b>CLO2</b>          | <b>PLO2</b> | Students will discuss with a group. To apply the knowledge to solve the different problems.          | #Interactive discussion<br>#Lecture discussion with white board.      |
| <b>Integral Calculus</b>      | <b>To explain</b> the methods of substitution, integration by parts, Integration of special trigonometric and rational functions, definite and indefinite integrals, applications of integrations.                                                             | <b>Week-4<br/>L-10-12</b>   | <b>CLO2</b>          | <b>PLO2</b> | Students will discuss with a group. To apply the knowledge to solve the different problems.          | Interactive discussion<br>#Lecture discussion with white board.       |
| <b>Exam Review</b>            | <b>To make an overview</b>                                                                                                                                                                                                                                     | <b>Week-5<br/>L-13-14</b>   |                      |             | Lectures, Questions and Answers                                                                      |                                                                       |
| <b>Exam.</b>                  |                                                                                                                                                                                                                                                                | <b>Week-5<br/>L-15</b>      |                      |             | Quiz and Tutorial Exam. Will be taken                                                                |                                                                       |
| <b>Differential equations</b> | <b>To introduce</b> to mathematical modeling of real problems<br><b>To solve</b> First-Order Ordinary Differential Equations, Exact Differential Equations and Integrating Factors, Separable Equations and Equations reducible to this form, Linear Equations | <b>Week-6-7<br/>L-16-21</b> | <b>CLO2<br/>CLO3</b> | <b>PLO3</b> | Students will identify, analyze and formulate complex engineering problems.                          | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam Review</b>            | <b>To make an overview</b>                                                                                                                                                                                                                                     | <b>Week-8<br/>L-22-24</b>   |                      |             | Lectures, Questions and Answers                                                                      |                                                                       |

|                                                                        |                                                                                                                                                                                          |                               |                      |                      |                                                                                           |                                                                |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|----------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Exam.</b>                                                           |                                                                                                                                                                                          | <b>Week-9<br/>L-25-27</b>     |                      |                      | Presentation and Tutorial Exam. Will be taken                                             |                                                                |
| <b>Explicit Solution of Higher-Order Linear Differential Equations</b> | <b>To discuss and explain</b> basic Theory, Homogeneous Linear Equation with Constant Coefficients, Method of Undetermined Coefficients, Variation of Parameters, Cauchy-Euler Equation. | <b>Week-10,11<br/>L-28-33</b> | <b>CLO4<br/>CLO5</b> | <b>PLO3<br/>PLO4</b> | Students will apply reasoning informed by contextual knowledge to understand the problems | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Series Solutions of Linear Differential Equations</b>               | <b>To analyze</b> Power Series Solutions about an Ordinary Point, Solutions about Singular Points                                                                                        |                               | <b>CLO4<br/>CLO5</b> | <b>PLO5</b>          | Students will identify, analyze and formulate complex engineering problems.               | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>                                                    | <b>To make an overview</b>                                                                                                                                                               | <b>Wee-12<br/>L-34-36</b>     |                      |                      | Lectures, Questions and Answers                                                           |                                                                |
| <b>Exam.</b>                                                           |                                                                                                                                                                                          | <b>Week-13<br/>L-37-39</b>    |                      |                      | Assignment and Tutorial Exam will be taken                                                |                                                                |
| <b>Recess before Exam.</b>                                             |                                                                                                                                                                                          | <b>Week-14,15</b>             |                      |                      |                                                                                           |                                                                |
| <b>Sem. End Exam.</b>                                                  | <b>Start</b>                                                                                                                                                                             | <b>Week-16</b>                |                      |                      |                                                                                           |                                                                |

### Assessment Pattern (According FET Ordinance)

#### CIE: (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

#### SMEE: (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

– Class Attendance & Participation (10%)

- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Edwards Differential Calculus*
2. *H T H Piaggio An Elementary Treatise on Differential Equations and Their Applications*

#### ***Reference Books and Other Materials***

3. *Paul Dawkins Calculus-I*
4. *Sapely L Ross Differential Equations*
5. *Lecture Sheet and Presentation Materials*

**HUM-1101: Communicative English**

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

**Prerequisite:** Basic English at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** To understand and demonstrate basic pattern of communication skills, process of communication and to develop oral, listening, reading, and writing skills.

**Course Objectives:** To develop skills on English reading, writing, speaking, and listening.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                        |
| <b>CLO2</b> | <b>Find</b> the basic concepts modals, phrases & idioms, prefixes & suffixes.                       |
| <b>CLO3</b> | <b>Find</b> the basic knowledge of sentence structures, Introduction to phonetic symbols, dialogue. |
| <b>CLO4</b> | <b>Deep</b> knowledge of Four Skills such as Reading, Writing, Listening and speaking.              |

| Course Content         |                                                                                                                                                                         | Time Frame              | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                         |                         | CLO            | PLO         |                                                                                                        |                                                                       |
| <b>Course Overview</b> | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course<br><b>To grasp</b> the content and policies for the class | <b>Week-1<br/>L-1-2</b> | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Grammar</b>         | <b>Discuss</b> about grammatical principles, modals, phrases and idioms prefixes and Suffixes, sentence structures, why and yes/no questions, conditional sentences.    | <b>Week-2<br/>L-3-4</b> | <b>CLO2</b>    | <b>PLO1</b> | Students will discuss with a group.                                                                    | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Vocabulary</b>      | <b>Explain</b> technical and scientific vocabulary, defining terms                                                                                                      | <b>Week-3<br/>L-5-6</b> | <b>CLO2</b>    | <b>PLO2</b> | Students will practice with a group.                                                                   | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Exam Review</b>     | <b>To make an overview</b>                                                                                                                                              | <b>Week-4<br/>L-7-8</b> |                |             | Lectures, Questions and Answers                                                                        |                                                                       |

|                                   |                                                                                                                                                   |                            |                      |             |                                                                    |                                                                                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------|-------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Exam.</b>                      |                                                                                                                                                   | <b>Week-5<br/>L-9-10</b>   |                      |             | Quiz and Tutorial Exam. Will be taken                              |                                                                                                                        |
| <b>Spoken English</b>             | <b>To introduce</b> to phonetic symbols, dialogue, responding to particular situations, extempore speech.                                         | <b>Week-6<br/>L-11-12</b>  | <b>CLO2<br/>CLO3</b> | <b>PLO3</b> | Students will practice with a group.                               | #White board discussion<br>#Lecture discussion with multimedia                                                         |
| <b>Reading</b>                    | <b>To discuss</b> about comprehension of technical & non technical materials-skimming, scanning, inferring & responding to context.               | <b>Week-7<br/>L-13-14</b>  | <b>CLO2<br/>CLO3</b> | <b>PLO3</b> | Students will practice with a group.                               | #White board discussion<br>#Lecture discussion with multimedia                                                         |
| <b>Exam Review</b>                | <b>To make an overview</b>                                                                                                                        | <b>Week-8<br/>L-15-16</b>  |                      |             | Lectures, Questions and Answers                                    |                                                                                                                        |
| <b>Exam.</b>                      |                                                                                                                                                   | <b>Week-9<br/>L-17-18</b>  |                      |             | Presentation and Tutorial Exam. Will be taken                      |                                                                                                                        |
| <b>Technical Writing</b>          | <b>To discuss</b> paragraph and composition writing on scientific and other themes, report writing, research paper writing, library references. . | <b>Week-10<br/>L-19-20</b> | <b>CLO3<br/>CLO4</b> | <b>PLO3</b> | Students will practice to write the paragraph on different topics. | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about writing paper to motivate students |
| <b>Professional Communication</b> | <b>To discuss</b> business letter, Job application, memos, quotations, tender notice.                                                             | <b>Week-11<br/>L-21-22</b> | <b>CLO3<br/>CLO4</b> |             | Students will practice to write the different letters              | #White board discussion<br>#Lecture discussion with multimedia                                                         |
| <b>Exam. Review</b>               | <b>To make an overview</b>                                                                                                                        | <b>Week-12<br/>L-23-24</b> |                      |             | Lectures, Questions and Answers                                    |                                                                                                                        |
| <b>Exam.</b>                      |                                                                                                                                                   | <b>Week-13<br/>L-25-26</b> |                      |             | Assignment and Tutorial Exam will be taken                         |                                                                                                                        |
| <b>Recess before Exam.</b>        |                                                                                                                                                   | <b>Week-14-15</b>          |                      |             |                                                                    |                                                                                                                        |
| <b>Sem. End Exam.</b>             | <b>Start</b>                                                                                                                                      | <b>Week-16</b>             |                      |             |                                                                    |                                                                                                                        |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                               |                            |                                    |                                |
| <b>Apply</b>                         |                                    |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                    |                                |



|          |      |      |  |  |
|----------|------|------|--|--|
| Evaluate | (05) |      |  |  |
| Create   |      | (05) |  |  |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. A.J. Thomas & A.V Martinet *A Practical English Grammar.*
2. John M. Lennon *Technical writing.*

**Reference Books and Other Materials**

- 3.A.Ashley *Oxford Handbook of Commercial Correspondence.*
- 4.J. Swales *Writing Scientific English.*
5. Robert J. Dixson *Complete Course in English.*
- 6 Rajendra Pal & J.S. Korlahalli *Essentials of Business Communications*

**HUM-1102: Communication English Laboratory**

Full Marks:100

Credit: 1.0

Contact Hours: 26

Exam. Duration: 6 Hours

**Prerequisite:** HUM-1101**Course Type:** Practical/Practice**Course Summary:** To understand and demonstrate basic pattern of communication skills, process of communication and to develop oral, listening, reading, and writing skills.**Course Objectives:** To develop skills on English reading, writing, speaking, and listening.**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                              |
|-------------|----------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course |
| <b>CLO2</b> | <b>Read</b> English fluently                 |
| <b>CLO3</b> | <b>Write</b> English fluently                |
| <b>CLO4</b> | <b>Speak</b> English fluently                |

| Course Content | Time | Alignment with | Suggested | Teaching |
|----------------|------|----------------|-----------|----------|
|----------------|------|----------------|-----------|----------|

| Outline                                                 | Specific Topics                                                                              | Frame                       | CLO                  | PLO                  | Activities                                                                                             | Strategy(s)                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------|----------------------|----------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Course Overview</b>                                  | To acquire general idea about the ICT Laboratory<br>To discuss the role for this Laboratory. | <b>Week-1<br/>L-1-2</b>     | <b>CLO1</b>          | <b>PLO1</b>          | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab. |
| <b>Vocabulary Test</b>                                  | To test Vocabulary                                                                           | <b>Week-2-3<br/>L-3-6</b>   | <b>CLO2</b>          | <b>PLO1<br/>POL2</b> | Students will discuss with a group.<br>To apply the knowledge to improve the vocabulary.               | #Interactive discussion<br>#Play the audio system to motivate the students                     |
| <b>Exam Review</b>                                      | To make an overview                                                                          | <b>Week-4<br/>L-7-8</b>     |                      |                      | Demonstrate, Questions and Answers                                                                     |                                                                                                |
| <b>Examination</b>                                      |                                                                                              | <b>Week-5<br/>L-9-10</b>    |                      |                      | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                                                |
| <b>Spoken English Test</b>                              | To test Spoken English                                                                       | <b>Week-6-7<br/>L-11-14</b> | <b>CLO2<br/>CLO3</b> | <b>PLO2<br/>PLO3</b> | Students will lecture /present any self selected topic                                                 | #Lecture discussion with multimedia                                                            |
| <b>Exam Review</b>                                      | To make an overview                                                                          | <b>Week-8<br/>L-15-16</b>   |                      |                      | Lectures, Questions and Answers                                                                        |                                                                                                |
| <b>Exam.</b>                                            |                                                                                              | <b>Week-9<br/>L-17-18</b>   |                      |                      | Presentation and Tutorial Exam. Will be taken                                                          |                                                                                                |
| <b>Reading, Writing, and Listening Test</b>             | To test Reading, Writing, and Listening efficiency.                                          | <b>W-10<br/>L-19-20</b>     | <b>CLO2<br/>CLO3</b> | <b>PLO3</b>          | Students will improve their reading, writing and listening efficiency.                                 |                                                                                                |
| <b>Technical writing and Professional Communication</b> | To practice technical writing and professional communication                                 | <b>Week-11<br/>L-21-22</b>  | <b>CLO2<br/>CLO3</b> | <b>PLO2<br/>PLO3</b> | Students will apply contextual knowledge to professional communication                                 | #Lecture discussion with multimedia<br>#Show video about different web technology.             |
| <b>Exam. Review</b>                                     | To make an overview                                                                          | <b>Wee-12<br/>L-23-24</b>   |                      |                      | Lectures, Questions and Answers                                                                        |                                                                                                |
| <b>Exam.</b>                                            |                                                                                              | <b>Week-13<br/>L-25-26</b>  |                      |                      | Presentation and Tutorial Exam. will be taken                                                          |                                                                                                |
| <b>Recess before Exam.</b>                              |                                                                                              | <b>Week-14-15</b>           |                      |                      |                                                                                                        |                                                                                                |
| <b>Sem. End Exam.</b>                                   | <b>Start</b>                                                                                 | <b>Week-16</b>              |                      |                      |                                                                                                        |                                                                                                |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| Bloom's Category<br>Marks:30 | Lab. Performance<br>(05) | Lab. Note Book<br>(05) | Continuous Viva-<br>Voce<br>(10) | Lab. Attendance<br>(10) |
|------------------------------|--------------------------|------------------------|----------------------------------|-------------------------|
| Remember                     |                          |                        | (10)                             |                         |
| Understand                   |                          |                        |                                  |                         |
| Apply                        |                          | (05)                   |                                  | (10)                    |
| Analyze                      |                          |                        |                                  |                         |
| Evaluate                     | (05)                     |                        |                                  |                         |
| Create                       |                          |                        |                                  |                         |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 10   |
| Understand          | 00   |
| Apply               | 15   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 00   |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. A.J. Thomas & A.V Martinet      *A Practical English Grammar.*
2. John M. Lennon                      *Technical writing.*

**Reference Books and Other Materials**

- 3.A.Ashley                                  *Oxford Handbook of Commercial Correspondence.*
- 4.J. Swales                                  *Writing Scientific English.*
5. Robert J. Dixon                        *Complete Course in English.*
- 6 Rajendra Pal & J.S. Korlahalli      *Essentials of Business Communications.*

**NCC-01: Bangladesh Studies****Full Marks: 100****Credit: 0.0****Topics Covered**

As per design of faculty and university

**NCC-02: Islamic Studies****Full Marks: 100****Credit: 0.0****Topics Covered**

As per design of faculty and university

**B.Sc. (Engg.) First Year-2022****1<sup>st</sup> Year 2<sup>nd</sup> Semester****ICT-1201: Electronic Devices and Circuits**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Physics, Chemistry and Mathematics at Higher Secondary Level**Course Type:** Theory**Course Summary:** This course provides a comprehensive introduction to the basic idea about semiconductor diodes and its application. This course introduces dc power supply, transistor and its applications. This course also focuses on integrated circuit.**Course Objectives:** To provide knowledge on semiconductor devices, their electronic characteristics, provide the knowledge of Diode circuits, Transistor circuits, Amplifier circuits, Oscillator circuits, and the functions of operational amplifier.**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                               |
| <b>CLO2</b> | <b>Understand</b> the properties of semiconductor,                                                         |
| <b>CLO3</b> | <b>Understand</b> the behavior of semiconductor devices such as diode, transistor, Field effect transistor |
| <b>CLO4</b> | <b>Model</b> and analysis of oscillator circuits.                                                          |
| <b>CLO5</b> | <b>Understand</b> the phenomena of Operational Amplifier.                                                  |

| Course Content                                         |                                                                                                                                                                                                                                                                                                                          | Time Frame                | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                            |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Outline                                                | Specific Topics                                                                                                                                                                                                                                                                                                          |                           | CLO            | PLO         |                                                                                                        |                                                                 |
| <b>Course Overview</b>                                 | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course communication systems<br><b>To grasp</b> the content and policies for the class                                                                                                                            | <b>Week-1<br/>L-1-3</b>   | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to the students    |
| <b>Semiconductor Diodes and Optoelectronic Devices</b> | <b>To discuss</b> about Energy bands in solids, Types of Semiconductor, Semiconductor diode, Forward and reverse biased p-n junction and their characteristics, junction breakdown.<br><b>To explain</b> the Zener Diode, Varactor Diode, Tunnel Diode, PIN Diode, LED, Photodiodes, Phototransistors, LCD, solar cells. | <b>Week-2,3<br/>L-4-9</b> | <b>CLO2</b>    | <b>PLO2</b> | Students will discuss with a group.<br>To apply the knowledge to explain the different diodes.         | #Interactive discussion<br>#Lecture discussion with multimedia. |
| <b>Exam. Review</b>                                    | <b>To make an overview</b>                                                                                                                                                                                                                                                                                               | <b>Week-4<br/>L-10-12</b> |                |             | Lectures, Questions and Answers                                                                        |                                                                 |

|                                            |                                                                                                                                                                                                                                                                                                                                                                     |                             |             |                      |                                                                                                           |                                                                                                          |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|----------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Exam.</b>                               |                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-5<br/>L-13-15</b>   |             |                      | Quiz and Tutorial Exam. Will be taken                                                                     |                                                                                                          |
| <b>DC Power Supply</b>                     | <b>To design and analyze</b> half-wave, full-wave and bridge rectifiers, General filter considerations, Capacitor filter, RC filter, Regulated power supply.                                                                                                                                                                                                        | <b>Week-6,7<br/>L-16-21</b> | <b>CLO3</b> | <b>PLO2</b>          | Students will design and analyze dc power supply circuits.                                                | #White board discussion<br>#Lecture discussion with multimedia                                           |
| <b>Exam. Review</b>                        | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                          | <b>Week-8<br/>L-22-24</b>   |             |                      | Lectures, Questions and Answers                                                                           |                                                                                                          |
| <b>Exam.</b>                               |                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-9<br/>L-25-27</b>   |             |                      | Presentation and Tutorial Exam. Will be taken                                                             |                                                                                                          |
| <b>Transistor and Transistor Amplifier</b> | <b>To classify</b> Bipolar Junction Transistor, construction, operation, biasing rule, CB, CE and CC configurations and their characteristics, Relations between $\alpha$ and $\beta$ , DC load line, Q-point and maximum undistorted output.<br><b>To explain</b> CB, CE and CC amplifiers, small signal low frequency model, Hybrid parameters, Power Amplifiers. | <b>Week-10<br/>L-28-30</b>  | <b>CLO3</b> | <b>PLO2<br/>PLO3</b> | Students will apply reasoning informed by contextual knowledge to characterize transistor                 | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about transmitter circuit. |
| <b>Oscillator</b>                          | <b>To discuss</b> Feedback, Feedback types, Conditions of oscillation, Types of oscillators and their operation, Application of Oscillators, Non-sinusoidal oscillation, Types of Multivibrators and their uses.                                                                                                                                                    | <b>Week-11<br/>L-31-33</b>  | <b>CLO4</b> | <b>PLO2<br/>PLO3</b> | Students will apply reasoning informed by contextual knowledge to understand about oscillator circuit.    | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about receiver             |
| <b>Operational amplifiers</b>              | <b>To explain</b> Differential and common mode operation, OP-amp basics, Inverting and non-inverting modes, OP-amp applications                                                                                                                                                                                                                                     | <b>Week-12<br/>L-34-36</b>  | <b>CLO5</b> | <b>PLO4</b>          | Students will apply reasoning informed by contextual knowledge to understand about operational amplifier. | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about receiver             |
| <b>Exam. Review</b>                        | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                          | <b>Week-13<br/>L-37-38</b>  |             |                      | Lectures, Questions and Answers                                                                           |                                                                                                          |
| <b>Exam.</b>                               |                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-13<br/>L-39</b>     |             |                      | Assignment and Tutorial Exam will be taken                                                                |                                                                                                          |

|                     |       |            |  |  |  |  |
|---------------------|-------|------------|--|--|--|--|
| Recess before Exam. |       | Week-14-15 |  |  |  |  |
| Sem. End Exam.      | Start | Week-16    |  |  |  |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. *Communication Systems*, S. Haykin
2. *Analog Communication*, V. Chandra Sekar

#### **Reference Books and Other Materials**

3. *Principle of Communication Engineering*, Anokh Singh
4. *Fundamentals of Communication Systems*, J. G. Proakis, M. Salehi

**ICT-1202:      Electronic Devices and Circuits Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-1201

**Course Type:** Practical/Laboratory work

**Course Summary:** To observe the characteristics of Diode, Different circuit configurations of Transistor, Design and implementations of rectifier circuits, Amplifier circuits, oscillator circuits. Observe different function of Operational Amplifiers

**Course Objectives:** Electronics Lab is helpful for the students to acquire the basic knowledge of electric circuits and its application. This course will prepare students to perform the analysis and design of various electronic circuits.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Know</b> about the general idea of this course and required instruments.                                                                                                                             |
| <b>CLO2</b> | <b>Acquire</b> knowledge about the properties of semiconductor materials, semiconductor diodes with their V-I characteristics, diode rectifier circuits with and without filters and voltage regulator. |
| <b>CLO3</b> | <b>Gain</b> the practical understanding of BJT biasing and different configurations.                                                                                                                    |
| <b>CLO4</b> | <b>Design and Implement</b> oscillator circuits and practical op-amp circuits.                                                                                                                          |

| Course Content                                         |                                                                                                                                                                                                                       | Time Frame                | Alignment with |              | Suggested Activities                                                                                                                                                        | Teaching Strategy(s)                                                                                             |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Outline                                                | Specific Topics                                                                                                                                                                                                       |                           | CLO            | PLO          |                                                                                                                                                                             |                                                                                                                  |
| <b>Course Overview</b>                                 | <b>To acquire</b> general idea about the Electronic Laboratory<br><b>To supply</b> the experimental list for this Laboratory.                                                                                         | <b>Week-1<br/>L-1-3</b>   | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details                                                                      | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab.                   |
| <b>Semiconductor Diodes and Optoelectronic Devices</b> | <b>To observe</b> the V-I characteristics of forward and reverse biased PN junction diode<br><b>To design</b> circuit diagram and observe the V-I characteristics<br><b>To use</b> zener diode as a voltage regulator | <b>Week-2,3<br/>L-4-9</b> | CLO2           | PLO1<br>POL2 | Students will discuss with a group.<br>To apply the knowledge to characterize the different semiconductor diodes. To prepare the laboratory report                          | #Interactive discussion<br>#Show the practical circuit diagram to observe the voltage regulation characteristics |
| <b>DC Power Supply</b>                                 | <b>To construct and analyze</b> half-wave and full-wave rectifiers without filters<br><b>To construct and analyze</b> full-wave bridge rectifiers without filters.                                                    | <b>Week-4<br/>L-10-12</b> | CLO2           | PLO2         | Students will perform group discussion.<br>To apply the knowledge, students will construct and analyze half-wave, full-wave and full-wave bridge rectifiers without filters | #Interactive discussion<br>#Show the practical circuit diagrams of rectifiers without filters.                   |
| <b>Exam. Review</b>                                    | <b>To make an overview</b>                                                                                                                                                                                            | <b>Week-5<br/>L-13-14</b> |                |              | Demonstrate, Questions and Answers                                                                                                                                          |                                                                                                                  |
| <b>Exam.</b>                                           |                                                                                                                                                                                                                       | <b>Week-5<br/>L-15</b>    |                |              | Quiz and Tutorial Exam. Will be taken                                                                                                                                       |                                                                                                                  |

|                                            |                                                                                                                                                                                                            |                               |      |              |                                                                                                      |                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------|--------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Transistor and Transistor Amplifier</b> | <b>To configure and observe</b> the input and output characteristics of CB, CE, and CC configurations of BJT<br><b>To observe</b> the input and output characteristics of CE, and CC transistor amplifiers | <b>Week-6,7<br/>L-16-21</b>   | CLO3 | PLO2         | Students will demonstrate /present the different transistor and transistor amplifier circuits.       | #White board discussion<br>#Lecture discussion with multimedia               |
| <b>Exam. Review</b>                        | <b>To make an overview</b>                                                                                                                                                                                 | <b>Week-8<br/>L-22-23</b>     |      |              | Lectures, Questions and Answers                                                                      |                                                                              |
| <b>Exam.</b>                               |                                                                                                                                                                                                            | <b>Week-8<br/>L-24</b>        |      |              | Presentation and Tutorial Exam. Will be taken                                                        |                                                                              |
| <b>Oscillator</b>                          | <b>To design</b> and calculate the frequency of different oscillator circuits.<br><b>To design</b> and show the output waveforms of multivibrators.                                                        | <b>W-9-10<br/>L-25-29</b>     | CLO3 | PLO3         | Students will design and observe the different oscillator and multivibrator circuits.                |                                                                              |
| <b>Operational Amplifier</b>               | <b>To show</b> op-amp used as a integrator, differentiator, adder and comparator.                                                                                                                          | <b>Week-10,11<br/>L-29-33</b> | CLO4 | PLO3<br>PLO4 | Students will apply contextual knowledge to understand and apply the op-amp. For different purposes. | #Lecture discussion with multimedia<br>#Show video about different circuits. |
| <b>Exam. Review</b>                        | <b>To make an overview</b>                                                                                                                                                                                 | <b>Week-12<br/>L-34-36</b>    |      |              | Lectures, Questions and Answers                                                                      |                                                                              |
| <b>Exam.</b>                               |                                                                                                                                                                                                            | <b>Week-13<br/>L-37-39</b>    |      |              | Presentation and Tutorial Exam. will be taken                                                        |                                                                              |
| <b>Recess before Exam.</b>                 |                                                                                                                                                                                                            | <b>Week-14,15</b>             |      |              |                                                                                                      |                                                                              |
| <b>Sem. End Exam.</b>                      | <b>Start</b>                                                                                                                                                                                               | <b>Week-16</b>                |      |              |                                                                                                      |                                                                              |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |



|        |  |  |  |  |
|--------|--|--|--|--|
| Create |  |  |  |  |
|--------|--|--|--|--|

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 10   |
| Understand       | 00   |
| Apply            | 15   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 00   |

#### Assessment and Evaluation

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

#### Learning Resources

##### **Text Books**

1. *Integrated Electronics-Analog and Digital Circuits and Systems*, J Millman, C. Halkias, Tata McGraw-Hill, India.
2. *Electrical Technology (Vol. IV)*, B. L. Theraja and A. K. Theraja, S. Chand & Co. Ltd, India.

##### **Reference Books and Other Materials**

3. *Digital Electronics*, W Gothmann William Gothmann, PHI (Pvt.) Ltd.
4. *Lab. Guide and Lab. Manuals*

#### **ICT-1203: Structured Programming Language**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Mathematics and ICT at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course will introduce the students to the fundamental concepts of programming language. This course focuses on Characters set, Tokens, Keywords, Identifiers, Control Flow, Functions and Program Structure, Pointers and Arrays, Structures & Union and File and Memory management of structured programming.

**Course Objectives:** : This course is designed to provide knowledge and expertise on structured programming language by solving various problems.

After completing this course a student would be able to:

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Understand</b> the concept of structured programming language                                                                 |
| <b>CLO2</b> | <b>Understand</b> the concepts of C programming language, its variables, constants, library functions, operators and expressions |
| <b>CLO3</b> | <b>Understand</b> the different structures of control flow of instructions.                                                      |
| <b>CLO4</b> | <b>Understand and use</b> of different data structures and functions.                                                            |
| <b>CLO5</b> | <b>Understand and use</b> of sub-functions, divide a complete solution into modules                                              |
| <b>CLO6</b> | <b>Handle</b> different format of data file and manage the run-time memory.                                                      |

| Course Content                         |                                                                                                                                                                                                                                                                                                                                                                                          | Time Frame           | Alignment With |                | Suggested Activities                                                     | Teaching Strategy(s)               |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|----------------|--------------------------------------------------------------------------|------------------------------------|
| Outlines                               | Specific Topics                                                                                                                                                                                                                                                                                                                                                                          |                      | CLO            | PLO            |                                                                          |                                    |
| Introduction                           | History of C, Importance of C, Basic structured of C programs, Programming style, and Executing a C program.                                                                                                                                                                                                                                                                             | Week 1, L-1-3        | CLO-1          | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>Type, Operators and Expression</b>  | Characters set, Tokens, Keywords, Identifiers, Constants, Variables, Data types, Declaration of Variables, Arithmetic Operators, Relational and Logical Operators, Assignment Operators, Increment and decrement Operators, Conditional Operator, Bitwise Operators, Special Operators, Conditional expressions, Precedence and Order of Evaluation, Type Conversions, Input and Output. | Week 2-3<br>L-4- 9   | CLO-2          | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Control Flow</b>                    | Statements and Blocks, IF-Else, Else-if, Switch, Loops-While and For, Loops-Do-While, Break and Continue, Goto and labels.                                                                                                                                                                                                                                                               | Week 4-5, L-10-15    | CLO-3          | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Functions and Program Structure</b> | Basics of Functions, Functions Returning Non-integers, External Variables, Scope Rules, Header Files, Static Variables, Register Variables, Recursion, The C Preprocessor, File Inclusion, Macro Substitution, Conditional Inclusion.                                                                                                                                                    | Week- 5-7<br>L-15-21 | CLO-4          | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Pointers and Arrays</b>             | Declaration and use, index types, multidimensional arrays, Pointers and Address ,Pointers and Function Arguments, Pointers and Arrays, Pointers to functions                                                                                                                                                                                                                             | Week 7- 9, L-21-27   | CLO5           | PLO3<br>PLO4   | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |

|                                   |                                                                                                                                                                                                        |                       |       |               |                                                                          |                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|---------------|--------------------------------------------------------------------------|------------------------------------|
| <b>Structures &amp; Union</b>     | Basics of Structures, Structures and Functions, Arrays of Structures, Pointer to Structure, Table Lookup, Unions.                                                                                      | Week10 -11<br>L-28-33 | CLO4  | PLO-3<br>PLO4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>File and Memory management</b> | Opening and Closing a file, Input /Output operations on Files, Error handling during I/O operations, Random access to Files, Command line arguments, Dynamic memory allocation, and pointer revisited. | Week 12-13<br>L-34-39 | CLO-6 | PLO4<br>PLO5  | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test         |
| Recess before Exam.               |                                                                                                                                                                                                        | Week 14- 15           |       |               |                                                                          |                                    |
| Sem. End Exam.<br>Start           |                                                                                                                                                                                                        | Week 16               |       |               |                                                                          |                                    |

#### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### **Learning Resources**

##### ***Text Books***

1. *Programming in ANSI C*, Balaguruswami, Tata McGraw-Hill publishing Co.
2. *The C programming language*, brian w. kernighandenis m. ritchie.prentice hall softwre series
3. *Object-Oriented Programming with C++*, E Balagurusamy, Tata McGraw Hill Publishing Company Ltd.

**Reference Books and Other Materials**

5. *Turbo C / C++ - The Complete Reference*, Herbert Schildt, McGraw-Hill.
6. *C++ Object-oriented Programming*, Mitchell, Macmillan, 1993.
7. *The Apprentice C++ Programmer*, Lee, Phillips, Thompson Computer Press, 1997.
8. *Programming with C*, Gottfried, Schaum's outline series.

**ICT-1204:      Structured Programming Language  
Laboratory**

Full Marks:100  
Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** ICT-1203

**Course Type:** Practical/Laboratory work

**Course Summary:** The practical knowledge on programming knowledge to help students to develop programming skills to solve different problems. This course assist student to implement various concepts and structures of C programming language

**Course Objectives:** This course is designed to improve skill and expertise on structured programming language by solving various problems.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Construct</b> algorithms and flow charts as the part of problem analysis.                              |
| <b>CLO2</b> | <b>Write, compile and debug</b> programs in C language.                                                   |
| <b>CLO3</b> | <b>Use</b> different data types, operators and expressions in a computer program.                         |
| <b>CLO4</b> | <b>Design and implement</b> programs involving decision structures, loops, arrays, structures and unions. |
| <b>CLO5</b> | <b>Understand and use</b> of functions, divide a complete solution into modules                           |
| <b>CLO6</b> | <b>Handle</b> different format of data file and manage the run-time memory.                               |

| Course Content                        |                                                                                                                                                                                                 | Time Frame         | Alignment with |                | Suggested Activities                                                                                                    | Teaching Strategy(s)                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|----------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Outlines                              | Specific Topics                                                                                                                                                                                 |                    | CLO            | PLO            |                                                                                                                         |                                                                      |
| <b>Course Overview</b>                | <b>To acquire</b> general idea about the ICT Laboratory and Software laboratory<br><b>To supply</b> the experimental list for this Laboratory.                                                  | Week 1<br>L-1- 3   | CLO-1          | PLO-1          | -Lecture<br>Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | Assignment<br><br>- Oral Presentation<br>-Laboratory trial output    |
| <b>Type, Operators and Expression</b> | Observe the characteristics of data types and operators, design and implement solutions of different problems using variables, identifiers, various operators and expressions, type conversions | Week 2- 3<br>L-4-9 | CLO-2          | PLO-2<br>PLO-3 | -Lecture<br>Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -- Assignment<br><br>- Oral Presentation<br>-Laboratory trial output |

|                                        |                                                                                                                                                                                                                                                                                       |                          |       |                |                                                                                                                      |                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Control Flow</b>                    | Observe the characteristics of control flow structures, design and implement solutions of different problems using Statements and Blocks, decision making statements, looping statements, jumping statements                                                                          | Week 4-5<br>L-10-15      | CLO-3 | PLO-2<br>PLO-3 | -Lecture Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br><br>-Laboratory trial output |
| <b>Functions and Program Structure</b> | Observe the characteristics of Functions, design and implement solutions of different problems using functions, user defined functions, Scope Rules, Static Variables, Register Variables, Recursion, nested blocks                                                                   | Week 5-7<br>L-15-21      | CLO-4 | PLO-3<br>PLO-4 | -Lecture Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br><br>-Laboratory trial output |
| <b>Pointers and Arrays</b>             | <b>Observe</b> the characteristics of pointers and arrays, design and implement solutions of different problems using arrays and pointers, Pointers to functions                                                                                                                      | Week 7-9<br>L-21-27      | CLO-5 | PLO-3<br>PLO-4 | -Lecture Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br><br>-Laboratory trial output |
| <b>Structures &amp; Union</b>          | <b>Observe</b> the characteristics of structure and union, design and implement solutions of different problems using Structures, Structures and Functions, Arrays of Structures, Pointer to Structure, Unions.                                                                       | Week10- 11<br>L-28-33    | CLO-4 | PLO-3<br>PLO-4 | -Lecture Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br><br>-Laboratory trial output |
| <b>File and Memory management</b>      | <b>Observe</b> the characteristics of file and memory management, design and implement solutions of different problems using file, Input /Output operations on Files, Error handling during I/O operations, Random access to Files, Dynamic memory allocation, and pointer revisited. | Week 12 - 13,<br>L-34-39 | CLO-6 | PLO-4<br>PLO-5 | -Lecture Presentation with video, Interactive Discussion<br><br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br><br>-Laboratory trial output |
| <b>Recess before Exam.</b>             |                                                                                                                                                                                                                                                                                       | Week 14-15               |       |                |                                                                                                                      |                                                                    |

|                             |  |         |  |  |  |  |
|-----------------------------|--|---------|--|--|--|--|
| <b>Sem. End Exam. Start</b> |  | Week 16 |  |  |  |  |
|-----------------------------|--|---------|--|--|--|--|

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

### Assessment and Evaluation

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### Learning Resources

#### **Text Books**

1. *Programming in ANSI C*, Balaguruswami, Tata McGraw-Hill publishing Co.
2. *The C programming language*, brian w. kernighandenis m. ritchie.prentice hall softwre series

#### **Reference Books and Other Materials**

3. *Object-Oriented Programming with C++*, E Balagurusamy, Tata McGraw Hill Publishing Company Ltd
4. *Lab. Guide and Lab. Manuals*

**ICT-1205: Numerical Methods and Analysis**

Full Marks: 100  
Credit: 2.0  
Contact Hours: 39

**Prerequisite:** Basic mathematics knowledge at B.Sc. (Engg.) Levels.  
**Course Type:** Theory

**Course Summary**

This course covers the different numerical methods of solving mathematical and scientific problems such as solving for the roots of nonlinear equations, numerical differentiation and integration, interpolating polynomials and ordinary differential equations.

**Course Objectives:** To provide knowledge of the basic concepts of The Numerical Methods, solution of Algebraic and Transcendental Equations, properties of Interpolation and Spline functions, Numerical Differentiation and Integration, understand Numerical solutions of Ordinary Differential Equation and Partial Differential Equation.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| <b>CLO1</b> | To acquire general idea about the course                                |
| <b>CLO2</b> | To Discuss about the solution of Algebraic and Transcendental Equations |
| <b>CLO3</b> | Analyze the properties of Interpolation and Spline functions            |
| <b>CLO4</b> | Understand about Numerical Differentiation and Integration              |
| <b>CLO5</b> | Explain and understand Numerical solutions of ODE and PDE               |

| Course Content                                            |                                                                                                                                                                                                                                                                  | Time Frame                   | Alignment with |                      | Suggested Activities                                                                                       | Teaching Strategy(s)                                                  |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|----------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                                   | Specific Topics                                                                                                                                                                                                                                                  |                              | CLO            | PLO                  |                                                                                                            |                                                                       |
| <b>Course Overview</b>                                    | To comprehend the mission and vision of the Department.<br>To acquire general idea about the course Fundamental and Numerical Methods.<br>To grasp the content and policies for the class.                                                                       | <b>Week-1<br/>L-1-3</b>      | CLO1           | PLO1                 | Students will give brief introduction about themselves<br>Course outline will be discussing in details     | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Solution of Algebraic and Transcendental Equations</b> | <b>To discuss</b> about introduction, Bisection Method, Method of False position and Iteration Method.<br><b>To analyze</b> Newton-Raphson Method, Secant Method and Muller's Method.<br><b>To explain</b> Lin-Bairstow's Method and Quotient-Difference Method. | <b>Week-2-3<br/>L-4-9</b>    | CLO2           | PLO1<br>PLO2<br>PLO3 | Students will discuss with a group.<br>To apply the knowledge to solve the different open method problems. | #Interactive discussion<br>#Lecture discussion with open method.      |
| <b>Exam Review</b>                                        | <b>To make an overview</b>                                                                                                                                                                                                                                       | <b>Week-4<br/>L-10-12</b>    |                |                      | Lectures, Questions and Answers                                                                            |                                                                       |
| <b>Examination</b>                                        |                                                                                                                                                                                                                                                                  | <b>Week-5<br/>L-13,14,15</b> |                |                      | Quiz and Tutorial Exam. Will be taken                                                                      |                                                                       |

|                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               |              |              |                                                                                                          |                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interpolation and Spline Functions</b>                                                                           | <p><b>To explain</b> Errors in Polynomial Interpolation, Forward and Backward Differences.</p> <p><b>To discuss</b> about Gauss Central Difference Formulae, Stirling's Formula, Bessel's Formula and Langrage's Interpolation Formula.</p> <p><b>To explain</b> Linear and Quadratic Splines, Cubic B-Splines.</p>                                                                                                                                                                                                                                                | <b>Week-6-7<br/>L-16-21</b>   | CLO3         | PLO2<br>PLO3 | Students will identify, analyze and formulate complex polynomial problems.                               | #White board discussion<br>#Lecture discussion with polynomial                                                                               |
| <b>Exam Review</b>                                                                                                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Week-8<br/>L-22-24</b>     |              |              | Lectures, Questions and Answers                                                                          |                                                                                                                                              |
| <b>Examination</b>                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-9<br/>L-25-27</b>     |              |              | Presentation and Tutorial Exam. Will be taken                                                            |                                                                                                                                              |
| <b>Numerical Differentiation and Integration, Numerical Solution of ordinary and Partial Differential Equations</b> | <p><b>To discuss numerical and integration:</b></p> <p>Errors in Numerical Differentiation and Differentiation Formulae with function values. Integration-trapezoidal Rule, Simpson's 1/3 and 3/8 Rule and Romberg Integration.</p> <p><b>To discuss about the solution of ordinary differential equations:</b></p> <p>Taylor's Series, Euler's Method and Runge-kutta Method, Milne's Method.</p> <p><b>To discuss about the solution of partial differential equations:</b></p> <p>Laplace's Equation, Jacobi's Methods, Gauss-Seidel Method and ADI Method.</p> | <b>Week-10-11<br/>L-28-33</b> | CLO4<br>CLO5 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand different types of methods. | #White board discussion<br>#Lecture discussion with numerical integration & differentiation<br>#Show video about different types of methods. |
| <b>Exam. Review</b>                                                                                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Wee-12<br/>L-34-36</b>     |              |              | Lectures, Questions and Answers                                                                          |                                                                                                                                              |
| <b>Exam.</b>                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-13<br/>L-37-39</b>    |              |              | Assignment and Tutorial Exam will be taken                                                               |                                                                                                                                              |
| <b>Recess before Exam.</b>                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-14-15</b>             |              |              |                                                                                                          |                                                                                                                                              |



|                       |              |                |  |  |  |  |
|-----------------------|--------------|----------------|--|--|--|--|
| <b>Sem. End Exam.</b> | <b>Start</b> | <b>Week-16</b> |  |  |  |  |
|-----------------------|--------------|----------------|--|--|--|--|

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. S.S. Sastry, "Introductory Methods of Numerical Analysis", Fifth Edition, PHI Publication.
2. Burden, Richard L. and Faires, J. Douglas, Numerical Analysis, Boston, MA: Brooks/Cole, Cengage Learning, 2011.

#### **References and other materials**

1. Chapra, Steven C. and Canale, Raymond P., Numerical Methods for Engineers, Boston: McGraw-Hill Higher Education 2010.
2. Chapra, Steven C., Applied Numerical Methods with MATLAB® for Engineers and Scientists, New York: McGraw-Hill, 2012.
3. Cheney, Ward and Kincaid, David, Numerical Mathematics and Computing (Seventh Edition), Boston, MA : Brooks/Cole, Cengage Learning, 2013.

Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** Basic Algebra & Calculus  
**Course Type:** Laboratory  
**Course Summary:** To provide knowledge to problem formulation, implementation with C-programming language of mathematical systems of equation, ordinary differential equations and partial differential equations.

**Course Objectives:** Ideas of mathematical logic, concepts of set theory and Boolean Algebra. To impart the mathematical concepts and numerical methods required to computer science. To know the fundamental ideas of mathematical logic, concepts of set theory and Boolean algebra

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Explore</b> different types of bracketing methods.                           |
| <b>CLO2</b> | <b>Understand</b> about open method and some algorithms.                        |
| <b>CLO3</b> | <b>Know</b> about how to use polynomial method for solving real life problems.  |
| <b>CLO4</b> | <b>Know</b> about different types of numerical integration and differentiation. |

| Course Content           |                                                                                                                                                                                                                                                                                                                                                                                                                                        | Time Frame                   | Alignment with |      | Suggested Activities                                                                                    | Teaching Strategy(s)                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                  | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                                                        |                              | CLO            | PLO  |                                                                                                         |                                                                       |
| <b>Bracketing Method</b> | <b>To understand</b> the algorithm and flow chart of Bisection method and Regula Falsi Method.<br><br><b>To implement</b> Bisection and Regula Falsi Method using C/C++/Python/Java etc.                                                                                                                                                                                                                                               | <b>Week-1<br/>L-1-3</b>      | CLO1           | PLO1 | Students will give brief introduction about themselves Course outline will be discussing in details     | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Open Method</b>       | <b>To understand</b> the algorithm and flow chart of fixed-point iteration method and secant method.<br><b>To implement</b> fixed point iteration method and secant method using C/C++/Python/Java etc.<br><b>To understand</b> the algorithm and flow chart of Newton Raphson method and modified Newton Raphson method.<br><b>To implement</b> Newton Raphson method and modified Newton Raphson method using C/C++/Python/Java etc. | <b>Week-2-3<br/>L-4-9</b>    | CLO2           | PLO2 | Students will discuss with a group. To apply the knowledge to solve the different open method problems. | #Interactive discussion<br>#Lecture discussion with open method.      |
| <b>Exam Review</b>       | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-4<br/>L-10-12</b>    |                |      | Lectures, Questions and Answers                                                                         |                                                                       |
| <b>Examination</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-5<br/>L-13,14,15</b> |                |      | Quiz and Tutorial Exam. Will be taken                                                                   |                                                                       |

|                                                    |                                                                                                                                                                                                   |                               |      |              |                                                                                                          |                                                                                                                                              |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------|--------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Polynomial Method</b>                           | <b>To understand</b> the algorithm and flow chart of Gauss elimination method and Jaccobi method.<br><b>To implement</b> Gauss elimination method and Jaccobi method using C/C++/Python/Java etc. | <b>Week-6-7<br/>L-16-21</b>   | CLO3 | PLO2         | Students will identify, analyze and formulate complex polynomial problems.                               | #White board discussion<br>#Lecture discussion with polynomial                                                                               |
| <b>Exam Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                        | <b>Week-8<br/>L-22-24</b>     |      |              | Lectures, Questions and Answers                                                                          |                                                                                                                                              |
| <b>Exam.</b>                                       |                                                                                                                                                                                                   | <b>Week-9<br/>L-25-27</b>     |      |              | Presentation and Tutorial Exam. Will be taken                                                            |                                                                                                                                              |
| <b>Numerical Integration &amp; Differentiation</b> | <b>To understand</b> the algorithm and flow chart of Simpson 1/3 rule and Runge Kutta method.<br><b>To implement</b> of Simpson 1/3 rule and Runge Kutta method using C/C++/Python/Java etc.      | <b>Week-10-11<br/>L-28-33</b> | CLO4 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand different types of methods. | #White board discussion<br>#Lecture discussion with numerical integration & differentiation<br>#Show video about different types of methods. |
| <b>Exam. Review</b>                                | <b>To make an overview</b>                                                                                                                                                                        | <b>Week-12<br/>L-34-36</b>    |      |              | Lectures, Questions and Answers                                                                          |                                                                                                                                              |
| <b>Exam.</b>                                       |                                                                                                                                                                                                   | <b>Week-13<br/>L-37-39</b>    |      |              | Assignment and Tutorial Exam will be taken                                                               |                                                                                                                                              |
| <b>Recess before Exam.</b>                         |                                                                                                                                                                                                   | <b>Week-14-15</b>             |      |              |                                                                                                          |                                                                                                                                              |
| <b>Sem. End Exam.</b>                              | <b>Start</b>                                                                                                                                                                                      | <b>Week-16</b>                |      |              |                                                                                                          |                                                                                                                                              |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

#### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### **Learning Resources**

##### **Text Books**

1. *Discrete Mathematics and Its Applications*, Kenneth H. Rosen, WCB/ McGraw-Hill
2. *Introductory Methods of Numerical Analysis*, S S Sastry

##### **Reference Books and Other Materials**

3. *Discrete Mathematical Structures*, Bernard Kolman, Robert C. Busby, Sharon Ross, Prentice Hall of India.
4. *Discrete Mathematical Structures with Applications to Computer Science*, J.P. Tremblay, R. Manohar, Tata McGraw Hill Publishing Company Ltd.
5. *Numerical Mathematical Analysis*, J B Scarborough.
6. *Object Oriented Numerical Methods*, V Rajaraman.

### **MATH-1201: Geometry and Complex Variables**

Full Marks: 100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** HSC Level Mathematics

**Course Type:** Theory

**Course Summary:** This course to develop expertise on representing 2D and 3D geometry, complex variable, function, differentiation and integration of complex variables.

**Course Objectives:** To provide knowledge on 2D and 3D geometry and complex variables.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                        |
| <b>CLO2</b> | <b>Understand</b> the 2D form of geometry.                          |
| <b>CLO3</b> | <b>Understand</b> the 3D form of geometry.                          |
| <b>CLO4</b> | <b>Understand</b> the complex number and variable                   |
| <b>CLO5</b> | <b>Understand</b> and apply the complex function                    |
| <b>CLO6</b> | <b>Understand</b> the differentiation of complex function           |
| <b>CLO7</b> | <b>Understand</b> the integration of complex functions              |
| <b>CLO8</b> | <b>Understand</b> the concepts of Conformal mapping                 |
| <b>CLO9</b> | <b>Apply</b> the concepts of Conformal mapping in physical problems |

| Course Content                               |                                                                                                                                                                                                                                                                                                                                                                                                         | Time Frame                   | Alignment with |              | Suggested Activities                                                                                             | Teaching Strategy(s)                                                  |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|--------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                      | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                         |                              | CLO            | PLO          |                                                                                                                  |                                                                       |
| <b>Course Overview</b>                       | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Vector Analysis and Linear Algebra<br><b>To grasp</b> the content and policies for the class                                                                                                                                                                                              | <b>Week-1<br/>L-1</b>        | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details           | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>2D Analytical geometry</b>                | <b>To discuss</b> Cartesian and polar coordinates, Change of axes, Pair of straight lines, The general second degree equation, Circle, System of circles, parabola, ellipse, hyperbola.                                                                                                                                                                                                                 | <b>Week-1-3<br/>L-2-7</b>    | CLO2           | PLO1<br>PLO2 | Students will discuss with a group.<br>To apply the knowledge to explain different form of 2D geometric problems | #White board discussion<br>#Lecture discussion with multimedia.       |
| <b>3D Analytical geometry</b>                | <b>To discuss</b> Rectangular coordinates, plane, straight line, sphere, central coincides and general equation of the second degree.                                                                                                                                                                                                                                                                   | <b>Week-3-4<br/>L-8-12</b>   | CLO3           | PLO2<br>PLO3 | Student will apply the knowledge to analyze different form of 3D geometric problems.                             | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam</b>                                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>Week-5<br/>L-13</b>       |                |              | Quiz and Tutorial Exam.                                                                                          |                                                                       |
| <b>Complex Variables</b>                     | <b>To analysis</b> The complex number system, graphical representation, fundamental operations, absolute value, polar form, De Moivre's theorem, roots of complex number, Euler's formula, polynomial equations, vector interpretation of complex number, Dot and cross product, complex conjugate, point set                                                                                           | <b>Week-5-6<br/>L-14-16</b>  | CLO4           | PLO1<br>PLO4 | Students will discuss with a group and understand the concepts of complex number and variables                   | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Complex Function Limit and continuity</b> | <b>To explain</b> Complex variable and functions, Single and multiple valued function, inverse function, transformations, curvilinear coordinates, elementary functions, branch points and branch lines, Riemann surfaces, limits, theorems on limits, infinity, continuity, theorems on continuity, uniform continuity, sequences, limit of sequences, theorems on limits of sequences infinite series | <b>Week-6,-7<br/>L-17-20</b> | CLO5           | PLO4         | Students will identify, analyze and formulate complex engineering problems.                                      | #White board discussion<br>#Lecture discussion with multimedia        |

| Exam                                                            | To make an overview                                                                                                                                                                                                                                                                                                                                                                                                              | Week-7<br>L-21                    |      |              | Quiz and<br>Tutorial Exam.                                                   |                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|--------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Complex Differentiation and the Cauchy–Riemann Equations</b> | <b>To discuss</b> Derivatives, Analytic functions, Cauchy-Riemann equations, Harmonic functions, Geometric interpretation of the derivative, differentials, rule of derivatives, derivatives of elementary functions, higher order derivatives, L'Hospital's rule, singular points, Orthogonal families, curves applications to geometry and mechanics, complex differential operators, gradient, divergence, curl and Laplacian | <b>Week-8,9<br/>L-22- 26</b>      | CLO6 | PLO4<br>PLO5 | Students will understand and apply the knowledge of complex differentiations | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Complex Integration and Cauchy's theorem</b>                 | <b>To analysis</b> Complex line integrals, real line integrals, connection between real and complex line integrals, properties of integrals, change of variables, Jordan curve theorem, Convention regarding traversal of a closed path, Green's theorem, Complex form of Green's theorem, Cauchy's theorem, Cauchy-Goursat theorem, Morera's theorem, indefinite integrals, integrals of special functions                      | <b>Weeks-9-10<br/>L-27-30</b>     | CLO7 | PLO4<br>PLO5 | Students will understand and apply the knowledge of complex Integrations     | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam.</b>                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Wee-11<br/>L-31</b>            |      |              | Lectures, Questions and Answers                                              |                                                                |
| <b>Conformal Mapping</b>                                        | <b>To discuss</b> Transformations, Jacobian of a transformation, Complex mapping function, conformal mapping, Riemann's mapping theorem, fixed or invariant points of a transform, general transform, successive transform, linear transform, bilinear transform, mapping of a half plane on to a circle, Schwarz-Christoffel transformation, transformation of Boundaries in parametric form, some special mapping              | <b>Weeks-11-12<br/>L-32-35</b>    | CLO8 | PLO4<br>PLO5 | Students will understand the concepts of Conformal mapping                   | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Physical Applications of Conformal Mapping</b>               | <b>To discuss</b> Boundary value problems, Harmonic and conjugate functions, Dirichlet and Neumann problems, Dirichlet problems for unit circle, Poisson's formula, half plan, solutions by conformal                                                                                                                                                                                                                            | <b>Weeks- 13- 14<br/>L-36- 38</b> | CLO9 | PLO4<br>PLO5 | Students will understand and apply the knowledge of Conformal mapping        | #White board discussion<br>#Lecture discussion with multimedia |

|                                |                                                                                                                                                                                |                        |  |  |                                       |  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--|--|---------------------------------------|--|
|                                | mapping, complex potential, sources and sinks, some special flows, Bernoulli's theorem, Coulomb's law, electric field and potential, line charges, conductance and capacitance |                        |  |  |                                       |  |
| <b>Exam.</b>                   |                                                                                                                                                                                | <b>Wee-14<br/>L-39</b> |  |  | Lectures,<br>Questions and<br>Answers |  |
| <b>Recess<br/>before Exam.</b> |                                                                                                                                                                                | <b>Week-15</b>         |  |  |                                       |  |
| <b>Sem. End<br/>Exam.</b>      | <b>Start</b>                                                                                                                                                                   | <b>Week-16</b>         |  |  |                                       |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-<br/>course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|-----------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                         |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                                    |                            |                                    |                                |
| <b>Apply</b>                         |                                         |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                         |                            |                                    |                                |
| <b>Evaluate</b>                      | (05)                                    |                            |                                    |                                |
| <b>Create</b>                        |                                         | (05)                       |                                    |                                |

**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

### **Text Book**

P. R. Vittal; Analytical Geometry 2D and 3D; Pearson

Murray R. Spiegel etl. Complex Variable; Schaum's Outlone

### **Reference Books and Other Materials**

*Barnett Rich and Christopher Thomas; . Geometry; Schaum's Outlone 4th Ed*

*Mark J. Ablowitz and A. S. Fokas; Complex Variable: Introduction and Applications; Cambridge Univ Press 2003, 2<sup>nd</sup> Ed.*

**HUM-1201: Sociology**

Full Marks:100  
Credit: 2.0  
Contact Hours: 26  
Exam. Duration: 4 Hours

**Prerequisite:** Basic Sociology at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course will introduce the students to the fundamental concepts of nature, scope, origin and development of sociology, properties of Social perspectives, Basic Concepts of society and cultural norms and values, Social process and type of society, Social structure and Institutions.

**Course Objectives:** To develop knowledge about inequality of sociology.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                 |
| <b>CLO2</b> | <b>Analyze</b> the characteristics of culture and socialization                              |
| <b>CLO3</b> | <b>Explain</b> the characteristics of pre-industrial, industrial and post-industrial society |
| <b>CLO4</b> | <b>Understand</b> the relation between Environment and Population                            |

| Course Content                    |                                                                                                                                                                                                                                                                                                                                                     | Time Frame                | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                           | Specific Topics                                                                                                                                                                                                                                                                                                                                     |                           | CLO            | PLO         |                                                                                                        |                                                                       |
| <b>Course Overview</b>            | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course<br><b>To grasp</b> the content and policies for the course.                                                                                                                                                                           | <b>Week-1<br/>L-1-2</b>   | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Basic Concept of Sociology</b> | <b>To define</b> , nature and scope, origin and development, relationship with other disciplines and natural sciences, scientific method and techniques for sociological investigation, objectivity in sociology.<br><b>To discuss</b> sociology and commonsense, sociological imaginations, functionalist, conflict, inter-actionist perspectives. | <b>Week-2-3<br/>L-3-6</b> | <b>CLO2</b>    | <b>PLO2</b> | Students will discuss with a group.                                                                    | #Interactive discussion<br>#Lecture discussion with multimedia.       |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |                      |             |                                                                                                         |                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------|-------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Society</b>                         | <b>To discuss</b> Hunter gatherer society to capitalist industrial society and late capitalism.<br><b>To explain</b> the role of Group, society, community, association, institution, organization, culture, material and nonmaterial culture, culture lag, civilization, norms, values, folkways, mores, cultural traits, cultural unity and diversity, acculturation, assimilation, diffusion, enculturation. | <b>Week-4<br/>L-7-8</b>    | <b>CLO2</b>          | <b>PLO2</b> | Students will discuss with a group.                                                                     | #Interactive discussion<br>#Lecture discussion with multimedia |
| <b>Exam Review</b>                     | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-5<br/>L-9-10</b>   |                      |             | Lectures, Questions and Answers                                                                         |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-6<br/>L-11-12</b>  |                      |             | Quiz and Tutorial Exam. Will be taken                                                                   |                                                                |
| <b>Social Process and Institutions</b> | <b>To discuss</b> Cooperation, competition and conflict, socialization and its agent, personality development. Family, marriage, kinship, property, religion, economy and political institutions, functionalist and conflict perspectives about institutions.                                                                                                                                                   | <b>Week-7<br/>L-13-14</b>  | <b>CLO2<br/>CLO3</b> | <b>PLO3</b> | Students will discuss with a group.                                                                     | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Social Structure</b>                | <b>To explain</b> Components of social structure, theories of social structure, social inequality and stratification, class structure, systems of stratification and social mobility, functionalist and conflict perspectives about social stratification.                                                                                                                                                      | <b>Week-8<br/>L-15-16</b>  | <b>CLO2<br/>CLO3</b> | <b>PLO3</b> | Students will discuss with a group.                                                                     | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam Review</b>                     | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-9<br/>L-17</b>     |                      |             | Lectures, Questions and Answers                                                                         |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-9<br/>L-18</b>     |                      |             | Presentation and Tutorial Exam. Will be taken                                                           |                                                                |
| <b>Population and Environment</b>      | <b>To discuss</b> Population growth, ecological balance, ecosystem, threats to global environment, environment as a social issue.                                                                                                                                                                                                                                                                               | <b>Week-10<br/>L-19-20</b> | <b>CLO4<br/>CLO5</b> | <b>PLO4</b> | Students will apply reasoning informed by contextual knowledge to understand population and environment | #White board discussion<br>#Lecture discussion with multimedia |

|                                      |                                                                                                                                                                                                                                                                       |                            |             |             |                                            |                                                                |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------|-------------|--------------------------------------------|----------------------------------------------------------------|
| <b>Social Change and Development</b> | <b>To explain</b> about Social change, factors of change, progress, evolution, development, theories of social change, planned change, technology and social change, social problems, cause and consequences of social problems, social disorganization and deviance. | <b>Week-11<br/>L-21-22</b> | <b>CLO5</b> | <b>PLO4</b> | Students will discuss with a group.        | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>                  | <b>To make an overview</b>                                                                                                                                                                                                                                            | <b>Week-12<br/>L-23-24</b> |             |             | Lectures, Questions and Answers            |                                                                |
| <b>Exam.</b>                         |                                                                                                                                                                                                                                                                       | <b>Week-13<br/>L-25-26</b> |             |             | Assignment and Tutorial Exam will be taken |                                                                |
| <b>Recess before Exam.</b>           |                                                                                                                                                                                                                                                                       | <b>Week-14-15</b>          |             |             |                                            |                                                                |
| <b>Sem. End Exam.</b>                | <b>Start</b>                                                                                                                                                                                                                                                          | <b>Week-16</b>             |             |             |                                            |                                                                |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### Learning Resources

##### **Text Books**

1. *Sociology by Anthony Giddens*
2. *Sociology: A Guide to Problems and Literature by Bottomore*

**Reference Books and Other Materials**

3. *Human Societies by Lenski et al*
4. *Social Change by Moore*
5. *Introduction to Sociology by Pascal Gisbert*
6. *Social Problems by Scarpitti & Anderson*
7. *Lecture Sheet and Presentation Materials*

**ICT-1000: Viva-Voce**  
Based on 1<sup>st</sup> Year 1<sup>st</sup> and 2<sup>nd</sup> Semester.

Full Marks:100  
Credit: 1.0

## B.Sc. (Engg.) Second Year-2023

### 2<sup>nd</sup> Year 1<sup>st</sup> Semester

**ICT-2101: Digital Electronics**

Full Marks:100  
Credit: 3.0  
Contact Hours: 39  
Exam. Duration: 4 Hours

**Prerequisite:** Physics, Mathematics and ICT at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** Digital Electronics course provides an introduction to the control of engineering systems using programmable logic, with specific focus on programmable logic devices. Within this context it introduces the fundamentals of digital logic, digital arithmetic, programmable logic and computer architecture. Research skills and aspects of professional practice are developed through group-based assignments. It emphasizes on techniques to design, analyze, plan, and implement complex digital systems.

**Course Objectives:** To provide knowledge on Digital Electronics including digital systems, different number systems, Boolean algebra, different logic gates, different simplification methods of Boolean expressions, designing techniques of combinational and sequential logic circuits and Digital ICs.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Impart</b> the knowledge on digital systems, different number systems and coding, error-checking in digital signals.          |
| <b>CLO2</b> | <b>Define</b> Boolean algebra and provide the knowledge of truth table, logic functions and different logic gates.               |
| <b>CLO3</b> | <b>Provide</b> the concepts of simplifying Boolean expression using K-map techniques and Quin-McCluskey minimization techniques. |
| <b>CLO4</b> | <b>Understand</b> the techniques of designing and analyzing combinational logic circuits.                                        |
| <b>CLO5</b> | <b>Impart</b> the design methods and analysis of sequential logic circuits.                                                      |
| <b>CLO6</b> | <b>Provide</b> the introductory concepts of digital ICs.                                                                         |

| Course Content      |                                                        | Time Frame       | Alignment with |      | Suggested Activities             | Teaching Strategy(s) |
|---------------------|--------------------------------------------------------|------------------|----------------|------|----------------------------------|----------------------|
| Outlines            | Specific Topics                                        |                  | CLO            | PLO  |                                  |                      |
| <b>Introduction</b> | Digital systems, Advantages of Digital System, Digital | Week -1<br>L-1-2 | CLO1           | PLO1 | -Lecture Presentation with video | -Assignment          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |       |                |                                                                          |                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------|----------------|--------------------------------------------------------------------------|------------------------------------|
|                                        | Computers, System block structure.                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |       |                | -Interactive Discussion<br>-Feedback                                     | - Oral Presentation                |
| <b>Number systems</b>                  | The decimal number system, Binary number system, Binary to Decimal Conversion, Decimal to Binary Conversion, Binary Addition and Subtraction, 1's complement and 2's complement Subtraction, Representation of Binary Numbers as Electrical Signals, Octal Number System, Decimal to Octal Conversion, Binary to Octal Conversion, Octal to Binary Conversion, Usefulness of Octal Number System, Hexadecimal Number System, Binary Coded Decimal (BCD) Code, Gray Code. | Week- 1 2<br>L-3-6   | CLO-1 | PLO-2          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Boolean algebra and logic gates</b> | Boolean algebra, Truth tables, logic functions, simple logic gates                                                                                                                                                                                                                                                                                                                                                                                                       | Week: 3-4<br>L-7-12  | CLO-2 | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Simplifying Boolean expressions</b> | Karnaugh map, Quine-McCluskey method.                                                                                                                                                                                                                                                                                                                                                                                                                                    | Week 5-6<br>L-13-18  | CLO-3 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Combinational logic circuits</b>    | Introduction, Threshold logic, Adder, Subtractor, Decoder, Encoder, Multiplexer                                                                                                                                                                                                                                                                                                                                                                                          | Week 7- 8<br>L-19-24 | CLO-4 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Sequential logic circuits</b>       | Synchronous and asynchronous operation, memory elements, Abstract model of sequential circuits, representation of sequential machines, analysis of sequential circuits.                                                                                                                                                                                                                                                                                                  | Week 9-10<br>L-25-30 | CLO-5 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |

|                                            |                                                                                                                     |                      |       |                |                                                             |                                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------|-------|----------------|-------------------------------------------------------------|------------------------------------|
| <b>Design of sequential logic circuits</b> | Introduction, problem formulation, state state minimization, assignment, circuit implementation, Counter, Register. | Week 1-12<br>L-31-36 | CLO-5 | PLO-4<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion | -Assignment<br>-Class test         |
| <b>Integrated Circuits</b>                 | IC Technology, Basic monolithic IC, Large and Medium scale integration, Digital ICs.                                | Week 13<br>L-37-39   | CLO-6 | PLO-4<br>PLO-5 | -Lecture Presentation<br>-Interactive Discussion            | -Assignment<br>- Oral Presentation |
| <b>Recess before Exam.</b>                 |                                                                                                                     | Week 14-15           |       |                |                                                             |                                    |
| <b>Semester End Exam</b>                   | Start                                                                                                               | Week 16              |       |                |                                                             |                                    |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. Digital Electronics, Malvino, Tata McGraw-Hill.
2. Digital Systems, Tocci Winder.

#### **Reference Books and Other Materials**

3. Digital Electronics, W Gothmann William Gothmann, PHI (Pvt.) Ltd.

4. Switching Theory and Digital Electronics, Dr. V K Jain, Khanna Publishers.

**ICT-2102: Digital Electronics and DLD Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-2101

**Course Type:** Practical/Laboratory work

**Course Summary:** Digital Electronics Lab is helpful for the students to acquire the basic knowledge of digital logic levels and its application to construct digital electronics circuits. This course will prepare students to perform the analysis and design of various digital electronic circuits.

**Course Objectives:** To introduce the basic idea about the fundamental digital logic gate design and construction using diode and transistor.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                            |
|-------------|------------------------------------------------------------|
| <b>CLO1</b> | Gain the practical understanding of different logic gates. |
| <b>CLO2</b> | Design and implement combinational logic circuits.         |
| <b>CLO3</b> | Design and implement sequential logic circuits.            |

| Course Content              |                                                                                                                                                                                                                                        | Time Frame          | Alignment with |                | Suggested Activities                                                                                                      | Teaching Strategy(s)                                       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Outline                     | Specific Topics                                                                                                                                                                                                                        |                     | CLO            | PLO            |                                                                                                                           |                                                            |
| <b>Logic Gates</b>          | <b>To design and verify</b> the AND/OR gates using Diode Logic (DL)<br><b>To design and verify</b> the AND/OR gates using Transistor Transistor Logic (TTL)<br><b>To design and verify</b> NAND and NOR as a universal gates using ICs | Week-1-3<br>L-1-9   | CLO-1          | PLO2<br>PLO3   | -Lecture<br>Interactive Discussion<br>Presentation with video<br><br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
| <b>Boolean Algebra</b>      | <b>To show and verify</b> transistor acts as NOT gate<br><b>Implementation</b> of the given Boolean function using logic gates.                                                                                                        | Week-4-5<br>L-10-15 | CLO-1          | PLO3,<br>PLO4  | Lecture<br>Interactive Discussion<br>Presentation with video<br><br>Experiment Implementation in the assigned Laboratory  | Oral Presentation<br>Assignment<br>Laboratory trial output |
| <b>Adder and Subtractor</b> | <b>To design and verify</b> operation of half-adder and Full-adder circuits using logic gates.<br><b>To design and verify</b> operation of half-subtractor and Full-                                                                   | Week-6-7<br>L-16-21 | CLO-2          | PLO-3<br>PLO-4 | -Lecture<br>Interactive Discussion<br>Presentation with video                                                             | Oral Presentation<br>Assignment<br>Laboratory trial output |

|                              |                                                                                                                                                                                                                                         |                       |       |                |                                                                                                                           |                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|                              | subtractor circuits using logic gates.                                                                                                                                                                                                  |                       |       |                | Experiment Implementation in the assigned Laboratory                                                                      |                                                            |
| <b>Combinational Circuit</b> | <b>Implementation and verification</b> of decoder using logic gates.<br><b>Implementation and verification</b> of encoder using logic gates.<br><b>Implementation and verification</b> of Multiplexer/De-multiplexer using logic gates. | Week-8-10<br>L-17-30  | CLO-2 | PLO-3<br>PLO-4 | -Lecture<br>Interactive Discussion<br>Presentation with video<br><br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
| <b>Sequential Circuit</b>    | <b>Verification</b> of state tables of RS, JK, T and D flip-flop using logic gates.<br><b>Design and verify</b> the 4-bit synchronous counter.                                                                                          | Week-11-12<br>L-31-36 | CLO-3 | PLO-3<br>PLO-4 | -Lecture<br>Interactive Discussion<br>Presentation with video<br><br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|                              | Recess before Exam.                                                                                                                                                                                                                     | Week-14, 15           |       |                |                                                                                                                           |                                                            |
|                              | Semester End Exam.                                                                                                                                                                                                                      | Week-16               | Start |                |                                                                                                                           |                                                            |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. Digital Electronics, Malvino, Tata McGraw-Hill.
2. Digital Systems, Tocci Winder.

**Reference Books and Other Materials**

3. Digital Electronics, W Gothmann William Gothmann, PHI (Pvt.) Ltd.
4. Switching Theory and Digital Electronics, Dr. V K Jain, Khanna Publishers
5. Lab. Guide and Lab. Manuals

**ICT-2103: Data Structure and Algorithm**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Basic C Programming**Course Type:** Theory

**Course Summary:** To provide knowledge on different data structure for computer programming language, complexity of algorithm design, basic operations of linked list, tree, graphs and those algorithms, string processing algorithms, sorting and searching algorithms and their complexity.

**Course Objectives:** Assess how the choice of data structures and algorithm design methods impacts the performance of programs.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                            |
| <b>CLO2</b> | <b>Discuss</b> about the fundamental of data structure                  |
| <b>CLO3</b> | <b>Analyze</b> the properties of linked list and algorithmic complexity |
| <b>CLO4</b> | <b>Understand</b> about STACK and Queue                                 |
| <b>CLO5</b> | <b>Explain and Design</b> process of searching and sorting              |

**Content of the Course**

| Course Content         |                                                                                                                                                                                                                             | Time Frame              | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                                                                             |                         | CLO            | PLO  |                                                                                                        |                                                                       |
| <b>Course Overview</b> | <p><b>To comprehend</b> the mission and vision of the Department.</p> <p><b>To acquire</b> general idea about the course Fundamental and Data Structure.</p> <p><b>To grasp</b> the content and policies for the class.</p> | <b>Week-1<br/>L-1-3</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |



|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                             |      |      |                                                                                                          |                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------|------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Fundamental of Data Structure &amp; Algorithm</b> | <b>To discuss</b> Basic Terminology; Elementary Data organization, Data structure, Data Structure operations, Data types, Logical, abstract view of data: Arrays, Records, Pointers, Structure.                                                                                                                                                                                                                                                     | <b>Week-2<br/>L-4-6</b>     | CLO2 | PLO2 | Students will give brief introduction about themselves<br>Course outline will be discussing in details   | #Socialized recitation<br>#Show presentation to motivate the students       |
| <b>Algorithms Complexity &amp; Linked Lists</b>      | <b>To explain algorithms complexity</b> –Definition and properties of algorithm, data flow-chart, pseudo code, Time-Space Trade off, Complexity Notations, and complexity calculations.<br><b>To discuss linked lists:</b> Introduction, Linked Lists Representation, linear and circular linked list, Traversing, Searching, Insertion and Deletion Algorithms of linked lists, Complexity calculation.                                            | <b>Week-3<br/>L-7-9</b>     | CLO3 | PLO3 | Students will discuss with a group.<br>To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with Linked List algorithms. |
| <b>Exam. Review</b>                                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-4<br/>L-10-12</b>   |      |      | Lectures, Questions and Answers                                                                          |                                                                             |
| <b>Exam.</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-5<br/>L-13-15</b>   |      |      | Quiz and Tutorial Exam. Will be taken                                                                    |                                                                             |
| <b>Stack Queue and Recursion, Trees &amp; Graph</b>  | <b>To discuss dtask queue and recursion:</b> Introduction to Stacks, Queue and recursion, Traversing, Searching, Insertion and Deletion Algorithms of stake, queue and recursion.<br><b>To discuss trees and graphs:</b> Introduction, Representation of different types of trees and graphs, binary search tree, Traversing, Searching, insertion, and Deleting algorithm of trees and graphs, Heap; heap sort, path lengths, Huffman's algorithm. | <b>Week-6-7<br/>L-16-21</b> | CLO4 | PLO4 | Students will identify, analyze and formulate Stack and queue related algorithmic problems.              | #White board discussion<br>#Lecture discussion with Stack and queue         |
| <b>Exam. Review</b>                                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-8<br/>L-22-24</b>   |      |      | Lectures, Questions and Answers                                                                          |                                                                             |
| <b>Exam.</b>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-9<br/>L-25-27</b>   |      |      | Presentation and Tutorial Exam. Will be taken                                                            |                                                                             |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |      |      |                                                                                                                                |                                                                                                                                         |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------|------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>String Processing, Searching &amp; Sorting Algorithm</b> | <b>To analyze string processing:</b> Introduction, basic terminology, storing strings, character data type, string operations, word processing, pattern matching algorithms.<br><b>To explain searching and sorting algorithms:</b> Introduction to searching and sorting, concepts of division and conquer, sorting algorithm: insertion sort, selection sort, quick sort, merge-sort, radix-sort, searching algorithm: binary search, binary search tree, data modification and hashing. | <b>Week-10,11<br/>L-28-33</b> | CLO5 | PLO5 | Students will apply reasoning informed by contextual knowledge to understand different types of searching & sorting algorithm. | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about searching & sorting algorithms to motivate students |
| <b>Exam. Review</b>                                         | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Wee-12<br/>L-34-36</b>     |      |      | Lectures, Questions and Answers                                                                                                |                                                                                                                                         |
| <b>Exam.</b>                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Week-13<br/>L-37-39</b>    |      |      | Assignment and Tutorial Exam will be taken                                                                                     |                                                                                                                                         |
| <b>Recess before Exam.</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Week-14-15</b>             |      |      |                                                                                                                                |                                                                                                                                         |
| <b>Sem. End Exam.</b>                                       | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Week-16</b>                |      |      |                                                                                                                                |                                                                                                                                         |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### Assessment and Evaluation

– Class Attendance & Participation (10%)

- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Data Structure and Algorithm*, Seymour Lipsuitz, Mc Graw Hill (Schaum's Outline Series).
2. *Data Structures using C and C++*, Y Langsam, M J Augenstein, A M Tanenbaum, Prentice Hall India Ltd.

#### **Reference Books and Other Materials**

3. *Java Gently*, J M Bishop, Addison Wesley, ISBN 0-201-59399-8.
4. *Fundamentals of Data structures* by E.Horowitz and S. Sahni
5. *Introduction to Algorithms* by T.H. Cormen, Leiserson

### **ICT-2104: Data Structure and Algorithm Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-2103

**Course Type:** Laboratory

**Course Summary:** To provide practical knowledge of different data structure for computer programming language, use of linked list, tree and graph in real problems, implementation of searching and sorting techniques.

**Course Objectives:** Assess how the choice of data structures and algorithm design methods impacts the performance of programs.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| <b>CLO1</b> | Students can apply Data Structure and Algorithm for solving real life problems. |
| <b>CLO2</b> | Analysis the performance and complexity of sorting and searching algorithm.     |
| <b>CLO3</b> | Implement Array and Linked List for solving real time problems.                 |
| <b>CLO4</b> | Understand different types of Stack and queue, trees and graphs.                |

| Course Content                                        |                                                                                                                                                                                                                              | Time Frame              | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                               | Specific Topics                                                                                                                                                                                                              |                         | CLO            | PLO  |                                                                                                        |                                                                       |
| <b>Introduction to Data Structure &amp; Algorithm</b> | To understand the algorithm of finding the largest element in array, understand the solutions of quadratic equation, find the avg of three numbers, procedure SWITCH to interchanges the values, pattern matching algorithm. | <b>Week-1<br/>L-1-3</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                             |      |              |                                                                                                       |                                                                            |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------|--------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Searching and Sorting</b> | <p><b>To understand</b> the algorithm of linear search, Binary Search and implement them using C/C++/Java/Python.</p> <p><b>To understand</b> the algorithm of selection sort, insertion sort, merge sort, quick sort, heap sort, Bubble sort and implement them using C/C++/Java/Python.</p>                                                                                                                                                                 | <b>Week-2-3<br/>L-4-9</b>   | CLO2 | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with searching and sorting. |
| <b>Exam Review</b>           | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-4<br/>L-10-12</b>   |      |              | Lectures, Questions and Answers                                                                       |                                                                            |
| <b>Examination</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Week-5<br/>L-13-15</b>   |      |              | Quiz and Tutorial Exam. Will be taken                                                                 |                                                                            |
| <b>Array and Linked List</b> | <p><b>To understand</b> the Principle of Recursion, Comparison between Recursion and Iteration, Tail Recursion, Factorial, Fibonacci Sequence, GCD, Tower of Hanoi (TOH) , singly linked list, doubly linked list, circular linked list, basic operation in linked list, node creation, node insertion and deletion from beginning, end and specified position, stack and queue as linked list.</p> <p><b>And Implement</b> them using C/C++/Java/Python.</p> | <b>Week-6-7<br/>L-16-21</b> | CLO3 | PLO1<br>PLO2 | Students will identify, analyze and formulate complex real-time algorithmic problems.                 | #White board discussion<br>#Lecture discussion with Array and Linked list  |
| <b>Exam Review</b>           | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-8<br/>L-22-24</b>   |      |              | Lectures, Questions and Answers                                                                       |                                                                            |
| <b>Exam.</b>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Week-9<br/>L-25-27</b>   |      |              | Presentation and Tutorial Exam. Will be taken                                                         |                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |      |              |                                                                                                                  |                                                                                                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------|--------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stack, Queue and Tree, Graphs</b> | <b>To discuss stack queue and recursion:</b> Introduction to Stacks, Queue and recursion, Traversing, Searching, Insertion and Deletion Algorithms of stack, queue and recursion.<br><b>To explain trees and graphs:</b> Introduction, Representation of different types of trees and graphs, binary search tree, Traversing, Searching, insertion, and Deleting algorithm of trees and graphs, Heap; heap sort, path lengths, Huffman's algorithm and implement above topics using C/C++/Java/Python. | <b>Week-10,11 L-28-33</b> | CLO4 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand different types of Tree and Graphs. | #White board discussion<br>#Lecture discussion with stack and Queue<br>#Show video about Tree and Graph algorithms to motivate students |
| <b>Exam. Review</b>                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Wee-12 L-34-36</b>     |      |              | Lectures, Questions and Answers                                                                                  |                                                                                                                                         |
| <b>Exam.</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-13 L-37-39</b>    |      |              | Assignment and Tutorial Exam will be taken                                                                       |                                                                                                                                         |
| <b>Recess before Exam.</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-14-15</b>         |      |              |                                                                                                                  |                                                                                                                                         |
| <b>Sem. End Exam.</b>                | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Week-16</b>            |      |              |                                                                                                                  |                                                                                                                                         |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Data Structure and Algorithm*, Seymour Lipsuitz, Mc Graw Hill (*Schaum's Outline Series*).
2. *Data Structures using C and C++*, Y Langsam, M J Augenstein, A M Tanenbaum, Prentice Hall India Ltd.

#### **Reference Books and Other Materials**

3. *Java Gently*, J M Bishop, Addison Wesley, ISBN 0-201-59399-8.
4. *Fundamentals of Data structures* by E.Horowitz and S. Sahni
5. *Introduction to Algorithms* by T.H. Cormen, Leiserson

### **ICT-2105: Object Oriented Programming I**

Full Marks:100  
Credit: 3.0  
Contact Hours: 39  
Exam. Duration: 4 Hours

**Prerequisite:** Basic ICT at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course investigates object-oriented methods including object-oriented programming, analysis and design. Current methodology is emphasized. The use of object-oriented features such as encapsulation, information hiding, inheritance and polymorphism is reinforced by class assignments and programming exercises.

**Course Objectives:** To provide knowledge Object oriented computer programming language concepts, data encapsulation, class and objects and their uses and scopes, function and operator overloading, virtual function and polymorphisms, C++ streams, I/O operations, and console managements.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| <b>CLO1</b> | Understand the object and classes relevant to the computer programming languages.  |
| <b>CLO2</b> | Understand the syntax and uses of function & operator overloading.                 |
| <b>CLO3</b> | Understand the syntax and uses of virtual function and polymorphism                |
| <b>CLO4</b> | Understand the Java programming language & its Methods                             |
| <b>CLO5</b> | Understand Standard Java Packages, Socket Programming & Java Database Connectivity |

| Course Content                                      |                                                                                                                                                                                                                                                                                                                                                           | Time Frame                    | Alignment with |             | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                             | Specific Topics                                                                                                                                                                                                                                                                                                                                           |                               | CLO            | PLO         |                                                                                                        |                                                                       |
| <b>Object-oriented concepts, Class, and Objects</b> | <b>To introduce C++.</b><br><b>To discuss</b> about classes, objects, methods, data abstraction and encapsulation, inheritance, polymorphism, dynamic binding and message passing, Declaration of class, Member functions, Defining the object of a class, Accessing a member of class, Array of class objects, Pointers and Classes, Unions and classes, | <b>Week-1</b><br><b>L-1-3</b> | <b>CLO1</b>    | <b>PLO1</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |             |      |                                                                                                       |                                                                                   |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                    | Nested class, Dynamic memory Allocations, Constructors, Destructors, Inline member functions, Static class members, Friend functions, Dynamic memory Allocations                                                                                                                                                                                                                                                                                                                       |                             |             |      |                                                                                                       |                                                                                   |
| <b>Overloading and Inheritance &amp; Pointers, Virtual Functions, Polymorphism</b> | <p><b>To Discuss</b> about Function Overloading, Operator Overloading, Overloading of binary operators, Overloading of Unary operators, Forms of inheritance, Derived Classes, Visibility of Inherited members, Member access control, Abstract base classes, Virtual base classes.(PLO2).</p> <p><b>To Analyze and Explain</b> Polymorphism, Early binding, Polymorphism with pointers, Virtual functions late binding, Pure virtual functions, Managing output with Manipulators</p> | <b>Week-2-3<br/>L-4-9</b>   | <b>CLO2</b> | PLO2 | Students will discuss with a group. To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with Inheritance and Polymorphism. |
| <b>Exam Review</b>                                                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-4<br/>L-10-12</b>   | <b>CLO3</b> | PLO2 | Lectures, Questions and Answers                                                                       |                                                                                   |
| <b>Exam.</b>                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-5<br/>L-13-15</b>   |             |      | Quiz and Tutorial Exam. Will be taken                                                                 |                                                                                   |
| <b>Introduction to Java &amp; Methods</b>                                          | <p><b>To introduce Java:</b> History of Java, Java Feature and advantages, creating classes with Java, Concept of constructors, Using JDK, Java application and Applet, Variables, Data Types, Arrays, Operators and Control Flow.</p> <p><b>To explain methods:</b> Using methods, declaring a class method, Implementation of Inheritance, calling a class method, Passing parameters, Local variables and variable scope.</p>                                                       | <b>Week-6,7<br/>L-16-21</b> | <b>CLO4</b> | PLO3 | Students will identify, analyze and formulate complex real-time programming problems.                 | #White board discussion<br>#Lecture discussion with Java and Methods              |
| <b>Exam Review</b>                                                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-8<br/>L-22-24</b>   |             |      | Lectures, Questions and Answers                                                                       |                                                                                   |
| <b>Exam.</b>                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-9<br/>L-25-27</b>   |             |      | Presentation and Tutorial Exam. Will be taken                                                         |                                                                                   |

|                                                                    |                                                                                                                                                                                                                                                                                                               |                                |      |      |                                                                                                   |                                                                                                                                   |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Using Standard Java Packages and Java Database Connectivity</b> | <b>To create</b> Graphical user interface with AWT, Managing graphics objects with GUI layout Managers, Event handling of various components.<br><b>To analyze and explain</b> JDBC, JDBC drivers, the JAVA SQL packages, SQL, JDBC connection and Executing SQL, the process of building a JAVA application. | <b>Week-10-11<br/>L-28-333</b> | CLO5 | PLO4 | Students will apply reasoning informed by contextual knowledge to understand Java and Mobile Apps | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about various use case of java to motivate students |
| <b>Exam. Review</b>                                                | <b>To make an overview</b>                                                                                                                                                                                                                                                                                    | <b>Week-12<br/>L-33-36</b>     |      |      | Lectures, Questions and Answers                                                                   |                                                                                                                                   |
| <b>Exam.</b>                                                       |                                                                                                                                                                                                                                                                                                               | <b>Week-13<br/>L-37-3</b>      |      |      | Assignment and Tutorial Exam will be taken                                                        |                                                                                                                                   |
| <b>Recess before Exam.</b>                                         |                                                                                                                                                                                                                                                                                                               | <b>Week-14-15</b>              |      |      |                                                                                                   |                                                                                                                                   |
| <b>Sem. End Exam.</b>                                              | <b>Start</b>                                                                                                                                                                                                                                                                                                  | <b>Week-16</b>                 |      |      |                                                                                                   |                                                                                                                                   |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |



|                 |    |
|-----------------|----|
| <b>Evaluate</b> | 05 |
| <b>Create</b>   | 05 |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. *Object-Oriented Programming with C++, E Balagurusamy, Tata McGraw-Hill Publishing Company Ltd Turbo C / C++ The Complete Reference, Herbert Schildt, McGraw-Hill.*
2. *Programming with Java, E Balagurusamy, Tata McGraw-Hill Publishing Company Ltd.*

**Reference Books and Other Materials**

3. *C++ Object-oriented Programming, Mitchell, Macmillan, 1993.*
4. *The Apprentice C++ Programmer, Lee, Phillips, Thompson Computer Press, 1997.*
5. *Mastering Java 2 by John Zukowski*

**ICT-2106: Object Oriented Programming Laboratory I**

Full Marks:100  
Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** Basic ICT at Higher Secondary Level

**Course Type:** Practical

**Course Summary:** This course investigates object-oriented methods including object-oriented programming, analysis and design. Current methodology is emphasized. The use of object-oriented features such as encapsulation, information hiding, inheritance and polymorphism is reinforced by class assignments and programming exercises.

**Course Objectives:** To provide knowledge Object oriented computer programming language concepts, data encapsulation, class and objects and their uses and scopes, function and operator overloading, virtual function and polymorphisms, C++ streams, I/O operations, and console managements.

**Course Learning Outcomes (CLO):**After completing this course a student would be able to

|             |                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Design and Implementation</b> of C++ codes with beginning with C++, Token, expression, and Function in C++.                                             |
| <b>CLO2</b> | <b>Design and Implement</b> of C++ codes with class and objects.                                                                                           |
| <b>CLO3</b> | Design and implementation of C++ codes with Constructor, Destructor, Operator overloading and inheritance.                                                 |
| <b>CLO4</b> | <b>Design and Implementation</b> of C++ codes with polymorphism and Virtual Function.                                                                      |
| <b>CLO5</b> | <b>Design and Implement</b> of Java codes with private and public objects and classes, function overloading, virtual functions and function polymorphisms. |

| Course Content            |                                                                                                                                                                                              | Time Frame | Alignment with |      | Suggested Activities                                   | Teaching Strategy(s)                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|------|--------------------------------------------------------|-------------------------------------------------|
| Outline                   | Specific Topics                                                                                                                                                                              |            | CLO            | PLO  |                                                        |                                                 |
| <b>Beginning with C++</b> | <b>Write and understand</b> a program that reflects scope resolution operator, manipulators, reference variable, type cast operator, different control structure, swapping function, pyramid |            | CLO1           | PLO1 | Students will give brief introduction about themselves | #Socialized recitation<br>#Show presentation to |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |              |              |                                                                                                        |                                                                                                                                   |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------|--------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                                                  | program, sinx and cosx function, different mathematical function.<br><b>To understand</b> a program having concept Function prototyping, call by reference, return by reference, inline function, default arguments, function overloading, friend and virtual function.                                                                                                                                                                                                                                                                                                         |  |              |              | Course outline will be discussing in details                                                           | motivate the students                                                                                                             |
| <b>Class and Objects</b>                         | <b>Write and understand</b> a program that reflects member function, nesting of member function, private member function.<br><b>To understand</b> a program having concept arrays within a class, static data members, static member function, arrays of objects, returning objects, objects as function arguments, pointers to members.                                                                                                                                                                                                                                        |  | CLO2         | PLO3         | Students will discuss with a group. To apply the knowledge to solve the different real-time problems.  | #Interactive discussion<br>#Lecture discussion with Class and objects.                                                            |
| <b>Exam Review</b>                               | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |              |              | Lectures, Questions and Answers                                                                        |                                                                                                                                   |
| <b>Exam.</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |              |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                                                                                   |
| <b>Constructor, Inheritance and Polymorphism</b> | <b>Write and understand</b> a program that reflects parameterize, copy and multiple constructor, Destructor, Overloading binary and unary operator, Single inheritance, multiple inheritance, multilevel inheritance and Hybrid Inheritance.<br><b>To understand</b> a program having concept Pointers to objects, Virtual function and Pure Virtual Function.                                                                                                                                                                                                                  |  | CLO3<br>CLO4 | PLO2<br>PLO3 | Students will identify, analyze and formulate complex real-time programming problems.                  | #White board discussion<br>#Lecture discussion with multimedia                                                                    |
| <b>Exam Review</b>                               | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |              |              | Lectures, Questions and Answers                                                                        |                                                                                                                                   |
| <b>Exam.</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |              |              | Presentation and Tutorial Exam. Will be taken                                                          |                                                                                                                                   |
| <b>Java Programming</b>                          | <b>Write and understand</b> a Java codes with private and public objects and classes, function overloading, virtual functions and function polymorphisms.<br><b>Run and understand</b> a Java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.<br><b>Write a Java program</b> that reads a file and displays the file on the screen, with a line number before each line. Write a Java program that |  | CLO5         | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand Java Programming Language | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about various use case of java to motivate students |

|                            |                                                                    |  |  |  |                                            |  |
|----------------------------|--------------------------------------------------------------------|--|--|--|--------------------------------------------|--|
|                            | displays the number of characters, lines and words in a text file. |  |  |  |                                            |  |
| <b>Exam. Review</b>        | <b>To make an overview</b>                                         |  |  |  | Lectures, Questions and Answers            |  |
| <b>Exam.</b>               |                                                                    |  |  |  | Assignment and Tutorial Exam will be taken |  |
| <b>Recess before Exam.</b> |                                                                    |  |  |  |                                            |  |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                                       |  |  |  |                                            |  |

### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Object-Oriented Programming with C++, E Balagurusamy, Tata McGraw-Hill Publishing Company Ltd Turbo C / C++ The Complete Reference, Herbert Schildt, McGraw-Hill.*
2. *Programming with Java, E Balagurusamy, Tata McGraw-Hill Publishing Company Ltd.*

#### **Reference Books and Other Materials**

3. *C++ Object-oriented Programming, Mitchell, Macmillan, 1993.*
4. *The Apprentice C++ Programmer, Lee, Phillips, Thompson Computer Press, 1997.*
5. *Mastering Java 2 by John Zukowski*  
*The Complete Reference of Java 2 by Herbert Schildt*

**ICT-2107: Signals and Systems**

Full Marks:100  
Credit: 3.0  
Contact Hours: 39  
Exam. Duration: 4 Hours

**Prerequisite:** ICT-1103, MATH-1101 and MATH-1201

**Course Type:** Theory

**Course Summary:** This course is designed to provide the knowledge on signals and systems. The course covers characterization of signals and its classification, LTI systems, Fourier series and Fourier transform representation of both continuous and discrete-time signals, application of Fourier representation, Laplace transform, z-transform and digital processing of continuous-time signal.

**Course Objectives:** To provide exposure on signals and systems concepts to the students, which is highly significant for communication engineering. Besides, the students will acquire sound knowledge for the modeling of signals to design the real systems.

**Course Learning Outcomes (CLO):** After completing this course, a student would be able to

|             |                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> knowledge about the signals and systems with their applications                                                                           |
| <b>CLO2</b> | <b>Know</b> about the LTI systems and its behavior and properties.                                                                                       |
| <b>CLO3</b> | <b>Understand</b> the representation and analysis of signals employing Fourier series and transformations.                                               |
| <b>CLO4</b> | <b>Know</b> about the Laplace and Z-transform of signals.                                                                                                |
| <b>CLO5</b> | <b>Attain</b> the knowledge about the application of Fourier representation and digital processing of Continuous Time Signals in the real system design. |

| Course Content         |                                                                                                                                                                                                                           | Time Frame            | Alignment with |      | Suggested Activities                                                                 | Teaching Strategy(s)                                                  |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------|------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                                                                           |                       | CLO            | PLO  |                                                                                      |                                                                       |
| <b>Course Overview</b> | <b>Comprehend</b> the mission and vision of the ICT Department<br><b>To acquire</b> general idea about the course objectives and fundamentals of signals and systems<br><b>To grasp</b> the content and learning policies | <b>Week-1<br/>L-1</b> | CLO1           | PLO1 | Students will introduce themselves.<br>Course outline will be discussing in details. | #Socialized recitation<br>#Show presentation to motivate the students |

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |              |              |                                                                                                                                                                                                                   |                                                                 |
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| <b>Introduction to Signals and Systems</b>                     | <p><b>To discuss</b> the characterization and classification of signals</p> <p><b>To understand</b> the transformations of the independent variable</p> <p><b>To explain</b> the representation of signals</p> <p><b>To know about</b> the systems and its basic properties</p>                                                                                                                                                                           | <b>Week-1-2<br/>L-2-5</b>    | CLO1<br>CLO2 | PLO2<br>PLO3 | Students will conduct group discussion. Student will understand the characterization and classification of signals, transformations of the independent variable and signals and systems and its basic properties. | #White board discussion<br>#Lecture discussion with multimedia. |
| <b>Linear Time-Invariant (LTI) System</b>                      | <p><b>To explain</b> the discrete and continuous-time LTI systems.</p> <p><b>To know</b> the properties of LTI systems</p> <p><b>To analyze</b> casual LTI systems described by differential and difference equations</p>                                                                                                                                                                                                                                 | <b>Week-2- 3<br/>L-6-8</b>   | CLO1<br>CLO2 | PLO3<br>PLO4 | Student will apply the knowledge to identify LTI system and investigate the properties and response of LTI systems.                                                                                               | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam. Review</b>                                            | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-3,-4<br/>L-9-10</b>  |              |              | Lectures, Questions and Answers                                                                                                                                                                                   |                                                                 |
| <b>Exam.</b>                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Week-4<br/>L-11-12</b>    |              |              | Quiz and Tutorial Exam. Will be taken.                                                                                                                                                                            |                                                                 |
| <b>Fourier Series (FS) Representati on of Periodic Signals</b> | <p><b>To know</b> the FS representation of continuous and discrete-time periodic signals</p> <p><b>To discuss</b> the convergence of the FS</p> <p><b>To acquire</b> knowledge about the properties of continuous and discrete-time FS</p> <p><b>To understand</b> the relation between Fourier series and LTI system</p>                                                                                                                                 | <b>Week-5<br/>L-13-15</b>    | CLO2<br>CLO3 | PLO2<br>PLO3 | Students will represent and analyze the continuous and discrete-time periodic signals in terms of FS. Furthermore, they will understand the FS properties and relation between the Fourier series and LTI system  | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Fourier Transform (FT) Representati on of Signals</b>       | <p><b>To explain</b> the representation of continuous and discrete-time aperiodic signals</p> <p><b>To analyze</b> the FT representation of continuous and discrete-time aperiodic signals</p> <p><b>To show</b> the FT representation of continuous and discrete-time periodic signals</p> <p><b>To understand</b> the properties of continuous and discrete-time FT</p> <p>To Explain the convolution properties of continuous and discrete-time FT</p> | <b>Week- 6-7<br/>L-16-21</b> | CLO2<br>CLO3 | PLO3<br>PLO4 | Students will know the representation of continuous and discrete-time Periodic and aperiodic signals in terms of FT. Furthermore, they will understand the FT properties.                                         | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam Review</b>                                             | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-8<br/>L-22-23</b>    |              |              | Lectures, Questions and Answers                                                                                                                                                                                   |                                                                 |
| <b>Exam.</b>                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Week-8, 9<br/>L-24-25</b> |              |              | Presentation and Tutorial Exam. will be taken                                                                                                                                                                     |                                                                 |

|                                                                   |                                                                                                                                                                                                                                                 |                                |      |              |                                                                                                                                                                                                                                                            |                                                                |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Time and Frequency Characterization of Signals and Systems</b> | <b>To analyze</b> the magnitude-phase representation of FT discuss<br><b>To discuss</b> about the first order and second order continuous and discrete-time systems.                                                                            | <b>Week-9<br/>L-26,-27</b>     | CLO3 | PLO2<br>PLO3 | Students will understand the magnitude-phase representation of FT and also know the first order and second order continuous and discrete-time systems                                                                                                      | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Laplace Transform (LT)</b>                                     | <b>To discuss</b> Laplace transform and its classification<br><b>To explain</b> about the solving of differential equations with initial conditions<br><b>To analyze</b> the LT of systems                                                      | <b>Week-10<br/>L-28-30</b>     | CLO4 | PLO4         | Students will know the LT and its classification, solve differential equations with initial conditions and analyze the LT of systems.                                                                                                                      | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>                                               | <b>To make an overview</b>                                                                                                                                                                                                                      | <b>Week-11<br/>L-31-32</b>     |      |              | Lectures, Questions and Answers                                                                                                                                                                                                                            |                                                                |
| <b>Exam.</b>                                                      |                                                                                                                                                                                                                                                 | <b>Week-11- 12<br/>L-33-34</b> |      |              | Assignment and Tutorial Exam will be taken                                                                                                                                                                                                                 |                                                                |
| <b>z-Transform</b>                                                | <b>To explain</b> z-transform properties of z-transform and ROC, and inversion of z-transform<br><b>To analyze</b> LTI system in terms of z-transform, computational structures for implementing DT systems, and unilateral z-transform.        | <b>Week-12-13<br/>L-35-37</b>  | CLO4 | PLO2<br>PLO3 | Students will understand z-transform, properties of z-transform and ROC, inversion of z-transform. Besides, they will <b>analyze</b> LTI system in terms of z-transform, computational structures for implementing DT systems, and unilateral z-transform. | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Digital Processing of Continuous Time Signals</b>              | <b>To discuss</b> the sampling process, sampling of continuous time and band pass signals<br><b>To design and analyze</b> analog low pass, high pass, band pass band stop filter, anti-aliasing filter design and signal reconstruction filter. | <b>Week-13<br/>L-38-39</b>     | CLO5 | PLO4         | Students will know the sampling process of continuous time and band pass signals and also design and analyze different types of filters.                                                                                                                   | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>                                               | <b>To make an overview</b>                                                                                                                                                                                                                      | <b>Wee-14<br/>L-40</b>         |      |              | Lectures, Questions and Answers                                                                                                                                                                                                                            |                                                                |
| <b>Exam.</b>                                                      |                                                                                                                                                                                                                                                 | <b>Week-14<br/>L-41, 42</b>    |      |              | Assignment and Tutorial Exam will be taken                                                                                                                                                                                                                 |                                                                |
| <b>Recess before Exam.</b>                                        |                                                                                                                                                                                                                                                 | <b>Week-15</b>                 |      |              |                                                                                                                                                                                                                                                            |                                                                |
| <b>Sem. End Exam.</b>                                             | <b>Start</b>                                                                                                                                                                                                                                    | <b>Week-16</b>                 |      |              |                                                                                                                                                                                                                                                            |                                                                |

**Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

#### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### **Learning Resources**

##### ***Text Books***

1. *Signals and Systems, Alan V. Oppenheim and Alan S. Willsky.*
2. *Digital Signal Processing, Sanjit K Mitra*

##### ***Reference Books and Other Materials***

1. *Signals and Systems, Simon Haykin & Barry Van Veen*
2. *Digital Signal Processing, J G Proakis & D G Manolakis*
3. *Lecture Sheet and Presentation Materials*

#### **ICT-2108: Signals and Systems Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-2107

**Course Type:** Practical/Laboratory work

**Course Summary:** This practical course is designed to develop expertise on signals and systems modeling. The practical knowledge on signals and systems modeling is highly

significant in the communication system design. This course covers characterization of signals and its classification, LTI systems, Fourier series and Fourier transform representation of both continuous and discrete-time signals, application of Fourier representation, Laplace transform, z-transform and digital processing of continuous-time signal.

**Course Objectives:** To provide practical knowledge about signals and systems modeling. Moreover, the students will know how to design transceiver model in terms of physical layer baseband signaling for a communication system.

**Course Learning Outcomes (CLO):** After completing this course, a student would be able to:

|             |                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> knowledge about the signals and systems with their applications.                                                                                    |
| <b>CLO2</b> | <b>Know</b> about the modeling of LTI systems.                                                                                                                     |
| <b>CLO3</b> | <b>Understand</b> the representation and analysis of signals employing Fourier series and transformations.                                                         |
| <b>CLO4</b> | <b>Know</b> about the Laplace and Z-transform of signals.                                                                                                          |
| <b>CLO5</b> | <b>Attain</b> the practical knowledge about the application of Fourier representation and digital processing of continuous time signals in the real system design. |

| Course Content            |                                                                                                                                                                                                      | Time Frame                  | Alignment with |              | Suggested Activities                                                                                                                                            | Teaching Strategy(s)                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Outline                   | Specific Topics                                                                                                                                                                                      |                             | CLO            | PLO          |                                                                                                                                                                 |                                                                                                                 |
| <b>Course Overview</b>    | <b>To Acquire</b> general idea about the signals and systems<br>Laboratory<br><b>To Supply</b> the experiment list for this Laboratory.<br>To know about the required instruments and software tools | <b>Week-1<br/>L-1- 2</b>    | CLO1           | PLO1         | Students will give brief introduction about themselves. Course outline will be discussed in details.                                                            | #Socialized recitation<br>#Show the required instruments and software tool to the students which uses this Lab. |
| <b>Signals Generation</b> | <b>To Generate</b> continuous and discrete-time signals using MATLAB<br><b>To Prepare</b> the laboratory report                                                                                      | <b>Week-1,-2<br/>L-3- 5</b> | CLO1           | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge, students will develop signal model and program to generate signals based on the given system model. | #Interactive discussion<br>#Show the sample signal model and familiarize with development tools                 |
| <b>Exam Review</b>        | <b>To make an overview</b>                                                                                                                                                                           | <b>Week-3<br/>L-8--9</b>    |                |              | Demonstrate, Questions and Answers                                                                                                                              |                                                                                                                 |



|                                                                         |                                                                                                                                                                                                                                                              |                                |              |              |                                                                                                                                                                                                                 |                                                                                                                          |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Exam.</b>                                                            |                                                                                                                                                                                                                                                              | <b>Week-4<br/>L-10,-11</b>     |              |              | Quiz and Tutorial Exam. will be taken                                                                                                                                                                           |                                                                                                                          |
| <b>Signal Processing Operations and System Properties Verifications</b> | <b>To Perform</b> basic signal processing operations on continuous and discrete-domain signals using MATLAB<br><b>To Verify</b> properties of a system in both continuous and discrete-domain<br><b>To Prepare</b> the laboratory report                     | <b>Week-4 -5<br/>L-12-14</b>   | CLO1<br>CLO2 | PLO2<br>PLO3 | Students will carry out group discussion. To apply the knowledge, students will perform basic signal processing operations and verify system properties on continuous and discrete-domain.                      | #Interactive discussion<br>#Show the basic signal processing operations<br># Demonstrate how to verify system properties |
| <b>Exam. Review</b>                                                     | <b>To make an overview</b>                                                                                                                                                                                                                                   | <b>Week-5-6<br/>L-15-16</b>    |              |              | Lectures, Questions and Answers                                                                                                                                                                                 |                                                                                                                          |
| <b>Exam.</b>                                                            |                                                                                                                                                                                                                                                              | <b>Week-6<br/>L- 17-18</b>     |              |              | Presentation and tutorial Exam. will be taken                                                                                                                                                                   |                                                                                                                          |
| <b>Convolution of Signals</b>                                           | <b>To Perform</b> convolution of signals in continuous and discrete-time using MATLAB<br><b>To Verify</b> the result of convolution by analytical calculation.<br><b>To Prepare</b> the laboratory report                                                    | <b>Week-7-8<br/>L-19-22</b>    | CLO2         | PLO2         | Students will perform group discussion. To apply the knowledge, students will perform convolution of signals in continuous and discrete-time and verify the result by analytical calculation.                   | #Interactive discussion<br>#Show the mathematical equations of convolution.                                              |
| <b>Fourier Series (FS) Representation of Signal</b>                     | <b>To Find</b> the trigonometric and exponential FS coefficients of a periodic signal.<br><b>To Plot</b> the discrete spectrum of the signal<br><b>To Reconstruct</b> the signal by combining the FS coefficients<br><b>To Prepare</b> the laboratory report | <b>Week-8<br/>L-23-24</b>      | CLO3         | PLO2<br>PLO3 | Students will apply contextual knowledge to calculate the trigonometric and exponential FS coefficients of a periodic signal.                                                                                   | #Interactive discussion<br>#Show the equations of FS coefficients calculation.                                           |
| <b>Exam. Review</b>                                                     | <b>To make an overview</b>                                                                                                                                                                                                                                   | <b>Wee-9<br/>L-25-26</b>       |              |              | Lectures, Questions and Answers                                                                                                                                                                                 |                                                                                                                          |
| <b>Exam.</b>                                                            |                                                                                                                                                                                                                                                              | <b>Week-9-10<br/>L-27-28</b>   |              |              | Presentation and tutorial Exam. will be taken                                                                                                                                                                   |                                                                                                                          |
| <b>Fourier, Laplace and z-Transformations of Signals</b>                | <b>To Perform</b> FT and IFT operations on continuous and discrete-time signals using MATLAB<br><b>To Analyze</b> the amplitude and phase spectrum.<br><b>To Perform</b> LT and z-transforms on signals<br><b>To Prepare</b> the laboratory report           | <b>Week-10-11<br/>L-29- 33</b> | CLO3<br>CLO4 | PLO2<br>PLO3 | Students will perform group discussion. To apply the knowledge, students will perform FT and IFT operations on continuous and discrete-time signals. Besides, they will perform LT and z-transforms on signals. | #Interactive discussion<br>#Show the equations of FT and IFT calculations.                                               |

|                                 |                                                                                                                                                                                                                                            |                          |      |              |                                                                                                                                                                                               |                                                                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Signal Sampling Process</b>  | <b>To generate</b> discrete time sequence by sampling a continuous time signal<br><b>To reconstruct</b> the signal with the generated discrete time sequence considering different Nyquist rate<br><b>To prepare</b> the laboratory report | <b>Week-12 L-34-35</b>   | CLO5 | PLO2<br>PLO3 | Students will perform group discussion. To apply the knowledge, students will apply sampling theorem to generated discrete time sequence by sampling a continuous time signal.                | #Interactive discussion<br>#Show the mathematical model of sampling process.                                            |
| <b>Exam. Review</b>             | <b>To make an overview</b>                                                                                                                                                                                                                 | <b>Wee-12-13 L-36-37</b> |      |              | Lectures, Questions and Answers                                                                                                                                                               |                                                                                                                         |
| <b>Exam.</b>                    |                                                                                                                                                                                                                                            | <b>Week-13 L-38-39</b>   |      |              | Presentation and Tutorial Exam. will be taken                                                                                                                                                 |                                                                                                                         |
| <b>Design of Analog Filters</b> | <b>To design first order low pass and high pass filters.</b><br><b>To observe</b> the response of the filters in logarithmic scale<br><b>To prepare</b> the laboratory report                                                              | <b>Week-14 L- 40-41</b>  | CLO5 | PLO2<br>PLO3 | Students will perform group discussion. To apply the knowledge, students will design first order low pass and high pass filters and observe the response of the filters in logarithmic scale. | #Interactive discussion<br>#Show the practical circuit/mathematical model of first order low pass and high pass filters |
| <b>Exam. Review</b>             | <b>To make an overview</b>                                                                                                                                                                                                                 | <b>Wee-14 L-42</b>       |      |              | Lectures, Questions and Answers                                                                                                                                                               |                                                                                                                         |
| <b>Exam.</b>                    |                                                                                                                                                                                                                                            | <b>Week-15 L-43-44</b>   |      |              | Presentation and Tutorial Exam. will be taken                                                                                                                                                 |                                                                                                                         |
| <b>Recess before</b>            |                                                                                                                                                                                                                                            |                          |      |              |                                                                                                                                                                                               |                                                                                                                         |
| <b>Sem. End Exam.</b>           | <b>Start</b>                                                                                                                                                                                                                               | <b>Week-16</b>           |      |              |                                                                                                                                                                                               |                                                                                                                         |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
|-------------------------|-------------|

|                   |    |
|-------------------|----|
| <b>Remember</b>   | 05 |
| <b>Understand</b> | 05 |
| <b>Apply</b>      | 10 |
| <b>Analyze</b>    | 00 |
| <b>Evaluate</b>   | 05 |
| <b>Create</b>     | 05 |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Signals and Systems, Alan V. Oppenheim and Alan S. Willsky.*
2. *Digital Signal Processing, Sanjit K Mitra*

#### ***Reference Books and Other Materials***

3. *Signals and Systems, Simon Haykin & Barry Van Veen*

## **MATH-2101: Vector Analysis and Linear Algebra**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** HSC Level Mathematics

**Course Type:** Theory

**Course Summary:** This course to develop expertise on representing various situational data in vector and matrix form. Linear relationship among various domain data. Different types of modeling, manipulation and analysis.

**Course Objectives:** To provide knowledge on vector analysis Linear Algebra:

**Course Learning Outcomes (CLO):** After completing this course, a student would be able to:

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                 |
| <b>CLO2</b> | <b>Define and experience</b> with the vector, matrix, vector space and functions.            |
| <b>CLO3</b> | <b>Understand</b> the various operations of vector and matrix, mapping and linear relations. |
| <b>CLO4</b> | <b>Analysis</b> of vector relations                                                          |
| <b>CLO5</b> | <b>Understand</b> the mapping functions.                                                     |
| <b>CLO6</b> | <b>Apply</b> for different data domain                                                       |

| Course content |                 | Time Frame | Alignment with |     | Suggested Activities | Teaching Strategy(s) |
|----------------|-----------------|------------|----------------|-----|----------------------|----------------------|
| Outline        | Specific Topics |            | CLO            | PLO |                      |                      |

|                        |                                                                                                                                                                                                            |                                 |              |              |                                                                                                           |                                                                       |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Course Overview</b> | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Vector Analysis and Linear Algebra<br><b>To grasp</b> the content and policies for the class | <b>Week-1<br/>L-1</b>           | CLO1         | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details    | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Vector Analysis</b> | Definition and Theorems on vector and scalar product of vectors, vector differentiation, vector differential operator, vector integration                                                                  | <b>Week-1,2<br/>L-2-4</b>       | CLO2         | PLO1<br>PLO2 | Students will discuss with a group.<br>To apply the knowledge to explain the different vector operations. | #White board discussion<br>#Lecture discussion with multimedia.       |
| <b>Linear Algebra</b>  | Vector space, subspace, sum and direct sum, Linear dependence and independence, basis and dimension, Linear transformation: Range, kernel, nullity, rank, singular and non-singular transformations        | <b>Week-3,4<br/>L-4-7</b>       | CLO3<br>CLO4 | PLO2<br>PLO3 | Student will apply the knowledge to analyze vector and vectorspace.                                       | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam Review</b>     | <b>To make an overview</b>                                                                                                                                                                                 | <b>Week-4<br/>L-8-9</b>         |              |              | Lectures, Questions and Answers                                                                           |                                                                       |
| <b>Examination</b>     |                                                                                                                                                                                                            | <b>Week-5<br/>L-10,11</b>       |              |              | Quiz and Tutorial Exam. Will be taken                                                                     |                                                                       |
| <b>Matrix</b>          | Matrix representation of a linear operator. Change of basis, similarity                                                                                                                                    | <b>Week-5, 6<br/>L-12-15</b>    | CLO3<br>CLO4 | PLO3         | Students will identify, analyze and formulate complex engineering problems.                               | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Mapping</b>         | Matrices and linear mappings, Characteristic roots and vectors                                                                                                                                             | <b>Week-6,7<br/>L-16-19</b>     | CLO5         | PLO3<br>PLO4 | Students will understand the sinusoidal signal and the calculate the equations.                           | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam Review</b>     | <b>To make an overview</b>                                                                                                                                                                                 | <b>Week-7, 8<br/>L-20,21,22</b> |              |              | Lectures, Questions and Answers                                                                           |                                                                       |
| <b>Exam.</b>           |                                                                                                                                                                                                            | <b>Week-8<br/>L-23-24</b>       |              |              | Presentation and Tutorial Exam. Will be taken                                                             |                                                                       |

|                            |                                                                                                         |                               |      |              |                                                                                         |                                                                |
|----------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------|------|--------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Linear Transform</b>    | Linear transformations, Theorems and problems Characteristic and Minimum Polynomials of square matrices | <b>Week-9,10<br/>L-25-29</b>  | CLO5 | PLO3<br>PLO4 | Students will apply reasoning informed by contextual knowledge to the linear transform. | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Linear function</b>     | Linear functions and dual vector spaces, Annihilators                                                   | <b>Week-10,11<br/>L-30-33</b> | CLO6 | PLO4         | Students understand linear function and vector space                                    | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam. Review</b>        | <b>To make an overview</b>                                                                              | <b>Wee-12<br/>L-34-36</b>     |      |              | Lectures, Questions and Answers                                                         |                                                                |
| <b>Exam.</b>               |                                                                                                         | <b>Week-13<br/>L-37-39</b>    |      |              | Assignment and Tutorial Exam will be taken                                              |                                                                |
| <b>Recess before Exam.</b> |                                                                                                         | <b>Week-14,15</b>             |      |              |                                                                                         |                                                                |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                                                                            | <b>Week-16</b>                |      |              |                                                                                         |                                                                |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                               |                            |                                    |                                |
| <b>Apply</b>                         |                                    |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                    |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                    |                                |
| <b>Create</b>                        |                                    | (05)                       |                                    |                                |

**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |

|        |    |
|--------|----|
| Create | 05 |
|--------|----|

**Text Book**

Elementary Linear Algebra; Larson & Falvo; 6<sup>th</sup> Ed. Huston Pub, Boston, New York 2009

**Reference Books and Other Materials**

*Vector Analysis and Tensor Calculus* by Spigel M R

*Linear Algebra* by Lipschutz, S

*Topics in Algebra* by Herstein, L.N

**B.Sc. (Engg.) Second Year-2023****2<sup>nd</sup> Year 2<sup>nd</sup> Semester****ICT-2201: Database Management Systems**

Full Marks: 100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1102

**Course Type:** Theory

**Course Summary:** This course is introduced about database management systems. This course covers Data Models, SQL, Data Storage & Indexing, Query Processing & Optimization, Transactions & Concurrency, Databases Recovery System, Database System Architecture.

**Course Objectives:** This course focuses on the fundamentals relational database management systems, and the current developments in database theory and their practice.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Understand</b> the different issues involved in the design and implementation of a database system.                                                              |
| <b>CLO2</b> | <b>Study</b> the physical and logical database designs, database modeling, relational, hierarchical, and network models                                             |
| <b>CLO3</b> | <b>Understand</b> and use data manipulation language to query, update, and manage a database                                                                        |
| <b>CLO4</b> | <b>Develop</b> an understanding of essential DBMS concepts such as: storage, database security, integrity, concurrency, distributed database, and Data Warehousing. |
| <b>CLO5</b> | <b>Understand</b> query processing and database recovery techniques.                                                                                                |
| <b>CLO6</b> | <b>Understand</b> different database architectures.                                                                                                                 |

| Course Content |                                                                                                                                                                                          | Time Frame        | Alignment with |       | Suggested Activities                                                     | Teaching Strategy(s)                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|-------|--------------------------------------------------------------------------|-----------------------------------------|
| Outlines       | Specific Topics                                                                                                                                                                          |                   | CLO            | PLO   |                                                                          |                                         |
| Introduction   | Database applications, database vs file systems, view of data, data models, database languages, database users and administrators, transaction management, database system architecture. | Week 1, L-1, 2, 3 | CLO-1          | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br><br>- Oral Presentation |

|                                            |                                                                                                                                                                                                                                                  |                                 |       |                |                                                                          |                                     |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------|----------------|--------------------------------------------------------------------------|-------------------------------------|
| <b>Data Models</b>                         | E-R model-Basic concepts, constraints, keys, E-R diagram, Relational model – structure of Relational database, relational algebra, modification of database and views.                                                                           | Week 2 & 3,<br>L-4,5, 6,7, 8, 9 | CLO-2 | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>SQL</b>                                 | Background, basic structure, set operations, aggregate functions, null values, nested sub-queries, derived relations, views, modification of the database, joined relations, data-definition language, embedded SQL, other SQL features          | Week 4-5<br>L-10-15             | CLO-3 | PLO-2 & PLO-3  | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Data Storage &amp; Indexing</b>         | Overview of physical storage media, tertiary media, file organization, indexing concepts, ordered indices, B <sup>+</sup> tree index files, static hashing, dynamic hashing.                                                                     | Week 6<br>L-16- 18              | CLO-4 | PLO-4 & PLO-5  | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Query Processing &amp; Optimization</b> | Overview, measures of query cost, selection operation, sorting, join, query optimization, choice of evaluation plans.                                                                                                                            | Week 7<br>L-19-21               | CLO-5 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Transactions &amp; Concurrency</b>      | Basic concepts, transaction states, implementation of atomicity and durability, concurrent execution and serializability, concurrency control protocols-lock based, time-stamp based, validation based, multi-version scheme, deadlock handling. | Week -8-9<br>L-22-27            | CLO-4 | PLO-3 & PLO-5  | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>- Oral Presentation |
| <b>Databases Recovery System</b>           | Failure classification, storage structure, Recovery and Atomicity, Log-based Recovery, Shadow Paging, Recovery with concurrent Transactions, Buffer management, Advanced Recovery Techniques.                                                    | Week 10- 11<br>L-28-33          | CLO-5 | PLO-4<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | - Assignment<br>-Class test         |
| <b>Database System Architecture</b>        | Introduction, Centralized and decentralized, Client-server architecture, server system architecture, distributed systems, distributed data storage; distributed transactions commit                                                              | Week 12-13<br>L-34-39           | CLO-6 | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion              | - Assignment<br>-Class test         |

|                            |                                                                                       |            |  |  |  |  |
|----------------------------|---------------------------------------------------------------------------------------|------------|--|--|--|--|
|                            | protocol, parallel systems, I/O parallelism, inter query and intra query parallelism. |            |  |  |  |  |
| <b>Recess before Exam.</b> |                                                                                       | Week 14-15 |  |  |  |  |
| <b>Semester End Exam.</b>  | Start                                                                                 | Week 16    |  |  |  |  |

### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Database System Concepts, A Silberschatz, H Korth, S Sudarshan, McGraw Hill.*
2. *Database Systems – Management and Design, Pratt and Adamski, Boyd and Fraser.*

#### ***Reference Books and Other Materials***

3. *An Introduction to Database System, C J Date, Addison-Wesley*
4. *Data Analysis for Database Design, D R Howe, Arnold.*

### **ICT-2202: Database Management Systems Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-2201

**Course Type:** Practical/Laboratory work



**Course Summary:** The practical knowledge on Database Management Systems.

**Course Objectives:** This course focuses on designing and implementing a database application, which will include a web-based user interface as its front-end and a supporting SQL Server database as its back end.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Have</b> a real hands-on experience by using a full-fledged database management system.       |
| <b>CLO2</b> | <b>Describe</b> fundamental elements of a relational database management systems and data models |
| <b>CLO3</b> | <b>Populate</b> a relational database and formulate SQL queries on the data                      |
| <b>CLO4</b> | <b>Learn</b> practical knowledge on data storage, indexing and query processing                  |
| <b>CLO5</b> | <b>Learn</b> practical knowledge on database recovery and security                               |
| <b>CLO6</b> | <b>Observe</b> practical knowledge on database system architecture                               |

| Course Content                     |                                                                                                                                                                                                                | Time Frame          | Alignment with |                | Suggested Activities                                                                                                | Teaching Strategy(s)                                           |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Outlines                           | Specific Topics                                                                                                                                                                                                |                     | CLO            | PLO            |                                                                                                                     |                                                                |
| Introduction / Course overview     | <b>To acquire</b> general idea about the ICT Laboratory and DBMS laboratory<br><b>To supply</b> the experimental list for this Laboratory.                                                                     | Week 1<br>L-1- 3    | CLO-1          | PLO-1          | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>Data Models</b>                 | Observe characteristics of various data models, Draw E-R diagram, Design Relational model using constraints, keys, relational algebra, modification of database and views.                                     | Week 2-3<br>L-4-9   | CLO-2          | PLO-1          | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>SQL</b>                         | Observe the basic structure of SQL, handling SQL queries specially in the case of set operations, aggregate functions, null values, nested sub-queries, derived relations, views, modification of the database | Week 4-5<br>L-10-15 | CLO-3          | PLO-2<br>PLO-3 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>Data Storage &amp; Indexing</b> | Observe the characteristics of physical storage media, tertiary media, file organization, database indexing                                                                                                    | Week 6<br>L-16-18   | CLO-4          | PLO-4<br>PLO-5 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | Assignment<br>- Oral Presentation<br>-Laboratory trial output  |

|                                            |                                                                                                                                                                                                                                                  |                       |       |                |                                                                                                                     |                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Query Processing &amp; Optimization</b> | Observe the characteristics of query processing and optimization, practical knowledge on query cost measurement, selection operation, sorting, join, query optimization, choice of evaluation plans.                                             | Week 7<br>L-19-21     | CLO-4 | PLO-3<br>PLO-4 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>Transactions &amp; Concurrency</b>      | Basic concepts, transaction states, implementation of atomicity and durability, concurrent execution and serializability, concurrency control protocols-lock based, time-stamp based, validation based, multi-version scheme, deadlock handling. | Week -8-9<br>L-22-27  | CLO-5 | PLO-3<br>PLO-5 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>Databases Recovery System</b>           | Understand practical knowledge on database recovery system, Failure classification, Log-based Recovery, Shadow Paging, Recovery with concurrent Transactions, Buffer management, Advanced Recovery Techniques.                                   | Week 10-11<br>L-28-33 | CLO-5 | PLO-4<br>PLO-5 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
| <b>Database System Architecture</b>        | Observe the basic characteristics of database system architecture like Centralized and decentralized, Client-server architecture, server system architecture, distributed systems, distributed data storage; parallel systems, I/O parallelism   | Week 12-13<br>L-34-39 | CLO-6 | PLO-2<br>PLO-3 | -Lecture<br>Presentation with video, Interactive Discussion<br>Experiment implementation in the assigned laboratory | -Assignment<br>- Oral Presentation<br>-Laboratory trial output |
|                                            | Recess before Exam.                                                                                                                                                                                                                              | Week 14-15            |       |                |                                                                                                                     |                                                                |
|                                            | Semester End Exam. Start                                                                                                                                                                                                                         | Week 16               |       |                |                                                                                                                     |                                                                |

**Assessment Pattern (According FET Ordinance)**

**CIE: (Continuous Internal Evaluation-30Marks)**

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 10          |
| <b>Understand</b>           | 00          |
| <b>Apply</b>                | 15          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 00          |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. *Database System Concepts*, A Silberschatz, H Korth, S Sudarshan, McGraw Hill.
2. *Database Systems – Management and Design*, Pratt and Adamski, Boyd and Fraser.

**Reference Books and Other Materials**

3. *An Introduction to Database System*, C J Date, Addison-Wesley
4. *Data Analysis for Database Design*, D R Howe, Arnold.
5. Lab. Manual

**ICT-2203: Web Technology**

Full Marks:100  
Credit: 3.0  
Contact Hours: 39  
Exam. Duration: 4 Hours

**Prerequisite:** Basic ICT at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course is an introduction to a broad range of topics in Web Technologies. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Web technology classes. The course covers HTML and HTML5, CSS and CSS3, JavaScript, PHP and MySQL, Mobile Apps Development.

**Course Objectives:** The course content enables students to: understand best technologies for solving web client/server problems, analyze and design real time web applications, use HTML5, JavaScript and CSS3 for dynamic effects and to validate form input entry, analyze to Use

appropriate Php applications, understand the use of mobile apps development platform and android studios.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                               |
|-------------|-----------------------------------------------|
| <b>CLO1</b> | To acquire general idea about the course      |
| <b>CLO2</b> | To Discuss about HTML and HTML5               |
| <b>CLO3</b> | Analyze the properties of JavaScript and CSS3 |
| <b>CLO4</b> | Understand about PHP and MySQL                |
| <b>CLO5</b> | Explain and design process of Mobile apps     |

### Content of the Course

| Course Content         |                                                                                                                                                                                                                                                                                                                            | Time Frame                | Alignment with |      | Suggested Activities                                                                                  | Teaching Strategy(s)                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                                                                                                                                                                            |                           | CLO            | PLO  |                                                                                                       |                                                                       |
| <b>Course Overview</b> | <b>To comprehend</b> the mission and vision of the Department.<br><b>To acquire</b> general idea about the course Fundamental and Web Technology.<br><b>To grasp</b> the content and policies for the class.                                                                                                               | <b>Week-1<br/>L-1-3</b>   | CLO1           | PLO1 | Students will give brief introduction about themselves Course outline will be discussing in details   | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>HTML and HTML5</b>  | <b>To discuss</b> about Introduction, Editors, Basic Elements, Attributes of HTML( <b>PLO1</b> ).<br><b>To analyze</b> Formatting, Links, Tables, Lists, Forms, Color of HTML( <b>PLO2</b> ).<br><b>To understand</b> HTML graphics( <b>PLO1</b> ).<br><b>To explain</b> HTML5: Elements, Forms, Graphics, Media and APIs. | <b>Week-2-3<br/>L-4-9</b> | CLO2           | PLO3 | Students will discuss with a group. To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Exam Review</b>     | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                 | <b>Week-4<br/>L-10-12</b> |                |      | Lectures, Questions and Answers                                                                       |                                                                       |
| <b>Exam.</b>           |                                                                                                                                                                                                                                                                                                                            | <b>Week-5<br/>L-13-15</b> |                |      | Quiz and Tutorial Exam. Will be taken                                                                 |                                                                       |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                               |              |              |                                                                                                        |                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>CSS and JavaScript</b>        | <b>To discuss CSS:</b> Introduction, Syntax, ID and Class, Texts, Fonts, Links, Lists, Tables, Selectors, Attributes Selectors, <b>CSS3:</b> Gradients, 2D Transform 3D Transform, Transition, Animations, and Media.<br><b>To explain JavaScript:</b> Introduction, Why JavaScript, Output, Statements, Comments, Variables, Data Types, Objectives, Functions, Operators, Comparisons, Conditions, Switch, Loops, Breaks, Regular Expressions, Ajax, JS HTML DOM . | <b>Week-6-7<br/>L-16-21</b>   | CLO2<br>CLO3 | PLO2<br>PLO3 | Students will identify, analyze and formulate complex real-time web development problems.              | #White board discussion<br>#Lecture discussion with multimedia                                                     |
| <b>Exam Review</b>               | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Week-8<br/>L-22-24</b>     |              |              | Lectures, Questions and Answers                                                                        |                                                                                                                    |
| <b>Exam.</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-9<br/>L-25-27</b>     |              |              | Presentation and Tutorial Exam. Will be taken                                                          |                                                                                                                    |
| <b>Php MySql and Mobile Apps</b> | <b>To discuss the</b> Basic Syntax, Types, Variables, Constants, Expressions, Operators, Control Structures, Functions, Classes and Objects, Exceptions of Php language, HTTP Authentication with PHP, Cookies, Sessions, Database, MySQL, Database Connections.<br><b>To discuss the</b> Introduction of Mobile apps development environments, android studio.                                                                                                      | <b>Week-10,11<br/>L-28-33</b> | CLO4<br>CLO5 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand Php MySql and Mobile Apps | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about Php MySql to motivate students |
| <b>Exam. Review</b>              | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Wee-12<br/>L-34-36</b>     |              |              | Lectures, Questions and Answers                                                                        |                                                                                                                    |
| <b>Exam.</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-13<br/>L-37-39</b>    |              |              | Assignment and Tutorial Exam will be taken                                                             |                                                                                                                    |
| <b>Recess before Exam.</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-14-15</b>             |              |              |                                                                                                        |                                                                                                                    |
| <b>Sem. End Exam.</b>            | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Week-16</b>                |              |              |                                                                                                        |                                                                                                                    |

#### Assessment Pattern (According FET Ordinance)

**CIE: (Continuous Internal Evaluation-30Marks)**

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. Web Technologies, Uttam Roy, OXFORD University press.
2. Web programming with HTML, XHTML and CSS, 2e, Jon Duckett, Wiley India

**Reference Books and Other Materials**

3. Web programming Bai, Michael Ekedahl, CENAGE Learning , India edition.
4. An Introduction to Web Design + Programming, Paul S.Wang, India Edition
5. *Lecture Sheet and Presentation Materials*

**ICT-2204: Web Technology Laboratory**

Full Marks:100  
Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** Basic ICT at Higher Secondary Level

**Course Type:** Laboratory

**Course Summary:** This course is an introduction to a broad range of topics in Web Technologies. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Web technology classes. The course covers HTML and HTML5, CSS and CSS3, JavaScript, PHP and MySQL, Mobile Apps Development.

**Course Objectives:** Understand the web technologies to create adaptive web pages for web application.

**Course Learning Outcomes (CLO):**After completing this course a student would be able to

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Integrate</b> frontend and backend web technologies in distributed systems. |
| <b>CLO2</b> | <b>Facilitate</b> interface between frontend and backend of a web application. |
| <b>CLO3</b> | <b>Debug, Test and Deploy</b> web applications in different web servers.       |
| <b>CLO4</b> | <b>Migrate</b> the web applications to the other platforms like .Net           |

### Content of the Course

| Course Content |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Time Frame                  | Alignment with |              | Suggested Activities                                                                                     | Teaching Strategy(s)                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|--------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline        | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             | CLO            | PLO          |                                                                                                          |                                                                       |
|                | <b>Experiment 1:</b> Design the following static web pages required for a Training and placement cell web site. <ul style="list-style-type: none"> <li>• Home Page</li> <li>• Login Page</li> <li>• Registration Page</li> </ul> <b>Experiment 2:</b> <ul style="list-style-type: none"> <li>• Company Details Page</li> <li>• Alumni Details Page</li> <li>• Placement Staff Details Page</li> </ul> <b>Experiment 3:</b> <ul style="list-style-type: none"> <li>• Student personal Info Page</li> <li>• Student Academic Page Semester Wise Percentage &amp; their Aggregate page</li> </ul> | <b>Week-1<br/>L-1-3</b>     | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details   | #Socialized recitation<br>#Show presentation to motivate the students |
|                | <b>Experiment 4:</b> Validate login page and registration page using regular expressions.<br><b>Experiment 5:</b> Apply different font styles, font families, font colors and other formatting styles to the above static web pages.<br><b>Experiment 6:</b> Install WAMPP sever and tomcat server, access above developed static web pages using these servers.                                                                                                                                                                                                                               | <b>Week-2-3<br/>L-4-9</b>   | CLO2           | PLO3         | Students will discuss with a group.<br>To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with multimedia.       |
|                | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-4<br/>L-10-12</b>   |                |              | Lectures, Questions and Answers                                                                          |                                                                       |
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-5<br/>L-13-15</b>   |                |              | Quiz and Tutorial Exam. Will be taken                                                                    |                                                                       |
|                | <b>Experiment7:</b> Write a servlet/PHP to connect to the database, Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration.                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-6-7<br/>L-16-21</b> | CLO2<br>CLO3   | PLO2<br>PLO3 | Students will identify, analyze and formulate complex real-time web development problems.                | #White board discussion<br>#Lecture discussion with multimedia        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                               |              |              |                                                                                                        |                                                                                                                    |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Experiment8:</b> Write a JSP/PHP to connect to the database, Insert the details of the student academic information with student academic info page.</p> <p><b>Experiment9:</b> User Authentication: Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a servlet for doing the following. 1. Create a Cookie and add these four-user id's and passwords to this Cookie. 2. Read the user id and passwords entered in the Login form (week1) and authenticate with the values (user id and passwords) available in the cookies. If he is a valid user (i.e., user-name and password match) you should welcome him by name (username) else you should display "You are not an authenticated user ". Use init-parameters to do this. Store the user-names and passwords in the webinf.xml and access them in the servlet by using the getInitParameters() method.</p> |                               |              |              |                                                                                                        |                                                                                                                    |
|  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-8<br/>L-22-24</b>     |              |              | Lectures, Questions and Answers                                                                        |                                                                                                                    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Week-9<br/>L-25-27</b>     |              |              | Presentation and Tutorial Exam. Will be taken                                                          |                                                                                                                    |
|  | <p><b>Experiment10:</b> Write a JSP which does the following job: Authenticate the user when he submits the login form using the user name and password from the database.</p> <p><b>Experiment11:</b> write a JSP to insert the student's semester wise percentages and calculate aggregate and insert into database.</p> <p><b>Experiment12:</b> write a JSP to search the students according to their aggregate and produce sorted list or according to their Enroll number.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Week-10,11<br/>L-28-33</b> | CLO4<br>CLO5 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand Php MySql and Mobile Apps | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about Php MySql to motivate students |



|                     |                     |                    |  |  |                                            |  |
|---------------------|---------------------|--------------------|--|--|--------------------------------------------|--|
|                     | To make an overview | Wee-12<br>L-34-36  |  |  | Lectures, Questions and Answers            |  |
|                     |                     | Week-13<br>L-37-39 |  |  | Assignment and Tutorial Exam will be taken |  |
| Recess before Exam. |                     | Week-14-15         |  |  |                                            |  |
| Sem. End Exam.      | Start               | Week-16            |  |  |                                            |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### Text Books

1. Web Technologies, Uttam Roy, OXFORD University press.
2. Web programming with HTML, XHTML and CSS, 2e, Jon Duckett, Wiley India

#### Reference Books and Other Materials

3. Web programming Bai, Michael Ekedahl, CENAGE Learning , India edition.
4. An Introduction to Web Design + Programming, Paul S.Wang, India Edition
5. Lecture Sheet and Presentation Materials

**ICT-2205: Communication Systems**

Full Marks:100  
Credit: 3.0  
Contact Hours: 39  
Exam. Duration: 4 Hours

**Prerequisite:** ICT-2107

**Course Type:** Theory

**Course Summary:** This course provides a comprehensive introduction to basic principles and techniques of communication systems. Lecture topics include: Analog to digital conversion, PCM, baseband transmission, power spectrum density analysis, inter-symbol interference, matched filters, noise analysis, different modulation techniques, coherent and non-coherent detections. Laboratory work is based on simulations in Matlab.

**Course Objectives:** To provide knowledge on communication systems, different types of modulation techniques, noise signals, transmitter system, and receiver system models.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                                             |
| <b>CLO2</b> | <b>Use</b> communication engineering knowledge to understand and design communication systems understand |
| <b>CLO3</b> | <b>Derive</b> mathematical equations for analog and digital signal                                       |
| <b>CLO4</b> | <b>Use</b> modulation technique's knowledge to understand analog and digital modulation techniques       |
| <b>CLO5</b> | <b>Explain</b> the transmitter and receiver circuits.                                                    |

| Course Content                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     | Time Frame                | Alignment with |                      | Suggested Activities                                                                                                              | Teaching Strategy(s)                                                  |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                          | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                                                     |                           | CLO            | PLO                  |                                                                                                                                   |                                                                       |
| <b>Course Overview</b>                           | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course communication systems<br><b>To grasp</b> the content and policies for the class                                                                                                                                                                                                                                       | <b>Week-1<br/>L-1-3</b>   | CLO1           | PLO1                 | Students will give brief introduction about themselves<br>Course outline will be discussing in details                            | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Analog Modulation Demodulation Techniques</b> | <b>To discuss</b> about Communication process, Communication Resources, Sources of information, Communication networks, Communication Channels, Modulation Process, Multiplexing, Basic signal processing operation in digital communication, Channels for digital communication.<br><b>To analyze</b> theory of AM, Mathematical Analysis of AM, Power Relations in AM Wave, Different Forms of AM, AM Modulators, AM Demodulators | <b>Week-2,3<br/>L-4-9</b> | CLO2           | PLO1<br>PLO2<br>PLO3 | Students will discuss with a group.<br>To apply the knowledge to analyze the different communication based mathematical equations | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Exam. Review</b>                              | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-4<br/>L-10-12</b> |                |                      | Lectures, Questions and Answers                                                                                                   |                                                                       |
| <b>Exam.</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-5<br/>L-13-15</b> |                |                      | Quiz and Tutorial Exam. Will be taken                                                                                             |                                                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                 |                             |              |              |                                                                                                 |                                                                                                          |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|--------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Digital Modulation Techniques</b> | <b>To discuss</b> the Digital modulation, Factors that influence the choice of digital modulation, Phase modulation, Pulse modulation: Types, PWM, PPM, Linear modulation: BPSK, DPSK, QPSK, $\pi/4$ QPSK, Offset QPSK, QAM; M-ary modulation techniques, Power spectrum, Bandwidth efficiency.<br><b>To explain</b> the angle modulation, pulse modulation and spread spectrum | <b>Week-6,7<br/>L-16-21</b> | CLO2<br>CLO3 | PLO2<br>PLO3 | Students will identify, analyze and formulate complex communication based engineering problems. | #White board discussion<br>#Lecture discussion with multimedia                                           |
| <b>Exam. Review</b>                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                      | <b>Week-8<br/>L-22-24</b>   |              |              | Lectures, Questions and Answers                                                                 |                                                                                                          |
| <b>Exam.</b>                         |                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-9<br/>L-25-27</b>   |              |              | Presentation and Tutorial Exam. Will be taken                                                   |                                                                                                          |
| <b>Transmitter</b>                   | <b>To classify</b> the transmitters, AM Transmitters, Carrier Frequency Requirement, Broadcast AM Transmitters, FM Transmitters, Comparison of Transmitters.                                                                                                                                                                                                                    | <b>Week-10<br/>L-28-30</b>  | CLO4<br>CLO5 | PLO3<br>PLO6 | Students will apply reasoning informed by contextual knowledge to understand about transmitter  | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about transmitter circuit. |
| <b>Receiver</b>                      | <b>To classify</b> the receivers, Salient Features and Basic Functions of receiver, TRF Receiver, Principle of Super heterodyne Receiver, AM Receivers, FM Receivers, Comparison of receivers.                                                                                                                                                                                  | <b>Week-11<br/>L-31-33</b>  | CLO4<br>CLO5 | PLO3<br>PLO6 | Students will apply reasoning informed by contextual knowledge to understand about receiver     | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about receiver             |
| <b>Exam. Review</b>                  | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                      | <b>Week-12<br/>L-34-36</b>  |              |              | Lectures, Questions and Answers                                                                 |                                                                                                          |
| <b>Exam.</b>                         |                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-13<br/>L-37-39</b>  |              |              | Assignment and Tutorial Exam will be taken                                                      |                                                                                                          |
| <b>Recess before Exam.</b>           |                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-14,15</b>           |              |              |                                                                                                 |                                                                                                          |
| <b>Sem. End Exam.</b>                | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>Week-16</b>              |              |              |                                                                                                 |                                                                                                          |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 05   |
| Understand          | 05   |
| Apply               | 10   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 05   |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources**

***Text Books***

1. *Communication Systems*, S. Haykin
2. *Analog Communication*, V. Chandra Sekar

***Reference Books and Other Materials***

3. *Principle of Communication Engineering*, Anokh Singh
4. *Fundamentals of Communication Systems*, J. G. Proakis, M. Salehi

**ICT-2206:      Communication Systems Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-2205

**Course Type:** Practical/Laboratory work

**Course Summary:** This course gives students deep knowledge in communication systems at the practical level. This lab focuses on the design and constructs transmitter circuit, to verify different analog and digital modulation techniques, to calculate bandwidth and power saving for different modulation techniques.

**Course Objectives:** This course will introduce the students to hands on knowledge on modeling and implementation of modulation techniques, analysis and evaluation of modulation techniques, antenna design and implementation.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                         |
| <b>CLO2</b> | <b>Verify</b> the different analog and digital modulation techniques |

|             |                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------|
| <b>CLO3</b> | <b>Able</b> to design and implement different demodulation techniques.                               |
| <b>CLO4</b> | <b>Calculate</b> Bandwidth and power saving of different modulation techniques by using MATLAB tools |
| <b>CLO5</b> | <b>Design and construct</b> transmitter and receiver circuit                                         |

### Content of the Course

| Course Content                                    |                                                                                                                                                                                                                                                                  | Time Frame                  | Alignment with |                      | Suggested Activities                                                                                                              | Teaching Strategy(s)                                                                           |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Outline                                           | Specific Topics                                                                                                                                                                                                                                                  |                             | CLO            | PLO                  |                                                                                                                                   |                                                                                                |
| <b>Course Overview</b>                            | <b>To acquire</b> general idea about the Communication Systems Laboratory.<br><b>To supply</b> the experimental list for this Laboratory.<br><b>To discuss</b> about the simulator tools                                                                         | <b>Week-1<br/>L-1-3</b>     | CLO1           | PLO1                 | Students will give brief introduction about themselves. Course outline will be discussing in details                              | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab. |
| <b>Analog Modulation /Demodulation Techniques</b> | <b>To generate</b> different analog modulation circuit and observe its properties<br><b>To generate</b> different analog demodulation circuit and observe its properties<br><b>To calculate</b> modulation index for different modulation techniques             | <b>Week-2,3<br/>L-4-9</b>   | CLO2<br>CLO3   | PLO2<br>PLO3<br>PLO4 | Students will discuss with a group. To apply the knowledge to calculate the different properties of analog modulation techniques. | #Interactive discussion<br>#Show the video to calculate modulation properties.                 |
| <b>Exam Review</b>                                | <b>To make an overview</b>                                                                                                                                                                                                                                       | <b>Week-4<br/>L-10-12</b>   |                |                      | Demonstrate, Questions and Answers                                                                                                |                                                                                                |
| <b>Exam.</b>                                      |                                                                                                                                                                                                                                                                  | <b>Week-5<br/>L-13-15</b>   |                |                      | Quiz and Tutorial Exam. Will be taken                                                                                             |                                                                                                |
| <b>Digital Modulation Techniques</b>              | <b>To generate</b> different digital modulation circuit and observe its properties<br><b>To generate</b> different digital demodulation circuit and observe its properties<br><b>To calculate</b> bandwidth and power saving for different modulation techniques | <b>Week-6,7<br/>L-16-21</b> | CLO3<br>CLO4   | PLO2<br>PLO3<br>PLO4 | Students apply the knowledge to calculate the different properties of digital modulation techniques.                              | #White board discussion<br>#Lecture discussion with multimedia                                 |
| <b>Exam Review</b>                                | <b>To make an overview</b>                                                                                                                                                                                                                                       | <b>Week-8<br/>L-22-23</b>   |                |                      | Lectures, Questions and Answers                                                                                                   |                                                                                                |
| <b>Exam.</b>                                      |                                                                                                                                                                                                                                                                  | <b>Week-8,9<br/>L-24-25</b> |                |                      | Presentation and Tutorial Exam. Will be taken                                                                                     |                                                                                                |

|                            |                                              |                               |      |              |                                                                                       |                                                                                    |
|----------------------------|----------------------------------------------|-------------------------------|------|--------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Transmitter</b>         | To design and construct transmitter circuit. | <b>W-9<br/>L-26-27</b>        | CLO5 | PLO4         | Students apply the theoretical knowledge to design and construct transmitter circuit. |                                                                                    |
| <b>Receiver</b>            | To design and construct transmitter circuit. | <b>Week-10,11<br/>L-28-33</b> | CLO5 | PLO3<br>PLO6 | Students will apply contextual knowledge to design and construct receiver circuit.    | #Lecture discussion with multimedia<br>#Show video about different web technology. |
| <b>Exam. Review</b>        | To make an overview                          | <b>Wee-12<br/>L-34,35,36</b>  |      |              | Lectures, Questions and Answers                                                       |                                                                                    |
| <b>Exam.</b>               |                                              | <b>Week-13<br/>L-37-39</b>    |      |              | Presentation and Tutorial Exam. will be taken                                         |                                                                                    |
| <b>Recess before Exam.</b> |                                              | <b>Week-14,15</b>             |      |              |                                                                                       |                                                                                    |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                 | <b>Week-16</b>                |      |              |                                                                                       |                                                                                    |

#### Assessment Pattern (According FET Ordinance)

##### CIE: (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

##### SMEE: (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 10          |
| <b>Understand</b>           | 00          |
| <b>Apply</b>                | 15          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 00          |

#### Assessment and Evaluation

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)

– Semester Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Communication Systems*, S. Haykin
2. *Analog Communication*, V. Chandra Sekar

#### ***Reference Books and Other Materials***

3. *Fundamentals of Communication Systems*, J. G. Proakis, M. Salehi
4. *Lab. Guide and Lab. Manuals*

### **STAT-2201: Statistics for Communication Engineering**

Full Marks 100

Credit 2.0

Contact Hours 39

Exam Duration 4 h

**Prerequisite:** HSC Level Mathematics

**Course Type:** Theory

**Course Summary:** This course to develop expertise on representing data, data uncertainty i.e. probability measurement, properties of random variable, probability models, hypothesis testing and parameter estimation

**Course Objectives:** To provide knowledge on probability, probability distribution, modeling of data and parameter estimation.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                           |
| <b>CLO2</b> | <b>Understand</b> the role of statistics in engineering                |
| <b>CLO3</b> | <b>Organize and representation</b> of Data                             |
| <b>CLO4</b> | <b>Handle</b> the uncertainty of model using probability               |
| <b>CLO5</b> | <b>Model</b> the random system using relevant probability distribution |
| <b>CLO6</b> | <b>Understand</b> Hypothesis and conclusion                            |
| <b>CLO7</b> | <b>Estimate</b> model parameters                                       |

### **Content of the Course**

| <b>Course Content</b>  |                                                                                                                                                                                                                  | <b>Time Frame</b> | <b>Alignment with</b> |            | <b>Suggested Activities</b>                                                                            | <b>Teaching Strategy(s)</b>                                           |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------|------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Outline</b>         | <b>Specific Topics</b>                                                                                                                                                                                           |                   | <b>CLO</b>            | <b>PLO</b> |                                                                                                        |                                                                       |
| <b>Course Overview</b> | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Statistics for Communication Engineering<br><b>To grasp</b> the content and policies for the class | <b>Week-1 L-1</b> | CLO1                  | PLO1       | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |              |              |                                                                                                                                                            |                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Data Collection, and Representation</b> | Meaning and Scope, Variables and Attributes, statistical data collection methods, sample space, Population, event, statistical data organization and presentations table and charts: Charts and Pareto Charts, Pie Charts, Histograms Outliers, Box-and-Whisker Plot.                                                                                                                                                                                                      | <b>Week-1-2<br/>L-2-6</b> | CLO2<br>CLO3 | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to explain the different format of data, frequency distribution and data representation graphs. | #White board discussion<br>#Lecture discussion with multimedia. |
| <b>Random Variable</b>                     | Discrete Random Variables, Continuous Random Variables, The Expectation of a Random Variable, The Variance of a Random Variable, Jointly Distributed Random Variables, Combinations and Functions of Random Variables<br><b>To make an overview</b>                                                                                                                                                                                                                        | <b>Week-3<br/>L-7-9</b>   | CLO3         | PLO2         | Students will understand estimate and analysis the relations of random variables.                                                                          | #White board discussion<br>#Lecture discussion with multimedia. |
| <b>Exam.</b>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Week-4<br/>L-10-12</b> |              |              | Quiz and Tutorial Exam. Will be taken                                                                                                                      |                                                                 |
| <b>Probability Distribution</b>            | Discrete Probability Distributions: The Binomial Distribution, Geometric and Negative Binomial Distributions, Poisson distribution, The Multinomial Distribution                                                                                                                                                                                                                                                                                                           | <b>Week-5<br/>L-13-15</b> | CLO4<br>CLO5 | PLO3         | Students will identify, analyze and formulate complex engineering problems.                                                                                | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Probability Distribution</b>            | Continuous Probability Distributions: The Uniform Distribution, The Exponential Distribution, The Gamma Distribution, The Beta Distribution, Normal Distribution: Probability Calculations Using the Normal Distribution, Linear Combinations of Normal Random Variables, Approximating Distributions with the Normal Distribution, Distributions Related to the Normal Distribution: Log-normal Distribution, Chi-square Distribution, T-Distribution and F-Distribution. | <b>Week-6<br/>L-16-18</b> | CLO4<br>CLO5 | PLO3<br>PLO4 | Students will understand the probability theories                                                                                                          | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam Review</b>                         | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-7<br/>L-19-20</b> |              |              | Lectures, Questions and Answers                                                                                                                            |                                                                 |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                |                               |              |              |                                                               |                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|---------------------------------------------------------------|----------------------------------------------------------------|
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-7<br/>L-21</b>        |              |              | Presentation and Tutorial Exam. Will be taken                 |                                                                |
| <b>Statistical Estimation</b>          | Statistical Estimation and Sampling Distributions: Point Estimates, unbiased and biased Point Estimates, Point Estimates of a population mean, Point Estimates of a population variance, Relative Efficiency, Sample Proportion, sample mean, sample variance, Constructing Parameter Estimates: The Method of Moments, Maximum Likelihood Estimates.                          | <b>Week-8-9<br/>L-22-27</b>   | CLO7         | PLO3<br>PLO4 | Students will apply reasoning informed statistical estimation | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Inferences on a Population Mean</b> | Confidence Interval Construction, Effect of the Sample Size on Confidence Intervals, One-Sided Confidence Intervals, z-Intervals, Hypothesis Testing: Rejection and Acceptance of the Null Hypothesis, Interpretation of p-Values, Calculation of p-Values, Significance Levels, Relationship between Confidence Intervals and Hypothesis Tests, One-side and two-side z-Tests | <b>Week-10-11<br/>L-28-33</b> | CLO6         | PLO4         | Students will understand the inference theories               | #White board discussion<br>#Lecture discussion with multimedia |
| <b>regression</b>                      | Simple Linear regression and correlations: model and parameter estimation                                                                                                                                                                                                                                                                                                      | <b>Week-12<br/>L-34-36</b>    | CLO4<br>CLO7 | PLO12        | Students will classify and apply the filter circuits.         | #White board discussion                                        |
| <b>Exam. Review</b>                    | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                     | <b>Week-13<br/>L-37-38</b>    |              |              | Lectures, Questions and Answers                               |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-13<br/>L-39</b>       |              |              | Assignment and Tutorial Exam will be taken                    |                                                                |
| <b>Recess before Exam.</b>             |                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-14,15</b>             |              |              |                                                               |                                                                |
| <b>Sem. End Exam.</b>                  | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                   | <b>Week-16</b>                |              |              |                                                               |                                                                |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                               |                            |                                    |                                |

|          |      |      |  |      |
|----------|------|------|--|------|
| Apply    |      |      |  | (10) |
| Analyze  |      |      |  |      |
| Evaluate | (05) |      |  |      |
| Create   |      | (05) |  |      |

**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

### Learning Resources

#### **Text Book**

Anthony Hayter, Probability and Statistics for Engineers and Scientists; Fourth Ed. Brooks/Cole, Cengage Learning, USA

Beginning Statistics, Schaum's Outline Series

#### **Reference Books and Other Materials**

LARRY J. STEPHENS, Ph.D. *Theory and problems of beginning statistics*, McGraw Hill, Schaum's series.

Douglas C. Montgomery, George C. Runger, *Applied statistics and probability for engineers*, Third Ed.

John Wiley & Sons, USA.

T.T. Soong, *Fundamentals of Probability and Statistics for Engineers*, 2004, John Wiley & Sons Ltd, England

#### **ICT-2207: Object Oriented Programming II**

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1203, ICT-2105

**Course Type:** Theory

**Course Summary:** This course explores advanced topics of contemporary object-oriented programming including programming paradigms, design patterns, generics, collection framework, threads, errors and exceptions handling, synchronization, networking programming.

**Course Objectives:** Students will have the opportunity to learn the object-oriented techniques used to develop programs and to attain software reliability. The goal of the course is to develop skills such program design, code reuse, collections, threads, error and exception handling, synchronize threads, and networking. All programs will be practiced in any contemporary programming languages (Java, Python, C++).

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>To understand and design</b> problems using various techniques                                                                                                                 |
| <b>CLO2</b> | <b>To understand and create</b> classes and collection of classes that work with different data types and support various ways to store and manage objects of any kind in memory. |
| <b>CLO3</b> | <b>To grasp, learn, and design</b> simultaneous execution of multiple processes.                                                                                                  |
| <b>CLO4</b> | <b>To understand and handle</b> errors and exceptions.                                                                                                                            |
| <b>CLO5</b> | <b>To learn</b> synchronization to manage dead locks.                                                                                                                             |
| <b>CLO6</b> | <b>To understand, design, and develop</b> socket programs.                                                                                                                        |

| Content of the Course                            |                                                                                                                                                                                                                                                                                                                                                                | Time Frame                | Alignment with       |                      | Suggested Activities                                                                                                                   | Teaching Strategy(s)                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                          | Specific Topics                                                                                                                                                                                                                                                                                                                                                |                           | CLO                  | PLO                  |                                                                                                                                        |                                                                       |
| <b>Course Overview and Programming paradigms</b> | To <b>comprehend</b> overall basics of this course.<br>To <b>acquire</b> idea programming paradigms, imperative programming paradigm, procedural programming paradigm, object-oriented programming, declarative programming paradigm, functional programming paradigms, database/data driven programming approach and <b>design</b> problems on those concepts | <b>Week-1<br/>L-1-2</b>   | <b>CLO1</b>          | <b>PLO1<br/>PLO3</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details                                 | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Design Patterns</b>                           | To <b>design</b> and <b>develop</b> problems based on creational pattern, structural pattern, behavioral pattern, intercepting pattern, front controller pattern.                                                                                                                                                                                              | <b>Week-1,2<br/>L-3-4</b> | <b>CLO1</b>          | <b>PLO3<br/>PLO5</b> | Students will discuss with a group.<br>To apply the knowledge of software design patterns to solve the different programming problems. | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Generics</b>                                  | To <b>acquire</b> the concepts of generic, generic types: generic methods and generic classes, type parameters in generics, generic interfaces, and <b>design</b> problems using the above concepts.                                                                                                                                                           | <b>Week-2<br/>L-5-6</b>   | <b>CLO1<br/>CLO2</b> | <b>PLO1<br/>PLO3</b> | Student will apply the knowledge to simplify programming problem with the concept of code reuse.                                       | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Collection Framework</b>                      | To <b>understand</b> the concepts of iterators, abstract class, collection classes and interfaces, using vectors and apply these to <b>design</b> and <b>develop</b> problems.                                                                                                                                                                                 | <b>Week-3<br/>L-7-9</b>   | <b>CLO1<br/>CLO2</b> | <b>PLO1<br/>PLO3</b> | Student will learn collections and use the concepts in programming.                                                                    | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Exam Review</b>                               | To <b>make an overview</b>                                                                                                                                                                                                                                                                                                                                     | <b>Week-4<br/>L-10-12</b> |                      |                      | Lectures, Questions and Answers                                                                                                        |                                                                       |
| <b>Exam.</b>                                     |                                                                                                                                                                                                                                                                                                                                                                | <b>Week-5<br/>L-13-15</b> |                      |                      | Quiz and Tutorial Exam. Will be taken                                                                                                  |                                                                       |
| <b>Threads and Multithreads</b>                  | To <b>understand</b> threads and multithreads and life cycle of a thread, threads priorities.<br>To <b>create</b> and <b>manage</b> threads.                                                                                                                                                                                                                   | <b>Week-6<br/>L-16-17</b> | <b>CLO1<br/>CLO3</b> | <b>PLO2<br/>PLO5</b> | Students will identify, analyze and formulate complex programming problems.                                                            | #Interactive discussion<br>#Lecture discussion with multimedia.       |

|                                |                                                                                                                                                                                                                    |                               |                      |                      |                                                                                 |                                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|----------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Errors and Exceptions</b>   | To <b>grasp</b> the types of exceptions, managing exceptions, exception objects, and exception chaining. To <b>raise</b> and <b>handle</b> multiple unrelated exceptions                                           | <b>Week-6,7<br/>L-18-19</b>   | <b>CLO4</b>          | <b>PLO2<br/>PLO4</b> | Students will identify, analyze and formulate complex programming problems.     | #Interactive discussion<br>#Lecture discussion with multimedia. |
| <b>Synchronization</b>         | To <b>understand</b> synchronization methods, synchronization block, static synchronization, deadlock, inter-thread communication, interrupting threads, reentrant monitors and <b>design</b> the problems.        | <b>Week-7<br/>L-20-21</b>     | <b>CLO5</b>          | <b>PLO2<br/>PLO3</b> | Students will analyze, learn, and formulate complex programming problems.       | #Interactive discussion<br>#Lecture discussion with multimedia. |
| <b>Exam Review</b>             | <b>To make an overview</b>                                                                                                                                                                                         | <b>Week-8<br/>L-22-24</b>     |                      |                      | Lectures, Questions and Answers                                                 |                                                                 |
| <b>Exam.</b>                   |                                                                                                                                                                                                                    | <b>Week-9<br/>L-25-27</b>     |                      |                      | Presentation and Tutorial Exam. Will be taken                                   |                                                                 |
| <b>Networking Program ming</b> | To <b>learn</b> networking concepts, common networking protocol networking interfaces, socket programming, socket class, InetAddress class, URL class, DatagramSocket class. To <b>design</b> networking problems. | <b>Week-10,11<br/>L-28-33</b> | <b>CLO1<br/>CLO6</b> | <b>PLO2<br/>PLO5</b> | Students will analyze, learn, design, and develop complex programming problems. | #Interactive discussion<br>#Lecture discussion with multimedia. |
| <b>Exam. Review</b>            | <b>To make an overview</b>                                                                                                                                                                                         | <b>Week-12<br/>L-34-36</b>    |                      |                      | Lectures, Questions and Answers                                                 |                                                                 |
| <b>Exam.</b>                   |                                                                                                                                                                                                                    | <b>Week-13<br/>L-37-39</b>    |                      |                      | Assignment and Tutorial Exam will be taken                                      |                                                                 |
| <b>Recess before Exam.</b>     |                                                                                                                                                                                                                    | <b>Week-14,15</b>             |                      |                      |                                                                                 |                                                                 |
| <b>Sem. End Exam.</b>          | <b>Start</b>                                                                                                                                                                                                       | <b>Week-16</b>                |                      |                      |                                                                                 |                                                                 |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |

|            |    |
|------------|----|
| Understand | 05 |
| Apply      | 10 |
| Analyze    | 00 |
| Evaluate   | 05 |
| Create     | 05 |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. Ivor Horton's *Beginning Java 2 JDK 5 Edition*, Wrox;

#### ***Reference Books and Other Materials***

3. <http://www.cs.umd.edu/class/fall2022/cmsc132-030X/>
4. <https://docs.python.org/3/tutorial/>
5. <https://sites.google.com/site/oopunyt/syllabus>

## **ICT-2208: Object Oriented Programming Laboratory II**

Full Marks:100

Credit: 1.0

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-1203, ICT-2105, ICT-2207

**Course Type:** Theory

**Course Summary:** This course explores advanced topics of contemporary object-oriented programming including programming paradigms, design patterns, generics, collection framework, threads, errors and exceptions handling, synchronization, networking programming.

**Course Objectives:** Students will have the opportunity to learn the object-oriented techniques used to develop programs and to attain software reliability. The goal of the course is to develop skills such program design, code reuse, collections, threads, error and exception handling, synchronize threads, and networking. All programs will be practiced in any contemporary programming languages (Java, Python, C++).

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Attain competence</b> on the use of Java/Python/C++ to develop programming problems.                                                                            |
| <b>CLO2</b> | <b>To design and develops</b> problems using programming paradigms and designs patterns and <b>grasp</b> basic concepts of object oriented programming.            |
| <b>CLO3</b> | <b>To create</b> classes and collection of classes that work with different data types and support various ways to store and manage objects of any kind in memory. |
| <b>CLO4</b> | <b>To design and solve</b> simultaneous execution of multiple processes.                                                                                           |
| <b>CLO5</b> | <b>To handle</b> errors and exceptions.                                                                                                                            |
| <b>CLO6</b> | <b>To learn and design</b> synchronization to manage dead locks.                                                                                                   |
| <b>CLO7</b> | <b>To design and develop</b> socket programs.                                                                                                                      |

| Content of the Course |                 | Time Frame | Alignment with |     | Suggested Activities | Teaching Strategy(s) |
|-----------------------|-----------------|------------|----------------|-----|----------------------|----------------------|
| Outline               | Specific Topics |            | CLO            | PLO |                      |                      |

|                                                                                                                        |                                                                                                                                                                                            |                                   |                            |                                           |                                                                                                                                        |                                                                               |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Exp #1:</b><br>Fundamentals of Java/Python/C++ and Installation                                                     | <b>To learn</b> Java/Python/C++ installation<br><b>To acquire</b> general idea about Java/Python/C++ programming                                                                           | <b>Week-1</b><br><b>L-1-2</b>     | <b>CLO1</b>                | <b>PLO1</b><br><b>PLO3</b>                | Students will give brief introduction about themselves<br>Course outline will be discussing in details                                 | #Socialized recitation<br>#Lecture with the ppt and online videos             |
| <b>Exp 2#:</b><br>Creational Patterns<br><b>Exp 3#:</b><br>Structural pattern<br><b>Exp 4#:</b><br>Behavioral patterns | To <b>learn</b> creational pattern, structural pattern, and behavioral pattern.<br><br>To <b>design</b> creational pattern, structural pattern, and behavioral pattern.                    | <b>Week-1,2</b><br><b>L-3-4</b>   | <b>CLO2</b>                | <b>PLO1</b><br><b>PLO2</b><br><b>PLO3</b> | Students will discuss with a group.<br>To apply the knowledge of software design patterns to solve the different programming problems. | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exp 5#:</b><br>Generic Methods<br><b>Exp 6#:</b><br>Generic Classes<br><b>Exp 7#:</b><br>Generic Interfaces         | To <b>understand</b> generic methods , generic classes, and generic interfaces.<br><br>To <b>design</b> problems using generic methods, generic classes, and generic interfaces.           | <b>Week-2</b><br><b>L-5-6</b>     | <b>CLO2</b><br><b>CLO3</b> | <b>PLO2</b><br><b>PLO3</b>                | Student will apply the knowledge to simplify programming problem with the concept of code reuse.                                       | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exp 8#:</b><br>Iterators<br><b>Exp 9#:</b><br>Interfaces<br><b>Exp 10#:</b><br>Vectors                              | To <b>understand</b> iterators, interfaces, and vectors.<br><br>To <b>design and develop</b> problems based on iterators, interfaces, and vectors.                                         | <b>Week-3</b><br><b>L-7-9</b>     | <b>CLO2</b><br><b>CLO3</b> | <b>PLO2</b><br><b>PLO3</b>                | Student will learn collections, design and develop complex programming problems.                                                       | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exam Review</b>                                                                                                     | <b>To make an overview</b>                                                                                                                                                                 | <b>Week-4</b><br><b>L-10-12</b>   |                            |                                           | Lectures, Questions and Answers                                                                                                        |                                                                               |
| <b>Exam.</b>                                                                                                           |                                                                                                                                                                                            | <b>Week-5</b><br><b>L-13-15</b>   |                            |                                           | Quiz and Tutorial Exam. Will be taken                                                                                                  |                                                                               |
| <b>Exp 11#:</b><br>Threads<br><b>Exp 12#:</b><br>Multithreads                                                          | To <b>understand</b> threads and multithreads, and threads priorities.<br><br>To <b>design and develop</b> programming problems based on threads and multithreads, and threads priorities. | <b>Week-6</b><br><b>L-16-17</b>   | <b>CLO2</b><br><b>CLO4</b> | <b>PLO2</b><br><b>PLO3</b><br><b>PLO5</b> | Students will identify, analyze, formulate, and develop complex programming problems.                                                  | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exp 13#:</b><br>Catching and throwing exceptions                                                                    | To <b>grasp</b> exceptions, managing exceptions, and exception objects.<br>To <b>handle</b> multiple unrelated exceptions.                                                                 | <b>Week-6,7</b><br><b>L-18-19</b> | <b>CLO2</b><br><b>CLO5</b> | <b>PLO1</b><br><b>PLO3</b>                | Students will identify, analyze, formulate, and develop complex programming problems.                                                  | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |

|                                            |                                                                                                                                        |                                     |                            |                                           |                                                                                    |                                                                               |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------|-------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Exp 14#:</b><br>Detect deadlock         | To <b>understand</b> synchronization methods, deadlock, and inter-thread communication.                                                | <b>Week-7</b><br><b>L-20-21</b>     | <b>CLO2</b><br><b>CLO6</b> | <b>PLO2</b><br><b>PLO3</b><br><b>PLO4</b> | Students will analyze, learn, formulate, and develop complex programming problems. | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exp 15#:</b><br>Synchronize processes   | To <b>design</b> and <b>develop</b> problem ensuring synchronization.                                                                  |                                     |                            |                                           |                                                                                    |                                                                               |
| <b>Exam Review</b>                         | <b>To make an overview</b>                                                                                                             | <b>Week-8</b><br><b>L-22-24</b>     |                            |                                           | Lectures, Questions and Answers                                                    |                                                                               |
| <b>Exam.</b>                               |                                                                                                                                        | <b>Week-9</b><br><b>L-25-27</b>     |                            |                                           | Presentation and Tutorial Exam. Will be taken                                      |                                                                               |
| <b>Exp 16#:</b><br>Client server algorithm | To <b>learn</b> networking concepts, common networking protocol networking interfaces, client server algorithm, and socket programming | <b>Week-10,11</b><br><b>L-28-33</b> | <b>CLO2</b><br><b>CLO7</b> | <b>PLO2</b><br><b>PLO3</b>                | Students will analyze, learn, design, and develop complex programming problems.    | #Interactive discussion<br>#Lecture discussion with multimedia.<br># Feedback |
| <b>Exp 17#:</b><br>Socket programming      | To <b>design</b> socket programming.                                                                                                   |                                     |                            |                                           |                                                                                    |                                                                               |
| <b>Exam Review</b>                         | <b>To make an overview</b>                                                                                                             | <b>Week-12</b><br><b>L-34-36</b>    |                            |                                           | Lectures, Questions and Answers                                                    |                                                                               |
| <b>Exam.</b>                               |                                                                                                                                        | <b>Week-13</b><br><b>L-37-39</b>    |                            |                                           | Assignment and Tutorial Exam will be taken                                         |                                                                               |
| <b>Recess before Exam.</b>                 |                                                                                                                                        | <b>Week-14,15</b>                   |                            |                                           |                                                                                    |                                                                               |
| <b>Sem. End Exam.</b>                      | <b>Start</b>                                                                                                                           | <b>Week-16</b>                      |                            |                                           |                                                                                    |                                                                               |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category</b><br><b>Marks:30</b> | <b>Tutorial/In-course</b><br><b>(10)</b> | <b>Assignment</b><br><b>(05)</b> | <b>Quiz/Presentation</b><br><b>(05)</b> | <b>Class Attendance</b><br><b>10</b> |
|--------------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|--------------------------------------|
| <b>Remember</b>                            |                                          |                                  | (05)                                    |                                      |
| <b>Understand</b>                          | (05)                                     |                                  |                                         |                                      |
| <b>Apply</b>                               |                                          |                                  |                                         | (10)                                 |
| <b>Analyze</b>                             |                                          |                                  |                                         |                                      |
| <b>Evaluate</b>                            | (05)                                     |                                  |                                         |                                      |
| <b>Create</b>                              |                                          | (05)                             |                                         |                                      |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. Ivor Horton's *Beginning Java 2 JDK 5 Edition*, Wrox;

**Reference Books and Other Materials**

3. <http://www.cs.umd.edu/class/fall2022/cmse132-030X/>
4. <https://docs.python.org/3/tutorial/>
5. <https://sites.google.com/site/oopunyt/syllabus>

**HUM-2201: Economics**

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

**Prerequisite:** Physics and Mathematics at Higher Secondary Level**Course Type:** Theory

**Course Summary:** This course provides a comprehensive introduction to the basic idea about economics. This course focuses on Theory of Demand, Supply and Consumer Behavior, Production and Cost, Theory of the Firm, Input-Output analysis, National income accounting and determination, Budgets of Bangladesh and Development Planning in Bangladesh.

**Course Objectives:** To provide knowledge on basic concepts of Economics, Microeconomics and Macroeconomics.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                          |
| <b>CLO2</b> | <b>Understand</b> the basic concepts of Economics, Microeconomics and Macroeconomics. |
| <b>CLO3</b> | <b>Understand</b> the Theory of Demand, Supply and Consumer Behavior.                 |
| <b>CLO4</b> | <b>Understand</b> the basis operations of Theory of the Firm, Input-Output analysis   |
| <b>CLO5</b> | <b>Understand</b> the Solution of National income accounting and determination.       |

| Course Content         |                                                                                                                                                                                               | Time Frame                    | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|------------------------|
| Outline                | Specific Topics                                                                                                                                                                               |                               | CLO            | PLO  |                                                                                                        |                        |
| <b>Course Overview</b> | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course communication systems<br><b>To grasp</b> the content and policies for the class | <b>Week-1</b><br><b>L-1-2</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation |



|                                                     |                                                                                                                                                                                                          |                             |      |             |                                               |                                                                 |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------|-------------|-----------------------------------------------|-----------------------------------------------------------------|
| <b>Basic Concepts of Economics</b>                  | <b>To define</b> the subject matter of Economics; Microeconomics Vs Macroeconomics; Law of Economics; Central economic problems of every society; Different economic systems; Economics and Engineering. | <b>Week-2,3<br/>L-3-6</b>   | CLO2 | PLO2        | Students will discuss with a group.           | #Interactive discussion<br>#Lecture discussion with multimedia. |
| <b>Exam. Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                               | <b>Week-4<br/>L-7-8</b>     |      |             | Lectures, Questions and Answers               |                                                                 |
| <b>Exam.</b>                                        |                                                                                                                                                                                                          | <b>Week-5<br/>L-9-10</b>    |      |             | Quiz and Tutorial Exam. Will be taken         |                                                                 |
| <b>Production and Cost and Theory of the Firm</b>   | <b>To discuss</b> Production and Cost and Theory of the Firm<br><b>To explain</b> Theory of the Firm                                                                                                     | <b>Week-6,7<br/>L-11-14</b> | CLO3 | PLO2        | Students will discuss with a group.           | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam. Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                               | <b>Week-8<br/>L-15-16</b>   |      |             | Lectures, Questions and Answers               |                                                                 |
| <b>Exam.</b>                                        |                                                                                                                                                                                                          | <b>Week-9<br/>L-17-18</b>   |      |             | Presentation and Tutorial Exam. Will be taken |                                                                 |
| <b>The Input-Output Analysis</b>                    | <b>To analysis</b> Meaning of input-output, Input-output analysis model; balance equation; coefficient matrix; determination of final demand vector.                                                     | <b>Week-10<br/>L-19-20</b>  | CLO4 | <b>PLO3</b> | Students will discuss with a group.           | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Basic Concepts of Macroeconomics</b>             | <b>To discuss</b> Growth; Unemployment; Inflation; Philips Curve, Business cycle; Circular flow of economics; two, three and four sector economics.                                                      | <b>Week-11<br/>L-21-22</b>  | CLO4 | <b>PLO3</b> | Students will discuss with a group..          | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>National Income Accounting and Determination</b> | <b>To explain</b> National income accounting and determination, Budgets of Bangladesh, Development Planning in Bangladesh                                                                                | <b>Week-12<br/>L-23-24</b>  | CLO5 | <b>PLO4</b> | Students will discuss with a group.           | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam. Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                               | <b>Week-13<br/>L-25</b>     |      |             | Lectures, Questions and Answers               |                                                                 |
| <b>Exam.</b>                                        |                                                                                                                                                                                                          | <b>Week-13<br/>L-26</b>     |      |             | Assignment and Tutorial Exam will be taken    |                                                                 |
| <b>Recess before Exam.</b>                          |                                                                                                                                                                                                          | <b>Week-14,15</b>           |      |             |                                               |                                                                 |

|                |       |         |  |  |  |  |
|----------------|-------|---------|--|--|--|--|
| Sem. End Exam. | Start | Week-16 |  |  |  |  |
|----------------|-------|---------|--|--|--|--|

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. *Economics by Samuelson and Nordhous.*
2. *Economics by Byrons and Stone.*

#### **Reference Books and Other Materials**

3. *Modern Economic Theory by Dewett, K.K*
4. *Advanced Economic Theory by Ahuja H.L*

**ICT-2000:**

**Viva-Voce**

**Based on 2<sup>nd</sup> Year 1<sup>st</sup> and 2<sup>nd</sup> Semester**

Full Marks:100

Credit: 1.0

**B.Sc. (Engg.) Third Year-2024**  
**3<sup>rd</sup> Year 1<sup>st</sup> Semester**

**ICT-3101: Wave Propagation and Antenna**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT:1103, MATH: 2101

**Course Type:** Theory

**Course Summary:** The Topic will be discussed in detail in class by the course teacher

**Course Objectives:** This course will introduce the students to the fundamental concepts underlying Electromagnetic theories and antenna, properties of EM wave, transmission and reflection of EM wave, transmission line theory and its characteristics, propagation theory of Radio waves, characteristics of antenna, types of antenna its application area etc. The emphasis is on studying and analyzing fundamental issues in antenna design and their impact on performance.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Learn</b> theory and properties of Electromagnetic waves.                                                                                                                                                                                                                                                                                                              |
| <b>CLO2</b> | <b>Emphasize</b> understanding of Maxwell's equations in differential and integral forms in solving electromagnetic problems.                                                                                                                                                                                                                                             |
| <b>CLO3</b> | <b>Understand</b> the concept of transmission and reflection of plane waves at boundaries.                                                                                                                                                                                                                                                                                |
| <b>CLO4</b> | <b>Learn</b> the basics of transmission lines (TL), propagation of harmonic signals on TL, TL parameters and equations. Be able to apply the knowledge in basic analysis and design problems. Learn the basics of the Smith Chart as a tool for transmission line calculations, and be able to use it for basic parameter calculation and analysis of transmission lines. |
| <b>CLO5</b> | <b>Predict</b> propagation effects of electromagnetic waves in the surface, space, sky, troposphere, ionosphere including MUF and Fading.                                                                                                                                                                                                                                 |
| <b>CLO6</b> | <b>Classify</b> the Guided Wave solutions -TE, TM, and TEM. Analyze and design rectangular waveguides and understand the propagation of electromagnetic waves.                                                                                                                                                                                                            |
| <b>CLO7</b> | <b>Learn</b> antenna fundamentals types of antennas. Characterize the radiation of an antenna in terms of its radiation pattern, directivity, gain, bandwidth and radiation resistance.                                                                                                                                                                                   |

| Course Content                                            |                                                                                                                                                                                                                                                                  | Time Frame | Alignment with |                | Suggested Activities                                                     | Teaching Strategy(s)                    |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|----------------|--------------------------------------------------------------------------|-----------------------------------------|
| Outlines                                                  | Specific Topics                                                                                                                                                                                                                                                  |            | CLO            | PLO            |                                                                          |                                         |
| <b>Theory &amp; Properties of Electromagnetic Waves</b>   | Potential in an electric field, potential in a changing magnetic field, displacement current.                                                                                                                                                                    | Week 1     | CLO-1          | PLO-1<br>PLO-2 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br><br>- Oral Presentation |
| <b>Equations of the Electromagnetic Fields, Radiation</b> | Potential in an electric field, Potential in a changing magnetic field, Faraday's law, The first electromagnetic field equation, The second electromagnetic field equation, Maxwell's equation rewritten, The plane wave in space-I, The plane wave in space-II, | Week 1-2   | CLO-2          | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test             |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |       |                |                                                                          |                                     |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|----------------|--------------------------------------------------------------------------|-------------------------------------|
|                                                                 | Energy of the radiated wave, Poynting's theorem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |       |                |                                                                          |                                     |
| <b>Transmission and Reflection of Plane Waves at Boundaries</b> | Conditions of field continuity at boundary surfaces, The analogous transmission line for plane wave propagation, Reflection from a plane conductor at normal incidence, Power flow in the reflected and transmitted waves, Current flow in the conductor, Depth of penetration, Skin effect, The perfect conductor concepts, Wave incident on a perfect conductor at an arbitrary angle, Wave incident at an arbitrary angle on a boundary between dielectrics, Index of reflection, Snell's law, Total reflection, Phase and group velocities, Elimination of reflections. | Week 3-4   | CLO-3 | PLO-2          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Transmission Lines</b>                                       | Introduction, Propagation through lines, Determination of secondary line constants, Underground cables, Wave velocity, Distortion of low frequency lines, General line equations, High frequency lines, Reflection, Line impedance, Impedance matching in high frequency lines, Smith chart.                                                                                                                                                                                                                                                                                | Week 5-6   | CLO-4 | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Propagation of Radio Wave</b>                                | Surface wave, space wave and sky wave, Tropospheric propagation, Ionospheric propagation, skip distance and MUF, fading.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Week 7-8   | CLO-5 | PLO-4          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| <b>Wave Guides</b>                                              | Rectangular Wave guides, Circular and other Wave guides, Cavity resonators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Week 9-10  | CLO-6 | PLO-5          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>- Oral Presentation |
| <b>Antenna Fundamentals</b>                                     | The radiation mechanism, Elementary doublet, Current and Voltage distribution, Resonant & Non-resonant antennas, Antenna gain, Bandwidth,                                                                                                                                                                                                                                                                                                                                                                                                                                   | Week 11-12 | CLO-7 | PLO-2          | -Lecture Presentation with video<br>-Interactive Discussion              | - Assignment<br>-Class test         |

|                                    |                                                                                                                                          |         |       |       |                                                  |                                     |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------|--------------------------------------------------|-------------------------------------|
|                                    | beam width and polarization, Effects of ground on antennas, Antenna coupling.                                                            |         |       |       |                                                  |                                     |
| <b>Different Types of Antennas</b> | Directional high-frequency antennas, Microwave antennas, Wideband and special -purpose antennas, Cell-Site Antennas and Mobile Antennas. | Week 13 | CLO-7 | PLO-4 | -Lecture Presentation<br>-Interactive Discussion | - Assignment<br>- Oral Presentation |

### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                               |                            |                                    |                                |
| <b>Apply</b>                         |                                    |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                    |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                    |                                |
| <b>Create</b>                        |                                    | (05)                       |                                    |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Networks, Lines and Fields*, John D Ryder, Prentice-Hall of India Private Ltd.
2. *Electronic Communications*, Dennis Roddy and John Coolen, Prentice-Hall of India Private Ltd.

#### **References and other related books**

3. *Hand Book of Electronics*, Gupta and Kumar, PragatiPrakasan.
4. *Radio Engineering*, J K Mithal, Khanna Publishers.
5. *Electrical Technology*, B.L Theraja and A.K Theraja.

### **ICT-3102: Wave Propagation and Antenna Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-3101

**Course Type:** Practical/Laboratory work

**Course Summary:** The topic will be discussed in class by the course teacher

**Course Objectives:** This course will introduce the students to hands on knowledge on Electromagnetic theories and antenna, properties of EM wave, transmission and reflection of EM wave, transmission line theory, propagation theory of Radio waves, characteristics of antenna, types of antenna.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Realize</b> Electromagnetic theories and antenna                           |
| <b>CLO2</b> | <b>Understand</b> properties of EM wave                                       |
| <b>CLO3</b> | <b>Comprehend</b> transmission line theory, propagation theory of Radio waves |
| <b>CLO4</b> | <b>Know</b> the characteristics of antenna                                    |

| Course Content                   |                                                                                                                                                | Time Frame            | Alignment with   |                 | Suggested Activities                                                                                                     | Teaching Strategy(s)                                          |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Outline                          | Specific Topics                                                                                                                                |                       | CLO              | PLO             |                                                                                                                          |                                                               |
| <b>Electromagnetic Wave</b>      | *Generation of Electromagnetic wave using Matlab.<br>* Visualizing Maxwell's equations using Matlab.                                           | Week-1-3<br>L-1-9     | CLO1             | PLO2<br>PLO3    | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
| <b>Electromagnetic Theory</b>    | * To study and analysis propagation of wave in Rectangular Waveguide using Matlab.<br>* Impedance Matching using Smith Chart using Matlab      | Week-4-5<br>L-10-15   | CLO-2<br>CLO-3   | PLO3<br>PLO4    | Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory  | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
| <b>Propagation of Radio Wave</b> | * Calculate phase & group velocity using Matlab.<br>* To study and analysis of Electromagnetic waves propagation characteristics using Matlab. | Week-6-7<br>L-16-21   | CLO-2 &<br>CLO-3 | PLO-3,<br>PLO-4 | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
| <b>Antenna</b>                   | *Plot radiation pattern of dipole antenna using Matlab.<br>*Plot the radiation pattern of Half wave dipole antenna using Matlab.               | Week-8-10<br>L-22-30  | CLO-4            | PLO-3<br>PLO-4  | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
| <b>Antenna</b>                   | * Plot radiation pattern of monopole antenna using Matlab.<br>*Plot of Radiation pattern of Uniform Linear Array using Matlab.                 | Week-11-13<br>L-31-39 | CLO-4            | PLO-3<br>PLO-4  | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |

|                            |       |            |  |  |  |  |
|----------------------------|-------|------------|--|--|--|--|
| <b>Recess before Exam.</b> |       | Week-14-15 |  |  |  |  |
| <b>Semester End Exam.</b>  | Start | Week-16    |  |  |  |  |

**Assessment Pattern (According FET Ordinance)****CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. Networks, Lines and Fields, John D Ryder, Prentice-Hall of India Private Ltd.
2. Electronic Communications, Dennis Roddy and John Coolen, Prentice-Hall of India Private Ltd..

**Reference Books and Other Materials**

3. Hand Book of Electronics, Gupta and Kumar, PragatiPrakasan.
4. Radio Engineering, J K Mithal, Khanna Publishers.
5. Lab. Guide and Lab. Manuals

**ICT-3103:      Microwave Engineering**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-2205, 3101**Course Type:** Theory

**Course Summary:** Electronics and Communications Engineering (ECE) involves researching, designing, developing and testing of electronic equipment used in various systems. Electronics and Communications engineers also conceptualize and oversee the manufacturing of communications and broadcast systems.

This stream of engineering deals with analogue transmission, basic electronics, microprocessors, solid-state devices, digital and analogue communication, analogue integrated circuits, microwave engineering, satellite communication, antennae and wave progression. It also deals with the manufacturing of microwave engineering, microwave tubes and resonators.

**Course Objectives:** To provide knowledge on television system, satellite communication, radar system and microwave engineering.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                  |
|-------------|--------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course     |
| <b>CLO2</b> | <b>Understand</b> the television system.         |
| <b>CLO3</b> | <b>Understand</b> the satellite system           |
| <b>CLO4</b> | <b>Understand</b> the Radar system               |
| <b>CLO5</b> | <b>Analysis</b> microwave communication networks |

| Course Content                 |                                                                                                                                                                                                                                             | Time Frame                  | Alignment with |              | Suggested Activities                                                                                     | Teaching Strategy(s)                                                  |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|--------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                        | Specific Topics                                                                                                                                                                                                                             |                             | CLO            | PLO          |                                                                                                          |                                                                       |
| <b>Course Overview</b>         | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Fundamental<br><b>To grasp</b> the content and policies for the class                                                         | <b>Week-1<br/>L-1-3</b>     | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details   | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Television System</b>       | <b>To discuss</b> Image Capturing, Image Continuity, Flicker, Interlaced Scanning, Composite Video Signal, Video Modulation, Sound Modulation, Channel bandwidth, TV Camera, TV Transmitter, Monochrome TV Receiver, Principle of Color TV. | <b>Week-2<br/>L-4-6</b>     | CLO2           | PLO1         | Students will discuss with a group.<br>To apply the knowledge to understand the basic Television system. | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Satellite Communication</b> | <b>To understand</b> General Structure, Merits & Demerits of Satellite Communication, Satellite Orbits, Linkages, Equipments on Satellite, Propagation of Signal, Ground Station, Digital Satellite Communication.                          | <b>Week-3<br/>L-7-9</b>     | CLO3           | PLO5         | Students will discuss with a group.<br>To apply the knowledge to understand the Satellite communication. | #Interactive discussion<br>#Lecture discussion with multimedia        |
| <b>Exam. Review</b>            | <b>To make an overview</b>                                                                                                                                                                                                                  | <b>Week-4<br/>L-10-12</b>   |                |              | Lectures, Questions and Answers                                                                          |                                                                       |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                             | <b>Week-5<br/>L-13-15</b>   |                |              | Quiz and Tutorial Exam. Will be taken                                                                    |                                                                       |
| <b>Radar</b>                   | <b>To discuss</b> the Radar Performance Factors, Basic Pulsed Radar System, Antennas & Scanning, Display Methods, Pulsed Radar Systems, Moving Target Indication, CW Doppler Radar, and Frequency Modulated CW Radar, Phased Array Radars.  | <b>Week-6-7<br/>L-16-21</b> | CLO4           | PLO2<br>PLO3 | Students will discuss and apply complex engineering knowledge to understand the Radar system.            | #White board discussion<br>#Lecture discussion with multimedia        |



|                              |                                                                                                                                                                                                                          |                               |      |              |                                                                                                     |                                                                                                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------|--------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Exam Review</b>           | <b>To make an overview</b>                                                                                                                                                                                               | <b>Week-8<br/>L-22-24</b>     |      |              | Lectures, Questions and Answers                                                                     |                                                                                                     |
| <b>Exam.</b>                 |                                                                                                                                                                                                                          | <b>Week-9<br/>L-25-27</b>     |      |              | Presentation and Tutorial Exam. Will be taken                                                       |                                                                                                     |
| <b>Microwave Engineering</b> | <b>To discuss</b> Short history of microwave engineering, Microwave frequencies, Microwave systems, Microwave applications, Microwave circuit Elements and analysis, Micro strip and Strip line, Smith Chart.            | <b>Week-10<br/>L-28,29,30</b> | CLO5 | PLO2<br>PLO3 | Students will apply reasoning informed by contextual knowledge to understand microwave engineering. | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about MW engineering. |
| <b>Microwave Components</b>  | <b>To analyze</b> Waveguide tees, E-plane tee, H-plane tee, Hybrid junctions, Cavity wave-meter, Hybrid ring, directional couplers, Isolators and circulators.<br><b>To discuss</b> about microwave resonators and tubes | <b>Week-11<br/>L-31,32,33</b> | CLO5 | PLO3<br>PLO6 | Students will apply reasoning informed by contextual knowledge to understand microwave components.  | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about MW components   |
| <b>Exam. Review</b>          | <b>To make an overview</b>                                                                                                                                                                                               | <b>Week-12<br/>L-34-36</b>    |      |              | Lectures, Questions and Answers                                                                     |                                                                                                     |
| <b>Exam.</b>                 |                                                                                                                                                                                                                          | <b>Week-13<br/>L-37-39</b>    |      |              | Assignment and Tutorial Exam will be taken                                                          |                                                                                                     |
| <b>Recess before Exam.</b>   |                                                                                                                                                                                                                          | <b>Week14-15</b>              |      |              |                                                                                                     |                                                                                                     |
| <b>Sem. End Exam.</b>        | <b>Start</b>                                                                                                                                                                                                             | <b>Week-16</b>                |      |              |                                                                                                     |                                                                                                     |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. *Electronic Communication, Roddy and Coolen, Prentice Hall of India, 4th Ed.*
2. *Radio Engineering, G K Mithal, Khanna Publishers.*

**Reference Books and Other Materials**

3. *Integrated Electronics, Millman and Halkias, McGraw-Hill*
4. *Microwave Engineering- David M. Pozar*
5. *Microwave Devices and Circuits- Samuel Y Laio*

**ICT-3104: Microwave Engineering Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-3103**Course Type:** Practical/Laboratory work

**Course Summary:** This course gives students deep knowledge in electronic communication and microwave engineering at the practical level. This lab focuses on the fundamental concepts on TDM, Pulse modulations, digital modulation techniques, source coding techniques and Error-control coding techniques.

**Course Objectives:** This course will introduce the students to hands on knowledge on image, Camera, principle of color TV and satellite communication, analysis and evaluation of microwave components and microwave resonators and tubes.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                              |
| <b>CLO2</b> | <b>Able</b> to understand Basics of conversions from analog signals into digital signals. |
| <b>CLO3</b> | <b>Able</b> to verify of microwave tube characteristics and gunn characteristics.         |
| <b>CLO4</b> | <b>Able</b> to verify of microwave tube characteristics and gunn characteristics          |
| <b>CLO5</b> | <b>Able</b> to analyze the frequency and wavelength of the different wave guides          |

| Course Content         |                                                                                                                                                                        | Time Frame              | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                                           |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Outline                | Specific Topics                                                                                                                                                        |                         | CLO            | PLO  |                                                                                                        |                                                                                                |
| <b>Course Overview</b> | <b>To acquire</b> general idea about the ECMW Laboratory<br><b>To supply</b> the experimental list for this Laboratory.<br><b>To discuss</b> about the simulator tools | <b>Week-1<br/>L-1-3</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab. |

|                                                    |                                                                                                                                                                                                                                                                           |                               |                      |              |                                                                                                                        |                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|--------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Frequency Spectrum</b>                          | To observe frequency spectrum of AM and FM waves<br>To obtain fidelity response of AM receiver.                                                                                                                                                                           | <b>Week-2-3<br/>L-4-9</b>     | CLO2                 | PLO2<br>PLO3 | Students will discuss with a group.<br>To apply the knowledge to observe the frequency response and fidelity response. | #Interactive discussion<br>#Student will apply the knowledge to analyze AM, FM signals. |
| <b>Exam Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                | <b>Week-4<br/>L-10-12</b>     |                      |              | Demonstrate, Questions and Answers                                                                                     |                                                                                         |
| <b>Exam.</b>                                       |                                                                                                                                                                                                                                                                           | <b>Week-5<br/>L-13-15</b>     |                      |              | Quiz and Tutorial Exam. Will be taken                                                                                  |                                                                                         |
| <b>Analog Modulation/ Demodulation Techniques</b>  | To generate amplitude, frequency and phase modulation signal<br>To extract modulating signal using Diode Detector.                                                                                                                                                        | <b>Week-6-7<br/>L-16-21</b>   | CLO2                 | PLO2<br>PLO3 | Students will demonstrate /present and solve the problems.                                                             | #White board discussion<br>#Demonstrate with multimedia                                 |
| <b>Exam Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                | <b>Week-8<br/>L-22-23</b>     |                      |              | Lectures, Questions and Answers                                                                                        |                                                                                         |
| <b>Exam.</b>                                       |                                                                                                                                                                                                                                                                           | <b>Week-8<br/>L-24</b>        |                      |              | Presentation and Tutorial Exam. Will be taken                                                                          |                                                                                         |
| <b>Digital Modulation/ Demodulation Techniques</b> | To generate and detect PCM<br>To generate DPCM<br>To generate Delta Modulation<br>To generate and Detect PSK, DPSK, FSK                                                                                                                                                   | <b>W-9<br/>L-25-27</b>        | CLO2                 | PLO3         | Students will simulate and observe the different modulated waves.                                                      | #Demonstrate with multimedia                                                            |
| <b>Microwave Engineering</b>                       | To characterize Reflex Klystron<br>To characterize Gunn Diode<br><b>measurement</b> of scattering parameters of magic tee.<br><b>Measurement</b> of Scattering Parameters of Circulator.<br><b>Attenuation</b> measurement.<br><b>Measurement</b> of microwave Frequency. | <b>Week-10-11<br/>L-28-33</b> | CLO3<br>CLO4<br>CLO5 | PLO4<br>PLO5 | Students will apply contextual knowledge to characterize the different microwave components                            | #Lecture discussion with multimedia<br>#Show video about different MW components        |
| <b>Exam. Review</b>                                | <b>To make an overview</b>                                                                                                                                                                                                                                                | <b>Week-12<br/>L-34-36</b>    |                      |              | Lectures, Questions and Answers                                                                                        |                                                                                         |
| <b>Exam.</b>                                       |                                                                                                                                                                                                                                                                           | <b>Week-13<br/>L-37-39</b>    |                      |              | Presentation and Tutorial Exam. will be taken                                                                          |                                                                                         |
| <b>Recess before Exam.</b>                         |                                                                                                                                                                                                                                                                           | <b>Week-14-15</b>             |                      |              |                                                                                                                        |                                                                                         |
| <b>Sem. End Exam.</b>                              | <b>Start</b>                                                                                                                                                                                                                                                              | <b>Week-16</b>                |                      |              |                                                                                                                        |                                                                                         |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |

|                   |      |      |  |      |
|-------------------|------|------|--|------|
| <b>Understand</b> |      |      |  |      |
| <b>Apply</b>      |      | (05) |  | (10) |
| <b>Analyze</b>    |      |      |  |      |
| <b>Evaluate</b>   | (05) |      |  |      |
| <b>Create</b>     |      |      |  |      |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

### **Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Electronic Communication, Roddy and Coolen, Prentice Hall of India, 4th Ed.*
2. *Radio Engineering, G K Mithal, Khanna Publishers.*

#### ***Reference Books and Other Materials***

3. *Microwave Engineering- David M. Pozar*
4. *Microwave Devices and Circuits- Samuel Y Laio*
5. *Lab. Guide and Lab. Manuals*

**ICT-3105:**

**Data Communication**

Full Marks: 100

Credit: 3.0

Contact Hours: 39

Exam. Time: 4 Hours

**Prerequisite:** Computer Fundamentals

**Course Type:** Theory

**Course Summary:** This Course gives a general overview of data communications and networking. This course introduces students to evolution trend of Data communication. It also helps students in understanding the procedure of transmitting data over the network and how to resolve the conflicting issues arising in the course of transmission. This course provides with practical knowledge and hands-on experience in transmitting data over the network.

This stream of engineering deals with analogue transmission, basic electronics, microprocessors, solid-state devices, digital and analogue communication, analogue integrated circuits, microwave engineering, satellite communication, antennae and wave progression. It also deals with the manufacturing of microwave engineering, microwave tubes and resonators.

**Course Objectives:** This is an introductory course in Data Communications and networking. It familiarizes the students with the basics of data communications, OSI model and techniques, applications and control of modern data communications networks. Topics included are network models,

digital and analog transmission, multiplexing, circuit and packet switching. This course will mainly focus to develop engineering skills in troubleshooting and designing data networks.

**Course Learning Outcomes (CLO): Upon successful completion of this course, students will be able to -**

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | Describe the concepts of data communication, and the general principles of network design and compare the different network topologies.                                                                                   |
| <b>CLO2</b> | Analyze digital and analog representations of data and channels, mechanisms and techniques of encoding and different technologies regarding guided and unguided media, their constructions and applications in real life. |
| <b>CLO3</b> | Critique error detection, correction and flow control techniques; multiple access control techniques.                                                                                                                     |

| Course Content             |                                                                                                                       | Time Frame                                           | Alignment with |               | Teaching-Learning Strategy                                               | Assessment techniques                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------|---------------|--------------------------------------------------------------------------|---------------------------------------------------|
| Outlines                   | Specific Topics                                                                                                       |                                                      | CLO            | PLO           |                                                                          |                                                   |
| <b>Course Overview</b>     | Socialization and Introduction to the course.                                                                         | Week - 1, Lecture – L-1                              | CLO-1          | PLO-1         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation                |
| <b>Introduction</b>        | Over view, Introduction to data communication, categories of network, Protocol and standards                          | Week – 3, Lecture - L-2, L-3, L-4                    | CLO-1          | PLO-2         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test<br>- Oral Presentation |
| <b>Transmission Layers</b> | Layered communication architecture, The OSI Model, TCP/IP Model, Addressing                                           | Week 4 Lecture - L-5, L-6, L-7                       | CLO-2          | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>Transmission Modes</b>  | Data and Signals, Periodic Analog Signals, Digital Signals, Transmission Impairment, Data rate limits and performance | Week 5, 6 Lecture - L-8, L-9, L-10, L-11, L-12, L-13 | CLO-3          | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>Transmission Media</b>  | Guided Media, Unguided Media                                                                                          | Week 7 Lecture - L-14, L-15, L-16,                   | CLO-2          | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>Data Conversion</b>     | Digital to digital conversion, Analog to digital conversion, Transmission Mode                                        | Week 8 Lecture - L-17, L-18, L-19                    | CLO-2          | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive                         | -Assignment<br>- Oral Presentation                |

|                        |                                                                                                                                                     |                                             |       |                  |                                                                                  |                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------|------------------|----------------------------------------------------------------------------------|----------------------------|
|                        |                                                                                                                                                     |                                             |       |                  | Discussion<br>-Feedback                                                          |                            |
| <b>Data Conversion</b> | Digital to Analog conversion<br>Analog to Analog conversion                                                                                         | Week 9<br>Lecture -<br>L-20, L-<br>21, L-22 | CLO-3 | PLO-4 &<br>PLO-5 | -Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion             | -Assignment<br>-Class test |
| Multiplexing           | Multiplexing<br>Spread Spectrum                                                                                                                     | Week 10<br>L-23, L-<br>24, L-25             | CLO-1 | PLO-4 &<br>PLO-5 | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| Network Switching      | Circuit-switched Network<br>Datagram Networks<br>Virtual-circuit Networks<br>Switch                                                                 | Week 11<br>L-26, L-<br>27, L-28             | CLO-1 | PLO-4 &<br>PLO-5 | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| Data Encoding          | Types of error in data<br>communication, Block<br>Coding, Cyclic Codes,<br>Checksum                                                                 | Week 12<br>L-29, L-<br>30, L-31             | CLO-3 | PLO-4 &<br>PLO-5 | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| Modulation Techniques  | Framing, Flow and Error<br>Control, Noisy and noiseless<br>channel, Multiple Access:<br>CSMA/CD, CSMA/CD,<br>Controlled Access,<br>FDMA, CDMA, TDMA | Week 13<br>L-32, L-<br>33, L-34             | CLO-3 | PLO-4 &<br>PLO-5 | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| Recess before Exam.    |                                                                                                                                                     |                                             |       |                  |                                                                                  |                            |
| Semester End Exam.     | Start                                                                                                                                               |                                             |       |                  |                                                                                  |                            |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 10          |
| <b>Understand</b>           | 00          |

|                 |    |
|-----------------|----|
| <b>Apply</b>    | 15 |
| <b>Analyze</b>  | 00 |
| <b>Evaluate</b> | 05 |
| <b>Create</b>   | 00 |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources:****Books Recommended**

1. Foruzan, B. A. (2013). Data communication and Networking. Science Engineering & Math Publications.
2. Blanchard, E. (2000). Introduction to Networking and Data Communications.
3. Tanenbaum, A. S., & Wetherall, D. J. (2011). Computer networks. Pearson.
4. Video Tutorials on Data communication for beginners.

**ICT-3106:****Data Communication Laboratory**

Full Marks: 100

Credit: 1.5

Contact Hours: 39

Exam. Time: 6 Hours

**Prerequisite:** ICT-3105**Course Type:** Practical

**Course Summary:** This Course gives a general overview of data communications laboratory. This course introduces students to evolution trend of Data communication. It also helps students in understanding the procedure of transmitting data over the network and how to resolve the conflicting issues arising in the course of transmission. This course provides with practical knowledge and hands-on experience in transmitting data over the network.

This stream of engineering deals with analogue transmission, basic electronics, microprocessors, solid-state devices, digital and analogue communication, analogue integrated circuits, microwave engineering, satellite communication, antennae and wave progression. It also deals with the manufacturing of microwave engineering, microwave tubes and resonators.

**Course Objectives:** This is an introductory course in data communications and networking. It familiarizes the students with the basics of data communications, OSI model and techniques, applications and control of modern data communications networks. Topics included are network models, digital and analog transmission, multiplexing, circuit and packet switching. This course will mainly focus to develop engineering skills in troubleshooting and designing data networks.

**Course Learning Outcomes (CLO):** Upon successful completion of this course, students will be able to

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | Bit stuffing, de-stuffing, channel encoding                                                                                                                                                                               |
| <b>CLO2</b> | Error detection, correction and flow control techniques; multiple access control techniques.                                                                                                                              |
| <b>CLO3</b> | Analyze digital and analog representations of data and channels, mechanisms and techniques of encoding and different technologies regarding guided and unguided media, their constructions and applications in real life. |
| <b>CLO4</b> | Describe the concepts of data communication, and the general principles of network design and compare the different network topologies.                                                                                   |

|             |                                                          |
|-------------|----------------------------------------------------------|
| <b>CLO5</b> | Analog to digital conversion and pulse code transmission |
|-------------|----------------------------------------------------------|

| Course Content                                                        |                                                                                                                                                                                                 | Time Frame                          | Alignment with |                | Teaching-Learning Strategy                                                                             | Assessment techniques                                                                          |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------|----------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Outlines                                                              | Specific Topics                                                                                                                                                                                 |                                     | CLO            | PLO            |                                                                                                        |                                                                                                |
| <b>Course Overview And Exp #1: Character Stuffing and De-stuffing</b> | <b>To acquire</b> general idea about the data communication laboratory<br><b>To supply</b> the experimental list for this laboratory.<br><b>Implementing</b> character stuffing and de-stuffing | <b>Week-1 L-1-3</b>                 | CLO1           | PLO1           | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab. |
| <b>Exp #2: Bit Stuffing and De-stuffing</b>                           | <b>Implementation</b> bit stuffing and de-stuffing                                                                                                                                              | <b>Week 2-3, Lecture - L-4-9</b>    | CLO-1          | PLO-2<br>PLO-3 | -Lecture<br>Presentation with video<br>-Interactive Discussion<br>-Feedback                            | -Assignment<br>-Class test<br>- Oral Presentation                                              |
| <b>Exp # 3: Channel Encoding</b>                                      | <b>Implementation</b> cyclic redundancy check and parity checker.                                                                                                                               | <b>Week 4 Lecture - L-10-13</b>     | CLO-2<br>CLO-3 | PLO-2<br>PLO-3 | -Lecture<br>Presentation with video<br>-Interactive Discussion<br>-Feedback                            | -Assignment<br>-Class test                                                                     |
| <b>Exp # 4: Error detection and correction</b>                        | <b>Implementation</b> of error detection and correction mechanism using Hamming code                                                                                                            | <b>Week 5-6 Lecture - L-14-25</b>   | CLO-2<br>CLO-3 | PLO-3<br>PLO-4 | -Lecture<br>Presentation with video<br>-Interactive Discussion<br>-Feedback                            | -Assignment<br>-Class test                                                                     |
| <b>Exp # 5: Encoding and decoding scheme</b>                          | <b>Implementation</b> of encoding and decoding scheme using NRZ-I                                                                                                                               | <b>Week 7-8 L-26-31</b>             | CLO-2<br>CLO-3 | PLO-3<br>PLO-4 | -Lecture<br>Presentation with video<br>-Interactive Discussion<br>-Feedback                            | -Assignment<br>-Class test                                                                     |
| <b>Exp # 6: IPv4 Implementation</b>                                   | <b>IPv4 implementation</b> of decimal to binary and vice versa                                                                                                                                  | <b>Week 9-10 Lecture - L-32-37</b>  | CLO-3          | PLO-3<br>PLO-4 | -Lecture<br>Presentation with video<br>-Interactive Discussion<br>-Feedback                            | -Assignment<br>- Oral Presentation                                                             |
| <b>Exp# 7: Encoding and decoding using Manchester</b>                 | <b>Implementation</b> of encoding and decoding scheme using Manchester                                                                                                                          | <b>Week 11-12 Lecture - L-38-43</b> | CLO-2<br>CLO-3 | PLO-3<br>PLO-4 | -Lecture<br>Presentation with video<br>-Interactive Discussion                                         | -Assignment<br>-Class test                                                                     |
| <b>Exp# 8: Encoding and decoding using Manchester</b>                 | <b>Implementation</b> of encoding and decoding scheme using AMI and Pseudo ternary                                                                                                              | <b>Week 13 L-44-46</b>              | CLO-2<br>CLO-3 | PLO-3<br>PLO-4 | Lecture<br>Presentation with video<br>-Interactive Discussion                                          | -Assignment<br>-Class test                                                                     |



|                                                                                             |                                                                                                                                                                                               |                            |                |                         |                                                                                  |                            |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------|-------------------------|----------------------------------------------------------------------------------|----------------------------|
|                                                                                             |                                                                                                                                                                                               |                            |                |                         | -Feedback                                                                        |                            |
| <b>Exp# 9:<br/>Network<br/>Topologies</b>                                                   | <b>Study</b> of network topologies using packet tracer software                                                                                                                               | <b>Week 14<br/>L-47-49</b> | CLO-3<br>CLO-4 | PLO-2<br>PLO-3<br>PLO-4 | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| <b>Exp# 10: LAN<br/>transmission,<br/>standard, cross-<br/>wired and<br/>straight cable</b> | <b>Study</b> of LAN transmission media's, topologies, interconnection devices & LAN standards and practically implement the cross-wired cable and straight through cable using clamping tool. | <b>Week 15<br/>L-50-52</b> | CLO-3<br>CLO-4 | PLO-4<br>PLO-5          | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| <b>Exp# 11:<br/>Analog to<br/>digital<br/>conversion and<br/>pulse code<br/>modulation</b>  | <b>Implementation</b> of analog to digital conversion and pulse coded transmission                                                                                                            | <b>Week 16<br/>L-53-55</b> | CLO-5          | PLO-4<br>PLO-5          | Lecture<br>Presentation with<br>video<br>-Interactive<br>Discussion<br>-Feedback | -Assignment<br>-Class test |
| Recess before<br>Exam.                                                                      |                                                                                                                                                                                               |                            |                |                         |                                                                                  |                            |
| Semester End<br>Exam.                                                                       | Start                                                                                                                                                                                         |                            |                |                         |                                                                                  |                            |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 10          |
| <b>Understand</b>           | 00          |
| <b>Apply</b>                | 15          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 00          |

### Assessment and Evaluation

– Lab. performance during Lab. hours (05%)

- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### Learning Resources:

### Books Recommended

1. Foruzan, B. A. (2013). *Data communication and Networking*. Science Engineering & Math Publications.
2. Blanchard, E. (2000). *Introduction to Networking and Data Communications*.

### References and other related books

3. Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer networks*. Pearson.
4. *Video Tutorials on Data communication for beginners*.

### ICT-3107:

### Computer Networking

Full Marks: 100

Credit: 3.0

Contact Hours: 39

**Prerequisite:** Basic ICT at Higher Secondary Level

**Course Type:** Theory

**Course Summary:** This course is an introduction to a broad range of topics in Computer Networking. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Computer Network classes. The course covers Network introduction and Network Models, Data and Signals, Digital and Analog Transmission, Multiplexing, Error Detection, Data Link Control, Media Access Control and Ethernet, Data Link and Network Layer Protocols, Unicast and Multicast Routing, Wireless Networks, Network Management and Firewalls.

**Course Objectives:** The main emphasis of this course is on the organization and management of local area networks (LANs). The course objectives include learning about computer network organization and implementation, obtaining a theoretical understanding of data communication and computer networks, and gaining practical experience in installation, monitoring, and troubleshooting of current LAN systems. Students are introduced to computer communication network design and its operations, and discuss the following topics: Open Systems Interconnection (OSI) communication model; error detection and recovery; local area networks; bridges, routers and gateways; network naming and addressing; and local and remote procedures. On completion of the course, students should be able, in part to design, implement and maintain a typical computer network (LAN).

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| <b>CLO1</b> | To acquire general idea about the course                          |
| <b>CLO2</b> | To Discuss about Computer Network, OSI Model and Physical Layer   |
| <b>CLO3</b> | Analyze the properties of Data Link Layer and Medium Access Layer |
| <b>CLO4</b> | Understand about Network Layer and Transport Layer                |
| <b>CLO5</b> | Explain and design process of Application Layer                   |

### Content of the Course

| Course Content                                |                                                                                                                                                                                                                                                                                                                                                                                                                 | Time Frame                  | Alignment with |                      | Suggested Activities                                                                                                 | Teaching Strategy(s)                                                              |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|----------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Outline                                       | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                                 |                             | CLO            | PLO                  |                                                                                                                      |                                                                                   |
| <b>Course Overview</b>                        | To comprehend the mission and vision of the Department.<br>To acquire general idea about the course Fundamental and Computer Networking.<br>To grasp the content and policies for the class.                                                                                                                                                                                                                    | <b>Week-1<br/>L-1-3</b>     | CLO1           | PLO1                 | Students will give brief introduction about themselves<br>Course outline will be discussing in details               | #Socialized recitation<br>#Show presentation to motivate the students             |
| <b>Introduction and The Physical Layer</b>    | <b>To discuss</b> about Types of Computer Networks, Network Topologies, Circuit and Packet Switching and Protocol Hierarchies.<br><b>To analyze</b> OSI Model, The TCP/IP Model and Uses of Computer Networks.<br><b>To understand</b> Transmission Media, Communication Channel and Wireless Transmission).<br><b>To explain</b> Telephone System, Narrowband ISDN Cellular Radio and Communication Satellite. | <b>Week-2,3<br/>L-4-9</b>   | CLO2           | PLO1<br>PLO2<br>PLO3 | Students will discuss with a group.<br>To apply the knowledge to solve the different real-time problems.             | #Interactive discussion<br>#Lecture discussion with OSI Model and Physical Layer. |
| <b>Exam Review</b>                            | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-4<br/>L-10-12</b>   |                |                      | Lectures, Questions and Answers                                                                                      |                                                                                   |
| <b>Exam.</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-5<br/>L-13-15</b>   |                |                      | Quiz and Tutorial Exam. Will be taken                                                                                |                                                                                   |
| <b>The Physical Layer and Data Link Layer</b> | <b>To explain</b> Telephone System, Narrowband ISDN Cellular Radio and Communication Satellite,<br><b>To discuss data link layer:</b> Design Issues, Error Detection and Correction, Data Communication Interface, Elementary Data Link Protocols, Example Data Link Protocols.                                                                                                                                 | <b>Week-6,7<br/>L-16-21</b> | CLO2<br>CLO3   | PLO2<br>PLO3         | Students will identify, analyze and formulate complex real-time error detection and correction development problems. | #White board discussion<br>#Lecture discussion with Physical and Data Link Layer  |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                             |                               |              |      |                                                                                                                       |                                                                                                                                             |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <b>To explain medium access layer:</b> The channel allocation problem, Multiple access protocols, IEEE standard for LANs and MANs, Bridges, High speed LANs.                                                                                                                                                                                                                                                |                               |              |      |                                                                                                                       |                                                                                                                                             |
| <b>Exam. Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>Week-8<br/>L-22-24</b>     |              |      | Lectures, Questions and Answers                                                                                       |                                                                                                                                             |
| <b>Exam.</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-9<br/>L-25-27</b>     |              |      | Presentation and Tutorial Exam. Will be taken                                                                         |                                                                                                                                             |
| <b>The Network, Transport and Application Layer</b> | <b>To discuss the network layer:</b> Design issues, Routing algorithms, Congestion control algorithms, Internetworking, The network layer in the internet.<br><b>To discuss the transport layer:</b> The transport service, Element of transport protocols, TCP and UDP, Performance issues.<br><b>To discuss the application layer:</b> Network security, DNS, SNMP, E-Mail, Usenet News, WWW, Multimedia. | <b>Week-10-11<br/>L-28-33</b> | CLO4<br>CLO5 | PLO3 | Students will apply reasoning informed by contextual knowledge to understand Network, Transport and Application Layer | #White board discussion<br>#Lecture discussion with network and Transport Layer<br>#Show video about Application Layer to motivate students |
| <b>Exam. Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>Week-12<br/>L-34-36</b>    |              |      | Lectures, Questions and Answers                                                                                       |                                                                                                                                             |
| <b>Exam.</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-13<br/>L-37-39</b>    |              |      | Assignment and Tutorial Exam will be taken                                                                            |                                                                                                                                             |
| <b>Recess before Exam.</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Week-14-15</b>             |              |      |                                                                                                                       |                                                                                                                                             |
| <b>Sem. End Exam.</b>                               | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-16</b>                |              |      |                                                                                                                       |                                                                                                                                             |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |

|          |      |      |  |  |
|----------|------|------|--|--|
| Analyze  |      |      |  |  |
| Evaluate | (05) |      |  |  |
| Create   |      | (05) |  |  |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. Computer Networks, Andrew. S. Tanenbum.
2. Data Communications and Networking, Behrouz. A Forouzan.

**Reference Books and Other Materials**

3. Understanding Data Communications, William A Shay, An International Thompson Publishing Company.
4. Introduction to Data Communications- A practical approach, Larry Hughes, Narosa Publishing House,

**ICT-3108:****Computer Networking Laboratory**

Full Marks: 100

Credit: 1.5

Contact Hours: 39

**Prerequisite:** Basic Networking knowledge at Higher Secondary Level**Course Type:** Laboratory

**Course Summary** This course is an introduction to a broad range of topics in Practical Computer Networking. It is meant to introduce the basic concepts of a range of topics, with the full details left to upper-level Computer Network classes. The course covers basic network topologies and network models, pair to pair network, Dynamic Host Configuration Protocol, DNS Server, Active directory, Cabling and packet sniffing, Basic Network Operation and Troubleshooting, IP Addressing Subnetting.

**Course Objectives:** The main emphasis of this course is on the organization and management of local area networks (LANs). The course objectives include learning about computer network organization and implementation, obtaining a theoretical understanding of data communication and computer networks, and gaining practical experience in installation, monitoring, and troubleshooting of current LAN systems. Students are introduced to computer communication network design and its operations, and discuss the following topics: local area networks; bridges, routers and gateways; network naming and addressing; and local and remote procedures. On completion of the course, students

should be able, in part to design, implement and maintain a typical computer network (LAN).

**Course Learning Outcomes (CLO):**After completing this course a student would be able to

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| <b>CLO1</b> | To acquire general idea about the LAN, Cabling and Different network devices |
| <b>CLO2</b> | To Discuss about Topologies, Printer and output device sharing               |
| <b>CLO3</b> | Analyze the properties ofDHCP, DNS and Directory Services                    |
| <b>CLO4</b> | Understand about Basic Network Operation and Troubleshooting                 |
| <b>CLO5</b> | Explain and testing basic Linux command                                      |

| Course Content                          |                                                                                                                                                                                                                                                                                                                  | Time Frame                | Alignment with |      | Suggested Activities                                                                                     | Teaching Strategy(s)                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                 | Specific Topics                                                                                                                                                                                                                                                                                                  |                           | CLO            | PLO  |                                                                                                          |                                                                       |
| <b>Basic Idea About Network Devices</b> | To acquire general idea about cabling, cross and straight through cabling, CAT 6 and CAT 7 cable identification.<br>To understand basic networking such as LAN, MAN and WAN.<br>To find out the characteristics of different network devices such as Hub, Switch, Router, Microtik Router other network devices. | <b>Week-1<br/>L-1-3</b>   | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details   | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Different Topologies and Sharing</b> | <b>To discuss</b> about Types of Networks Topologies and analysis their performance based on PDR, E2E and other parameters.<br><b>To share</b> printer and scanner and MODEM through multiple PCs.<br><b>To understand</b> Hybrid Topologies and implement them in real time.                                    | <b>Week-2,3<br/>L-4-9</b> | CLO2           | PLO3 | Students will discuss with a group.<br>To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture on Topologies                     |
| <b>Exam Review</b>                      | <b>To make an overview</b>                                                                                                                                                                                                                                                                                       | <b>Week-4<br/>L-10-12</b> |                |      | Lectures, Questions and Answers                                                                          |                                                                       |
| <b>Exam.</b>                            |                                                                                                                                                                                                                                                                                                                  | <b>Week-5<br/>L-13-15</b> |                |      | Quiz and Tutorial Exam. Will be taken                                                                    |                                                                       |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |              |              |                                                                                                                           |                                                                                                  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Network Protocols</b>   | <p><b>To explain</b> DHCP and implement how DHCP work and understand it logically.</p> <p><b>To discuss</b> about domain name Server, how DNS work, explain Subnetting and super netting.</p> <p><b>To explain</b> basic directory services, active directory and passive directory services.</p> <p>To configure basic router and Mikrotik Router.</p>                                                                                                                                             | <b>Week-6,7<br/>L-16-21</b>   | CLO3         | PLO2<br>PLO3 | Students will identify, analyze and formulate complex real-time DHCP and DNS problems.                                    | #White board discussion<br>#Lecture discussion with DHCP and DNS                                 |
| <b>Exam Review</b>         | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-8<br/>L-22-24</b>     |              |              | Lectures, Questions and Answers                                                                                           |                                                                                                  |
| <b>Exam.</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-9<br/>L-25-27</b>     |              |              | Presentation and Tutorial Exam. Will be taken                                                                             |                                                                                                  |
| <b>Troubleshooting</b>     | <p><b>To discuss</b> how to set up windows using disk or bootable pen-drive, Setting NIC driver for basic networking.</p> <p><b>To discuss</b> about several network related errors such as NIC adapter missing, IP address missing and other driver related problem findings and their solutions.</p> <p><b>To discuss</b> about network security related issues and try to solve these problems practically.</p> <p><b>To run</b> several Linux commands to test the security of the network.</p> | <b>Week-10,11<br/>L-28-33</b> | CLO4<br>CLO5 | PLO3         | Students will apply reasoning informed by contextual knowledge to understand network related problems and their solutions | #White board discussion<br>#Lecture discussion with network related problems and their solutions |
| <b>Exam. Review</b>        | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-12<br/>L-34-36</b>    |              |              | Lectures, Questions and Answers                                                                                           |                                                                                                  |
| <b>Exam.</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-13<br/>L-37-39</b>    |              |              | Assignment and Tutorial Exam will be taken                                                                                |                                                                                                  |
| <b>Recess before Exam.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-14,15</b>             |              |              |                                                                                                                           |                                                                                                  |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Week-16</b>                |              |              |                                                                                                                           |                                                                                                  |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |

|                   |      |      |  |      |
|-------------------|------|------|--|------|
| <b>Understand</b> | (05) |      |  |      |
| <b>Apply</b>      |      |      |  | (10) |
| <b>Analyze</b>    |      |      |  |      |
| <b>Evaluate</b>   | (05) |      |  |      |
| <b>Create</b>     |      | (05) |  |      |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. Computer Networks, Andrew. S. Tanenbum.
2. Data Communications and Networking, Behrouz. A Forouzan.
3. Lab. Manual



## B.Sc. (Engg.) Third Year-2024

### 3<sup>rd</sup> Year 2<sup>nd</sup> Semester

**ICT-3201:**
**Microprocessor and Microcontroller**

Full Marks: 100

Credit: 3.0

Contact Hours: 39

**Prerequisite:** ICT-1201

**Course Type:** Theory

**Course Summary:** Microprocessor: microcontroller and microcomputer, evaluation of microprocessor and application, introduction to 8-bit, 16-bit, and 32-bit microprocessors; addressing modes: absolute addressing, 8086 internal architecture, PIN diagram of 8086, Max-Min mode, register structure; memory read write cycle; Instruction set; pipeline concept: interrupts, programmed I/O, memory mapped I/O, interrupt driven I/O, direct memory access; block transfer; cycle stealing; interleaved; multi-tasking and virtual memory; memory interface; bus interface; arithmetic co-processor; assembly language programming of 8086 microprocessors.

**Course Objectives:** The purpose of this course is to teach students the fundamentals of microprocessor and microcontroller systems. The student will be able to incorporate these concepts into their electronic designs for other courses where control can be achieved via a microprocessor/controller implementation. Microprocessor is the course used to provide an understanding of microprocessor hardware and software. Students completing this course will work with microprocessor-based equipment, and be capable of distinguishing hardware from software faults. The superior students will also be capable of participating in product development efforts, including support and development of assembly language code.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Compare</b> accepted standards and guidelines to select appropriate Microprocessor and Microcontroller to meet specified performance requirements.                                                                          |
| <b>CLO2</b> | <b>Assess and solve</b> basic binary math operations using the microprocessor and explain the microprocessor's and Microcontroller's internal architecture and its operation within the area of manufacturing and performance. |
| <b>CLO3</b> | <b>Apply</b> knowledge and demonstrate programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller.                                             |
| <b>CLO4</b> | <b>Design</b> electrical circuitry to the Microprocessor I/O ports in order to interface the processor to external devices.                                                                                                    |
| <b>CLO5</b> | <b>Formulate</b> appropriate computing solution for processor or controller-based application.                                                                                                                                 |

### Content of the Course

| Course Content |                 | Time Frame | Alignment with |     | Suggested Activities | Teaching Strategy(s) |
|----------------|-----------------|------------|----------------|-----|----------------------|----------------------|
| Outline        | Specific Topics |            | CLO            | PLO |                      |                      |

|                                                                                           |                                                                                                                                                                                                                                                                        |                              |              |              |                                                                                                      |                                                                                  |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------|--------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Microcontroller &amp; microcomputer, evolution of microprocessor &amp; application</b> | <b>To compare</b> microprocessors and microcontroller.<br><b>To review</b> the evolution of microprocessor and its application.<br><b>To explain</b> that a microcomputer system consists of CPU, RAM and ROM memories, input /output interfaces and peripheral units. | <b>Week-1<br/>L-1-3</b>      | CLO1<br>CLO2 | PLO1         | Students will give brief introduction about themselves. Course outline will be discussing in details | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Introduction to 8-bit, 16-bit, and 32-bit microprocessors</b>                          | <b>To understand</b> the different instruction bits 8086 microprocessor.                                                                                                                                                                                               | <b>Week-2<br/>L-4-5</b>      | CLO1<br>CLO2 | PLO1         | Students will discuss with a group and try to understand different types of microprocessor           | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Addressing modes: absolute addressing</b>                                              | <b>To understand</b> the different addressing modes of 8086 microprocessor.                                                                                                                                                                                            | <b>Week-2,3<br/>L-6-7</b>    | CLO3         | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>8086 internal architecture, PIN diagram of 8086</b>                                    | <b>To identify</b> 8086 internal architecture and its PIN diagram.                                                                                                                                                                                                     | <b>Week-3<br/>L-8-9</b>      | CLO2         | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Max-Min mode, register structure</b>                                                   | <b>To know</b> max and min mode and registers structure.                                                                                                                                                                                                               | <b>Week-4<br/>L-10-11</b>    | CLO4         | PLO2         | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Memory read write cycle, Instruction set</b>                                           | <b>To understand</b> how to read and write into memory and different instruction sets.                                                                                                                                                                                 | <b>Week-4, 5<br/>L-12-14</b> | CLO4         | PLO2         | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam. Review</b>                                                                       | <b>To make an overview</b>                                                                                                                                                                                                                                             | <b>Week-6<br/>L-15-17</b>    |              |              | Lectures, Questions and Answers                                                                      |                                                                                  |
| <b>Exam.</b>                                                                              |                                                                                                                                                                                                                                                                        | <b>Week-7<br/>L-18-20</b>    |              |              | Quiz and Tutorial Exam. Will be taken                                                                |                                                                                  |
| <b>Pipeline concept: interrupts</b>                                                       | <b>To understand</b> pipelining concepts.<br><br><b>To analyze</b> interrupts.                                                                                                                                                                                         | <b>Week-8<br/>L-21-22</b>    | CLO4         | PLO2<br>PLO4 | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |

|                                                                                      |                                                                                                                                                                                              |                                |      |              |                                                                                             |                                                                                   |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|--------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Programmed I/O, memory mapped I/O, interrupt driven I/O, direct memory access</b> | <b>To design</b> electrical circuitry to the Microprocessor I/O ports in order to interface the processor to external devices.                                                               | <b>Week-8,9<br/>L-23-24</b>    | CLO5 | PLO3         | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback  |
| <b>Exam Review</b>                                                                   | <b>To make an overview</b>                                                                                                                                                                   | <b>Week-9<br/>L-25-26</b>      |      |              | Lectures, Questions and Answers                                                             |                                                                                   |
| <b>Exam.</b>                                                                         |                                                                                                                                                                                              | <b>Week-10<br/>L-27-29</b>     |      |              | Presentation and Tutorial Exam. Will be taken                                               |                                                                                   |
| <b>Block transfer; cycle stealing</b>                                                | <b>To understand</b> data transfer in microprocessor.                                                                                                                                        | <b>Week-11<br/>L-30-31</b>     | CLO5 | PLO2         | Students will apply reasoning informed by contextual knowledge to understand the problem    | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback  |
| <b>Interleaved; multi-tasking and virtual memory</b>                                 | <b>To understand</b> concurrency and <b>review</b> virtual memory.                                                                                                                           | <b>Week-11,12<br/>L-32-33</b>  | CLO5 | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to solve the different problems. | #-Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Memory interface; bus interface; arithmetic co-processor</b>                      | <b>To apply</b> knowledge to interface peripherals with the microprocessor.                                                                                                                  | <b>Week-12<br/>L-34,35</b>     | CLO5 | PLO3         | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback  |
| <b>Assembly language programming of 8086 microprocessors</b>                         | <b>To apply</b> knowledge and <b>demonstrate</b> programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller. | <b>Week-13<br/>L-36-38</b>     | CLO5 | PLO3<br>PLO4 | Students will identify, analyze and formulate complex engineering problems.                 | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback  |
| <b>Exam. Review</b>                                                                  | <b>To make an overview</b>                                                                                                                                                                   | <b>Week-14<br/>L-39-40</b>     |      |              | Lectures, Questions and Answers                                                             |                                                                                   |
| <b>Exam.</b>                                                                         |                                                                                                                                                                                              | <b>Week-14,-15<br/>L-42-43</b> |      |              | Assignment and Tutorial Exam will be taken                                                  |                                                                                   |
| <b>Recess before Exam.</b>                                                           |                                                                                                                                                                                              | <b>Week-15</b>                 |      |              |                                                                                             |                                                                                   |
| <b>Sem. End Exam.</b>                                                                | <b>Start</b>                                                                                                                                                                                 | <b>Week-16</b>                 |      |              |                                                                                             |                                                                                   |

#### Assessment Pattern (According FET Ordinance)

CIE: (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|

|                   |      |      |      |      |
|-------------------|------|------|------|------|
| <b>Remember</b>   |      |      | (05) |      |
| <b>Understand</b> | (05) |      |      |      |
| <b>Apply</b>      |      |      |      | (10) |
| <b>Analyze</b>    |      |      |      |      |
| <b>Evaluate</b>   | (05) |      |      |      |
| <b>Create</b>     |      | (05) |      |      |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books**

1. Douglas V Hall – Microprocessors and Interfacing, 3rd Edition, Tata Mcgraw Hill, 2019-2020

**Reference Books and Other Materials**

2. Barry B. Brey (2009), The Intel Microprocessors, Eighth Edition. Prentice Hall
3. Godse (2007-2008), Microprocessor and Microcontroller System, First Edition, Technical Publications.

**ICT-3202:      Microprocessor and Microcontroller Lab**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-1201**Course Type:** Practical/Laboratory work

**Course Summary:** This course introduces the programming, architecture and interfacing of the Intel 80x86 microprocessors for the students who had previous knowledge in both computer architecture: hardware and software. This course helps students to understand how the microcomputer system works and how to build a microcomputer system using a microprocessor or microcontroller using assembly language. This course focuses on programming language for machine code, rules, syntax and how to write programs to perform any task with microprocessor.

**Course Objectives:** This course is intended for students who have some prior programming experience. The course will introduce the basics of the microprocessor instructions, assembly language and writing different programs in assembly language using EMU 8086.

**Course Learning Outcomes (CLO):**After completing this course a student would be able to

|             |                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Discuss</b> different concepts of Microprocessors.                                                                                                              |
| <b>CLO2</b> | <b>Analyze</b> and <b>identify</b> the major component of a computer system and steps involving in assembling, linking, and executing a program to run the system. |
| <b>CLO3</b> | <b>Create</b> programs for microcontroller using assembly language and programming knowledge.                                                                      |
| <b>CLO4</b> | <b>Assess</b> and <b>solve</b> basic binary math operations using the microprocessor.                                                                              |
| <b>CLO5</b> | <b>Apply</b> knowledge and <b>demonstrate</b> programming proficiency.                                                                                             |

| Course content                                                                    |                                                                                                          | Time Frame                | Alignment with |      | Suggested Activities                                                                                       | Teaching Strategy(s)                                                             |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------|----------------|------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Outline                                                                           | Specific Topics                                                                                          |                           | CLO            | PLO  |                                                                                                            |                                                                                  |
| <b>Exp #1: Introduction to Assembly Language and EMU 8086.</b>                    | <b>To be familiar</b> with the assembly programming language and the 8086 microprocessor.                | <b>Week-1<br/>L-1-3</b>   | CLO1<br>CLO2   | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details     | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exp #2: Program structure and familiarization with different instructions.</b> | <b>To identify</b> program structure.<br><b>To understand</b> different program structure.               | <b>Week-2<br/>L-4-5</b>   | CLO1<br>CLO2   | PLO1 | Students will discuss with a group.<br>To apply the knowledge to learn program structure and instructions. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exp #3: Arithmetic operations.</b>                                             | <b>To learn</b> arithmetic operations.<br><br><b>To practice</b> some examples of arithmetic operations. | <b>Week-2,3<br/>L-6-7</b> | CLO3<br>CLO4   | PLO3 | Students will discuss with a group.<br>To apply the knowledge to solve the different problems.             | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exp #4: Understanding Advanced 8086 I/O Instructions</b>                       | <b>To understand</b> how to read and write into memory and different instruction sets.                   | <b>Week-3<br/>L-8-9</b>   | CLO2           | PLO2 | Students will discuss with a group.<br>To apply the knowledge to solve the different problems.             | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exp #5: Use of Loop.</b>                                                       | <b>To apply</b> knowledge and <b>demonstrate</b> different loops of assembly language.                   | <b>Week-4<br/>L-10-12</b> | CLO2<br>CLO3   | PLO2 | Students will discuss with a group.<br>To apply the knowledge to solve the different problems              | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam Review</b>                                                                | <b>To make an overview</b>                                                                               | <b>Week-5<br/>L-13-15</b> |                |      | Demonstrate, Questions and Answers                                                                         |                                                                                  |

|                                                                                         |                                                                                                                        |                               |              |              |                                                                                             |                                                                                  |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|--------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Examination</b>                                                                      |                                                                                                                        | <b>Week-6<br/>L-16-18</b>     |              |              | Quiz and Tutorial Exam. Will be taken                                                       |                                                                                  |
| <b>Exp #6: Use of Array.</b>                                                            | <b>To apply</b> knowledge and <b>demonstrate</b> array of assembly language.                                           | <b>Week-7<br/>L-19-21</b>     | CLO5         | PLO2         | Students will demonstrate /present and solve the problems.                                  | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam Review</b>                                                                      | <b>To make an overview</b>                                                                                             | <b>Week-8<br/>L-22,23</b>     |              |              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                                                                            |                                                                                                                        | <b>Week-8,9<br/>L-24-25</b>   |              |              | Presentation and Tutorial Exam. Will be taken                                               |                                                                                  |
| <b>Exp #7: Logic operations in assembly language.</b>                                   | <b>To learn</b> logic of assembly language.<br><br><b>To write</b> a program in assembly language to illustrate logic. | <b>W-9<br/>L-26-27</b>        | CLO3<br>CLO5 | PLO2<br>PLO3 | Students will discuss with a group. To apply the knowledge to solve the different problems  | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exp #8: Familiarizing with MDA 8086 trainer Kit Commands in Serial Monitor Mode.</b> | <b>To apply</b> with the MDA 8086 microprocessor kit and its operations.                                               | <b>Week-10,11<br/>L-28-33</b> | CLO3<br>CLO5 | PLO3<br>PLO5 | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam. Review</b>                                                                     | <b>To make an overview</b>                                                                                             | <b>Wee-12<br/>L-34-36</b>     |              |              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                                                                            |                                                                                                                        | <b>Week-13<br/>L-37-39</b>    |              |              | Presentation and Tutorial Exam. will be taken                                               |                                                                                  |
| <b>Recess before Exam.</b>                                                              |                                                                                                                        | <b>Week-14,15</b>             |              |              |                                                                                             |                                                                                  |
| <b>Sem. End Exam.</b>                                                                   | <b>Start</b>                                                                                                           | <b>Week-16</b>                |              |              |                                                                                             |                                                                                  |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's Category | Test |
|------------------|------|
| Remember         | 10   |
| Understand       | 00   |
| Apply            | 15   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 00   |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. Mazidi, M. A., Naimi, S., & Naimi, S. (2011). The AVR Microcontroller and Embedded Systems. August.
2. Hall, D. V., & Rood, A. L. (1986). Microprocessors and interfacing: programming and hardware (pp. 442-487). McGraw-Hill.
3. Yu, Y. Y., & Marut, C. (1992). Assembly Language Programming and Organization IBM PC. McGraw-Hill Higher Education.

**Reference Books and Other Materials**

4. Irvine, K. R. (1990). Assembly Language for the IBM-PC. Prentice Hall PTR.
5. <https://kader05cuett.wordpress.com/lecture-materials-2/microprocessor-and-interfacing-sessional/>

**ICT-3203: Optical Communications**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-2205**Course Type:** Theory**Course Summary:** The topic will be discussed in class by the course teacher.**Course Objectives:** To realize the significance of optical fiber communications.**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|              |                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO-1</b> | Understand and analyze the constructional parameters of optical fibers including the reason Light is used Instead of Electricity. |
| <b>CLO-2</b> | Understand Ray theory transmission and different modes fiber.                                                                     |
| <b>CLO-3</b> | Estimate the losses due to attenuation, absorption, scattering and bending.                                                       |
| <b>CLO-4</b> | Be aware of how optical fiber cable is made in Industry                                                                           |
| <b>CLO-5</b> | Compare various optical detectors and choose suitable one for different applications.                                             |
| <b>CLO-6</b> | Be able to design the optical system.                                                                                             |
| <b>CLO-7</b> | Gaining practical knowledge on how to measure optical fiber.                                                                      |
| <b>CLO-8</b> | Be able to know the real applications of optical fiber and future development of it.                                              |

| Course Content | Time Frame | Alignment with | Suggested Activities | Teaching Strategy(s) |
|----------------|------------|----------------|----------------------|----------------------|
|----------------|------------|----------------|----------------------|----------------------|

| Outlines                                             | Specific Topics                                                                                                                                                                                                                                                      |             | CLO   | PLO            |                                                                          |                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|----------------|--------------------------------------------------------------------------|----------------------------------------|
| <b>Introduction</b>                                  | Historical development, The general system, Advantages of optical fiber communication.                                                                                                                                                                               | Week 1      | CLO-1 | PLO-1<br>PLO-2 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br><br>- Oral Presentation |
| <b>Fundamentals of Optics</b>                        | Light, Electromagnetic spectrum, Propagation of light, Geometric and physical optics, Snell's law of reflection and refraction, Total internal reflection, Optical energy losses, Specular and diffuse reflection, Diffraction and polarization, Optical components. | Week 2- 3   | CLO-1 | PLO-2          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Optical fiber wave guides</b>                     | Introduction, Ray theory transmission, cylindrical fiber, Single-mode fibers.                                                                                                                                                                                        | Week 4 - 5  | CLO-2 | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Transmission characteristics of optical fiber</b> | Introduction, Attenuation, Losses, Dispersion, Polarization, Nonlinear phenomena.                                                                                                                                                                                    | Week 6 - 7  | CLO-3 | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Optical cables and connection</b>                 | Introduction, Preparation of optical fiber, Optical fiber cables, Stability of the fiber transmission characteristics, Cable design. Fiber connections.                                                                                                              | Week 8 -9   | CLO-4 | PLO-3          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Optical sources and detectors</b>                 | Introduction, Optical sources- LASER and Light Emitting Diode, Optical detectors.                                                                                                                                                                                    | Week 10 -11 | CLO-5 | PLO-4          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation     |
| <b>Optical fiber measurements</b>                    | Introduction, Fiber attenuation measurements, Fiber dispersion measurements, Fiber                                                                                                                                                                                   | Week 12 -13 | CLO-7 | PLO-4<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test             |



|                                             |                                                                                                                                                                                                                                                      |         |                |  |                                                  |                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|--|--------------------------------------------------|------------------------------------|
|                                             | refractive index profile measurements, Fiber cutoff wavelength measurements, Fiber numerical aperture measurements, Fiber diameter measurements, Mode-field diameter for single mode fiber, Reflectance and optical return loss, Field measurements. |         |                |  |                                                  |                                    |
| <b>Applications and future developments</b> | Introduction, Public network applications, Military applications, Civil, consumer and industrial applications. Optical sensor systems, Computer applications, Local area networks.                                                                   | Week 14 | CLO-6<br>CLO-8 |  | -Lecture Presentation<br>-Interactive Discussion | -Assignment<br>- Oral Presentation |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                               |                            |                                    |                                |
| <b>Apply</b>                         |                                    |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                    |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                    |                                |
| <b>Create</b>                        |                                    | (05)                       |                                    |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources****Text Books****Books Recommended**

1. *Optical Fiber Communications John M Senior, Prentice Hall of India..*
2. *Optical Communication Systems, John Gower, Prentice Hall of India.*

**References and other related books**

3. *Fiber Optic Communication, D.C. Agarwal, S CHAND & Company Ltd.*
4. *Fiber Optic Theory, Heathkit Educational System.*
5. *Fiber Optic Communications, Joseph C. Palais*

**ICT-3204: Optical Communication Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-3203**Course Type:** Practical/Laboratory work**Course Summary:** The topic will be discussed in class by the course teacher.**Course Objectives:** To learn the basic elements of optical fiber transmission link, fiber modes configurations and structures.**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| <b>CLO1</b> | Understand the properties of Step index fibers and Graded index fibers          |
| <b>CLO2</b> | Know the mode field diameter of Step index fibers                               |
| <b>CLO3</b> | Achieve the knowledge on Material dispersion and waveguide dispersion           |
| <b>CLO4</b> | Understand Rayleigh Scattering Loss, Infrared Absorption Loss, Ultraviolet Loss |
| <b>CLO5</b> | Gain the knowledge on WDM                                                       |

| Course Content |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Time Frame           | Alignment with |              | Suggested Activities                                                                                                     | Teaching Strategy(s)                                          |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|--------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Outline        | Specific Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      | CLO            | PLO          |                                                                                                                          |                                                               |
|                | *Calculate for Step Index Fibers (using MATLAB/Optical ETB): (i) Numerical aperture (ii) Acceptance angle (iii) Solid acceptance angle (iv) Propagation constant<br>* Calculate for Step Index Fibers (using MATLAB/ Optical ETB): (i) Propagation constant (ii) Normalized propagation constant (iii) V number (iv) Check whether the fiber is SM or MM (v) Graph between normalized propagation constant and V number (vi) Cut off wavelength (vii) Number of modes traveling in fibers | Week 1-3<br>L-1-9    | CLO-1          | PLO2<br>PLO3 | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
|                | * Calculate for Graded Index Fibers (using MATLAB/ Optical ETB): (i) Graphical representation of core refractive index w.r.t. radius of core for                                                                                                                                                                                                                                                                                                                                          | Week-4- 5<br>L-10-15 | CLO-1          | PLO3<br>PLO4 | Lecture Interactive<br>Discussion<br>Presentation with video                                                             | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |                |                |                                                                                                                    |                                                            |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------|----------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|  | different profile parameters. (ii) Graphical representation of NA w.r.t. radius of core for different profile parameters (iii) Acceptance angle for given value of radius.<br>* Calculate for Graded Index Fibers (using MATLAB/ Optical ETB): (i) NA, (ii) V number (iii) Check whether the fiber is SM or MM (iv) Graph between V number and wavelength (v) Cut off wavelength (vi) Number of modes traveling in fiber.                                                                 |                       |                |                | Experiment Implementation in the assigned Laboratory                                                               |                                                            |
|  | * Calculate for Step Index Fiber (using MATLAB/ Optical ETB) of Mode Field Diameter.<br>* Calculate for pure Silica fibers (using MATLAB/Optical ETB): (i) Graph for refractive index with wavelength (ii) Graph for $dn/d\lambda$ with wavelength $\lambda$ (iii) Graph for $d^2n/d\lambda^2$ with wavelength $\lambda$ (iv) Graph for material dispersion (Dm) with wavelength (v) Calculate material dispersion (Dm) at given wavelength (vi) Find zero material dispersion wavelength | Week-6-7<br>L-16-21   | CLO-2<br>CLO-3 | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|  | *Calculate for Step Index Fibers (using MATLAB/ Optical ETB): (i) V number (ii) Cut off wavelength (iii) Waveguide dispersion at given wavelength (iv) Plot of waveguide dispersion with wavelength<br>*Calculate (using MATLAB/ Optical ETB): (i) The Rayleigh Scattering Loss for Silica Fibers (ii) Plot a graph for Loss with wavelength                                                                                                                                              | Week-8-10<br>L-22-30  | CLO-3<br>CLO-4 | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|  | *Calculate the Infrared absorption loss for silica fibers and plot a graph with wavelength (using MATLAB/ Optical ETB).<br>*Wavelength Division Multiplexing and De-multiplexing (using MATLAB/ Optical ETB).                                                                                                                                                                                                                                                                             | Week-11-13<br>L-31-39 | CLO-4<br>CLO-5 | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|  | Recess before Exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Week-14, 15           |                |                |                                                                                                                    |                                                            |

|  |                          |         |  |  |  |  |
|--|--------------------------|---------|--|--|--|--|
|  | Semester End Exam. Start | Week-16 |  |  |  |  |
|--|--------------------------|---------|--|--|--|--|

### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-20Marks)

| Bloom's Category<br>Marks:30 | Lab. Performance<br>(05) | Lab. Note Book<br>(05) | Continuous Viva-<br>Voce<br>(10) | Lab. Attendance<br>(10) |
|------------------------------|--------------------------|------------------------|----------------------------------|-------------------------|
| Remember                     |                          |                        | (10)                             |                         |
| Understand                   |                          |                        |                                  |                         |
| Apply                        |                          | (05)                   |                                  | (10)                    |
| Analyze                      |                          |                        |                                  |                         |
| Evaluate                     | (05)                     |                        |                                  |                         |
| Create                       |                          |                        |                                  |                         |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 10   |
| Understand          | 00   |
| Apply               | 15   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 00   |

### **Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### **Learning Resources**

#### **Books Recommended**

1. *Optical Fiber Communications* John M Senior, Prentice Hall of India.

#### **Related other books**

2. *Optical Communication Systems*, John Gower, Prentice Hall of India.
3. *Fiber Optic Communication*, D.C. Agarwal, S CHAND & Company Ltd.
4. *Fiber Optic Theory*, Heathkit Educational System.
5. *Fiber Optic Communications*, Joseph C. Palais

### **ICT-3205: Computer Architecture and Organization**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** None

**Course Type:** Theory

**Course Summary:** This course aims to provide a strong foundation to understand modern computer system architecture and to apply these insights and principles to future computer designs. The course is structured around the three primary building blocks of general-purpose computing systems: Microprocessors, memories, and parallel processing.

**Course Objectives:** course will introduce the students to the fundamental concepts underlying modern computer organization and architecture. Main objective of the course is to familiarize students about hardware design including logic design, basic structure and behavior of the various functional modules of the computer and how they interact to provide the processing needs of the user. It will also provide knowledge on microprocessor and interfacing devices, 8085/8086 microprocessor architecture, assembly language, and interfaces.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|              |                                                                                 |
|--------------|---------------------------------------------------------------------------------|
| <b>CLO-1</b> | Understand the basics of computer hardware and Register Transfer Logic          |
| <b>CLO-2</b> | Understand the basics of processors organization and logic design of processors |
| <b>CLO-3</b> | Understand the basics of control logic design of microcomputer                  |
| <b>CLO-4</b> | Understand the fundamental knowledge on system organization of computer system  |
| <b>CLO-5</b> | Understand the basic concept on assembly language programming                   |
| <b>CLO-6</b> | Understand the basic concepts on 8085/8086 Instructions                         |

| Course Content                                                 |                                                                                                                                                                                                          | Time Frame       | Alignment with |                | Suggested Activities                                                     | Teaching Strategy(s)                   |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|----------------|--------------------------------------------------------------------------|----------------------------------------|
| Outlines                                                       | Specific Topics                                                                                                                                                                                          |                  | CLO            | PLO            |                                                                          |                                        |
| <b>Register Transfer Logic</b>                                 | Inter-register transfer, arithmetic, logic and shift micro-operations, conditional control statements, fixed point binary data, overflow, arithmetic shifts, floating point data, and instruction codes. | Week 1 L-1-3     | CLO-1          | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br><br>- Oral Presentation |
| <b>Processor Logic Design</b>                                  | Processor organization, design of arithmetic and logic unit, status register, design of shifter, processor unit, design of accumulator.                                                                  | Week 2-3 L-4-9   | CLO-2          | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Control Logic Design</b>                                    | Control organization, hardware control, micro-program control, control of processor unit, PLA control.                                                                                                   | Week-4-5 L-10-15 | CLO-3          | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>System Organization</b>                                     | Communication methods, IO and system control, parallel processing- processor level parallelism, multiprocessors, fault tolerance.                                                                        | Week 6-7 L-16-21 | CLO-4          | PLO-4<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |
| <b>Introduction to 8085/8086 assembly language programming</b> | The 8085/8086 programming model, Instruction classification, Instruction, Data formats, Storage, How to write assemble and execute a simple program, overview of the instruction set.                    | Week -8 L-22-27  | CLO-5          | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test             |

|                                                                         |                                                                                                                                                                                                                                                                         |                        |       |                |                                                                          |                                    |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------|----------------|--------------------------------------------------------------------------|------------------------------------|
| <b>8085/8086 Microprocessor Architecture and Memory I/O Interfacing</b> | Microprocessor operations, 8085/8086 MPU, Example of an 8085/8086-based microcomputer, Memory interfacing, Basic interfacing concepts, Interfacing output displays, Interfacing input devices, Memory mapped I/O, Testing and troubleshooting I/O interfacing circuits. | Week 10 -11<br>L-28-33 | CLO-5 | PLO-3<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>Introduction to 8085/8086 Instructions</b>                           | Data transfer (copy) operations, Arithmetic Operations, Logic operations, Branch operations, Writing assembly language program, debugging a program, stack and subroutines.                                                                                             | Week 12-13<br>L-34-39  | CLO-6 | PLO-3<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test         |
| Recess before Exam.                                                     |                                                                                                                                                                                                                                                                         | Week 14, 15            |       |                |                                                                          |                                    |
| Semester End Exam.                                                      | Start                                                                                                                                                                                                                                                                   | Week 16                |       |                |                                                                          |                                    |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quizz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|--------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                           |                            |
| <b>Understand</b>                | (05)                           |                        |                                |                            |
| <b>Apply</b>                     |                                |                        |                                | (10)                       |
| <b>Analyze</b>                   |                                |                        |                                |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                                |                            |
| <b>Create</b>                    |                                | (05)                   |                                |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### Learning Resources

**Text Books**

*Computer Architecture and Logic Design*, T C Bartee, McGraw Hill.

**Reference Books and Other Materials**

*Digital Logic and Computer Design*, M. Morris Mano, Prentice-Hall, New Delhi.

*Computer Architecture & Organization*, J P Hayes, McGraw-Hill.

*Architecture of Computer Hardware and Software*, I Englander, Wiley Eastern.

*Microprocessor Architecture, Programming and Applications with 8085*, Ramesh S. Goankar.

*The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, and Pentium pro Processors*

**ICT-3207: Operating Systems**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1203, ICT-2104, ICT-2106

**Course Type:** Theory

**Course Summary:** Covers the classical internal algorithms and structures of operating systems, including CPU scheduling, memory management, and device management. Considers the unifying concept of the operating system as a collection of cooperating sequential processes. Covers topics including file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity.

**Course Objectives:** To provide knowledge on operating systems of computer and mobile phone systems, structure of operating systems, CPU scheduling, deadlocks, and memory management of digital systems.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>understand</b> the goals of OS concepts, design of operating systems, and extrapolate the interactions among the various components of computing systems. |
| <b>CLO2</b> | <b>understand</b> the structure and organization of the file system.                                                                                         |
| <b>CLO3</b> | <b>understand</b> what a process is and how processes are synchronized and scheduled.                                                                        |
| <b>CLO4</b> | <b>understand</b> different approaches to deadlock handling, dead lock prevention, avoidance, detection, and recovery and memory management.                 |
| <b>CLO5</b> | <b>use</b> system calls for managing processes, memory and the file system.                                                                                  |

| Course Content                       |                                                                                                                                                     | Time Frame                      | Alignment with |              | Suggested Activities                                                                                 | Teaching Strategy(s)                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------|--------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Outline                              | Specific Topics                                                                                                                                     |                                 | CLO            | PLO          |                                                                                                      |                                                                                  |
| <b>Operating System Introduction</b> | <b>To acquire</b> knowledge of OS, its design, and different operating system.<br><b>To understand</b> parallel, real-time, and distributed systems | <b>Week-1</b><br><b>L-1,2,3</b> | CLO1           | PLO1         | Students will give brief introduction about themselves. Course outline will be discussing in details | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Operating System Structures</b>   | <b>To understand</b> structures and design of operating systems                                                                                     | <b>Week-2</b><br><b>L-4,5,6</b> | CLO2           | PLO2<br>PLO3 | Students will discuss with a group. To apply the knowledge to solve the different problems.          | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |

|                                |                                                                                                                                    |                                     |      |              |                                                                                             |                                                                                  |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------|--------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Processes</b>               | <b>To understand</b> process, its state, threads, and inter-process communication.                                                 | <b>Week-3</b><br><b>L-7,8,9</b>     | CLO3 | PLO2         | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>CPU Scheduling</b>          | <b>To acquire</b> knowledge of CPU scheduling.<br><b>To learn</b> CPU various scheduling algorithms.                               | <b>Week-4</b><br><b>L-10,11,12</b>  | CLO3 | PLO4         |                                                                                             | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam Review</b>             | <b>To make an overview</b>                                                                                                         | <b>Week-5</b><br><b>L-13,14,15</b>  |      |              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Examination</b>             |                                                                                                                                    | <b>Week-6</b><br><b>L-16,17,18</b>  |      |              | Quiz and Tutorial Exam. Will be taken                                                       |                                                                                  |
| <b>Process Synchronization</b> | <b>To understand</b> process synchronization, and semaphores.<br><b>To understand</b> the algorithms used to synchronize hardware. | <b>Week-7</b><br><b>L-19,20,21</b>  | CLO3 | PLO2<br>PLO4 | Students will identify, analyze and formulate complex engineering problems.                 | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam Review</b>             | <b>To make an overview</b>                                                                                                         | <b>Week-8</b><br><b>L-22,23,24</b>  |      |              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                   |                                                                                                                                    | <b>Week-9</b><br><b>L-25,26,27</b>  |      |              | Presentation and Tutorial Exam. Will be taken                                               |                                                                                  |
| <b>Deadlocks</b>               | <b>To discuss and explain</b> dead lock and deadlock handling, dead lock prevention, avoidance, detection, and recovery.           | <b>Week-10</b><br><b>L-28,29</b>    | CLO4 | PLO2<br>PLO4 | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Memory Management</b>       | <b>To understand</b> logical and physical memory address space.<br><b>To learn</b> different approaches memory management.         | <b>Week-10,11</b><br><b>L-30,31</b> | CLO4 | PLO3         | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |



|                            |                                                            |                               |      |      |                                                                                             |                                                                                  |
|----------------------------|------------------------------------------------------------|-------------------------------|------|------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>File System</b>         | To understand file access methods and protection strategy. | <b>Week-11<br/>L-32,33</b>    | CLO5 | PLO4 | Students will discuss with a group. To apply the knowledge to solve the different problems. | -Lecture with the ppt and online videos<br>-interactive discussion<br>- Feedback |
| <b>Exam. Review</b>        | To make an overview                                        | <b>Wee-12<br/>L-34,35,36</b>  |      |      | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>               |                                                            | <b>Week-13<br/>L-37,38,39</b> |      |      | Assignment and Tutorial Exam will be taken                                                  |                                                                                  |
| <b>Recess before Exam.</b> |                                                            | <b>Week-14,15</b>             |      |      |                                                                                             |                                                                                  |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                               | <b>Week-16</b>                |      |      |                                                                                             |                                                                                  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### Text Books

1. Silberschatz, A., Galvin, P. B., & Gagne, G. (2014). *Operating system concepts essentials*. John Wiley & Sons, Inc. 8th Edition.
2. Andrew S. Tanenbaum and Albert S. Woodhull. *Operating Systems: Design and Implementation*, Prentice-Hall, (2nd Ed.).

**Reference Books and Other Materials**

3. Stallings, W. (2005). *Operating Systems: Internals and Design Principles—Edition: 5*. Pearson.
4. <https://www.geeksforgeeks.org/operating-systems/>
5. [https://www.tutorialspoint.com/operating\\_system/](https://www.tutorialspoint.com/operating_system/)
6. <https://www.studytonight.com/operating-system/>
7. <https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/>

**ICT-3208: Operating Systems Laboratory**

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-1203, ICT-2104, ICT-2106**Course Type:** Practical/Laboratory work

**Course Summary:** Covers the classical internal algorithms and structures of operating systems, including CPU scheduling, memory management, and device management. Considers the unifying concept of the operating system as a collection of cooperating sequential processes. Covers topics including file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity.

**Course Objectives:** This course will provide hand-on training on operating systems of computer and mobile phone systems, structure of operating systems, CPU scheduling, deadlocks, and memory management of digital systems.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Apply</b> fundamental programming concepts, algorithms, theorems and methods in operating system to solve substantial problems (e.g. CPU scheduling algorithm, page replacement algorithms, process and file system management using system calls) on Ubuntu platforms. |
| <b>CLO2</b> | <b>Experiment</b> with Ubuntu commands and shell programming on Linux platforms.                                                                                                                                                                                           |
| <b>CLO3</b> | <b>Understand</b> what a process is and how processes are synchronized and scheduled.                                                                                                                                                                                      |
| <b>CLO4</b> | <b>Understand</b> different approaches to deadlock handling, dead lock prevention, avoidance, detection, and recovery and memory management.                                                                                                                               |
| <b>CLO5</b> | <b>Use</b> system calls for managing processes, memory and the file system.                                                                                                                                                                                                |

| Topics                                                                                                                                          | Specific Outcome(s)                                                                  | Time Frame          | Alignment with |              | Suggested Activities                                                                      | Teaching Strategy(s)                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------|----------------|--------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                                                                                 |                                                                                      |                     | CLO            | PLO          |                                                                                           |                                                                                   |
| <b>Exp #1:</b> Non-preemptive CPU Scheduling algorithms to find turnaround time and waiting time. (1) FCFS (2) SJF (3) Round Robin (4) Priority | <b>To learn</b> calculating computational cost of various CPU scheduling algorithms. | <b>Week-1 L-1,2</b> | CLO3           | PLO1<br>PLO2 | Student will bring your notebook.<br><br>Code Block software is installed in OS (Ubuntu). | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |

|                                                                                                                                                                                                                                                                                                            |                                                                                |                             |              |              |                                                            |                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------|--------------|--------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Exp #2:</b> Multi-level queue scheduling algorithm considering the following scenario: All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user. (1) FCFS (2) SJF (3) Round Robin (4) Priority | <b>To learn</b> various CPU scheduling algorithms.                             | <b>Week-1,2<br/>L-3,4</b>   | CLO3         | PLO1<br>PLO2 | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exp #3:</b> File allocation strategies: (1) Sequential (2) Indexed (3) Linked                                                                                                                                                                                                                           | <b>To learn</b> the interactions among the various components of file systems. | <b>Week-2<br/>L-5,6</b>     | CLO4<br>CLO5 | PLO2         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exp #4:</b> Memory management Techniques: (1) MVT (2) MFT                                                                                                                                                                                                                                               | <b>To learn</b> memory management techniques.                                  | <b>W-3<br/>L-7,8</b>        | CLO4<br>CLO5 | PLO2         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exp #5:</b> Contiguous memory allocation techniques: (1) Worst-fit (2) Best-fit (3) First-fit                                                                                                                                                                                                           | <b>To learn</b> memory allocation techniques.                                  | <b>W-3,4<br/>L-9,10</b>     | CLO4         | PLO2         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exp #6:</b> Open source operating system: (1) Basic Linux command                                                                                                                                                                                                                                       | <b>To learn</b> basic Linux command.                                           | <b>W-4<br/>L-11,12</b>      | CLO1<br>CLO2 | PLO1<br>PLO2 | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exam Review</b>                                                                                                                                                                                                                                                                                         | <b>To make an overview</b>                                                     | <b>Week-5<br/>L-13,14</b>   |              |              | Demonstrate, Questions and Answers                         |                                                                                   |
| <b>Examination</b>                                                                                                                                                                                                                                                                                         |                                                                                | <b>Week-5,6<br/>L-15,16</b> |              |              | Quiz and Tutorial Exam. Will be taken                      |                                                                                   |
| <b>Exp #7:</b> File organization techniques: (1) Single level directory (2) Two level directory (3) Hierarchical                                                                                                                                                                                           | <b>To learn</b> how to organize files.                                         | <b>W-6<br/>L-17,18</b>      | CLO1<br>CLO5 | PLO2         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |

|                                                                                                                          |                                                                                                             |                                |              |              |                                                            |                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------|--------------|--------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Exp #8:</b> (1) Implemented Banker's algorithm for the purpose of deadlock avoidance. (2) Dining-Philosophers problem | <b>To learn the</b> deadlock avoidance algorithms.<br><br><b>To understand</b> Dining-Philosophers problem. | <b>W-7<br/>L-19,20</b>         | CLO4         | PLO2<br>PLO4 | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br><br>- interactive discussion<br><br>- Feedback |
| <b>Exam Review</b>                                                                                                       | <b>To make an overview</b>                                                                                  | <b>Week-7,8<br/>L-21,22</b>    |              |              | Lectures, Questions and Answers                            |                                                                                           |
| <b>Exam.</b>                                                                                                             |                                                                                                             | <b>Week-8,9<br/>L-23,24,25</b> |              |              | Presentation and Tutorial Exam. Will be taken              |                                                                                           |
| <b>Exp #9:</b> Disc scheduling algorithms: (1) FCFS (2) SCAN (3) C-SCAN                                                  | <b>To understand</b> disc scheduling algorithms.                                                            | <b>W-9<br/>L-26,27</b>         | CLO4<br>CLO5 | PLO2         |                                                            | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback         |
| <b>Exp #10:</b> Page Replacement algorithms: (1) FIFO (2) LRU (3) LFU                                                    | <b>To understand</b> page replacement algorithms.                                                           | <b>Week-10<br/>L-28,29,30</b>  | CLO5         | PLO4         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback         |
| <b>Exp #11:</b> Open source operating system: (1) Shell Scripting                                                        | <b>To learn</b> shell scripting in Ubuntu OS.                                                               | <b>W-11<br/>L-31,32,33</b>     | CLO1<br>CLO2 | PLO2         | Students will demonstrate /present and solve the problems. | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback         |
| <b>Exam. Review</b>                                                                                                      | <b>To make an overview</b>                                                                                  | <b>Week-12<br/>L-34,35,36</b>  |              |              | Lectures, Questions and Answers                            |                                                                                           |
| <b>Exam.</b>                                                                                                             |                                                                                                             | <b>Week-13<br/>L-37,38,39</b>  |              |              | Presentation and Tutorial Exam. will be taken              |                                                                                           |
| <b>Recess before Exam.</b>                                                                                               |                                                                                                             | <b>Week-14,15</b>              |              |              |                                                            |                                                                                           |
| <b>Sem. End Exam.</b>                                                                                                    | <b>Start</b>                                                                                                | <b>Week-16</b>                 |              |              |                                                            |                                                                                           |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |

|          |      |  |  |  |
|----------|------|--|--|--|
| Analyze  |      |  |  |  |
| Evaluate | (05) |  |  |  |
| Create   |      |  |  |  |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 10   |
| Understand       | 00   |
| Apply            | 15   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 00   |

### **Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. Silberschatz, A., Galvin, P. B., & Gagne, G. (2014). Operating system concepts essentials. John Wiley & Sons, Inc. 8th Edition.
2. Stallings, W. (2005). Operating Systems: Internals and Design Principles—Edition: 5. Pearson.

#### ***Reference Books and Other Materials***

3. Andrew S. Tanenbaum and Albert S. Woodhull. *Operating Systems: Design and Implementation*, Prentice-Hall, (2nd Ed.).
4. <https://www.geeksforgeeks.org/operating-systems/>
5. [https://www.tutorialspoint.com/operating\\_system/](https://www.tutorialspoint.com/operating_system/)
6. <https://www.studytonight.com/operating-system/>
7. <https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/>

**HUM-3201: Ethics and Cyber Law**

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

#### **Prerequisite:**

**Course Type:** Theory

**Course Summary:** This course to develop an understanding on basic concepts ICT policy in Bangladesh, cyber crime, ICT acts, Intellectual property law, patent, trademarks

**Course Objectives:** To provide knowledge on cyber law, ICT policy in Bangladesh, Different types of cyber crime, Human rights, ICT act, Intellectual property law, patent, trademarks.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                              |
|-------------|----------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course |
| <b>CLO2</b> | <b>Understand</b> cyber law                  |
| <b>CLO3</b> | <b>Understand</b> ICT policy in Bangladesh   |
| <b>CLO4</b> | <b>Understand</b> Cybercrime and its penalty |
| <b>CLO5</b> | <b>Understand</b> ICT act and Human rights   |

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| <b>CLO6</b> | <b>Understand</b> Intellectual property law, patents and trademarks |
|-------------|---------------------------------------------------------------------|

| Course Content                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Time Frame                         | Alignment with |              | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------|--------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                    | CLO            | PLO          |                                                                                                        |                                                                       |
| <b>Course Overview</b>          | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Statistics for Communication Engineering<br><b>To grasp</b> the content and policies for the class                                                                                                                                                                                                                                                                                                | <b>Week-1</b><br><b>L-1</b>        | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Cyber Law</b>                | Definition Nature, Nature, Scope, Utility of Cyber Law, origin and Development of Cyber Law and Internet.                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Week-1, 2</b><br><b>L-2,3,4</b> | CLO2<br>CLO3   | PLO1<br>PLO2 | Students will discuss with a group and understand the cyber law.                                       | #White board discussion<br>#Lecture discussion with multimedia.       |
| <b>Exam.</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-3</b><br><b>L-5</b>        |                |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                       |
| <b>ICT Policy in Bangladesh</b> | Internet Service Providers (ISP)- Domain Name, Present Legal Basis of (ISP)- Domain Name, Present Legal Basis of ISP in Bangladesh; e-Readiness in Bangladesh-e-Commerce in Bangladesh, e-Governance in Bangladesh, e-Learning/Education in Bangladesh, e-Journal in Bangladesh, e-Voting in Bangladesh; Electronic Evidence- Digital signature, The Evidence Act of 1872 Vs. ICT act-2006, Electronic Evidence in Bangladesh, Legal Effects of electronic evidence, UNCITRAL Model Law on Electronic Evidence. | <b>Week-3, 4</b><br><b>L-6,7</b>   | CLO2<br>CLO3   | PLO2<br>PLO3 | Student will understand about the ICT policy of Bangladesh                                             | #White board discussion<br>#Lecture discussion with multimedia        |
| <b>Exam.</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-4</b><br><b>L-8</b>        |                |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                       |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                    |              |       |                                                                       |                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------|-------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Cybercrime</b>              | Jurisdiction and Cybercrime, Criminal Justice in Bangladesh and Implications on Cyber Crime; cyber vandalism, Hacking, Malicious spreading in Viruses, Password fraud, cheating, Cyber Pornography, Child Pornography, Protection of Copyrights and Intellectual Property right. Invasion of Privacy, Constitutional basis of Privacy, Unsolicited e-Mail, Defamation, Harassment and e-Mail Abuse, Present Legal Protection | <b>Week-5<br/>L-9, 10</b>          | CLO2<br>CLO3 | PLO4  | Student will understand about cybercrime                              | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Week-6<br/>L-11</b>             |              |       | Quiz and Tutorial Exam. Will be taken                                 |                                                                |
| <b>Human Rights</b>            | Human Rights violation and Internet; the Information and Communication Technology Act, 2006-Objectives, Strengths & weaknesses of the ICT Law, Regulation of Cryptography;                                                                                                                                                                                                                                                   | <b>Week – 6,7<br/>L-12, 13, 14</b> | CLO4<br>CLO5 | PLO12 | Student will know and understand about human rights                   | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Week-8<br/>L-15</b>             |              |       | Quiz and Tutorial Exam. Will be taken                                 |                                                                |
| <b>International Cyber Law</b> | India, Sri Lanka, Japan Malaysia, Australia and the USA, International Conventions on Cyber Law & Crime.                                                                                                                                                                                                                                                                                                                     | <b>Week-8, 9<br/>L-16, 17</b>      | CLO5<br>CLO6 | PLO3  | Students will understand and inform about the international cyber law | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Electronic Commerce</b>     | Electronic Money, Online credit card Payments and Electronic bills of Lading, UNCITRAL Model Law on Electronic Commerce.                                                                                                                                                                                                                                                                                                     | <b>Week-9, 10<br/>L-18, 19, 20</b> | CLO4         | PLO12 | Students will understand and apply the Electronic commerce methods    | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam.</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Week-11<br/>L-21</b>            |              |       | Quiz and Tutorial Exam. Will be taken                                 |                                                                |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                    |      |       |                                                                    |                                                                |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|-------|--------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Intellectual Property Law</b> | Basic Concepts of IP Law, Nature of IPR, Computer-related Intellectual property rights: Copyright- Original and development of Copyright Law, subject matter of copyright protection, Rights protected by copyright, Neighboring rights, Limitations of Copyright Protecting, Piracy and infringement, Remedies Computer Program, New technology and copyright, Software Patents Vs. Copyright, International Convention copy right.                                                                                  | <b>Week-12<br/>L-22, 23</b>        | CLO5 | PLO12 | Students will understand and apply the intellectual property law   | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Patent and Trademarks</b>     | Patents and technological development, Requirements for patentability and ownership of patents, Scope of exclusive rights and duration of protection, patents infringement, defenses and remedies, Legal arrangement for the transfer of technology. Types of intellectual property licenses<br>Reasons for the protection of trademarks, Acquisition of trademark right, Registration procedure, Duration of protection and renewal, termination. Trademarks in Cyberspace, Domain name and Meta- tag Controversies. | <b>Week-13<br/>14<br/>L-24, 25</b> |      |       | Students will classify and apply the patent and trademarks policy. | #White board discussion                                        |
| <b>Exam. Review</b>              | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Wee-15<br/>L-,26</b>            |      |       | Lectures, Questions and Answers                                    |                                                                |
| <b>Sem. End Exam.</b>            | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-16</b>                     |      |       |                                                                    |                                                                |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quizz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|--------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                           |                            |
| <b>Understand</b>                | (05)                           |                        |                                |                            |
| <b>Apply</b>                     |                                |                        |                                | (10)                       |
| <b>Analyze</b>                   |                                |                        |                                |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                                |                            |
| <b>Create</b>                    |                                | (05)                   |                                |                            |



**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

**Reference Books and Other Materials**

*Cryptography and Network Security: Principles and Practice, William Stallings*  
*Cryptography and Network Security, AtulKahate*  
*Applied Cryptograph, Bruce Schneier*

**ICT-3000: Viva-Voce**  
**Based on 3<sup>rd</sup> Year 1<sup>st</sup> and 2<sup>nd</sup> Semester**

Full Marks:100  
 Credit: 1.0

**B.Sc. Engg. Fourth Year-2025**  
**4<sup>th</sup> Year 1<sup>st</sup> Semester**

**ICT-4101: Information Theory and Coding**

Full Marks:100  
 Credit: 3.0  
 Contact Hours: 39  
 Exam. Duration: 4 Hours

**Prerequisite:** ICT-2205  
**Course Type:** Theory  
**Course Summary:**  
**Course Objectives:** To provide the knowledge of different information sources

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| <b>CLO1</b> | Understand the notion of information and information sources in the quantitative sense. |
| <b>CLO2</b> | Know entropy, relative entropy and mutual information.                                  |
| <b>CLO3</b> | Understand the notion of entropy rates of stochastic process.                           |
| <b>CLO4</b> | Understand information rates.                                                           |
| <b>CLO5</b> | Know the different data compression techniques as it applied to information.            |
| <b>CLO6</b> | Understand the notion on channels and codes.                                            |
| <b>CLO7</b> | Understand different source coding theorems.                                            |
| <b>CLO8</b> | Understand channel coding for noisy channels.                                           |

| Course Content             |                                                                                                                                   | Time Frame | Alignment with |       | Suggested Activities             | Teaching Strategy(s)               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|----------------|-------|----------------------------------|------------------------------------|
| Outlines                   | Specific Topics                                                                                                                   |            | CLO            | PLo   |                                  |                                    |
| <b>Information Sources</b> | Introduction, Probability Spaces and Random Variables, Random Processes and Dynamical Systems, Distributions, Standard Alphabets, | Week 1     | CLO-1          | PLO-1 | -Lecture Presentation with video | -Assignment<br>- Oral Presentation |

|                                                         |                                                                                                                                                                                                                                                                                                                                                   |              |       |               |                                                                          |                                    |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------|--------------------------------------------------------------------------|------------------------------------|
|                                                         | Expectation, Asymptotic Mean Stationarity, Ergodic Properties.                                                                                                                                                                                                                                                                                    |              |       |               | -Interactive Discussion<br>-Feedback                                     |                                    |
| <b>Entropy, Relative Entropy and Mutual Information</b> | Entropy and Entropy Rate, Basic Properties of Entropy, Joint Entropy and Conditional Entropy, Relative Entropy and Mutual Information, Relationship between Entropy and Mutual Information, Chain Rules for Entropy, Relative Entropy and Mutual Information.                                                                                     | Week 2 & 3   | CLO-2 | PLO-2         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Entropy Rates of a Stochastic Process</b>            | Markov Chains, Entropy Rate, Example: Entropy Rate of a Random Walk on a Weighted Graph, Second Law of Thermodynamics Functions of Markov Chains.                                                                                                                                                                                                 | Week 4 & 5   | CLO-3 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Information Rates</b>                                | Introduction, Stationary Codes and Approximation, Information Rate of Finite Alphabet Processes, Conditional Relative Entropy, Limiting Entropy Densities, Information for General Alphabets, Some Convergence Results. Information Rates for General Alphabets, A Mean Ergodic, Theorem for Densities, Information Rates of Stationary Processes | Week 6 & 7   | CLO-4 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Data Compression</b>                                 | Examples of Codes, Kraft Inequality, Optimal Codes, Bounds on the Optimal Code Length, Kraft Inequality for Uniquely Decodable Codes, Huffman Codes, Some Comments on Huffman Codes, Optimality of Huffman Codes, Shannon–Fano–Elias Coding, Competitive Optimality of the Shannon Code, Generation of Discrete Distributions from Fair Coins.    | Week 8 & 9   | CLO-5 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Channels and Codes</b>                               | Introduction, Channels, Stationarity Properties of Channels, Examples of Channels, The Rohlin-Kakutani Theorem, Distortion and Fidelity Criteria, Performance, The rho-bar distortion, d-bar Continuous Channels, The Distortion-Rate Function.                                                                                                   | Week 10 & 11 | CLO-6 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>Source Coding Theorems</b>                           | Source Coding and Channel Coding, Block Source Codes for AMS Sources, Block Coding Stationary Sources, Block Coding AMS Ergodic Sources, Subadditive Fidelity Criteria, Asynchronous Block Codes, Sliding Block Source                                                                                                                            | Week 12 & 13 | CLO-7 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test         |

|                                  |                                                                                                                                                                                                                                                            |         |       |               |                                                  |                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------------|--------------------------------------------------|---------------------------------|
|                                  | Codes, A Geometric Interpretation of OPTA's.                                                                                                                                                                                                               |         |       |               |                                                  |                                 |
| <b>Coding for noisy channels</b> | Noisy Channels, Feinstein's Lemma, Feinstein's Theorem, Channel Capacity, Robust Block Codes, Block Coding Theorems for Noisy Channels, Joint Source and, Channel Block Codes, Synchronizing Block Channel Codes, Sliding Block Source and Channel Coding. | Week 14 | CLO-8 | PLO-3 & PLO-4 | -Lecture Presentation<br>-Interactive Discussion | -Assignment - Oral Presentation |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| <b>Remember</b>              |                            |                    | (05)                      |                        |
| <b>Understand</b>            | (05)                       |                    |                           |                        |
| <b>Apply</b>                 |                            |                    |                           | (10)                   |
| <b>Analyze</b>               |                            |                    |                           |                        |
| <b>Evaluate</b>              | (05)                       |                    |                           |                        |
| <b>Create</b>                |                            | (05)               |                           |                        |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category  | Test |
|-------------------|------|
| <b>Remember</b>   | 05   |
| <b>Understand</b> | 05   |
| <b>Apply</b>      | 10   |
| <b>Analyze</b>    | 00   |
| <b>Evaluate</b>   | 05   |
| <b>Create</b>     | 05   |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. Simon Haykin, *Communication Systems*

#### **Books Recommended**

2. TM Cover and J. M Tomas, *Elements of Information Theory*
3. Carlson, *Communication system.*
4. "Information Theory and Coding" by N Abramson
5. "Information Theory" by R B Ash

6. "Error Control Coding" by Shu Lin and D J Costello
7. "Information theory and Coding Basics and Practices" by Veluswamy S
8. "Information Theory and Coding" by Muralidhar Kulkarni and K S Shivaprakasha
9. "Fundamentals in Information Theory and Coding" by Borda

### ICT-4102: Information Theory and Coding Laboratory

Full Marks:100

Credit: 1.5

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-4101

**Course Type:** Practical/Laboratory work

#### Course Summary:

**Course Objectives:** This course will introduce the students to the practical concepts underlying information theory, mutual information, information gain, design and Analyze channel capacity, noise control coding techniques.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to:

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| <b>CLO1</b> | Understand the Entropy and its types                                               |
| <b>CLO2</b> | Know different source coding techniques                                            |
| <b>CLO3</b> | Understand different channel coding techniques in noiseless and noisy environments |
| <b>CLO4</b> | Understand Channel coding with different codes                                     |

| Outline | Name of the Experiments                                                                                                                                                                                                                                        | Time Frame          | Alignment with |               | Suggested Activities                                                                                                     | Teaching Strategy(s)                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|         |                                                                                                                                                                                                                                                                |                     | CLO            | PLO           |                                                                                                                          |                                                               |
|         | *Find Entropy, Joint Entropy, Conditional Entropy, Relative Entropy (KL divergence), Mutual Information, Normalized Mutual Information, and Normalized Variation Information (using MATLAB).<br>*Compute the entropy of an entered text string (using MATLAB). | Week-1-3<br>L-1-9   | CLO-1          | PLO2<br>PLO3  | -Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |
|         | *Huffman Coding, Run Length Encoding, Shanon-Fano Encoding (using MATLAB)                                                                                                                                                                                      | Week-4-5<br>L-10-15 | CLO-2          | PLO3,<br>PLO4 | Lecture Interactive<br>Discussion<br>Presentation with video<br>Experiment<br>Implementation in the assigned Laboratory  | Oral<br>Presentation<br>Assignment<br>Laboratory trial output |

|                            |                                                                                                                                                                            |                       |               |                |                                                                                                                    |                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|----------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|                            | Find Entropy and Mutual Information of Noiseless Channel (using MATLAB).<br>Channel Coding for Noisy Channel (using MATLAB).                                               | Week-6-7<br>L-16-21   | CLO-3         | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|                            | Channel Coding using Hamming (7,4) Block Code (using MATLAB).                                                                                                              | Week-8-10<br>L-22-30  | CLO-3 & CLO-4 | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
|                            | Channel Coding (Forward error correction codes and Convolutional Codes both are used for Error correction and which will perform better in AWGN environment) using MATLAB. | Week-11-13<br>L-31-39 | CLO-4         | PLO-3<br>PLO-4 | -Lecture Interactive Discussion<br>Presentation with video<br>Experiment Implementation in the assigned Laboratory | Oral Presentation<br>Assignment<br>Laboratory trial output |
| <b>Recess before Exam.</b> |                                                                                                                                                                            | Week-14, 15           |               |                |                                                                                                                    |                                                            |
| <b>Semester End Exam.</b>  | Start                                                                                                                                                                      | Week-16               |               |                |                                                                                                                    |                                                            |

**Assessment Pattern (According FET Ordinance)****CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Simon Haykin, Communication Systems*

### **Books Recommended**

2. *TM Cover and J. M Tomas, Elements of Information Theory*
3. *Carlson, Communication system.*
4. *“Information Theory and Coding” by N Abramson*
5. *“Information Theory” by R B Ash*

### **ICT-4103: Digital Image Processing**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1203, MATH-2101, STAT-2201, ICT-4103

**Course Type:** Theory

**Course Summary:** In this course students will learn digital image processing techniques including representation, sampling and quantization, image acquisition, image transforms, image enhancement, and image smoothing and sharpening. More advanced topics include image filtering, color image processing, and image segmentation.

**Course Objectives:** Interpret image storage, sampling, and frequency domain processing operations.

**Course Learning Outcomes (CLO):** After completing this course, a student will be able to

|             |                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Achieve</b> knowledge of a broad range of fundamental image processing and image analysis techniques and concepts.           |
| <b>CLO2</b> | <b>Learn</b> different techniques employed for the enhancement of images.                                                       |
| <b>CLO3</b> | <b>Understand</b> the need for image transforms different types of image transforms and their properties.                       |
| <b>CLO4</b> | <b>Understand</b> the need for image compression and to learn the spatial and frequency domain techniques of image compression. |
| <b>CLO5</b> | <b>Learn</b> different post-processing operations.                                                                              |
| <b>CLO6</b> | <b>Develop</b> any image processing application.                                                                                |

### **Content of the Course**

| <b>Course Content</b>     |                                                                                                                                                                                                                                                                      | <b>Time Frame</b>     | <b>Alignment with</b> |            | <b>Suggested Activities</b>                                                                           | <b>Teaching Strategy(s)</b>                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Outline</b>            | <b>Specific Topics</b>                                                                                                                                                                                                                                               |                       | <b>CLO</b>            | <b>PLO</b> |                                                                                                       |                                                                                  |
| <b>Image Fundamentals</b> | <b>To understand</b> human vision system.<br><b>To learn</b> image capturing, digital image generating, image sampling, quantization techniques.<br><b>To acquire</b> general idea of pixel relationships, image region, image boundary, and some linear operations. | <b>Week-1 L-1,2,3</b> | CLO1                  | PLO1       | Students will give brief introduction about themselves. Course outline will be discussing in details. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |

|                                             |                                                                                                                                                                                                                                                  |                                               |      |                              |                                                                                             |                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------|------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Image Enhancement in Spatial Domain.</b> | <b>To learn</b> intensity (gray level) transformation and spatial filtering.<br><b>To understand</b> image enhancement techniques and pixel-level arithmetic operations.                                                                         | <b>Week-2</b><br><b>L-4,5,6</b>               | CLO2 | PLO1<br>PLO2                 | Students will discuss with a group. To apply the knowledge to solve the different problems. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Color Image Processing</b>               | <b>To understand</b> color different model for color image processing.<br><b>To learn</b> addition color generation and transformation.                                                                                                          | <b>Week-3</b><br><b>L-7,8,9</b>               | CLO3 | PLO2<br>PLO4                 | Students will discuss with a group. To apply the knowledge to solve the different problems. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam. Review</b>                         | <b>To make an overview</b>                                                                                                                                                                                                                       | <b>Week-4</b><br><b>L-10,11,12</b>            |      |                              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                                |                                                                                                                                                                                                                                                  | <b>Week-5</b><br><b>L-13,14,15</b>            |      |                              | Quiz and Tutorial Exam. Will be taken                                                       |                                                                                  |
| <b>Image Compression</b>                    | <b>To understand</b> image compression, image compression standards, and benefits of image compression.<br><b>To learn</b> JPEG and JPEG 2000 image compression standards.                                                                       | <b>Week-6,7</b><br><b>L-16,17,18,19,20,21</b> | CLO4 | PLO2<br>PLO3                 | Students will identify, analyze and formulate complex engineering problems.                 | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam Review</b>                          | <b>To make an overview</b>                                                                                                                                                                                                                       | <b>Week-8</b><br><b>L-22,23,24</b>            |      |                              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                                |                                                                                                                                                                                                                                                  | <b>Week-9</b><br><b>L-25,26,27</b>            |      |                              | Presentation and Tutorial Exam. Will be taken                                               |                                                                                  |
| <b>Morphological Image Processing</b>       | <b>To explain</b> mathematical morphological tools to represent and describe region's shape such as boundaries, skeletons, and the convex hull.<br><b>To define</b> several theoretic operations and discuss binary sets and logical operations. | <b>Week-10</b><br><b>L-28,29,30</b>           | CLO5 | PLO2<br>PLO3<br>PLO4         | Students will apply reasoning informed by contextual knowledge to understand the problems.  | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Segmentation</b>                         | <b>To detect</b> intensity discontinuities, such as points, lines, and edges.<br><b>To learn</b> image segmentation                                                                                                                              | <b>Week-11</b><br><b>L-31,32,33</b>           | CLO6 | PLO2<br>PLO3<br>PLO4<br>PLO5 | Students will identify, analyze and formulate complex engineering problems.                 | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam. Review</b>                         | <b>To make an overview</b>                                                                                                                                                                                                                       | <b>Week-12</b><br><b>L-34,35,36</b>           |      |                              | Lectures, Questions and Answers                                                             |                                                                                  |
| <b>Exam.</b>                                |                                                                                                                                                                                                                                                  | <b>Week-13</b><br><b>L-37,38,39</b>           |      |                              | Assignment and Tutorial Exam will be taken                                                  |                                                                                  |

|                            |              |                   |  |  |  |  |
|----------------------------|--------------|-------------------|--|--|--|--|
| <b>Recess before Exam.</b> |              | <b>Week-14,15</b> |  |  |  |  |
| <b>Sem. End Exam.</b>      | <b>Start</b> | <b>Week-16</b>    |  |  |  |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### Text Books

1. *Digital Image Processing* by Rafael C. Gonzalez, 2nd edition, Pearson Education, 2009. ISBN-81-7808-625-

#### Reference Books and Other Materials

2. *Digital Image Processing: PIKS Scientific Inside* by William K. Pratt, 3rd edition, Wiley, 2010.
4. *Sapely L Ross Differential Equations*
3. <https://www.tutorialspoint.com/dip/>
4. <https://www.geeksforgeeks.org/digital-image-processing-basics>
5. <https://homepages.inf.ed.ac.uk/rbf/HIPR2/glossary.htm>

**ICT-4104: Digital Image Processing Lab**

Full Marks:100  
Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** ICT-1203, MATH-2101, STAT-2201, ICT-4103

**Course Type:** Practical/Laboratory work



**Course Summary:** In this course students will learn digital image processing techniques including representation, sampling and quantization, image acquisition, image transforms, image enhancement, and image smoothing and sharpening. More advanced topics include image filtering, color image processing, and image segmentation.

**Course Objectives:** Interpret image storage, sampling, and frequency domain processing operations.  
Evaluate current technologies and issues that are specific to image processing systems.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Achieve</b> knowledge of a broad range of fundamental image processing and image analysis techniques and concepts.           |
| <b>CLO2</b> | <b>Learn</b> different techniques employed for the enhancement of images.                                                       |
| <b>CLO3</b> | <b>Understand</b> the need for image transforms different types of image transforms and their properties.                       |
| <b>CLO4</b> | <b>Understand</b> the need for image compression and to learn the spatial and frequency domain techniques of image compression. |
| <b>CLO5</b> | <b>Learn</b> different post-processing operations.                                                                              |
| <b>CLO6</b> | <b>Develop</b> any image processing application.                                                                                |

| Topics                                                                                                                                                                                 | Specific Outcome(s)                                                                                                                | Time Frame                              | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                                                                                                                                                                                        |                                                                                                                                    |                                         | CLO            | PLO  |                                                                                                        |                                                                                           |
| <b>Exp #1:</b> Fundamentals of Python and OpenCV with python                                                                                                                           | <b>To learn</b> Anaconda and OpenCV with Python installation<br><b>To acquire</b> general idea about OpenCV and Python programming | <b>Week-1</b><br><b>L-1,2,3</b>         | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | -Lecture with the ppt and online videos<br><br>- interactive discussion<br><br>- Feedback |
| <b>Exp #2:</b> Point Processing and Pixel Operations.<br><b>Exp #3:</b> Image Arithmetic, Histogram calculation and Equalization.<br><b>Exp #4:</b> Linear transformation of an image. | <b>To learn</b> point processing, arithmetic operations.<br><b>To study</b> image enhancement techniques.                          | <b>Week-2,3</b><br><b>L-4,5,6,7,8,9</b> | CLO2<br>CLO3   | PLO2 | Students will discuss with a group, apply the knowledge to learn this operations.                      | -Lecture with the ppt and online videos<br><br>- interactive discussion<br><br>- Feedback |
| <b>Exam Review</b>                                                                                                                                                                     | <b>To make an overview</b>                                                                                                         | <b>Week-4</b><br><b>L-10,11,12</b>      |                |      | Demonstrate, Questions and Answers                                                                     |                                                                                           |
| <b>Examination</b>                                                                                                                                                                     |                                                                                                                                    | <b>Week-5</b><br><b>L-13,14,15</b>      |                |      | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                                           |

|                                                                                                                                                                   |                                                                                                                                    |                                                 |              |              |                                                                                                                                                          |                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Exp #5:</b> Spatial filtering operations.<br><b>Exp #6:</b> Edge, corner, and boundary detection                                                               | <b>To learn</b> image filtering techniques<br><b>To learn</b> edge, corner, boundary detection operation                           | <b>Week-6,7</b><br><b>L-16,17,18,19,20,21</b>   | CLO2<br>CLO6 | PLO2<br>PLO3 | Students will demonstrate and solve the problems.                                                                                                        | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exam Review</b>                                                                                                                                                | <b>To make an overview</b>                                                                                                         | <b>Week-8</b><br><b>L-22,23</b>                 |              |              | Lectures, Questions and Answers                                                                                                                          |                                                                                   |
| <b>Exam.</b>                                                                                                                                                      |                                                                                                                                    | <b>Week-8,9</b><br><b>L-24,25</b>               |              |              | Presentation and Tutorial Exam. Will be taken                                                                                                            |                                                                                   |
| <b>Exp # 7:</b> Color model transformation.<br><b>Exp #8 :</b> Morphological operations.                                                                          | <b>To learn</b> color transformation.<br><b>To learn</b> post-processing operations                                                | <b>W-9</b><br><b>L-26,27</b>                    | CLO4<br>CLO5 | PLO3         | Students will discuss and learn the operations                                                                                                           |                                                                                   |
| <b>Exp #9:</b> Compress image into JPEG and JPEG 200 format.<br><b>Exp #10:</b> Segment background/ moving foreground.<br><b>Exp #11:</b> Wavelet Transformation. | <b>To learn</b> image compression techniques.<br><b>To learn</b> video segmentation<br><b>To understand</b> wavelet transformation | <b>Week-10,11</b><br><b>L-28,29,30,31,32,33</b> | CLO4<br>CLO6 | PLO2<br>PLO3 | Students will apply contextual knowledge to understand the compression and segmentations. Student will also learn creating image processing applications | -Lecture with the ppt and online videos<br>- interactive discussion<br>- Feedback |
| <b>Exam. Review</b>                                                                                                                                               | <b>To make an overview</b>                                                                                                         | <b>Week-12</b><br><b>L-34,35,36</b>             |              |              | Lectures, Questions and Answers                                                                                                                          |                                                                                   |
| <b>Exam.</b>                                                                                                                                                      |                                                                                                                                    | <b>Week-13</b><br><b>L-37,38,39</b>             |              |              | Presentation and Tutorial Exam. will be taken                                                                                                            |                                                                                   |
| <b>Recess before Exam.</b>                                                                                                                                        |                                                                                                                                    | <b>Week-14,15</b>                               |              |              |                                                                                                                                                          |                                                                                   |
| <b>Sem. End Exam.</b>                                                                                                                                             | <b>Start</b>                                                                                                                       | <b>Week-16</b>                                  |              |              |                                                                                                                                                          |                                                                                   |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category</b><br><b>Marks:30</b> | <b>Lab. Performance</b><br><b>(05)</b> | <b>Lab. Note Book</b><br><b>(05)</b> | <b>Continuous Viva-Voce</b><br><b>(10)</b> | <b>Lab. Attendance</b><br><b>(10)</b> |
|--------------------------------------------|----------------------------------------|--------------------------------------|--------------------------------------------|---------------------------------------|
| <b>Remember</b>                            |                                        |                                      | (10)                                       |                                       |
| <b>Understand</b>                          |                                        |                                      |                                            |                                       |

|          |      |      |  |      |
|----------|------|------|--|------|
| Apply    |      | (05) |  | (10) |
| Analyze  |      |      |  |      |
| Evaluate | (05) |      |  |      |
| Create   |      |      |  |      |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's Category | Test |
|------------------|------|
| Remember         | 10   |
| Understand       | 00   |
| Apply            | 15   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 00   |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources****Text Books**

1. *Digital Image Processing* by Rafael C. Gonzalez, 4th edition, Pearson Education, 2009. ISBN-13: 978-0131687288.

**Reference Books and Other Materials**

2. [https://docs.opencv.org/4.x/d6/d00/tutorial\\_py\\_root.html](https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html)
3. *Digital Image Processing: PIKS Scientific Inside* by William K. Pratt, 3rd edition, Wiley, 2010.
4. <https://www.tutorialspoint.com/dip/>
5. <https://www.geeksforgeeks.org/digital-image-processing-basics>
6. <https://homepages.inf.ed.ac.uk/rbf/HIPR2/glossary.htm>

**ICT-4105: System Analysis and Software Engineering**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1201, 2105**Course Type:** Theory

**Course Summary:** In this course students will learn system analysis and software engineering including basic idea about system and system environment, role of system analyst, system design and testing. This course also focuses on software engineering and CASE study.

**Course Objectives:** To provide knowledge on information system, system life-cycle, role of system analyst, system design, system testing and quality, software engineering, CASE, and re-engineering.

**Course Learning Outcomes (CLO):** After completing this course, a student will be able to

|             |                                          |
|-------------|------------------------------------------|
| <b>CLO1</b> | <b>Understand</b> the system life-cycle, |
|-------------|------------------------------------------|

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| <b>CLO2</b> | <b>Understand</b> the challenges and issues to develop a system          |
| <b>CLO3</b> | <b>Test and analysis</b> the quality of a system                         |
| <b>CLO4</b> | <b>Understand</b> the roll of a system analyst                           |
| <b>CLO5</b> | <b>Understand</b> the planning and decision making level of organization |
| <b>CLO6</b> | <b>Understand</b> the CASE and software reuse.                           |

| Course Content                                                 |                                                                                                                                                       | Time Frame            | Alignment with |              | Suggested Activities                                                                                  | Teaching Strategy(s)                                                             |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------|--------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Outline                                                        | Specific Topics                                                                                                                                       |                       | CLO            | PLO          |                                                                                                       |                                                                                  |
| <b>System Concepts and the Information Systems Environment</b> | <b>To idea about</b> Introduction, Definition, Characteristics of a system, Elements of a system, Types of Systems.                                   | <b>Week-1 L-1,2,3</b> | CLO1           | PLO1         | Students will give brief introduction about themselves. Course outline will be discussing in details. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Role of System Analyst</b>                                  | <b>To discuss</b> Historical perspective, Skills and Qualifications of a system analyst, Multifaceted role of an analyst, The analyst/user interface. | <b>Week-2 L-4,5,6</b> | CLO2           | PLO1<br>PLO2 | Students will discuss with a group. To apply the knowledge to solve the different problems.           | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>The process and Stages of System Design</b>                 | <b>To explain</b> Logical and Physical design, Design methodologies, Major development activities.                                                    | <b>Week-3 L-7,8,9</b> | CLO3           | PLO2<br>PLO4 | Students will discuss with a group. To apply the knowledge to solve the different problems.           | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam. Review</b>                                            | <b>To make an overview</b>                                                                                                                            | <b>Week-4 L-10,11</b> |                |              | Lectures, Questions and Answers                                                                       |                                                                                  |
| <b>Exam.</b>                                                   |                                                                                                                                                       | <b>Week-4 L-12</b>    |                |              | Quiz and Tutorial Exam. Will be taken                                                                 |                                                                                  |
| <b>System Testing and Quality Assurance</b>                    | <b>To discuss</b> System testing, Nature of test data, Test plan, Quality assurance, Trends in testing.                                               | <b>Week-5 L-13-15</b> | CLO4           | PLO2<br>PLO3 | Students will identify, analyze and formulate complex engineering problems.                           | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Introduction to Software Engineering</b>                    | <b>Definition</b> of software Engineering, Why engineering software is different? Characteristics of software product, Software life cycle.           | <b>Week-6 L-16-18</b> | CLO4           | PLO4<br>PLO5 | Students will identify, analyze and formulate complex engineering problems.                           | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |

|                            |                                                                                                            |                                |      |                              |                                                                                            |                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------|------|------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>CASE</b>                | <b>To study</b> Computer-aided Software Engineering, CASE workbenches, Software Engineering Environments.  | <b>Week-7,8<br/>L-19-24</b>    | CLO4 | PLO4<br>PLO5                 | Students will study and analyze the CASE                                                   | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam Review</b>         | <b>To make an overview</b>                                                                                 | <b>Week-9<br/>L-25-26</b>      |      |                              | Lectures, Questions and Answers                                                            |                                                                                  |
| <b>Exam.</b>               |                                                                                                            | <b>Week-9<br/>L-27</b>         |      |                              | Presentation and Tutorial Exam. Will be taken                                              |                                                                                  |
| <b>Software Management</b> | <b>To explain</b> managing People, Software Cost Estimation, Process Improvement.                          | <b>Week-10<br/>L-28,29,30</b>  | CLO5 | PLO2<br>PLO3<br>PLO4         | Students will apply reasoning informed by contextual knowledge to understand the problems. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Software Evolution</b>  | <b>To discuss</b> Software Reuse, Software Maintenance, Configuration Management, Software Re engineering. | <b>Week-11<br/>L-31,32,33</b>  | CLO6 | PLO2<br>PLO3<br>PLO4<br>PLO5 | Students will identify, analyze and formulate complex engineering problems.                | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam. Review</b>        | <b>To make an overview</b>                                                                                 | <b>Week-12<br/>L-34,35,36</b>  |      |                              | Lectures, Questions and Answers                                                            |                                                                                  |
| <b>Exam.</b>               |                                                                                                            | <b>Week-13<br/>L- 37,38,39</b> |      |                              | Assignment and Tutorial Exam will be taken                                                 |                                                                                  |
| <b>Recess before Exam.</b> |                                                                                                            | <b>Week-14,15</b>              |      |                              |                                                                                            |                                                                                  |
| <b>Sem. End Exam.</b>      | <b>Start</b>                                                                                               | <b>Week-16</b>                 |      |                              |                                                                                            |                                                                                  |

**Assessment Pattern (According FET Ordinance)****CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |

|                   |    |
|-------------------|----|
| <b>Understand</b> | 05 |
| <b>Apply</b>      | 10 |
| <b>Analyze</b>    | 00 |
| <b>Evaluate</b>   | 05 |
| <b>Create</b>     | 05 |

### **Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### **Learning Resources**

#### ***Text Books***

1. *Modern Structured Analysis*, E Yourdon, Prentice-Hall
2. *SSADM Version 4 a Practical Approach*, M Goodland and C Slater, Mc Graw-Hill.
3. *System Analysis and Development*, P J Layzell and P Loucopolus, Chartwell-Bratt.

#### ***Reference Books and Other Materials***

4. *System Analysis and Design*, S Y Wu and M S Wu, West.
5. *System Analysis and Design*, Elias M Awad, Galgotia Publications.
6. *Software Engineering, A Practitioner's Approach*, R S Pressman, McGraw-Hill.
7. *Software Engineering I Sommerville*, 4th Edition, Addisi-Wesley.
8. *Software Engineer's Reference Book*, J McDermit, Butterworth Heinemann.
9. *CASE Using Software Development Tools*, A S Fisher, Wiley.

### **ICT-4106: System Analysis and Software Engineering Laboratory**

Full Marks:100

Credit: 1

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-4105

**Course Type:** Laboratory

**Course Summary:** In this course students will learn system analysis and software engineering including basic idea about system and system environment, role of system analyst, system design and testing. This course also focuses on software engineering and CASE study.

**Course Objectives:** To provide knowledge on information system, system life-cycle, role of system analyst, system design, system testing and quality, software engineering, CASE, and re-engineering.

**Course Learning Outcomes (CLO):** After completing this course, a student will be able to

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Understand</b> various information and different level of management            |
| <b>CLO2</b> | <b>Understand</b> the roll of a system analyst and <b>learn</b> data flow diagrams |
| <b>CLO3</b> | <b>Understand</b> the feasibility of system                                        |
| <b>CLO4</b> | <b>Understand</b> the system development life cycle and data flow diagram          |
| <b>CLO5</b> | <b>Understand</b> the input, output, planning, decision, and system evaluation     |
| <b>CLO6</b> | <b>Understand</b> the system security, cyber-crime, E-commerce, and E-governance   |

| Course Content                                         |                                                                                                                                                                                                                                                                                                                   | Time Frame                | Alignment with |                      | Suggested Activities                                                                                                   | Teaching Strategy(s)                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|----------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Outline                                                | Specific Topics                                                                                                                                                                                                                                                                                                   |                           | CLO            | PLO                  |                                                                                                                        |                                                                                                |
| <b>Course Overview and Requirement Collection</b>      | <b>To acquire</b> general idea about this laboratory.<br><b>To supply</b> the experimental list for this Laboratory.<br><b>Content:</b> Data, information - Quality, Need, Application, system - Attribute, Element and Type, plan to collect the data properly from the target user of the proposed, and system. | <b>Week-1<br/>L-1-3</b>   | CLO1           | PLO1                 | Students will give brief introduction about themselves<br>Course outline will be discussing in details                 | #Socialized recitation<br>#Show the different instruments to the students which uses this Lab. |
| <b>Information Gathering and Process Specification</b> | Various Sources of Information, information gathering - strategies and tools, Process Specification Tools, Structured English, Decision Table, Decision Tree, Conversion of technical tool from specification                                                                                                     | <b>Week-2<br/>L-4-6</b>   | CLO2           | PLO1<br>PLO2         | Students will discuss with a group.<br>To apply the knowledge to solve the different problems.                         | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback               |
| <b>Feasibility Study</b>                               | Rest of Process Specification Topics, Feasibility Study – Steps, Evaluation of Alternative Solution, Cost and benefit Calculation and Analysis, Feasibility Report Creation, System Proposal Structure                                                                                                            | <b>Week-3<br/>L-7-9</b>   | CLO3           | PLO2<br>PLO4<br>PLO7 | Students will discuss with a group.<br>To apply the knowledge of feasibility to real-life software/ system development | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback               |
| <b>Exam. Review</b>                                    | <b>To make an overview</b>                                                                                                                                                                                                                                                                                        | <b>Week-4<br/>L-10,11</b> |                |                      | Lectures, Questions and Answers                                                                                        |                                                                                                |
| <b>Exam.</b>                                           |                                                                                                                                                                                                                                                                                                                   | <b>Week-4<br/>L-12</b>    |                |                      | Quiz and Tutorial Exam. Will be taken                                                                                  |                                                                                                |

|                                                                                 |                                                                                                                                                                                                                                                                            |                             |      |                      |                                                                             |                                                                                  |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------|----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Design the proposed solution and Data flow diagram</b>                       | System Development Life Cycle (SDLC), Roles and action of System Analyst, Tools used by System Analysts, Design the solution for the proposed system, Flow oriented Model, DFD - convention, advantage, use, levels and layers, comparisons, Tools used in developing DFDs | <b>Week-5<br/>L-13-15</b>   | CLO4 | PLO2<br>PLO3         | Students will identify, analyze and formulate complex engineering problems. | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Data Input and Output Method and Program evaluation and review technique</b> | Input Design, Output Design, File and Database (with Data Dictionary) Design, File Architecture, File Maintenance, Project Management - Importance, Activities and Phases, Representation, Scheduling project plans, and Project Management Software                       | <b>Week-6<br/>L-16-18</b>   | CLO5 | PLO5                 | Students will identify, analyze and review the system                       | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>System Testing</b>                                                           | Project Management - Importance, Activities and Phases, Representation, Scheduling project plans, and Project Management Software, Differentiate among failure, Bug and Error, Testing, Manual System Testing, Dynamic System Testing, and Objectives of Quality Assurance | <b>Week-7,8<br/>L-19-24</b> | CLO5 | PLO2<br>PLO4         | Students will study, analyze, and test the developed system                 | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam Review</b>                                                              | <b>To make an overview</b>                                                                                                                                                                                                                                                 | <b>Week-9<br/>L-25-26</b>   |      |                      | Lectures, Questions and Answers                                             |                                                                                  |
| <b>Exam.</b>                                                                    |                                                                                                                                                                                                                                                                            | <b>Week-9<br/>L-27</b>      |      |                      | Presentation and Tutorial Exam. Will be taken                               |                                                                                  |
| <b>Information Security and Cyber Crime</b>                                     | Security Challenges and vulnerabilities, Various Computer Crime, Malware, Different Cyber Crime, Information Security, Tools for Protecting Information Resources, Safe and Ethical Uses of Computers                                                                      | <b>Week-10<br/>L-28-30</b>  | CLO6 | PLO2<br>PLO4<br>PLO8 | Students will apply the knowledge of security and cyber-crime in a system   | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |



|                                      |                                                                                                                                                                                                                                                                                       |                            |      |                                      |                                                                                 |                                                                                  |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------|--------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>E-Commerce: Digital Marketing</b> | E-commerce: Digital Marketing and Digital Goods, Electronic Commerce and the Internet, The Growth of E-Commerce, Eight unique features of E-Commerce technology, The Benefits of Disintermediation to the Consumer, Types of Electronic Commerce, Types of electronic payment systems | <b>Week-11 L-31-33</b>     | CLO6 | PLO2<br>PLO3<br>PLO4<br>PLO5<br>PLO6 | Students will identify, analyze, formulate, develop, and test E-commerce system | #Lecture with the ppt and online videos<br>#interactive discussion<br># Feedback |
| <b>Exam. Review</b>                  | <b>To make an overview</b>                                                                                                                                                                                                                                                            | <b>Wee-12 L-34,35,36</b>   |      |                                      | Lectures, Questions and Answers                                                 |                                                                                  |
| <b>Exam.</b>                         |                                                                                                                                                                                                                                                                                       | <b>Week-13 L- 37,38,39</b> |      |                                      | Assignment and Tutorial Exam will be taken                                      |                                                                                  |
| <b>Recess before Exam.</b>           |                                                                                                                                                                                                                                                                                       | <b>Week-14,15</b>          |      |                                      |                                                                                 |                                                                                  |
| <b>Sem. End Exam.</b>                | <b>Start</b>                                                                                                                                                                                                                                                                          | <b>Week-16</b>             |      |                                      |                                                                                 |                                                                                  |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### Learning Resources

##### **Text Books**

1. *Modern Structured Analysis, E Yourdon, Prentice-Hall*
2. *SSADM Version 4 a Practical Approach, M Goodland and C Slater, Mc Graw-Hill.*

3. *System Analysis and Development*, P J Layzell and P Loucopolus, Chartwell-Bratt.

### Reference Books and Other Materials

4. *System Analysis and Design*, S Y Wu and M S Wu, West.
5. *System Analysis and Design*, Elias M Awad, Galgotia Publications.
6. *Software Engineering, A Practitioner's Approach*, R S Pressman, McGraw-Hill.
7. *Software Engineering I Sommerville*, 4th Edition, Addisi-Wesley.
8. *Software Engineer's Reference Book*, J McDermit, Butterworth Heinemann.
9. *CASE Using Software Development Tools*, A S Fisher, Wiley.

### ICT-4107: Computer Graphics and Visualization

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Vector and Geometry, C/C++

**Course Type:** Theory

**Course Summary:** Computer graphics are an intrinsic component of many modern software applications and are often essential to the success of these applications. The objective of this course is to familiarize the student with fundamental algorithms and data structures that are used in today's interactive graphics systems as well as programming and architecture of high-resolution graphics computers. The principles and practice of computer graphics are described from their mathematical foundations to the modern applications domains of scientific visualization, virtual reality, computer games and film animation. The course will include some practical experience of graphical software environments such as Java Graphics 2D and OpenGL.

**Course Objectives:** The course introduces the basic concepts of computer graphics. It provides the necessary theoretical background and demonstrates the application of computer science to graphics. The course further allows students to develop programming skills in computer graphics through programming assignments.

**Course Learning Outcomes (CLO):** Upon successful completion of this course, students will be able to

|             |                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Explain</b> the application areas, graphic pipeline, and different display technologies.                                                                    |
| <b>CLO2</b> | <b>Demonstrate</b> the concepts of color models, lighting and shading models, textures, ray tracking, hidden surface elimination, anti-aliasing and rendering. |
| <b>CLO3</b> | <b>Design</b> various algorithms for scan conversion and filling of basic objects and their comparative analysis.                                              |

| Course Content |                 | Time Frame | Alignment with |     | Teaching-Learning Strategy | Assessment techniques |
|----------------|-----------------|------------|----------------|-----|----------------------------|-----------------------|
| Outlines       | Specific Topics |            | CLO            | PLO |                            |                       |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |       |               |                                                                          |                                                   |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------|---------------|--------------------------------------------------------------------------|---------------------------------------------------|
| <b>Introduction:</b>                                      | Introduction to Computer Graphics, Graphics hardware, Display devices and Input devices                                                                                                                                                                                                                                                                                                                                                                                                                              | Week - 1, 2<br>Lecture – L-1, L-2, L-3, L-4, L-5, L-6        | CLO-1 | PLO-1         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation                |
| <b>Image Representation:</b>                              | Scan Conversion, Basic raster graphics algorithms for drawing 2D, Line Drawing Algorithms, Circle Drawing Algorithms, Recursively Define Drawing: C Curve, Koch Curve, Aliasing and Anti-Aliasing Technique, Super-sampling                                                                                                                                                                                                                                                                                          | Week – 3, 4<br>Lecture - L-7, L-8, L-9, L-10, L-11, L-12     | CLO-1 | PLO-2         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test<br>- Oral Presentation |
| <b>2D Viewing and Clipping</b>                            | Introduction to two-dimensional viewing, clipping and transformations, Line clipping and polygon clipping algorithms.                                                                                                                                                                                                                                                                                                                                                                                                | Week 5, 6<br>Lecture - L-13, L-14, L-15, L-16, L-17, L-18    | CLO-2 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>3D Viewing, Clipping, and Geometric Representation</b> | Three-dimensional viewing, 3D object representation, B-spline curves and surfaces, BSP trees, Octrees                                                                                                                                                                                                                                                                                                                                                                                                                | Week 7, 8<br>Lecture - L-19, L-20, L-21, L-22, L-23, L-24    | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>Hidden Surface</b>                                     | Fractal geometry methods, Visible surface detection methods                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Week 9, 10<br>L-25-30                                        | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |
| <b>Visualization:</b>                                     | Visualization of 2D/3D scalar fields: color mapping, iso surfaces. Direct volume data rendering: ray-casting, transfer functions, segmentation. Visualization of: Vector fields and flow data, Time-varying data, High-dimensional data: dimension reduction, parallel coordinates, Non-spatial data: multi-variate, tree/graph structured, text Perceptual and cognitive foundations, Evaluation of visualization methods, Applications of visualization, Basic Animation Techniques like traditional, key framing. | Week 11 & 12<br>Lecture - L-31, L-32, L-33, L-34, L-35, L-36 | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation                |

|                               |            |                                              |       |                  |                                                                      |                            |
|-------------------------------|------------|----------------------------------------------|-------|------------------|----------------------------------------------------------------------|----------------------------|
| <b>Exam. Review</b>           |            | Week 13<br>Lecture -<br>L-37, L-<br>38, L-39 | CLO-5 | PLO-4 &<br>PLO-5 | -Lecture<br>Presentation<br>with video<br>-Interactive<br>Discussion | -Assignment<br>-Class test |
| <b>Recess before<br/>Exam</b> |            | Week 14<br>& 15                              |       |                  |                                                                      |                            |
| <b>Semester End<br/>Exam</b>  | .<br>Start | Week 16                                      |       |                  |                                                                      |                            |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### Learning Resources:

##### **Books Recommended**

1. *Schaum's Outlines Computer Graphics*, Zhigang Xiang, Roy Plastock, McGrawHill,
2. *Computer Graphics C Version*, D Hearn and M P Baker, Prentice-Hall.

##### **References and other related materials**

3. *Computer Graphics Principles and Practice*, Foley and co-authors, Addison Wesley.
4. *From Pixels to Pictures*, Sun Microsystem Inc, Addison Wesley.

**ICT-4108: Computer Graphics and Visualization Laboratory**

Full Marks:100

Credit: 1

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** Vector and Geometry, C/C++**Course Type:** Theory

**Course Summary:** Computer graphics are an intrinsic component of many modern software applications and are often essential to the success of these applications. The objective of this course is to familiarize the student with fundamental algorithms and data structures that are used in today's interactive graphics systems as well as programming and architecture of high-resolution graphics computers. The principles and practice of computer graphics are described from their mathematical foundations to the modern applications domains of scientific visualization, virtual reality, computer games and film animation. The course will include some practical experience of graphical software environments such as Java Graphics 2D and OpenGL.

**Course Objectives:** The course introduces the basic concepts of computer graphics. It provides the necessary theoretical background and demonstrates the application of computer science to graphics. The course further allows students to develop programming skills in computer graphics through programming assignments.

**Course Learning Outcomes (CLO):** Upon successful completion of this course, students will be able to -

|             |                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Explain</b> the application areas, graphic pipeline, and different display technologies.                                                                    |
| <b>CLO2</b> | <b>Demonstrate</b> the concepts of color models, lighting and shading models, textures, ray tracking, hidden surface elimination, anti-aliasing and rendering. |
| <b>CLO3</b> | <b>Design</b> various algorithms for scan conversion and filling of basic objects and their comparative analysis.                                              |

| Course Content               |                                                                                                                                                                                                                             | Time Frame                                               | Alignment with |       | Teaching-Learning Strategy                                               | Assessment techniques                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------|-------|--------------------------------------------------------------------------|---------------------------------------------------|
| Outlines                     | Specific Topics                                                                                                                                                                                                             |                                                          | CLO            | PLO   |                                                                          |                                                   |
| <b>Introduction:</b>         | Introduction to Computer Graphics, Graphics hardware, Display devices and Input devices                                                                                                                                     | Week - 1, 2<br>Lecture – L-1, L-2, L-3, L-4, L-5, L-6    | CLO-1          | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation                |
| <b>Image Representation:</b> | Scan Conversion, Basic raster graphics algorithms for drawing 2D, Line Drawing Algorithms, Circle Drawing Algorithms, Recursively Define Drawing: C Curve, Koch Curve, Aliasing and Anti-Aliasing Technique, Super-sampling | Week – 3, 4<br>Lecture - L-7, L-8, L-9, L-10, L-11, L-12 | CLO-1          | PLO-2 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test<br>- Oral Presentation |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |       |               |                                                                          |                                    |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------|---------------|--------------------------------------------------------------------------|------------------------------------|
| <b>2D Viewing and Clipping</b>                            | Introduction to two-dimensional viewing, clipping and transformations, Line clipping and polygon clipping algorithms.                                                                                                                                                                                                                                                                                                                                                                                                | Week 5, 6<br>Lecture - L-13, L-14, L-15, L-16, L-17, L-18    | CLO-2 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>3D Viewing, Clipping, and Geometric Representation</b> | Three-dimensional viewing, 3D object representation, B-spline curves and surfaces, BSP trees, Octrees                                                                                                                                                                                                                                                                                                                                                                                                                | Week 7, 8<br>Lecture - L-19, L-20, L-21, L-22, L-23, L-24    | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Hidden Surface</b>                                     | Fractal geometry methods, Visible surface detection methods                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Week 9, 10<br>L-25-30                                        | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| <b>Visualization:</b>                                     | Visualization of 2D/3D scalar fields: color mapping, iso surfaces. Direct volume data rendering: ray-casting, transfer functions, segmentation. Visualization of: Vector fields and flow data, Time-varying data, High-dimensional data: dimension reduction, parallel coordinates, Non-spatial data: multi-variate, tree/graph structured, text Perceptual and cognitive foundations, Evaluation of visualization methods, Applications of visualization, Basic Animation Techniques like traditional, key framing. | Week 11 & 12<br>Lecture - L-31, L-32, L-33, L-34, L-35, L-36 | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>Exam. Review</b>                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Week 13<br>Lecture - L-37, L-38, L-39                        | CLO-5 | PLO-4 & PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test         |
| <b>Recess before Exam.</b>                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Week 14 & 15                                                 |       |               |                                                                          |                                    |
| <b>Semester End Exam.</b>                                 | Start                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Week 16                                                      |       |               |                                                                          |                                    |

Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 05   |
| Understand          | 05   |
| Apply               | 10   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 05   |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### **Learning Resources:**

#### **Books Recommended**

1. *Schaum's Outlines Computer Graphics*, Zhigang Xiang, Roy Plastock, McGraw-Hill,
2. *Computer Graphics C Version*, D Hearn and M P Baker, Prentice-Hall.

#### **References and other related materials**

3. *Computer Graphics Principles and Practice*, Foley and co-authors, Addison Wesley.
4. *From Pixels to Pictures*, Sun Microsystems Inc, Addison Wesley.

**ICT-4109:      Cryptography and Network Security**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-1203, MATH-2101

**Course Type:** Theory

#### **Course Summary:**

**Course Objectives:** To provide knowledge on what network management is all about, functions and applications of network management, architecture and organization of it, exchange of management information, fundamental issues of network security, different ciphering techniques.

#### **Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | Define network management and understand the necessity & the functions of network management system.            |
| <b>CLO2</b> | Learn about the elements of network management system and architectural organization of it.                     |
| <b>CLO3</b> | Understand the structure of management information and the exchange of information between management entities. |
| <b>CLO4</b> | Distinguish between information & network security and know about information security architecture.            |
| <b>CLO5</b> | Analyze different symmetric and asymmetric encryption techniques.                                               |
| <b>CLO6</b> | Learn about different data integrity and authentication mechanism.                                              |
| <b>CLO7</b> | Understand the behavior of intruders & viruses and learn about different measures of network security.          |

| Course Content                         |                                                                                                                                                                                                                      | Time Frame | Alignment With |                      | Suggested Activities                                                     | Teaching Strategy(s)               |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|----------------------|--------------------------------------------------------------------------|------------------------------------|
| Outlines                               | Specific Topics                                                                                                                                                                                                      |            | CLO            | PLO                  |                                                                          |                                    |
| Network Management Basics              | Challenges of IT Managers, Need for Network Management System, Network Management; Definition, Goals and Functions, Different Functional Groups, Management Applications, Need for Standard Management System.       | Week 1-2   | <b>CLO-1</b>   | <b>PLO-1</b>         | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| Network Management System Organization | Architecture of NMS, Network Management Organization Model, Distributed Network Management System, Comparison of OSI & Internet Management Model.                                                                    | Week 3 & 4 | <b>CLO-2</b>   | <b>PLO1</b>          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test 1       |
| Management Information Communication   | SMI, Management Information Base, Managed Objects in OSI and Internet Perspective, Application Layer Communication, Commands & Responses in CMIP & SNMP.                                                             | Week 5 & 6 | <b>CLO-3</b>   | <b>PLO2<br/>PLO3</b> | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| Security Fundamentals                  | Information Security, Threats & Attacks, OSI security architecture, Security Services & Mechanisms, Placement of Encryption Function, A Model for Network Security, Stream Cipher & Block Cipher, Key Distribution . | Week 7 & 8 | <b>CLO-4</b>   | <b>PLO4<br/>PLO6</b> | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test 2       |



|                                 |                                                                                                                                                                                                        |              |              |             |                                                                          |                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-------------|--------------------------------------------------------------------------|------------------------------------|
| Modern Encryption Techniques    | Symmetric & Asymmetric Encryption, Substitution Cipher, Transposition Cipher, Modern Block Cipher, Block Chain Ciphering.                                                                              | Week 9 & 10  | <b>CLO-5</b> | <b>PLO4</b> | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test 3       |
| Data Integrity & Authentication | Hash Function, MAC, Digital Signature, Entity Authentication, Mutual & One-way Authentication.                                                                                                         | Week 11 & 12 | <b>CLO-6</b> | <b>PLO5</b> | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| Network & System Security       | General Model of Network Access Control, Elements of Network Access Control System, Network Access Control Enforcement Methods, Intrusion Detection, Viruses & Related Threats, Virus Countermeasures. | Week 13 & 14 | <b>CLO-7</b> | <b>PLO6</b> | -Lecture Presentation<br>-Interactive Discussion<br>-Feedback            | -Assignment<br>-Class test 4       |

**Assessment Pattern (According FET Ordinance)****CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category Marks:30</b> | <b>Lab. Performance (05)</b> | <b>Lab. Note Book (05)</b> | <b>Continuous Viva-Voce (10)</b> | <b>Lab. Attendance (10)</b> |
|----------------------------------|------------------------------|----------------------------|----------------------------------|-----------------------------|
| <b>Remember</b>                  |                              |                            | (10)                             |                             |
| <b>Understand</b>                |                              |                            |                                  |                             |
| <b>Apply</b>                     |                              | (05)                       |                                  | (10)                        |
| <b>Analyze</b>                   |                              |                            |                                  |                             |
| <b>Evaluate</b>                  | (05)                         |                            |                                  |                             |
| <b>Create</b>                    |                              |                            |                                  |                             |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 10          |
| <b>Understand</b>       | 00          |
| <b>Apply</b>            | 15          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 00          |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

**Learning Resources:****Recommendation Books:**

1. *Cryptography & Network Security -- William Stalling*
2. *Cryptography & Network Security—Behrouz Forouzan*

#### References and other related materials

3. *SNMP, SNMPV2, RMON -- William Stalling*
4. *Network Management Standards – Subramanian*

#### ICT-4110: Cryptography and Network Security Laboratory

Full Marks:100

Credit: 1

Contact Hours: 39

Exam. Duration: 6 Hours

**Prerequisite:** ICT-4109

**Course Type:** Theory

#### Course Summary:

**Course Objectives:** To provide knowledge on how network management technically be developed and implemented by product-based applications, functions and development of network management applications, architecture and organization of technical support, practical knowledge about exchange of information with security measure and proper management, fundamental practical issues of network security comprising different ciphering techniques with applications and real-time implications as well.

**Course Learning Outcomes (CLO):** After completing this course, a student would be able to

|             |                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | Develop network management infrastructure with critical understanding of the necessity & the functions of network management system.                  |
| <b>CLO2</b> | Earn practical knowledge about the elements of network management system and architectural organization of it.                                        |
| <b>CLO3</b> | Configure the structure of management information and the secure exchange of information between management entities.                                 |
| <b>CLO4</b> | Determine the technical differences about information & network security and deep practical expertise about information security architecture design. |
| <b>CLO5</b> | Develop and analyze different symmetric and asymmetric encryption techniques.                                                                         |
| <b>CLO6</b> | Comprehend about different data integrity and authentication mechanism regarding applications.                                                        |
| <b>CLO7</b> | Understand the behavior of intruders & viruses to take different measures of network security.                                                        |

| Topics                                     | Specific Outcome(s)                                                                                                                                                                                                                                                                                                                                                                                                               | Time Frame                                               | Suggested Activities                                                                                   | Teaching Strategy(s)                                                                                                                        | Alignment with CLO |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Beginning with Encryption Technique</b> | <p><b>Write and understand</b> a program that Encrypt a message with Caesar cipher and other basic Encipherment algorithm using python.</p> <p><b>To understand</b> a program having concept of basic encryption techniques with Python and MatLab programming Language. Analyzing the pros and cons of the techniques and improvement of the code to upgrade the application of fundamental and practical technique as well.</p> | <b>Week-1, 2, 3, 4</b><br><b>L-1,2, 3, 4, 5, 6, 7, 8</b> | Students will give brief introduction about themselves<br>Course outline will be discussing in details | <p>#Socialized recitation</p> <p>#Show presentation to motivate the students</p> <p>Technical knowledge delivery to develop the program</p> | <b>CLO1</b>        |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                       |                                                                                                       |                                                                        |                   |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------|
| <b>Advanced Encryption Algorithm and Analysis</b> | <p><b>Write and understand</b> a program that Encrypt a message with Reverse cipher algorithm using python. Additionally, Practice of Parity Bit checking technique to analyze message with Python and MatLab.</p> <p><b>To understand</b> a program having concept of basic encryption techniques with Python and MatLab programming Language. Analyzing the pros and cons of the techniques and improvement of the code to upgrade the application of fundamental and practical technique as well.</p> | <b>Week-5, 6, 7, 8, 9<br/>L-9, 10, 11, 12, 13, 14, 15, 16, 17, 18</b> | Students will discuss with a group. To apply the knowledge to solve the different real-time problems. | #Interactive discussion<br>#Lecture discussion with Class and objects. | <b>CLO2</b>       |
| <b>Exam Review</b>                                | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Week-10, 11<br/>L-19,20,21,22</b>                                  | Lectures, Questions and Answers                                                                       |                                                                        |                   |
| <b>Exam.</b>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-12</b>                                                        | Quiz and Tutorial Exam. Will be taken                                                                 |                                                                        |                   |
| <b>Brute Force Attack</b>                         | <p><b>Write and understand</b> a program to analyze Cryptanalysis of Caesar Cipher and other basic technique with brute force attack using python and Matlab.</p> <p><b>To understand</b> a program having concept of basic encryption techniques with Python and MatLab programming Language. Analyzing the pros and cons of the techniques and improvement of the code to upgrade the application of fundamental and practical technique as well.</p>                                                  | <b>Week-13, 14, 15,<br/>L-23, 24, 25, 26, 27, 28</b>                  | Students will identify, analyze and formulate complex real-time programming problems.                 | #White board discussion<br>#Lecture discussion with multimedia         | <b>CLO3, CLO4</b> |
| <b>Exam Review</b>                                | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Week-16<br/>L-29, 30</b>                                           | Lectures, Questions and Answers                                                                       |                                                                        |                   |
| <b>Exam.</b>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Week-17</b>                                                        | Presentation and Tutorial Exam. Will be taken                                                         |                                                                        |                   |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                   |                                                                                                                                                 |                                                                                                                                        |             |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Transposition Technique and CRC method</b> | <b>Write and understand</b> a program to implement Transposition Cipher Encryption Technique using python and MatLab. Moreover, implementation of CRC (cyclic Redundancy Checking) using python and MatLab for advanced analysis of Message.<br><b>To understand</b> a program having concept of basic encryption techniques with Python and MatLab programming Language. Analyzing the pros and cons of the techniques and improvement of the code to upgrade the application of fundamental and practical technique as well. | <b>Week-18, 19, 20<br/>L-31,32,33, 34, 35, 36</b> | Students will apply reasoning informed by contextual knowledge to understand Transposition Cipher Technique CRC for analyzing a message content | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about various use case of Transposition Cipher Technique | <b>CLO5</b> |
| <b>Exam. Review</b>                           | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Week-21<br/>L-37,38</b>                        | Lectures, Questions and Answers                                                                                                                 |                                                                                                                                        |             |
| <b>Exam.</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-22</b>                                    | Assignment and Tutorial Exam will be taken                                                                                                      |                                                                                                                                        |             |
| <b>Recess before Exam.</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Week-23</b>                                    |                                                                                                                                                 |                                                                                                                                        |             |
| <b>Sem. End Exam.</b>                         | <b>Start</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Week-24</b>                                    |                                                                                                                                                 |                                                                                                                                        |             |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category Marks:30</b> | <b>Tutorial/In-course (10)</b> | <b>Assignment (05)</b> | <b>Quiz/Presentation (05)</b> | <b>Class Attendance 10</b> |
|----------------------------------|--------------------------------|------------------------|-------------------------------|----------------------------|
| <b>Remember</b>                  |                                |                        | (05)                          |                            |
| <b>Understand</b>                | (05)                           |                        |                               |                            |
| <b>Apply</b>                     |                                |                        |                               | (10)                       |
| <b>Analyze</b>                   |                                |                        |                               |                            |
| <b>Evaluate</b>                  | (05)                           |                        |                               |                            |
| <b>Create</b>                    |                                | (05)                   |                               |                            |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's Category</b> | <b>Test</b> |
|-------------------------|-------------|
| <b>Remember</b>         | 05          |
| <b>Understand</b>       | 05          |
| <b>Apply</b>            | 10          |
| <b>Analyze</b>          | 00          |
| <b>Evaluate</b>         | 05          |
| <b>Create</b>           | 05          |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources:**

**Recommendation Books:**

1. *Cryptography & Network Security -- William Stalling*
2. *Cryptography & Network Security—Behrouz Forouzan*

**References and other related materials**

3. *SNMP, SNMPV2, RMON -- William Stalling*
4. *Network Management Standards – Subramanian*
5. *MatLab and Python*

**ICT-4201: Artificial Intelligence and Machine Learning**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** Discrete Mathematics, Data Structures**Course Type:** Theory

**Course Summary:** Artificial intelligence is an extremely broad field with applications to many disciplines and many subfields. This course gives a broad survey of artificial intelligence, as opposed to focusing on any particular subfield of AI. The course offered by the department of CSE, will cover methods from search, optimization, probabilistic reasoning, and learning, among other topics. Of course, these topics are closely related with each other. For example, the knowledge acquired through learning can be used both for problem solving and for reasoning. In fact, the skill for problem solving itself should be acquired through learning. Also, methods for problem solving are useful both for reasoning and planning. Further, both natural language understanding and computer vision can be solved using methods developed in the field of pattern recognition

**Course Objectives:** To provide knowledge on Artificial intelligence, behavior of intelligent agents, problem solving by search algorithm, knowledge representation approaches, PROLOG programming, machine learning, data driven and statistical approaches of machine learning, expert system models and expert system shells.

**Course Learning Outcomes (CLO):** Upon successful completion of this course, students will be able to

|              |                                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b>  | <b>Define</b> the terms Artificial Intelligence, rationality, autonomy & different types of AI agents.                                                                                                                             |
| <b>CLO2</b>  | <b>Analyze</b> various AI search algorithms (uninformed, informed, heuristic, constraint satisfaction, genetic algorithms) in relation to the benefit and behavior of the human society.                                           |
| <b>CLO3</b>  | <b>Implement</b> the fundamentals of knowledge representation (logic-based, frame-based, semantic nets), inference and theorem proving and building procedure of simple knowledge-based systems, concerning strong morality issues |
| <b>CLO 4</b> | <b>Apply</b> fundamental knowledge of machine learning algorithms to solve analytical problems within the fields related to machine learning.                                                                                      |

| Course Content            |                                                                                                                                                                                                                        | Time Frame           | Alignment with |                | Teaching-Learning Strategy                                               | Assessment techniques                             |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|----------------|--------------------------------------------------------------------------|---------------------------------------------------|
| Outlines                  | Specific Topics                                                                                                                                                                                                        |                      | CLO            | PLO            |                                                                          |                                                   |
| <b>Introduction</b>       | Introduction to AI definition, Foundations of Artificial Intelligence, History of Artificial Intelligence, and The State of the Art.                                                                                   | Week – 1- 2<br>L-1-6 | CLO-1          | PLO-1          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br><br>- Oral Presentation            |
| <b>Intelligent Agents</b> | Intelligent agents, How Agents Should Act, Structure of Intelligent Agents, and Environments.                                                                                                                          | Week – 3-4<br>L-7-12 | CLO-1          | PLO-2          | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test<br>- Oral Presentation |
| <b>Problem Solving</b>    | Problem-Solving Agents, Formulating Problems, Data structures for search trees Uninformed Search Strategies: BFS, DFS, DLS, Uniform cost search, Iterative deepening search, Bidirectional search and Comparing search | Week 5-6<br>L-13-18  | CLO-2          | PLO-2<br>PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test                        |

|                                                    |                                                                                                                                                                                                                                               |                       |       |                |                                                                          |                                    |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|----------------|--------------------------------------------------------------------------|------------------------------------|
|                                                    | strategies. Informed Search: Best-First Search, Heuristic functions, Iterative deepening A* search (IDA*) Hill-climbing search and Simulated annealing.                                                                                       |                       |       |                |                                                                          |                                    |
| <b>Knowledge and Reasoning</b>                     | Building a Knowledge Base Agent, Types of Knowledge, Knowledge representations, Reasoning, Propositional logic, First order logic, Inference in First order Logic, Logical Reasoning Systems (Languages for AI Problem Solving: PROLOG, LISP) | Week 7-8<br>L-19-24   | CLO-3 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| Constraint Satisfaction Problems (CSP)             | Constraint Satisfaction Problems<br>Constraint Propagation<br>Backtracking Search<br>Local Search, Optimization Problem                                                                                                                       | Week 9-10<br>L-25-30  | CLO-4 | PLO-3 & PLO-4  | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>-Class test         |
| Natural Language Processing (NLP) and Game Playing | Natural Language Processing Introduction, Syntactic Processing<br>Semantic Analysis, Discourse & Pragmatic Processing, Game Playing Overviews, The Minimax search procedure, Adding Alpha-Beta Cut-offs, Iterative Deepening                  | Week 11-12<br>L-31-36 | CLO-5 | PLO-3<br>PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | -Assignment<br>- Oral Presentation |
| <b>Uncertain Knowledge and Reasoning:</b>          | PCM, SVM, Uncertainty, Probabilistic Reasoning Systems, Bayes' Rule and Its Use, Semantics of Belief Networks, Inference in Belief Networks, Clustering methods, Fuzzy sets and fuzzy logic, and Decision Networks.                           | Week 13<br>L-37-39    | CLO-5 | PLO-4<br>PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | -Assignment<br>-Class test         |
|                                                    | Recess before Exam.                                                                                                                                                                                                                           | Week 14 -15           |       |                |                                                                          |                                    |
|                                                    | Semester End Exam. Start                                                                                                                                                                                                                      | Week 16               |       |                |                                                                          |                                    |

#### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| <b>Remember</b>              |                            |                    | (05)                      |                        |
| <b>Understand</b>            | (05)                       |                    |                           |                        |
| <b>Apply</b>                 |                            |                    |                           | (10)                   |
| <b>Analyze</b>               |                            |                    |                           |                        |

|          |      |      |  |  |
|----------|------|------|--|--|
| Evaluate | (05) |      |  |  |
| Create   |      | (05) |  |  |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 05   |
| Understand       | 05   |
| Apply            | 10   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 05   |

#### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

#### Learning Resources:

##### **Books Recommended**

1. *Artificial Intelligence: A Modern Approach*, Stuart Russell and Peter Norving, Prentice Hall,
2. *Introduction to Artificial Intelligence and Expert Systems*, Patterson, D. W, Prentice Hall,

##### **References and other related books**

3. *Artificial Intelligence*, Rich, E. et al, McGraw Hill.
4. “*Machine Learning For Absolute Beginners: A Plain English Introduction (Second Edition)*” by Oliver Theobald

**ICT-4202: Artificial Intelligence and Machine Learning Laboratory**

Full Marks:100  
Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:** ICT-4201

**Course Type:** Theory

**Course Summary:** Artificial intelligence is an extremely broad field with applications to many disciplines and many subfields. This course gives a broad survey of artificial intelligence, as opposed to focusing on any particular subfield of AI. The course offered by the department of CSE, will cover methods from search, optimization, probabilistic reasoning, and learning, among other topics. Of course, these topics are closely related with each other. For example, the knowledge acquired through learning can be used both for problem solving and for reasoning. In fact, the skill for problem solving itself should be acquired through learning. Also, methods for problem solving are useful both for reasoning and planning. Further, both natural language understanding and computer vision can be solved using methods developed in the field of pattern recognition.

**Course Objectives:** To provide knowledge on Artificial intelligence, behavior of intelligent agents, problem solving by search algorithm, knowledge representation approaches, PROLOG programming, machine learning, data driven and statistical approaches of machine learning, expert system models and expert system shells.



**Course Learning Outcomes (CLO): Upon successful completion of this course, students will be able to -**

|             |                                                                                                                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Define</b> the terms Artificial Intelligence, rationality, autonomy & different types of AI agents.                                                                                                                             |
| <b>CLO2</b> | <b>Analyze</b> various AI search algorithms (uninformed, informed, heuristic, constraint satisfaction, genetic algorithms) in relation to the benefit and behavior of the human society.                                           |
| <b>CLO3</b> | <b>Implement</b> the fundamentals of knowledge representation (logic-based, frame-based, semantic nets), inference and theorem proving and building procedure of simple knowledge-based systems, concerning strong morality issues |
| <b>CLO4</b> | <b>Apply</b> fundamental knowledge of machine learning algorithms to solve analytical problems within the fields related to machine learning.                                                                                      |

|       | Topic Outline                                                                                                                                                      |                |     |                                   |                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----------------------------------|--------------------------------------------|
| Class | Topics or Assignments                                                                                                                                              | Alignment with |     | Reading Reference                 | Activities                                 |
|       |                                                                                                                                                                    | CLO            | PLO |                                   |                                            |
| 1     | Intelligent agents: Lab task on what Artificial Intelligence is about and different types of AI agents.                                                            | 1              |     | Lab. Manual, Experiment No. 1     | Laboratory Experiment                      |
| 2     | Knowledge-based agents and logical problem solving: introduction to knowledge representation and propositional logic.                                              | 1              |     | Lab. Manual, Experiment No. 2     | Laboratory Experiment                      |
| 3     | First-order logic as a basis for building intelligent agents capable of acting and reacting in a complex environment.                                              | 2              |     | Lab. Manual, Experiment No. 3 (a) | Laboratory Experiment                      |
| 4     | Searching as a problem-solving technique: a review of "conventional" searching methods including breadth-first, depth-first, bi-directional and best-first search. | 2              |     | Lab. Manual, Experiment No. 3 (b) | Laboratory Experiment                      |
| 5     | Searching as a problem-solving technique: depth limited search and iterative depth limited search.                                                                 | 2-3            |     | Lab. Manual, Experiment No. 4     | Laboratory Experiment                      |
| 6     | Searching as a problem-solving technique: Heuristic functions and their effect on performance of search algorithms.                                                | 2-3            |     | Lab. Manual, Experiment No. 5     | Laboratory Experiment                      |
|       | <b>Mid Term Exam</b>                                                                                                                                               |                |     |                                   |                                            |
| 7     | Introduction to WEKA.                                                                                                                                              | 2-3            |     | Lab. Manual, Experiment No. 6     | Laboratory Experiment                      |
| 8     | Supervised learning: regression - Attributes and target variable, Linear regression by Maximum Likelihood.                                                         | 2-3            |     | Lab. Manual, Experiment No. 7     | Laboratory Experiment                      |
| 9     | Supervised learning: classification - Nearest neighbors(kNNs), Decision trees                                                                                      | 2-3            |     | Lab. Manual, Experiment No. 8 (a) | Laboratory Experiment                      |
| 10    | Unsupervised learning: Introduction to clustering, K-means                                                                                                         | 2-3            |     | Lab. Manual, Experiment No. 8 (b) | Laboratory Experiment                      |
| 11    | Project Presentation                                                                                                                                               | 2-3            |     | Lab. Manual, Experiment No. 9     | Laboratory Experiment                      |
| 12    | Final Term Examination (Quiz, Lab test, Viva, Project Presentation)                                                                                                |                |     |                                   | Quiz, Lab test, Viva, Project Presentation |

**Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-course<br>(10) | Assignment<br>(05) | Quiz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|----------------------------|--------------------|---------------------------|------------------------|
| Remember                     |                            |                    | (05)                      |                        |
| Understand                   | (05)                       |                    |                           |                        |
| Apply                        |                            |                    |                           | (10)                   |
| Analyze                      |                            |                    |                           |                        |
| Evaluate                     | (05)                       |                    |                           |                        |
| Create                       |                            | (05)               |                           |                        |

**SMEE: (Semester Mid and End Examination-30 Marks)**

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 05   |
| Understand          | 05   |
| Apply               | 10   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 05   |

**Assessment and Evaluation**

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

**Learning Resources:****Books Recommended**

1. *Artificial Intelligence: A Modern Approach*, Stuart Russell and Peter Norving, Prentice Hall,
2. *Introduction to Artificial Intelligence and Expert Systems*, Patterson, D. W, Prentice Hall,

**References and other related books**

3. *Artificial Intelligence*, Rich, E. et al, McGraw Hill.
4. *"Machine Learning For Absolute Beginners: A Plain English Introduction (Second Edition)"* by Oliver Theobald

**ICT-4203: Wireless Communications**

Full Marks:100

Credit: 3.0

Contact Hours: 39

Exam. Duration: 4 Hours

**Prerequisite:** ICT-2205, 3101**Course Type:** Theory

**Course Summary:** To provide knowledge on mobile and cellular system, design of cellular systems, cell coverage of signal and traffics, cell frequency management, operational techniques and techniques, switching techniques, intelligent cell concepts and applications

**Course Objectives:** To understand the theory of wireless propagation environment and wireless communication systems, to be able to design and analyze optimum receivers for various wireless communication in fading channel, to understand the theories and practice of diversity, multiple input multiple

output (MIMO) system, space time coding, spread spectrum modulation, orthogonal frequency division multiple access, to be able to simulate and design wireless communication systems.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                                    |
| <b>CLO2</b> | <b>Differentiate</b> the mobile and cellular system,                            |
| <b>CLO3</b> | <b>Calculate</b> coverage area and traffics of a cell                           |
| <b>CLO4</b> | <b>Understand</b> frequency management techniques under a cell                  |
| <b>CLO5</b> | <b>Discuss</b> the switching techniques, Intelligent cells and its applications |

#### Content of the Course

| Course Content                                  |                                                                                                                                                                                                 | Time Frame                | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                  |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Outline                                         | Specific Topics                                                                                                                                                                                 |                           | CLO            | PLO  |                                                                                                        |                                                                       |
| <b>Course Overview</b>                          | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the Wireless Communication Systems<br><b>To grasp</b> the content and policies for the class | <b>Week-1<br/>L-1-3</b>   | PLO1           | CLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to motivate the students |
| <b>Introduction to Cellular Mobile Systems</b>  | <b>To discuss</b> Basic Cellular System, Operation of Cellular Systems, Planning a Cellular Systems, Analog and Digital Cellular Systems.                                                       | <b>Week-2,3<br/>L-4-9</b> | CLO2           | PLO2 | Students will discuss with a group.<br>To apply the knowledge to solve the different problems.         | #Interactive discussion<br>#Lecture discussion with multimedia.       |
| <b>Elements of Cellular Radio System Design</b> | General description, Frequency reuse channels, Co-channel Interference, Handoff Mechanism, Cell splitting, and Components of Cellular Systems.                                                  |                           | CLO2           | PLO2 |                                                                                                        | #Interactive discussion<br>#Lecture discussion with multimedia        |
| <b>Exam Review</b>                              | <b>To make an overview</b>                                                                                                                                                                      | <b>Week-4<br/>L-10-12</b> |                |      | Lectures, Questions and Answers                                                                        |                                                                       |
| <b>Examination</b>                              |                                                                                                                                                                                                 | <b>Week-5<br/>L-13-15</b> |                |      | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                       |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |              |              |                                                                                             |                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------|--------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Cell Coverage for Signal and Traffic</b>        | <b>To Discuss</b> Obtaining the Mobile Point-to-Point Model Propagation over Water or Flat Open Area, Foliage Loss, Propagation in Near in Distance, Long Distance Propagation, Obtain Path Loss from a Point-to-Point Prediction Model, Form of a Point-to-Point Model, Computer Generation of a Point-to-Point Prediction, Cell-Site Antenna Heights and Signal Coverage Cells, Mobile-to-Mobile Propagation. | <b>Week-6<br/>L-16-18</b>  | CLO2<br>CLO3 | PLO2<br>PLO3 | Students will identify, analyze and formulate complex engineering problems.                 | #White board discussion<br>#Lecture discussion with multimedia                                                          |
| <b>Frequency Management and Channel Assignment</b> | Frequency management, Frequency spectrum utilization, Setup channels, Definition of Channel assignment, Fixed channel assignment, Non-fixed channel assignment.                                                                                                                                                                                                                                                 | <b>Week-7<br/>L-19-21</b>  | CLO3<br>CLO4 | PLO4<br>PLO6 |                                                                                             | White board discussion<br>#Lecture discussion with multimedia                                                           |
| <b>Operational Techniques and Technologies</b>     | Adjusting the Parameters of a System, Coverage-Hole Filler, Leaky Feeder, Cell Splitting, Small Cells (Microcells), Narrow-beam Concept, Separation between Highway Cell Sites, Low-Density Small-Market Design.                                                                                                                                                                                                | <b>Week-8<br/>L-22-24</b>  | CLO4         | PLO5         |                                                                                             | White board discussion<br>#Lecture discussion with multimedia                                                           |
| <b>Exam Review</b>                                 | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Week-9<br/>L-25-26</b>  |              |              | Lectures, Questions and Answers                                                             |                                                                                                                         |
| <b>Exam.</b>                                       |                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Week-9<br/>L-27</b>     |              |              | Presentation and Tutorial Exam. Will be taken                                               |                                                                                                                         |
| <b>Switching and Traffic</b>                       | <b>To Analyze</b> General Description, Cellular Analog Switching Equipment, Cellular Digital Switching Equipment, Special Features for Handling Traffic, MTSO Interconnection, Small Switching Systems, System Enhancement.                                                                                                                                                                                     | <b>Week-10<br/>L-28-30</b> | CLO4<br>CLO5 | PLO6<br>PLO7 | Students will apply reasoning informed by contextual knowledge to understand modern physics | #White board discussion<br>#Lecture discussion with multimedia<br>#Show video about modern physics to motivate students |
| <b>Intelligent Cell Concept and Applications</b>   | Intelligent Cell Concept, Applications of Intelligent Microcell Systems, In-building Communication, CDMA Cellular Radio Network.                                                                                                                                                                                                                                                                                | <b>Week-11<br/>L-31-33</b> | CLO5         | PLO7         |                                                                                             | White board discussion<br>#Lecture discussion with multimedia                                                           |

|                                |                            |                            |  |  |                                                  |  |
|--------------------------------|----------------------------|----------------------------|--|--|--------------------------------------------------|--|
| <b>Exam. Review</b>            | <b>To make an overview</b> | <b>Wee-12<br/>L-34-36</b>  |  |  | Lectures,<br>Questions and<br>Answers            |  |
| <b>Exam.</b>                   |                            | <b>Week-13<br/>L-37-39</b> |  |  | Assignment and<br>Tutorial Exam<br>will be taken |  |
| <b>Recess before<br/>Exam.</b> |                            | <b>Week-14-15</b>          |  |  |                                                  |  |
| <b>Sem. End<br/>Exam.</b>      | <b>Start</b>               | <b>Week-16</b>             |  |  |                                                  |  |

### Assessment Pattern (According FET Ordinance)

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quiz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|------------------------------------|----------------------------|-----------------------------------|--------------------------------|
| <b>Remember</b>                      |                                    |                            | (05)                              |                                |
| <b>Understand</b>                    | (05)                               |                            |                                   |                                |
| <b>Apply</b>                         |                                    |                            |                                   | (10)                           |
| <b>Analyze</b>                       |                                    |                            |                                   |                                |
| <b>Evaluate</b>                      | (05)                               |                            |                                   |                                |
| <b>Create</b>                        |                                    | (05)                       |                                   |                                |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

### Assessment and Evaluation

- Class Attendance & Participation (10%)
- Quiz/Presentation (10%)
- In-course/Tutorial (10%)
- Final Exam (70%)

### Learning Resources

#### **Text Books**

1. *Mobile Cellular Telecommunications, Analog and Digital Systems*, William C. Y. Lee, McGraw Hill.
2. *Mobile Radio Communications*, Raymond Steele, IICE Press, New York.

#### **Reference Books and Other Materials**

3. *Telecommunication System Engineering*, Roger L Freeman, John Wiley & Sons.

Credit: 1.5  
Contact Hours: 39  
Exam. Duration: 6 Hours

- Prerequisite:** ICT-4203
- Course Type:** Practical/Laboratory work
- Course Summary:** This course gives students deep knowledge in wireless communication systems at the practical level. This lab focuses on the propagation path-loss, mobile velocity, transmitting and receiving power, distance between cell site and mobile unit.
- Course Objectives:** This course will introduce the students to the practical concepts of mobile and cellular system, cell coverage and traffics, frequency management, quality of services, operation techniques, switching techniques, intelligent cell design.

**Course Learning Outcomes (CLO):** After completing this course a student would be able to

|             |                                                               |
|-------------|---------------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course                  |
| <b>CLO2</b> | <b>Simulate</b> and analyze cell's coverage area and traffics |
| <b>CLO3</b> | <b>Calculate</b> different losses by using MATLAB tools       |
| <b>CLO4</b> | <b>Estimate</b> transmitting power by using MATLAB tools      |
| <b>CLO5</b> | <b>Simulate</b> and configure switches                        |

#### Content of the Course

| Course Content                                  |                                                                                                                                                                                                                              | Time Frame                      | Alignment with |      | Suggested Activities                                                                                   | Teaching Strategy(s)                                                                                    |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------|------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Outline                                         | Specific Topics                                                                                                                                                                                                              |                                 | CLO            | PLO  |                                                                                                        |                                                                                                         |
| <b>Course Overview</b>                          | <b>To acquire</b> general idea about the Wireless Communication Laboratory<br><b>To supply</b> the experimental list for this Laboratory.<br><b>To Discuss</b> about simulator software.                                     | <b>Week-1</b><br><b>L-1,2,3</b> | CLO1           | PLO1 | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Discuss about the wireless trainer kits to the students which uses this Lab. |
| <b>Elements of Cellular Radio System Design</b> | <b>To calculate</b> maximum velocity of Mobile unit using MATLAB tools<br><b>To calculate</b> coherence bandwidth using MATLAB tools<br><b>To calculate</b> the distance between transmitter and receiver using MATLAB tools | <b>Week-2</b><br><b>L-4,5,6</b> | CLO2           | PLO2 | Students will discuss with a group.<br>To apply the knowledge to solve the different problems.         | #Interactive discussion                                                                                 |

|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                       |                                               |              |              |                                                                                                |                                                                                     |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|--------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Study of wireless Communications using Communication Trainer Kits</b>                       | <b>To design</b> Adaptive Linear Equalize<br>Code Division Multiple Access (CDMA) – Multipath<br>Code Division Multiple Access (CDMA) – Multiuser<br>Global System for Mobile Communication (GSM) (Using WiCOMM-T - Wireless Digital Communication Training system – SDR Platform )<br>Spread Spectrum – DSSS Modulation & Demodulation (Using Emona 101 Trainer Kit) | <b>Week-3</b><br><b>L-7,8,9</b>               | CLO3         | PLO3         | Students will discuss with a group.<br>To apply the knowledge to solve the different problems. | #Interactive discussion                                                             |
| <b>Exam Review</b>                                                                             | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                            | <b>Week-4</b><br><b>L-10,11,12</b>            |              |              | Demonstrate, Questions and Answers                                                             |                                                                                     |
| <b>Examination</b>                                                                             |                                                                                                                                                                                                                                                                                                                                                                       | <b>Week-5</b><br><b>L-13,14,15</b>            |              |              | Quiz and Tutorial Exam. Will be taken                                                          |                                                                                     |
| <b>Study of Propagation Path loss Models : Indoor &amp; Outdoor( Using Matlab Programming)</b> | <b>To calculate</b> Free Space Propagation – Path Loss Model<br>Link Budget Equation for Satellite Communication<br>Carrier to Noise Ratio in Satellite Communication<br>Outdoor Propagation – Okumura Model.                                                                                                                                                         | <b>Week-6,7</b><br><b>L-16,17,18,19,20,21</b> | CLO3<br>CLO4 | PLO3<br>PLO4 | Students will demonstrate /present and solve the problems.                                     | #White board discussion<br>#Lecture discussion with multimedia                      |
| <b>Exam Review</b>                                                                             | <b>To make an overview</b>                                                                                                                                                                                                                                                                                                                                            | <b>Week-8</b><br><b>L-22,23</b>               |              |              | Lectures, Questions and Answers                                                                |                                                                                     |
| <b>Exam.</b>                                                                                   |                                                                                                                                                                                                                                                                                                                                                                       | <b>Week-8,9</b><br><b>L-24,25</b>             |              |              | Presentation and Tutorial Exam. Will be taken                                                  |                                                                                     |
| <b>Antenna Design Concept (using 4NEC2)</b>                                                    | <b>To design and construct</b> Dipole Antennas<br>Yagi – Uda Antenna – 3 element<br>Yagi – Uda Antenna – 5 element<br>Yagi – Uda Antenna – 7 elements.                                                                                                                                                                                                                | <b>W-9,10</b><br><b>L-26,27,28,29,30</b>      | CLO4         | PLO5         | Students will Design and construct different antennas                                          |                                                                                     |
| <b>Switching and Traffic</b>                                                                   | <b>To configure</b> the Cellular Switching System                                                                                                                                                                                                                                                                                                                     | <b>Week-11</b><br><b>L-31,32,33</b>           | CLO5         | PLO7         | Students will apply contextual knowledge to configure the cellular switching system.           | #Lecture discussion with multimedia<br>#Show video about different Cellular Switch. |

|                                    |                            |                               |  |  |                                                        |  |
|------------------------------------|----------------------------|-------------------------------|--|--|--------------------------------------------------------|--|
| <b>Exam. Review</b>                | <b>To make an overview</b> | <b>Wee-12<br/>L-34,35,36</b>  |  |  | Lectures,<br>Questions and<br>Answers                  |  |
| <b>Exam.</b>                       |                            | <b>Week-13<br/>L-37,38,39</b> |  |  | Presentation<br>and Tutorial<br>Exam. will be<br>taken |  |
| <b>Recess<br/>before<br/>Exam.</b> |                            | <b>Week-<br/>14,15</b>        |  |  |                                                        |  |
| <b>Sem. End<br/>Exam.</b>          | <b>Start</b>               | <b>Week-16</b>                |  |  |                                                        |  |

### **Assessment Pattern (According FET Ordinance)**

**CIE:** (Continuous Internal Evaluation-20Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Lab. Performance<br/>(05)</b> | <b>Lab. Note Book<br/>(05)</b> | <b>Continuous Viva-<br/>Voce<br/>(10)</b> | <b>Lab. Attendance<br/>(10)</b> |
|--------------------------------------|----------------------------------|--------------------------------|-------------------------------------------|---------------------------------|
| <b>Remember</b>                      |                                  |                                | (10)                                      |                                 |
| <b>Understand</b>                    |                                  |                                |                                           |                                 |
| <b>Apply</b>                         |                                  | (05)                           |                                           | (10)                            |
| <b>Analyze</b>                       |                                  |                                |                                           |                                 |
| <b>Evaluate</b>                      | (05)                             |                                |                                           |                                 |
| <b>Create</b>                        |                                  |                                |                                           |                                 |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 10          |
| <b>Understand</b>           | 00          |
| <b>Apply</b>                | 15          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 00          |

### **Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)
- Semester Final Exam (70%)

### **Learning Resources**

#### **Text Books**

1. *Mobile Cellular Telecommunications, Analog and Digital Systems*, William C. Y. Lee, McGraw Hill.
2. *Mobile Radio Communications*, Raymond Steele, IICE Press, New York.

#### **Reference Books and Other Materials**

3. *Telecommunication System Engineering*, Roger L Freeman, John Wiley & Sons.
4. *Lab. Guide and Lab. Manuals*



Contact Hours: 39  
Exam. Duration: 6 Hours

**Prerequisite:****Course Type:** Theory

**Course Summary:** This course gives students deep knowledge in Telecommunication Engineering. This course focuses on the basic telephony, local and long distance networks, signaling techniques, transmission systems, and digital and analog switching.

**Course Objectives:** provide knowledge on telephone networks, tele-trafficking, local and long distance telephone networks, analog and digital signaling, analog and digital switching, Digital Networks.

**Course Learning Outcomes (CLO):**

After completing this course a student would be able to

|             |                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | Define and explain the meaning/function of telecommunication & telephone network elements and the fundamental principles of tele-traffic engineering. |
| <b>CLO2</b> | Understand the fundamental principles of subscriber loop & long distance network design and locating local & toll exchange.                           |
| <b>CLO3</b> | Distinguish different types of analog & digital signaling system and understand the effects of numbering on signaling.                                |
| <b>CLO4</b> | Analyze the basic impairments and understand different modes of analog & digital transmission.                                                        |
| <b>CLO5</b> | Understand the functions of switching system and distinguish different switching systems.                                                             |
| <b>CLO6</b> | Analyze the working principle of different types of digital switching.                                                                                |
| <b>CLO7</b> | Understand the behavior Protocol Architecture and Transmission Structure of digital Networks.                                                         |

| Course Content                 |                                                                                                                                                                                         | Time Frame | Alignment with |       | Suggested Activities                                                     | Teaching Strategy(s)                |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|-------|--------------------------------------------------------------------------|-------------------------------------|
| Outlines                       | Specific Topics                                                                                                                                                                         |            | CLO            | PLO   |                                                                          |                                     |
| Basics of Telephony            | Elements of Telecommunication System, Simple Telephone Network and Terminologies, Essentials of Tele-traffic: Unit of Traffic, Grade of Service, Availability & Handling of Lost calls. | Week 1 & 2 | CLO-1          | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>- Oral Presentation |
| Local & Long Distance Networks | Introduction, Subscriber Loop Design Constraints, Exchange Location, National Networks, Toll Exchange Location, Network Design Procedure, Traffic Routing in National Network.          | Week 3 & 4 | CLO-2          | PLO-3 | -Lecture Presentation<br>-Interactive Discussion<br>-Feedback            | - Assignment<br>-Class test         |
| Signaling Techniques           | Introduction, Signaling in the Subscriber Loop, Different Types of Signaling, Effects of Numbering on Signaling, Signaling System No7, and Network architecture of SS7.                 | Week 5 & 6 | CLO-3          | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |

|                        |                                                                                                                                                                             |              |       |       |                                                                          |                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------|--------------------------------------------------------------------------|-------------------------------------|
| Transmission System    | Basic Impairments to Voice Channel Transmission, Two-Wire/Four-Wire Transmission, Multiplexing, Digital Transmission, Modes of Digital Transmission, Practical Application. | Week 7 & 8   | CLO-4 | PLO-2 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| Evolution of Switching | Basic Switching Function, Electro-mechanical Switching, Pulse Dialing, Step-by-Step Switching, Principle of Crossbar Switching, Stored Program Control Switching.           | Week 9 & 10  | CLO-5 | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| Digital Switching      | Introduction, Advantages & Issues of PCM Switching, Space Switch, Time Switch, Time Division Switching, Two-dimensional Digital Switching.                                  | Week 11 & 12 | CLO-6 | PLO-1 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>- Oral Presentation |
| Digital Networks       | Integrated Services Digital Network: Introduction, ISDN Network, Protocol Architecture, Transmission Structure, User-Network Interfaces, CATV, IDLC, xDSL, SONET/SDH.       | Week 13 & 14 | CLO-7 |       | -Lecture Presentation<br>-Interactive Discussion                         | - Assignment<br>-Class test         |

**Assessment Pattern (According FET Ordinance)****CIE:** (Continuous Internal Evaluation-20Marks)

| Bloom's Category<br>Marks:30 | Lab. Performance<br>(05) | Lab. Note Book<br>(05) | Continuous Viva-<br>Voce<br>(10) | Lab. Attendance<br>(10) |
|------------------------------|--------------------------|------------------------|----------------------------------|-------------------------|
| Remember                     |                          |                        | (10)                             |                         |
| Understand                   |                          |                        |                                  |                         |
| Apply                        |                          | (05)                   |                                  | (10)                    |
| Analyze                      |                          |                        |                                  |                         |
| Evaluate                     | (05)                     |                        |                                  |                         |
| Create                       |                          |                        |                                  |                         |

**SMEE:** (Semester Mid and End Examination-30 Marks)

| Bloom's Category | Test |
|------------------|------|
| Remember         | 10   |
| Understand       | 00   |
| Apply            | 15   |
| Analyze          | 00   |
| Evaluate         | 05   |
| Create           | 00   |

**Assessment and Evaluation**

- Lab. performance during Lab. hours (05%)
- Lab. attendance (10%)
- Lab. Note book on experiment (05%)
- Viva-Voce on experiment (10%)

– Semester Final Exam (70%)

### Learning Resources:

#### Recommended Books:

*Telecommunication Switching Systems and Networks - Viswanathan.*

**Telecommunication System Engineering - R.L. Freeman.**

*Telecommunication Switching and Networks – P. Ganasivam.*

### HUM-4207: Software Project Management

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

**Prerequisite:** System Analysis and Software Engineering, Computer Fundamentals, Programming concepts

**Course Type:** Theory

**Course Summary:** An introduction to the concept and techniques of project management for a broad range of systems, including Web-based application development. Topics include resource management, organizational factors, project manager responsibilities, team building, and risk management. Tools and techniques for project estimating and scheduling will be presented. Case study and group projects.

**Course Objectives:** This course will introduce the students with software project management, motivation in project management, client management, risk management, software quality assurance, software testing, software upgradation and maintenance, international project management.

#### Course Learning Outcomes (CLO): Upon successful completion of this course, students will be able to -

|             |                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | <b>Apply</b> project management concepts and techniques to an IT project through working in a group as team leader or active team member. |
| <b>CLO2</b> | <b>Identify</b> issues that could lead to IT project success or failure and the responsibilities of IT project managers.                  |
| <b>CLO3</b> | <b>Explain</b> project management in terms of the software development process.                                                           |

| Course Content                |                                                                      | Time Frame                                               | Alignment With CLO | Alignment with PLO | Teaching-Learning Strategy                                               | Assessment techniques                              |
|-------------------------------|----------------------------------------------------------------------|----------------------------------------------------------|--------------------|--------------------|--------------------------------------------------------------------------|----------------------------------------------------|
| Outlines                      | Specific Topics                                                      |                                                          |                    |                    |                                                                          |                                                    |
| Course Overview<br>Chapter 1: | Introduction to Project Management                                   | Week - 1, 2<br>Lecture – L-1, L-2, L-3, L-4, L-5, L-6    | CLO-1              | PLO-1              | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br><br>- Oral Presentation            |
| Chapter 2, 3:                 | Project Management and IT Context, Project Management Process Groups | Week – 3, 4<br>Lecture - L-7, L-8, L-9, L-10, L-11, L-12 | CLO-1              | PLO-2              | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test<br>- Oral Presentation |

|                |                                                                        |                                                              |       |               |                                                                          |                                     |
|----------------|------------------------------------------------------------------------|--------------------------------------------------------------|-------|---------------|--------------------------------------------------------------------------|-------------------------------------|
| Chapter 4,5:   | Project Integration Management, Project Scope Management               | Week 5, 6<br>Lecture - L-13, L-14, L-15, L-16, L-17, L-18    | CLO-2 | PLO-2 & PLO-3 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| Chapter 6,7:   | Project Time Management, Project Cost Management                       | Week 7, 8<br>Lecture - L-19, L-20, L-21, L-22, L-23, L-24    | CLO-3 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| Chapter 8,9:   | Project Quality Management, Project Human Resource Management          | Week 9, 10<br>Lecture - L-25, L-26, L-27, L-28, L-29, L-30   | CLO-4 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>-Class test         |
| Chapter 10, 11 | Project Communications Management, Project Risk Management             | Week 11 & 12<br>Lecture - L-31, L-32, L-33, L-34, L-35, L-36 | CLO-5 | PLO-3 & PLO-4 | -Lecture Presentation with video<br>-Interactive Discussion<br>-Feedback | - Assignment<br>- Oral Presentation |
| Chapter 12, 13 | Project Procurement Management Chapter, Project Stakeholder Management | Week 13<br>Lecture - L-37, L-38, L-39                        | CLO-5 | PLO-4 & PLO-5 | -Lecture Presentation with video<br>-Interactive Discussion              | - Assignment<br>-Class test         |
|                | Recess before Exam.                                                    | Week 14 & 15                                                 |       |               |                                                                          |                                     |
|                | Semester End Exam. Start                                               | Week 16                                                      |       |               |                                                                          |                                     |

**Assessment Pattern (According FET Ordinance)**

– Class Attendance &amp; Participation 10 marks

**CIE:** (Continuous Internal Evaluation-30Marks)

| Bloom's Category<br>Marks:30 | Tutorial/In-<br>course<br>(10) | Assignment<br>(05) | Quizz/Presentation<br>(05) | Class Attendance<br>10 |
|------------------------------|--------------------------------|--------------------|----------------------------|------------------------|
| Remember                     |                                |                    | (05)                       |                        |
| Understand                   | (05)                           |                    |                            |                        |
| Apply                        |                                |                    |                            | (10)                   |
| Analyze                      |                                |                    |                            |                        |
| Evaluate                     | (05)                           |                    |                            |                        |
| Create                       |                                | (05)               |                            |                        |

**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| Bloom's<br>Category | Test |
|---------------------|------|
| Remember            | 05   |
| Understand          | 05   |
| Apply               | 10   |
| Analyze             | 00   |
| Evaluate            | 05   |
| Create              | 05   |

### Learning Resources:

#### Books Recommended

1. Schwalbe, Kathy (2016) *Information Technology Project Management* Edition: 8<sup>th</sup>  
ISBN-13:978-1285452340, ISBN-10: 1285452348
2. *System Analysis and Development*, P J Layzell and P Loucopolus, Chartwell-Bratt.

#### References and other related books

3. *System Analysis and Design*, S Y Wu and M S Wu, West.
4. *System Analysis and Design*, Elias M Awad, Galgotia Publications.
5. *Software Engineering, A Practitioner's Approach*, R S Pressman, McGraw-Hill.
6. *Software Engineering* ISommerville, 4th Edition, Addisi-Wesley.

**HUM-4201: Industrial Management and Accountancy**

Full Marks:100

Credit: 2.0

Contact Hours: 26

Exam. Duration: 4 Hours

#### Prerequisite:

**Course Type:** Theory

**Course Summary:** This course to develop an understanding on basic concepts of management, accounting, business planning and integration with Information systems

**Course Objectives:** To provide knowledge on the role of Information system in Business, MIS for organization, business strategies, Information system planning, decision making concepts, decision support systems, enterprise resource planning systems

**Course Learning Outcomes (CLO):**After completing this course a student would be able to

|             |                                                           |
|-------------|-----------------------------------------------------------|
| <b>CLO1</b> | <b>Acquire</b> general idea about the course              |
| <b>CLO2</b> | Understand the concepts of information system in business |

|             |                                                  |
|-------------|--------------------------------------------------|
| <b>CLO3</b> | Understand the concepts of MIS for organizations |
| <b>CLO4</b> | Understand business strategies                   |
| <b>CLO5</b> | Understand the concepts of IS planning           |
| <b>CLO6</b> | Evaluate decision support systems                |
| <b>CLO7</b> | Understand enterprise resource planning          |

### Content of the Course

| Course Content                                |                                                                                                                                                                                                                                  | Time Frame                               | Alignment with |              | Suggested Activities                                                                                   | Teaching Strategy(s)                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------|--------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Outline                                       | Specific Topics                                                                                                                                                                                                                  |                                          | CLO            | PLO          |                                                                                                        |                                                                 |
| <b>Course Overview</b>                        | <b>Comprehend</b> the mission and vision of the Department<br><b>To acquire</b> general idea about the course Statistics for Communication Engineering<br><b>To grasp</b> the content and policies for the class                 | <b>Week-1</b><br><b>L-1</b>              | CLO1           | PLO1         | Students will give brief introduction about themselves<br>Course outline will be discussing in details | #Socialized recitation<br>#Show presentation to the students    |
| <b>Basic Management</b>                       | Basic Industrial Management: Industry, Type of industry, Types of business, Management, Principles of management, Functions of management, Levels of management, Roles of Management, Scientific management and Core management. | <b>Week-1, 2</b><br><b>L-2,3,4</b>       | CLO2<br>CLO3   | PLO1<br>PLO2 | Students will discuss with a group and understand basic management.                                    | #White board discussion<br>#Lecture discussion with multimedia. |
| <b>Exam.</b>                                  |                                                                                                                                                                                                                                  | <b>Week-3</b><br><b>L-5</b>              |                |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                 |
| <b>Accounting Basic</b>                       | Scope, Nature and purpose of accounting, Basic accounting principles, Transactions, Journal, Ledgers and Trial Balance, Cash book, Preparation of financial statement. Cost accounts.                                            | <b>Week-3, 4</b><br><b>L-6,7</b>         | CLO2<br>CLO3   | PLO2<br>PLO3 | Student will understand the basic concepts of accounting                                               | #White board discussion<br>#Lecture discussion with multimedia  |
| <b>Exam.</b>                                  |                                                                                                                                                                                                                                  | <b>Week-4</b><br><b>L-8</b>              |                |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                 |
| <b>IS in business Org.</b>                    | Role of IS in Business, Early views & Models of IS, Impact of MIS, The three Era Model, Strategic Uses of Information System, Strategic Information System.                                                                      | <b>Week-5</b><br><b>L-9, 10</b>          | CLO2<br>CLO3   | PLO3         | Student will apply the knowledge of Business organization                                              | #White board discussion<br>#Lecture discussion with multimedia. |
| <b>Exam.</b>                                  |                                                                                                                                                                                                                                  | <b>Week-6</b><br><b>L-11</b>             |                |              | Quiz and Tutorial Exam. Will be taken                                                                  |                                                                 |
| <b>Organizations &amp; Strategic Planning</b> | Organization Structure, MIS for Organization Planning, Strategic Framework, Generic Business Strategies, Planning Tools & Techniques, Supply Chain & Value Chain Analysis.                                                       | <b>Week – 6,7</b><br><b>L-12, 13, 14</b> | CLO4<br>CLO5   | PLO4         | Student will apply the knowledge of MIS systems                                                        | #White board discussion<br>#Lecture discussion with multimedia. |

|                                        |                                                                                                                                                                                                                                                                                            |                                    |              |              |                                                                  |                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------|--------------|------------------------------------------------------------------|----------------------------------------------------------------|
|                                        |                                                                                                                                                                                                                                                                                            |                                    |              |              |                                                                  |                                                                |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                            | <b>Week-8<br/>L-15</b>             |              |              | Quiz and Tutorial Exam. Will be taken                            |                                                                |
| <b>Development of MIS</b>              | Development of Long Range Plans for MIS, Ascertaining the Class of Information, Determining the Information Requirement, Development and Implementation of MIS, Management of Quality in MIS, Organization for Development of the MIS, Factors of Success and Failures.                    | <b>Week-8, 9<br/>L-16, 17</b>      | CLO4<br>CLO5 | PLO3         | Students will identify, analyze and formulate MIS system.        | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Strategic IS Planning</b>           | Information system Planning, Purpose & Stimuli Driving IS Planning, Assessment of Current Environment, Scope, Expectations & Objectives, Planning Framework & Approaches, Understanding Current Situation, Analyzing Techniques to Determine IS Strategy, Business Process Re-Engineering. | <b>Week-9, 10<br/>L-18, 19, 20</b> | CLO7         | PLO3<br>PLO4 | Students will apply reasoning of IS planning                     | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Exam.</b>                           |                                                                                                                                                                                                                                                                                            | <b>Week-11<br/>L-21</b>            |              |              | Quiz and Tutorial Exam. Will be taken                            |                                                                |
| <b>Strategic Management of IS</b>      | Requirements and Sub-functions, Application Portfolio Management, Information Management, Technology Infrastructure Management, Determining IS Strategy.                                                                                                                                   | <b>Week-12<br/>L-22, 23</b>        | CLO6         | PLO4         | Students understand the strategies to manage information systems | #White board discussion<br>#Lecture discussion with multimedia |
| <b>Decision Making and Information</b> | Decision Making Concepts, Decision Methods, Tools and Procedures, Organizational Decision Makings, MIS and Decision Making Concepts, Information Concepts, Quality of Information, Classification of Information, Methods of Data and Information Collection, Value of Information.        | <b>Week-13<br/>14<br/>L-24, 25</b> | CLO4<br>CLO7 |              | Students will understand the policies of decision making         | #White board discussion                                        |
| <b>Exam. Review</b>                    | <b>To make an overview</b>                                                                                                                                                                                                                                                                 | <b>Wee-15<br/>L-,26</b>            |              |              | Lectures, Questions and Answers                                  |                                                                |

|                       |              |                |  |  |  |  |
|-----------------------|--------------|----------------|--|--|--|--|
| <b>Sem. End Exam.</b> | <b>Start</b> | <b>Week-16</b> |  |  |  |  |
|-----------------------|--------------|----------------|--|--|--|--|

**Assessment Pattern (According FET Ordinance)**

– Class Attendance &amp; Participation 10 marks

**CIE:** (Continuous Internal Evaluation-30Marks)

| <b>Bloom's Category<br/>Marks:30</b> | <b>Tutorial/In-<br/>course<br/>(10)</b> | <b>Assignment<br/>(05)</b> | <b>Quizz/Presentation<br/>(05)</b> | <b>Class Attendance<br/>10</b> |
|--------------------------------------|-----------------------------------------|----------------------------|------------------------------------|--------------------------------|
| <b>Remember</b>                      |                                         |                            | (05)                               |                                |
| <b>Understand</b>                    | (05)                                    |                            |                                    |                                |
| <b>Apply</b>                         |                                         |                            |                                    | (10)                           |
| <b>Analyze</b>                       |                                         |                            |                                    |                                |
| <b>Evaluate</b>                      | (05)                                    |                            |                                    |                                |
| <b>Create</b>                        |                                         | (05)                       |                                    |                                |

**SMEE:** (Semester Mid and End Examination-30+30=60 Marks)

| <b>Bloom's<br/>Category</b> | <b>Test</b> |
|-----------------------------|-------------|
| <b>Remember</b>             | 05          |
| <b>Understand</b>           | 05          |
| <b>Apply</b>                | 10          |
| <b>Analyze</b>              | 00          |
| <b>Evaluate</b>             | 05          |
| <b>Create</b>               | 05          |

**Text Book***Strategic Planning for Information Systems, John Ward & Pat Griffith***Reference Books and Other Materials***Schaums Easy Outline of Accounting, Joel Lerner**Management Information System, W. S. Jawdekar**Management Information System, D. P. Goyal**Management Information System, Laudon&Laudon**Management Information System, James A. O'Brien***ICT-4208: Field Work/Industrial Attachment**

Full Marks:100

Credit: 2.0

**ICT-4002: Project**

Full Marks:100

Credit: 3.0

**ICT-4000: Viva-Voce  
Based on 4<sup>th</sup> Year 1<sup>st</sup> and 2<sup>nd</sup> Semester**

Full Marks:100

Credit: 1.0



| <b>List of Faculty Members</b><br><b>Department of Information and Communication Technology</b><br><b>Islamic University, Kushtia-7003, Bangladesh</b> |                                                                                                                                                                                                                                                                                                                                               |
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| Name:<br>Designation:<br>Academic Qualifications:<br>Research Interest:<br>Contact Detail:                                                             | <b>Md. Ashek Raihan Mahmud</b><br>Associate Professor<br>M.Sc. (RMIT, Australia), M.Phil. (IU)<br>Mobile Ad-hoc Network, Telecommunication<br>Cell: 01818-464047<br>e-mail:                                                                                                                                                                   |
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| Name:<br>Designation:<br>Academic Qualifications:<br>Research Interest:<br>Contact Detail: | <b>Dr. Md. Sadek Ali</b> (Study Leave)<br>Associate Professor<br>M.Sc. (IU), PhD (On going, Chaina)<br>Mobile Communication, Optical Communication, Digital Signal Processing<br>Cell: 01923-579420<br>e-mail: <a href="mailto:sadek_ice@yahoo.com">sadek_ice@yahoo.com</a>                                                                        |
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